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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

DEVELOPMENT OF PERIODIC TABLE

1. Mendeleev periodic table is based on :-
(1) Atomic number
(2) Increasing order of number of protons
(3) Electronic configuration
(4) None of the above
CPT001

2. Which of the following sets of elements follows Newland's octave rule :-
(1) Be, Mg, Ca (2) Na, K, Rb
(3) F, Cl, Br (4) B, Al, Ga
CPT003

3. The places that were left empty by Mendeleev were, for:-
(1) Aluminium and silicon
(2) Gallium and germanium
(3) Arsenic and antimony
(4) Molybdenum and tungsten
CPT006

4. Which is not anomalous pair of elements in the Medeleev's periodic table:-
(1) Ar and K (2) Co and Ni
(3) Te and I (4) Al and Si
CPT007

5. The law of triads is applicable to :-
(1) Os, Ir, Pt (2) Ca, Sr, Ba
(3) Fe, Co, Ni (4) Ru, Rh, Pt
CPT008

6. Elements which occupied position in the lother meyer curve, on the peaks, were :–
(1) Alkali metals
(2) Highly electro positive elements
(3) Elements having large atomic volume
(4) All
CPT009

7. In a period, the elements are arranged in :–
(1) Decreasing order of nuclear charge
(2) Decreasing order of No. of electrons
(3) Increasing order of nuclear charge
(4) In order of same nuclear charge
CPT010

8. Which of the following statement is wrong :-
(1) 4th period contains 18 elements
(2) 3rd period contains 18 elements
(3) 1st period contains two gaseous elements
(4) In p–block, metal, non-metal and metalloids are present
CPT012

9. The discovery of which of the following group of elements gave a death blow to the Newlands Law:-
(1) Inert gases
(2) Alkali metals
(3) Transuranic element
(4) Halogens
CPT013

10. In a group, reason of periodicity is :-
(1) Same number of electrons in their atom
(2) Same ionisation potential
(3) Same valence shell electronic configuration
(4) Electrons present in the same subshells
CPT016

11. Which of the following element was absent in the Mendeleev's periodic table :-
(1) Tc (2) Si (3) B (4) F
CPT017

12. Which statement is wrong for the long form of periodic table :-
(1) Number of periods are 7 and groups 18
(2) No. of valence shell electrons in a period are same
(3) 3rd group contains 32 elements
(4) Lanthanides and actinides are placed in same group
CPT019

13. The elements which are cited as an example to prove the validity of mendeleev's periodic law are
(1) H, He (2) Ga, Sc (3) Co, Ni (4) Zr, Hf
CPT020

14. According to the latest IUPAC recommendation the inert gases are in :–
(1) Zero group (2) Eighth group
(3) Eighteenth group (4) All are correct
CPT023

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15. Elements having atomic number 112 is named as:-
(1) Roentgenium (2) Copernicium
(3) Flerovium (4) Livermorium

CPT024

PERIOD, GROUP AND BLOCK

16. Which of the following set of elements belongs to same period :-
(1) Zn, Cd, Hg (2) Fr, Ra, U
(3) K, Ca, Ag (4) None

CPT026

17. The element with atomic number Z = 115 will be placed in :-
(1) 7th period, IA group
(2) 8th period, IVA group
(3) 7th period, VA group
(4) 6th period, VB group

CPT027

18. The number of elements in 5th and 6th period of periodic table are respectively :-
(1) 8, 18 (2) 18, 18
(3) 18, 32 (4) 18, 28

CPT029

19. The eighteenth element of periodic table resembles with :-
(1) Eleventh element
(2) Second element
(3) Tenth element
(4) Second and tenth both element

CPT030

20. The IUPAC name of the element which is placed after Db_{105} is the periodic table, will be:-
(1) Un nil pentium
(2) Un un nilium
(3) Un nil hexium
(4) Un nil quadium

CPT032

21. The element with atomic number Z=118 will be :–
(1) Noble gas
(2) Transition metal
(3) Alkali metal
(4) Alkaline earth metal

CPT033

22. The atom having the valence shell electronic configuration $4s^2\ 4p^2$ would be in:-
(1) Group II A and period 3
(2) Group II B and period 4
(3) Group IV A and period 4
(4) Group IV A and period 3

CPT034

23. In the periodic table, where are non–metals located
(1) Between groups II A and III A
(2) On the lower left hand side
(3) On the upper left hand side
(4) On the upper right hand side

CPT035

24. Elements with similar chemical properties :-
(1) Occur only within the same period
(2) Have identical atomic mass (weight)
(3) Have identical number of neutrons
(4) Have the same number of electrons in the outer most shell.

CPT036

25. The chemistry of lithium is very similar to that of magnesium even though they are placed in different groups. Its reason is :-
(1) Both are found together in nature
(2) Both have nearly the same size
(3) Both have similar electronic configurations
(4) The ratio of their charge to size is nearly the same

CPT037

26. The element having electronic configuration $4f^{14}\ 5d^0\ 6s^2$ belongs to :-
(1) d–block, 12th group
(2) f–block, III B group
(3) f–block, 14th group
(4) s–block, 2nd group

CPT038

27. The fourteen elements collectively placed in 3rd group and 7th period are called :-
(1) Typical elements
(2) Representative element
(3) Actinides
(4) Lanthanones

CPT039

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28. Which of the following statement is wrong for the transition elements :-
(1) Transition elements are placed from 3rd to 6th period.
(2) Last electron enters in (n –1)d orbital
(3) All are solids
(4) General electronic configuration is $(n-1)d^{1-10}\ ns^{1-2}$

CPT040

29. Element with the electronic configuration given below, belong to which group in the periodic table $1s^2$, $2s^22p^6$, $3s^23p^63d^{10}$, $4s^24p^64d^{10}$, $5s^25p^3$
(1) 3rd (2) 5th (3) 15th (4) 17th

CPT041

30. Which of the following electronic configuration belongs to inert gas elements :-
(1) $ns^2\ (n-1)d^{10}$ (2) $ns^2\ (n-1)s^2p^6$
(3) $ns^2\ np^6$ (4) None

CPT043

31. From atomic number 58 to 71, elements are placed in ;-
(1) 5th period and III A group
(2) 6th period and III B group
(3) separate period and group
(4) 7th period and IV B group

CPT044

32. True statement is :-
(1) All the transuranic elements are synthetic elements
(2) Elements of third group are called bridge elements
(3) Element of $1s^2$ configuration is placed in IIA group
(4) Electronic configuration of elements of a period is same

CPT045

33. An element recently discovered is Uub. Expected electronic configuration of the element is :-
(1) [Kr] $5s^24d^{10}$ (2) [Xe] $6s^24f^{14}5d^8$
(3) [Rn] $7s^25f^{14}6d^{10}$ (4) [Rn] $7s^25f^{10}d^0$

CPT046

34. Elements having $ns^2\ np^6$ valence shell electronic configuration lies in :-
(1) Zero group and 1st–7th period
(2) 18th group and 2nd–7th period
(3) 18th group and 1st–6th period
(4) All are correct

CPT047

35. Certain regular intervals after which the outer electronic configuration of the elements occur are called :–
(1) Mass number (2) Atomic numbers
(3) Magic numbers (4) Avogadro number

CPT048

36. The electronic configuration of elements X and Z are $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ and $1s^2\ 2s^2\ 2p^5$ respectively. What is the position of element X with respect to position of Z in the periodic table -
(1) Just below Z (2) Just above Z
(3) Left to the Z (4) Right to the Z

CPT050

37. Which of the following sequence contains atomic number of only representative elements
(1) 55, 12, 24, 53 (2) 13, 33, 59, 83
(3) 3, 33, 53, 87 (4) 22, 33, 55, 66

CPT052

38. Uranium (At No. - 92) is the last natural element in the periodic table. If the last element of the periodic table which is recently discovered is Uub. What will be the total number of transuranic elements in the periodic table :-
(1) 21 (2) 20 (3) 11 (4) 12

CPT054

39. Which of the following statement is not correct for given electronic configuration
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 6s^2$
(1) It belongs to IIB group and 6th period
(2) It is liquid at room temperature
(3) It is a transition element
(4) It is used in thermometres

CPT055

Zeff, SCREENING CONSTANT & ATOMIC RADIUS

40. Which statement is false:-
(1) Screening effect increases down the group
(2) Z_{eff} decreases down the group
(3) Z_{eff} increases in a period
(4) All

CPT056

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41. Effective nuclear charge in hydrogen is :-
(1) more than nuclear charge
(2) less than nuclear charge
(3) equal to nuclear charge
(4) can't predict
CPT058

42. In sodium atom, the screening is due to :-
(1) $3s^2$, $3p^6$ (2) $2s^1$
(3) $1s^2$, $2s^2$ $2p^6$ (4) $1s^2$, $2s^2$
CPT059

43. The screening effect of d- electrons is :-
(1) Equal to the p - electrons
(2) Much more than p - electrons
(3) Same as f - electrons
(4) Less than p - electrons
CPT060

44. The correct order of size would be:-
(1) Ni < Pd $\simeq$ Pt (2) Pd < Pt < Ni
(3) Pt > Ni > Pd (4) Pd > Pt > Ni
CPT061

45. Which of the following order of radii is correct
(1) Li < Be < Mg (2) $H^+ < Li^+ < H^-$
(3) O < F < Ne (4) $Na^+ > F^- > O^{-2}$
CPT062

46. K^+, Ar, Ca^{2+} and S^{2-} contains-
(1) Same electronic configuration and atomic volume
(2) Different electronic configuration but same IP.
(3) Same electronic configuration but different atomic volume
(4) None
CPT063

47. Which of the following is not different for an atom and its corresponding ion :-
(1) Number of electrons
(2) Nuclear charge
(3) Size
(4) None of these
CPT064

48. Which of the following is not isoelectronic series :-
(1) Cl^-, P^{3-}, Ar (2) N^{3-}, Ne, Mg^{+2}
(3) B^{+3}, He, Li^+ (4) N^{3-}, S^{2-}, Cl^-
CPT065

49. Which group of atoms have nearly same atomic radius:-
(1) Na, K, Rb, Cs (2) Li, Be, B, C
(3) Fe, Co, Ni (4) F, Cl, Br, I
CPT066

50. Which of the following has largest radius :-
(1) $1s^2$, $2s^2$ $2p^6$, $3s^2$
(2) $1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^1$
(3) $1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^3$
(4) $1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^5$
CPT067

51. Which of the following order of atomic/ionic radius is not correct :-
(1) $I^- > I > I^+$ (2) $Mg^{+2} > Na^+ > F^-$
(3) $P^{+5} < P^{+3}$ (4) Li > Be > B
CPT068

52. In the lithium atom, screening effect of valence shell electron is caused by-
(1) Electrons of K and L shell
(2) Electrons of K shell
(3) Two electrons of 1st and one of 2nd shell
(4) None
CPT069

53. Correct order of ionic radii is
(1) $Ti^{4+} < Mn^{7+}$ (2) $^{37}Cl^- < ^{35}Cl^-$
(3) $K^+ > Cl^-$ (4) $P^{3+} > P^{5+}$
CPT070

54. S^{-2} is not isoelectronic with :-
(1) Ar (2) Cl^- (3) HS^- (4) Ti^{+3}
CPT071

55. The best reason to account for the general tendency of atomic diameters to decrease as the atomic number increases within a period of the periodic table is the fact that
(1) Outer electrons repel inner electrons
(2) Closer packing among the nuclear particles is achieved
(3) The number of neutrons increases
(4) The increasing nuclear charge exerts a greater attractive force on the electrons
CPT072

56. In an anion :-
(1) Number of proton decreases
(2) Protons are more than electrons
(3) Effective nuclear charge is more
(4) Radius is larger than neutral atom
CPT073

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57. Maximum size of first member of a period is due to
(1) Maximum number of shells
(2) Maximum screening effect
(3) Minimum Z_{eff}
(4) All
CPT074

58. Which of the following ion has largest size :-
(1) F^- (2) Al^{+3} (3) Cs^+ (4) O^{-2}
CPT075

59. Spot the incorrect order of atomic radii :-
(1) $r_{Cu > Zn}$ (2) $r_{Cl > F}$ (3) $r_{P > S}$ (4) $r_{Sc > Ti}$
CPT076

60. Incorrect statement is :-
(1) Atomic size $\propto Z_{eff}$
(2) Atomic size $\propto$ Negative charge
(3) Bond length $\propto$ Size of atoms
(4) Atomic size $\propto$ Screening effect
CPT077

61. Which of the following orders of atomic radii are correct :-
(a) Li < Be < Na (b) Ni < Cu < Zn
(c) Ti > V > Cr (d) Ti > Zr $\simeq$ Hf
Correct answer is :-
(1) All (2) a, b (3) b, c (4) b, d
CPT078

62. Decreasing order of size of ions is :-
(1) $Br^- > S^{-2} > Cl^- > N^{-3}$
(2) $N^{3-} > S^{-2} > Cl^- > Br^-$
(3) $Br^- > Cl^- > S^{-2} > N^{-3}$
(4) $N^{-3} > Cl^- > S^{-2} > Br^-$
CPT079

63. Which of the following statement is wrong :-
(1) Z_{eff} in group remains constant
(2) In a period atomic size decreases
(3) Screening effect in a period remains constant
(4) In a period atomic radius of inert gas element is maximum
CPT080

64. Which of the following sequence is correct for decreasing order of ionic radius :–
(1) Se^{-2}, I^-, Br^-, O^{-2}, F^- (2) I^-, Se^{-2}, O^{-2}, Br^-, F^-
(3) Se^{-2}, I^-, Br^-, F^-, O^{-2} (4) I^-, Se^{-2}, Br^-, O^{-2}, F^-
CPT081

65. According to Slater's rule, order of effective nuclear charge for last electron in case of Li, Na and K :-
(1) Li > Na > K (2) K > Na > Li
(3) Na > Li > K (4) Li < Na = K
CPT082

IONISATION POTENTIAL

66. Correct orders of 1st I.P. are :-
(a) Li < B < Be < C (b) O < N < F
(c) Be < N < Ne
(1) a, b (2) b, c (3) a, c (4) a, b, c
CPT083

67. The ionisation potential of isotopes of an element will be :-
(1) Same
(2) Different
(3) Depends on atomic masses
(4) Depends on number of neutrons
CPT084

68. Compared to the first ionisation potential, the value of second ionisation potential of an element is :-
(1) Negligible (2) Smaller
(3) Greater (4) Double
CPT085

69. In which of the following pairs, the ionisation energy of the first species is less than that of the second :-
(1) O^-, O^{2-} (2) S, P (3) N, P (4) Be^+, Be
CPT086

70. The correct order of stability of oxidation state of Al^+, Al^{+2}, Al^{+3} is :-
(1) $Al^{+3} > Al^{+2} > Al^+$ (2) $Al^{+2} > Al^{+3} > Al^+$
(3) $Al^{+2} < Al^+ > Al^{+3}$ (4) $Al^{+3} > Al^+ > Al^{+2}$
CPT087

71. Least ionisation potential will be of :-
(1) Be^{3+} (2) H (3) Li^{+2} (4) He^+
CPT088

72. Ionisation energy increases in the order :-
(1) Be, B, C, N (2) B, Be, C, N
(3) C, N, Be, B (4) N, C, Be, B
CPT089

73. IP is influenced by -
(1) Size of atom
(2) Effective nuclear charge
(3) Electrons present in inner shell
(4) All
CPT090

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74. Highest ionisation potential in a period is shown by
(1) Alkali metals
(2) Noble gases
(3) Halogens
(4) Representative elements
CPT091

75. Which of the following decreases in going down the halogen group :-
(1) Ionic radius
(2) Atomic radius
(3) Ionisation potential
(4) None of these
CPT092

76. Minimum first ionisation energy is shown by which electronic configuration:-
(1) $1s^2, 2s^2\ 2p^5$ (2) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^2$
(3) $1s^2, 2s^2\ 2p^6, 3s^1$ (4) $1s^2, 2s^2\ 2p^6$
CPT093

77. With reference to ionisation potential which one of the following set is correct :-
(1) Li > K > B (2) B > Li > K
(3) Cs > Li > K (4) Cs < Li < K
CPT094

78. Successive ionisation energies of an element 'X' are given below (in kcal)

IP_1	IP_2	IP_3	IP_4
165	195	556	595

Electronic configuration of the element 'X' is:-
(1) $1s^2, 2s^2 2p^6, 3s^2\ 3p^2$
(2) $1s^2, 2s^1$
(3) $1s^2, 2s^2 2p^2$
(4) $1s^2, 2s^2 2p^6, 3s^2$
CPT095

79. Second IP of which of the element is maximum–
(1) Lithium (2) Oxygen
(3) Nitrogen (4) Fluorine
CPT096

80. Which has the lowest IE :-
(1) $3d^2$ (2) $4s^1$ (3) $3p^3$ (4) $2p^6$
CPT097

81. The energy needed to remove one electron from unipositive ion is abbreviated as :-
(1) 1st I.P. (2) 3rd I.P. (3) 2nd I.P. (4) 1st E.A.
CPT098

82. Among the following elements (Whose electronic configuration is given below) the one having the highest ionisation energy is
(1) [Ne] $3s^2\ 3p^3$ (2) [Ne] $3s^2\ 3p^4$
(3) [Ne] $3s^2 3p^5$ (4) [Ar] $3d^{10} 4s^2 4p^2$
CPT099

83. The correct order of decreasing first ionisation energy is :-
(1) Si > Al > Mg > Na (2) Si > Mg > Al > Na
(3) Al > Si > Mg > Na (4) Mg > Li > Al > Si
CPT100

84. Out of Na^+, Mg^{+2}, O^{-2} and N^{-3}, the pair of species showing minimum and maximum IP would be.
(1) Na^+, Mg^{+2} (2) Mg^{+2}, N^{-3}
(3) N^{-3}, Mg^{+2} (4) O^{-2}, N^{-3}
CPT101

85. The element having highest I.P. in the two series C, N, O and Si, P, S :–
(1) P (2) N (3) S (4) O
CPT102

86. Factor which does not affects the ionisation potential
(1) Atomic size
(2) Atomic mass
(3) Effective nuclear charge
(4) Shielding effect
CPT103

87. Lowest IP will be shown by the element having the configuration :–
(1) [He] $2s^2$ (2) $1s^2$
(3) [He] $2s^2\ 2p^2$ (4) [He] $2s^2\ 2p^5$
CPT104

88. Which ionisation potential (IP) in the following equations involves the greatest amount of energy:-
(1) $K^+ \rightarrow K^{+2} + e^-$ (2) $Li^+ \rightarrow Li^{+2} + e^-$
(3) $Fe \rightarrow Fe^+ + e^-$ (4) $Ca^+ \rightarrow Ca^{+2} + e^-$
CPT105

89. Values of first four ionisation potential of an elements are 68, 370, 400, 485. It belongs to which of the following electronic configuration:-
(1) $1s^2, 2s^1$ (2) $1s^2, 2s^2\ 2p^1$
(3) $1s^2, 2s^2\ 2p^6, 3s^1$ (4) (1) and (3) both
CPT106

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90. (a) $M^-_{(g)} \rightarrow M_{(g)}$ (b) $M_{(g)} \rightarrow M^+_{(g)}$
(c) $M^+_{(g)} \rightarrow M^{+2}_{(g)}$ (d) $M^{+2}_{(g)} \rightarrow M^{+3}_{(g)}$
Minimum and maximum I.P. would be of :-
(1) a, d (2) b, c (3) c, d (4) d, a

CPT107

91. In the plot of the first ionisation energy against atomic number the peaks are occupied by :-
(1) Inert gases
(2) Alkali metals
(3) Halogens
(4) Transition elements

CPT108

92. Which one of the following has highest ionisation potential :–
(1) Li^+ (2) Mg^+ (3) He (4) Ne

CPT109

93. In which of the following pairs, the ionisation energy of the first species is less than that of the second
(1) N, P (2) Be^+, Be
(3) N, N^- (4) Ne, Ne^+

CPT110

94. Consider the following ionisation reactions
$A(g) \longrightarrow A^+(g) + e^-$ IE in (kJ/mol) is A_1
$A^+(g) \longrightarrow A^{+2}(g) + e^-$ IE in (kJ/mol) is A_2
$A^{+2}(g) \longrightarrow A^{+3}(g) + e^-$ IE in (kJ/mol) is A_3
then correct order of IE is :-
(1) $A_1 > A_2 > A_3$ (2) $A_1 = A_2 = A_3$
(3) $A_1 < A_2 < A_3$ (4) $A_3 = A_2 < A_1$

CPT111

95. Select the correct order of I.E. :-
(1) $Cl^- > Cl > Cl^+$ (2) $Cl^+ > Cl > Cl^-$
(3) $Cl > Cl^+ > Cl^-$ (4) $Cl^- > Cl^+ > Cl$

CPT112

ELECTRON AFFINITY

96. In which case the energy released is minimum:-
(1) $Cl \rightarrow Cl^-$ (2) $B \rightarrow B^-$
(3) $N \rightarrow N^-$ (4) $C \rightarrow C^-$

CPT113

97. In the process $Cl_{(g)} + e^- \xrightarrow{\Delta H} Cl^-_{(g)}$, ΔH is
(1) Positive (2) Negative
(3) Zero (4) None

CPT114

98. Process in which maximum energy is released:-
(1) $O \rightarrow O^{-2}$ (2) $Mg^+ \rightarrow Mg^{+2}$
(3) $Cl \rightarrow Cl^-$ (4) $F \rightarrow F^-$

CPT115

99. Which of the following is energy releasing process
(1) $X^- \rightarrow X(g) + e^-$
(2) $O^-(g) + e^- \rightarrow O^{2-}$
(3) $O(g) \rightarrow O^+(g) + e^-$
(4) $O(g) + e^- \rightarrow O^-(g)$

CPT116

100. In which of the following process energy is liberated:-
(1) $Cl \rightarrow Cl^+ + e$ (2) $HCl \rightarrow H^+ + Cl^-$
(3) $Cl + e \rightarrow Cl^-$ (4) $O^- + e \rightarrow O^{-2}$

CPT117

101. Element of which atomic number has highest electron affinity:-
(1) 35 (2) 17 (3) 9 (4) 53

CPT118

102. Second electron affinity of an element is :–
(1) Always exothermic
(2) Endothermic for few elements
(3) Exothermic for few elements
(4) Always endothermic

CPT119

103. The electron affinity
(1) Of carbon is greater than oxygen
(2) Of fluorine is less than iodine
(3) Of fluorine is less than Chlorine
(4) Of sulphur is less than oxygen

CPT120

104. Which of the following configuration will have least electron affinity.
(1) ns^2np^5 (2) ns^2np^2
(3) ns^2np^3 (4) ns^2np^4

CPT121

105. Which of the following element will form most stable bivalent anion.
(1) Fluorine (2) Oxygen
(3) Chlorine (4) Nitrogen

CPT122

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106. Energy absorbed in second electron addition in an atom is called.
(1) 1st IP (2) 2nd EA
(3) 1st EA (4) 2nd IP

CPT123

107. The amount of energy released for the process $X_{(g)} + e^- \rightarrow X^-_{(g)}$ is minimum and maximum respectively for :–
(a) F (b) Cl (c) N (d) B
Correct answer is :–
(1) c & a (2) d & b
(3) a & b (4) c & b

CPT124

108. Which of the following electronic configuration is expected to have highest electron affinity:-
(1) $2s^2\ 2p^0$ (2) $2s^2\ 2p^2$
(3) $2s^2\ 2p^3$ (4) $2s^2\ 2p^1$

CPT125

ELECTRONEGATIVITY

109. Electronegativity scale of pauling is based upon :-
(1) Bond length
(2) Bond energy
(3) Atomic radius
(4) Covalent radius

CPT126

110. Polarity of a bond can be explained by :–
(1) Electron affinity
(2) Ionisation potential
(3) Electronegativity
(4) All of the above

CPT129

111. Outermost electronic configuration of the most electronegative element is :–
(1) ns^2np^3 (2) ns^2np^6
(3) ns^2 (4) ns^2np^5

CPT130

112. Electronegativity values for elements are useful in predicting :-
(1) Bond energy of a molecule
(2) Polarity of a bond
(3) Nature of an oxide
(4) All

CPT132

113. Mulliken scale of electronegativity uses the concept of :-
(1) E. A. and EN of pauling
(2) E. A. and atomic size
(3) E.A. and I.P.
(4) E.A. and bond energy

CPT133

114. The pair with minimum difference in electronegativity is :-
(1) F, Cl (2) C, H
(3) P, H (4) Na, Cs

CPT134

115. Least electronegative element is :-
(1) I (2) Br
(3) C (4) Cs

CPT135

116. In which of the following pair of elements the electronegativity of first element is less than that of second element :-
(1) Zr, Hf
(2) K, Rb
(3) Cl, S
(4) None of the above

CPT136

117. The nomenclature of ICl is iodine chloride because
(1) Size of I < Size of Cl
(2) Atomic number of I > Atomic number of Cl
(3) E.N. of I < E.N. of Cl
(4) E. A. of I < E. A. of Cl

CPT137

118. Among the following least and most polar bonds are respectively :-
(a) C – I (b) N – O
(c) C – F (d) P – F
(1) d and c (2) a and d
(3) b and d (4) b and c

CPT138

119. If the ionisation potential is IP, electron affinity is EA and electronegativity is X then which of the following relation is correct :-
(1) 2X – EA – IP = 0
(2) 2EA – X – IP = 0
(3) 2IP – X – EA = 0
(4) All of the above

CPT139

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120. The properties which are not common to both groups 1 and 17 elements in the periodic table are :-
(1) Electropositive character increases down the groups
(2) Reactivity decreases from top to bottom in these groups
(3) Atomic radii increases as the atomic number increases
(4) Electronegativity decreases on moving down the group

CPT140

121. Electronegativity of an element can be measured using :-
(1) Pauling's scale (2) Mulliken's scale
(3) Both (4) None

CPT141

122. As we proceed across the period in periodic table, we find there is a decrease in :-
(1) Ionisation energy (2) Electron affinity
(3) Electronegativity (4) Atomic radii

CPT142

123. Which compound strongly absorb CO_2 ?
(1) BeO (2) K_2O
(3) H_3PO_4 (4) P_4O_6

CPT143

124. The electronegativities of the following elements: H, O, F, S and Cl increase in the order :-
(1) H < O < F < S < Cl (2) Cl < H < O < F < S
(3) H < S < O < Cl < F (4) H < S < Cl < O < F

CPT144

125. Which of the following is different from other three oxides :-
(1) MgO (2) SnO
(3) PbO (4) ZnO

CPT145

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**1**	**2**	**4**	**2**	**4**	**3**	**2**	**1**	**3**	**1**	**2**	**2**	**3**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**2**	**3**	**3**	**4**	**3**	**1**	**3**	**4**	**4**	**4**	**2**	**3**	**1**	**3**	**3**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**2**	**1**	**3**	**2**	**3**	**1**	**3**	**2**	**3**	**2**	**3**	**3**	**4**	**1**	**2**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**3**	**2**	**4**	**3**	**1**	**2**	**2**	**4**	**4**	**4**	**4**	**3**	**3**	**1**	**1**
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	**3**	**1**	**3**	**4**	**4**	**4**	**1**	**3**	**2**	**4**	**2**	**2**	**4**	**2**	**3**
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	**3**	**2**	**4**	**1**	**2**	**3**	**3**	**2**	**3**	**2**	**2**	**1**	**2**	**3**	**1**
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	**1**	**1**	**4**	**3**	**2**	**3**	**2**	**3**	**4**	**3**	**2**	**4**	**3**	**3**	**2**
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	**2**	**4**	**2**	**2**	**3**	**4**	**4**	**3**	**3**	**4**	**1**	**3**	**2**	**1**	**2**
Question	121	122	123	124	125										
Answer	**3**	**4**	**2**	**4**	**1**										

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which of the following orders of ionic radii is correctly represented ?

(1) $H^- > H^+ > H$ (2) $Na^+ < F^- < O^{2-}$

(3) $F^- > O^{2-} > Na^+$ (4) $Al^{3+} > Mg^{2+} > N^{3-}$

CPT156

2. Be^{2+} is isoelectronic with which of the following ions?

(1) H^+ (2) Li^+ (3) Na^+ (4) Mg^{2+}

CPT157

3. Acidity of diprotic acids in aqueous solutions increases in the order :-

(1) $H_2S < H_2Se < H_2Te$

(2) $H_2Se < H_2S < H_2Te$

(3) $H_2Te < H_2S < H_2Se$

(4) $H_2Se < H_2Te < H_2S$

CPT158

4. Reason of lanthanoid contraction is :-

(1) Negligible screening effect of 'f' orbitals

(2) Increasing nuclear charge

(3) Decreasing nuclear charge

(4) Decreasing screening effect

CPT159

AIPMT 2015

5. The species Ar, K^+ and Ca^{2+} contain the same number of electrons. In which order do their radii increase ?

(1) $Ca^{2+} < Ar < K^+$ (2) $Ca^{2+} < K^+ < Ar$

(3) $K^+ < Ar < Ca^{2+}$ (4) $Ar < K^+ < Ca^{2+}$

CPT160

6. The number of d-electrons in Fe^{2+} (Z = 26) is not equal to the number of electrons in which one of the following?

(1) p-electrons in Cl (Z = 17)

(2) d-electrons in Fe (Z = 26)

(3) p-electrons in Ne (Z = 10)

(4) s-electrons in Mg (Z = 12)

CPT161

7. Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii ? (Numbers in the brackets are atomic numbers).

(1) Zr (40) and Nb (41)

(2) Zr (40) and Hf (72)

(3) Zr (40) and Ta (73)

(4) Ti (22) and Zr (40)

CPT162

Re-AIPMT 2015

8. The formation of the oxide ion, O^{2-} (g), from oxygen atom requires first an exothermic and then an endothermic step as shown below :

$O(g) + e^- \rightarrow O^-_{(g)}$; $\Delta_f H^- = -141$ kJ mol^{-1}

$O^-(g) + e^- \rightarrow O^{2-}_{(g)}$; $\Delta_f H^- = +780$ kJ mol^{-1}

Thus process of formation of O^{2-} in gas phase is unfavourable even thought O^{2-} is isoelectronic with neon. It is due to the fact that,

(1) Oxygen is more electronegative

(2) Addition of electron in oxygen results in larger size of the ion

(3) Electron repulsion outweighs the stability gained by achieving noble gas configuration

(4) O^- ion has comparatively smaller size than oxygen atom

CPT163

9. Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ?

(At. no. Z = 22)

(1) 3s 3p 3d 4s

(2) 3s 3p 4s 3d

(3) 3s 4s 3p 3d

(4) 4s 3s 3p 3d

CPT164

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NEET-I 2016

10. In which of the following options the order of arrangement does not agree with the variation of property indicated against it ?

(1) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$ (increasing ionic size)

(2) B < C < N < O (increasing first ionisation enthalpy)

(3) I < Br < Cl < F (increasing electron gain enthalpy)

(4) Li < Na < K < Rb (increasing metallic radius)

CPT165

NEET(UG) 2017

11. The element Z = 114 has been discovered recently. It will belong to which of the following family/group and electronic configuration ?

(1) Carbon family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^2$

(2) Oxygen family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^4$

(3) Nitrogen family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^6$

(4) Halogen family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^5$

CPT166

NEET(UG) 2018

12. Which of the following oxides is most acidic in nature?

(1) MgO (2) BeO

(3) BaO (4) CaO

CPT167

13. The correct order of atomic radii in group 13 elements is

(1) B < Al < In < Ga < Tl

(2) B < Al < Ga < In < Tl

(3) B < Ga < Al < Tl < In

(4) B < Ga < Al < In < Tl

CPT168

NEET(UG) 2019

14. Which of the following is an amphoteric hydroxide?

(1) $Sr(OH)_2$ (2) $Ca(OH)_2$

(3) $Mg(OH)_2$ (4) $Be(OH)_2$

CPT169

15. For the second period elements the **correct** increasing order of first ionisation enthalpy is:-

(1) Li < Be < B < C < N < O < F < Ne

(2) Li < B < Be < C < O < N < F < Ne

(3) Li < B < Be < C < N < O < F < Ne

(4) Li < Be < B < C < O < N < F < Ne

CPT170

16. 4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The **correct** option is :-

(1) 5f > 6p > 5p > 4d

(2) 6p > 5f > 5p > 4d

(3) 6p > 5f > 4d > 5p

(4) 5f > 6p > 4d > 5p

CPT171

NEET(UG) 2019 (ODISHA)

17. Match the oxide given in column A with its property given in column B:

Column-A	**Column-B**
(i) Na_2O	(a) Neutral
(ii) Al_2O_3	(b) Basic
(iii) N_2O	(c) Acidic
(iv) Cl_2O_7	(d) Amphoteric

Which of the following options has all correct pairs?

(1) (i)-(b), (ii)-(a), (iii)-(d), (iv)-(c)

(2) (i)-(c), (ii)-(b), (iii)-(a), (iv)-(d)

(3) (i)-(a), (ii)-(d), (iii)-(b), (iv)-(c)

(4) (i)-(b), (ii)-(d), (iii)-(a), (iv)-(c)

CPT172

NEET(UG) 2020

18. Identify the **incorrect** match :

Name	**IUPAC Official Name**
(a) Unnilunium	(i) Mendelevium
(b) Unniltrium	(ii) Lawrencium
(c) Unnilhexium	(iii) Seaborgium
(d) Unununnium	(iv) Darmstadtium

(1) (d), (iv) (2) (a), (i)

(3) (b), (ii) (4) (c), (iii)

CPT173

19. Match the following :

Oxide	**Nature**
(a) CO	(i) Basic
(b) BaO	(ii) Neutral
(c) Al_2O_3	(iii) Acidic
(d) Cl_2O_7	(iv) Amphoteric

Which of the following is **correct** option?

(a)	**(b)**	**(c)**	**(d)**
(1) (iv)	(iii)	(ii)	(i)
(2) (i)	(ii)	(iii)	(iv)
(3) (ii)	(i)	(iv)	(iii)
(4) (iii)	(iv)	(i)	(ii)

CPT174

NEET(UG) 2020(COVID-19)

20. Match the element in column I with that in column II.

Column-I	**Column-II**
(a) Copper	(i) Non-metal
(b) Fluorine	(ii) Transition metal
(c) Silicon	(iii) Lanthanoid
(d) Cerium	(iv) Metalloid

Identify the correct match :

(1) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

CPT175

21. Which of the following oxide is amphoteric in nature?

(1) SnO_2 (2) SiO_2 (3) GeO_2 (4) CO_2

CPT176

NEET(UG) 2021

22. **Statement-I :** Acid strength increases in the order given as HF << HCl << HBr << HI.

Statement-II : As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement-I and Statement-II are true.
(2) Both Statement-I and Statement-II are false.
(3) Statement-I is correct but Statement-II is false.
(4) Statement-I is incorrect but Statement-II is true.

CPT177

23. Zr (Z =40) and Hf (Z =72) have similar atomic and ionic radii because of :

(1) belonging to same group
(2) diagonal relationship
(3) lanthanoid contraction
(4) having similar chemical properties

CPT179

24. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?

(1) HF < HCl < HBr < HI :
Increasing acidic strength
(2) $H_2O < H_2S < H_2Se < H_2Te$:
Increasing pK_a values
(3) $NH_3 < PH_3 < AsH_3 < SbH_3$:
Increasing acidic character
(4) $CO_2 < SiO_2 < SnO_2 < PbO_2$:
Increasing oxidizing power

CPT180

NEET(UG) 2022

25. The IUPAC name of an element with atomic number 119 is

(1) unnilennium (2) unununnium
(3) ununoctium (4) ununennium

CPT181

26. If first ionization enthalpies of element X and Y are 419 kJ mol^{-1} and 590 kJ mol^{-1}, respectively and second ionization enthalpies of X and Y are 3069 kJ mol^{-1} and 1145 kJ mol^{-1}, respectively.

Then correct statement is :-

(1) X is an alkali metal and Y is an alkaline earth metal.
(2) X is an alkaline earth metal and Y is an alkali metal.
(3) Both X and Y are alkali metals.
(4) Both X and Y are alkaline earth metals.

CPT182

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27. The correct order of first ionization enthalpy for the given four element is :
(1) C < N < F < O (2) C < N < O < F
(3) C < O < N < F (4) C < F < N < O
CPT183

NEET(UG) 2023

28. The element expected to form largest ion to achieve the nearest noble gas configuration is
(1) F (2) N (3) Na (4) O
CPT210

29. Which of the following is **correctly** matched?
(1) Basic oxides ⇒ In_2O_3, K_2O, SnO_2
(2) Neutral oxides ⇒ CO, NO_2, N_2O
(3) Acidic oxides ⇒ Mn_2O_7, SO_2, TeO_3
(4) Amphoteric oxides ⇒ BeO, Ga_2O_3, GeO
CPT211

30. The **correct** sequence given below containing neutral, acidic, basic and amphoteric oxide each, respectively, is
(1) NO, ZnO, CO_2, CaO (2) ZnO, NO, CaO, CO_2
(3) NO, CO_2, ZnO, CaO (4) NO, CO_2, CaO, ZnO
CPT212

NEET(UG) 2024

31. Arrange the following elements in increasing order of electronegativity.
N, O, F, C, Si
Choose the correct answer from the options given below :
(1) Si < C < N < O < F (2) Si < C < O < N < F
(3) O < F < N < C < Si (4) F < O < N < C < Si
CPT213

32. Arrange the following elements in increasing order of first ionization enthalpy :
Li, Be, B, C, N
Choose the correct answer from the options given below :
(1) Li < Be < B < C < N
(2) Li < B < Be < C < N
(3) Li < Be < C < B < N
(4) Li < Be < N < B < C
CPT214

NEET(UG) 2024 (Re-Examination)

33. The **correct** decreasing order of atomic radii (pm) of Li, Be, B and C is :
(1) Be > Li > B > C (2) Li > Be > B > C
(3) C > B > Be > Li (4) Li > C > Be > B
CPT215

34. Match List-I with List-II

List-I (Block/group in periodic table)	**List-II (Element)**
(A) Lanthanoid	I. Ce
(B) d-block element	II. As
(C) p-block element	III. Cs
(D) s-block element	IV. Mn

Choose the **correct** answer from the options given below :
(1) A-I, B-II, C-IV, D-III
(2) A-I, B-IV, C-III, D-II
(3) A-I, B-IV, C-II, D-III
(4) A-IV, B-I, C-II, D-III
CPT216

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**2**	**1**	**1**	**2**	**1**	**2**	**3**	**2**	**2,3**	**1**	**2**	**4**	**4**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**4**	**1**	**3**	**2**	**1**	**1**	**3**	**2**	**4**	**1**	**3**	**2**	**3**	**4**
Question	31	32	33	34											
Answer	**1**	**2**	**2**	**3**											

EXERCISE – III (Analytical Questions) Master Your Understanding

1. Given below are two statements:
Statement-I : F, atom has less negative electron gain enthalpy than Cl, atom
Statement-II : Addition of electron is more effective in 3p orbital of Cl atom than 2p orbital of F atom.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I & II both are correct.
(2) Statement-I is correct but II are incorrect.
(3) Statement-I is incorrect but II is correct.
(4) Both statement-I & II are incorrect.
CPT184

2. Given below are two statements:
Statement-I : In Lanthenoid series 14 elements are present
Statement-II : f orbital can consist maximum 14 electrons.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I & II both are correct.
(2) Statement-I is correct but II are incorrect.
(3) Statement-I is incorrect but II is correct.
(4) Both statement-I & II are incorrect.
CPT186

3. Given below are two statements:
Statement-I : Atomic radius of Ga is less than Al
Statement-II : Effective nuclear charge of Ga is greater than Al.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I & II both are correct.
(2) Statement-I is correct but II are incorrect.
(3) Statement-I is incorrect but II is correct.
(4) Both statement-I & II are incorrect.
CPT187

4. Given below are two statements:
Statement-I : The radius of 5d series element is nearly similar to the 4d series element.
Statement-II : Due to poor shielding effect of 4f e^- Effective nuclear charge of 5d series elements is greater than that of 4d series elements.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I & II both are correct.
(2) Statement-I is correct but II are incorrect.
(3) Statement-I is incorrect but II is correct.
(4) Both statement-I & II are incorrect.
CPT189

5. Given below are two statements:
Statement-I : The formation of O^{-2} is endothermic
Statement-II : Ionisation energy of S^- is more than O^-.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I & II both are correct.
(2) Statement-I is correct but II are incorrect.
(3) Statement-I is incorrect but II is correct.
(4) Both statement-I & II are incorrect.
CPT191

6. Which of the following are true :-
(a) $EA_1(F) > EA_1(Cl)$
(b) $EA_1(Br) > EA_1(I)$
(c) $EA_1(Br) > EA_1(F)$
(d) $EA_1(Cl) > EA_1(F)$
(1) If (a) (b) and (c) are correct.
(2) If (a) & (b) both options are correct.
(3) If (b) and (d) both options are correct.
(4) If (a) and (c) both options are correct.
CPT192

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7. **Assertion (A) :** Generally, ionisation enthalpy increase from left to right in a period
Reason (R) : In a Period, Z_{eff} Increases from left to right.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CPT197

8. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Generelly electron gain enthalpy become's more negative as we go down a group
Reason (R) : Chlorine has maximum value of electron gain enthalpy in periodic Table.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CPT198

9. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : In Mg, Al and Si First ionisation enthalpy of Mg is more then Al but less then Si
Reason (R) : Z_{eff} of Mg > Al.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CPT200

10. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : In Ionisation energy, Electron afinity and Electronegativity only EN value regularly increase in a period.
Reason (R) : EN value of carbon atom are same in all compound of carbon.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CPT201

11. Match the column :

Column-I Element		**Column-II Atomic radius (pm)**	
(1)	Be	(a)	88
(2)	N	(b)	70
(3)	O	(c)	111
(4)	C	(d)	74

(1) 1 – d, 2 – c, 3 – a, 4 – b
(2) 1 – b, 2 – c, 3 – a, 4 – d
(3) 1 – c, 2 – d, 3 – b, 4 – a
(4) 1 – b, 2 – a, 3 – c, 4 – d

CPT204

12. Match the column :

Column-I Electronic Configuration		**Column-II (ΔH_{eg}) in kJ mol^{-1}**	
(1)	$1s^2\ 2s^2\ 2p^6$	(a)	–53
(2)	$1s^2\ 2s^2\ 2p^6\ 3s^1$	(b)	–333
(3)	$1s^2\ 2s^2\ 2p^5$	(c)	–141
(4)	$1s^2\ 2s^2\ 2p^4$	(d)	+116

(1) 1 – d, 2 – b, 3 – a, 4 – c
(2) 1 – d, 2 – a, 3 – b, 4 – c
(3) 1 – c, 2 – d, 3 – b, 4 – a
(4) 1 – b, 2 – c, 3 – a, 4 – d

CPT205

13. Match the column :

Column-I (Order)		Column-II (Properties)	
(1)	B > Tl > Ga > Al > In	(a)	Electron affinity (EA)
(2)	Cl > F > Br > I	(b)	Electronagativity (EN)
(3)	F > Cl > Br > I	(c)	Ionisation energy (IP)
(4)	B < Ga < Al < In < Tℓ	(d)	Atomic radius

(1) 1 – a, 2 – b, 3 – d, 4 – c
(2) 1 – a, 2 – b, 3 – c, 4 – d
(3) 1 – c, 2 – a, 3 – b, 4 – d
(4) 1 – d, 2 – c, 3 – b, 4 – a

CPT208

14. Match the column :

Column-I Electronic Configuration		Column-II [Group No.]	
(1)	$(n-1)\,d^5\,ns^1$	(a)	3rd
(2)	$(n-1)\,f^{14}$ $(n-1)d^0\,ns^2$	(b)	15th
(3)	$(n-1)d^{10}\,ns^2\,np^3$	(c)	11th
(4)	$(n-1)d^{10}\,ns^1$	(d)	6th

(1) 1 – d, 2 – b, 3 – a, 4 – c
(2) 1 – d, 2 – a, 3 – b, 4 – c
(3) 1 – c, 2 – d, 3 – b, 4 – a
(4) 1 – b, 2 – c, 3 – a, 4 – d

CPT209

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	**1**	**2**	**1**	**1**	**1**	**3**	**4**	**3**	**2**	**2**	**3**	**2**	**3**	**2**	

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Classification of elements and Periodicity in Properties - Periodic Table

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (4)
Mendeleev's periodic table → The physical & chemical properties of the elements are the periodic function of their atomic weight.

2. Ans. (1)
Be, Mg, Ca because difference of atomic number b/w Be, Mg and Mg, Ca is 8.

3. Ans. (2)
The place that were left empty by Mendeleev's were for :-
Eka aluminium — Gallium
Eka Silicon — Germanium

4. Ans. (4)
Anomalous pairs

Ar & K	Te & I	Th & Pa	Co & Ni
39.9 39.1	127.5 127	232 231	58.9 58.6
AW↓	AW↓	AW↓	AW↓

5. Ans. (2)
Law of triads is applicable to

Ca	Sr	Ba
40	88.5	137

$x = \frac{40+137}{3} = 88.5$

6. Ans. (4)
On the peak → Alkali metal
→ Highly electropositive
→ Large atomic volume

7. Ans. (3)
$Z_{(\text{atomic number})}$ = number of proton = Nuclear charge
In each period L → R
Z ↑ hence nuclear charge ↑

8. Ans. (2)
3rd period contains only 8 elements.

9. Ans. (1)
The discovery of inert gas gave a death blow to the Newlands law because after the discovery of inert gases the Newlands octave Law become invalid.

10. Ans. (3)
The reason of periodicity is repetition of same valence shell electronic configuration.

11. Ans. (1)
Technetium was absent in M.P.T. which was named as Eka manganese by Mendeleev's.

12. Ans. (2)
(1) Period = 7 & group = 18
(2) number of valence shell electrons in a group are same.
(3) III–B group → 32 elements
14 + 14 + 4 = 32
Lanthanoids actinoids (Sc, Y, La, Ac)
(4) Lanthanides and actinides are placed in III–B group

13. Ans. (2)
Mendeleev's left blank spaces in periodic table & predicted their property.
(Ga, Sc, Ge, Tc)

14. Ans. (3)
Inert gases → Eighteenth group

15. Ans. (2)
112 → Copernicium

16. Ans. (2)
(a) Zn, Cd, Hg — Same group not period
(b) 7th period → Fr_{87} Ra_{88} U_{92}
(c) 4th period → K_{19} Ca_{10}
5th period → Ag_{47}

17. Ans. (3)
Z = 104 to 118 (period number = 7)
Group number = last 2 digit in atomic number of element.
Z = 115 V–A group

18. Ans. (3)

5th period	18 elements	5s, 4d, 5p
6th period	32 elements	6s, 4f, 5d, 6p

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19. Ans. (4)

Elements of same group possess similar property

18th element will resemble with 2nd & 10th elements. (Inert gases)

20. Ans. (2)

Sg_{106} → Unnilhexium

21. Ans. (1)

Z = 118

18th group ⇒ noble gas.

22. Ans. (3)

$4s^2\ 4p^2$

For p-block

Group number = 12 + number of outermost p-electrons

= 12 + 2 = 14th/(IV – A) group.

Period = Highest value of principal Quantum number = 4th period.

23. Ans. (4)

Non-metals → In periodic table non-metals exist on upper hand right side.

24. Ans. (4)

Elements with similar chemical properties are placed in same group i.e. same number of electrons in outermost shell.

25. Ans. (4)

Li & Mg shows diagonal relationship due to nearly same ionic potential $\phi = \frac{\text{charge}}{\text{size}}$

26. Ans. (2)

$Yb_{70} = [Xe]\ 6s^2\ 5d^0\ 4f^{14}$

Atomic number

$$\left.\begin{array}{l} 58 \le Z \le 71 \Rightarrow \text{Lanthanoids } (6^{th}\text{ period}) \\ 90 \le Z \le 103 \Rightarrow \text{Actinoids } (7^{th}\text{ period}) \end{array}\right] \rightarrow (\text{III} - \text{B})\ \text{group}$$

27. Ans. (3)

3rd group & 7th period → Actinoids.

28. Ans. (1)

d-block elements are known as transition elements (except group-12 i.e. Zn, Cd, Hg, Uub)

transition elements starts from 4th period in periodic table

29. Ans. (3)

In p block

For group = 12 + number of outermost p- electrons

= 12 + 3

= 15th group.

30. Ans. (3)

Inert gas means $8e^-$ in outermost shell

$\underbrace{ns^2\ np^6}_{8e^-}$

31. Ans. (2)

$58 \le Z \le 71 \Rightarrow$ Lanthanoids (6th period)

(III – B group)

32. Ans. (1)

Transuranic elements → atomic number more than 92

• all are radioactive & artificial.

33. Ans. (3)

Uub = 112

EC = $[Rn]_{86}\ 7s^2\ 5f^{14}\ 6d^{10}$

34. Ans. (2)

$ns^2\ np^6$

18th group & 2nd → 7th period

35. Ans. (3)

Magic number (2, 8, 8, 18, 18, 32)

36. Ans. (1)

$X = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ = Cl

$Z = 1s^2\ 2s^2\ 2p^5$ = F

Just below Z

37. Ans. (3)

Representative elements → s-block & p-block elements except inert gas are known as representative elements.

Atomic number→	3	33	53	87
Elements→	Li	As	I	Fr
	(s-block)	(p-block)	(p-block)	(s-block)

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38. Ans. (2)

Uub $\Rightarrow$ 112 (last element)

$\therefore$ Total transuranic elements = 112 – 92

= 20

39. Ans. (3)

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 6s^2$ = Hg

- $_{80}Hg \rightarrow$ belongs to II-B group & 6th period
- Hg is liquid at room temperature
- Hg is used in thermometers
- D–block elements are known as transition elements except II–B group.

40. Ans. (2)

From top to bottom in a group Z_{eff} remain constant.

41. Ans. (3)

$_1H^1 \rightarrow$ For single electron species $\boxed{\sigma = 0}$

$z_{eff} = z - \sigma$

$\boxed{z_{eff} = z}$

$\therefore$ effective nuclear charge is equal to nuclear charge.

42. Ans. (3)

$_{11}Na = \underbrace{1s^2\ 2s^2\ 2p^6}\ 3s^{\textcircled{1}}$ ← Test electron

$\Downarrow$

Screening effect is due to inner shell electrons.

43. Ans. (4)

The screening effect of d–electrons is less than p–electrons. In general, d & f electrons have a poor shielding effect compared to s & p electrons. This is because s & p electrons are close to the nucleus whereas d & f electrons are more diffused.

44. Ans. (1)

Group – 4

Ni
∧
Pd
≈
Pt

→ due to lanthanoid contraction

Size of 4d series ≈ 5d series

45. Ans. (2)

(1) Correct order Be < Li < Mg

(2) $\underset{\text{(Only nucleus no radius)}}{H^+} < \underbrace{Li^+ < H^-}_{\text{(Isoelectronic species)}}$ $\quad (2e^-)\ \ (2e^-)$

$$\text{Atomic radii} \propto \frac{1}{Z_{eff}} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

(3) $\underbrace{(r_C > (r_F)}_{\text{Covalent radii}} <<< \underbrace{(r_{Ne})}_{\text{Vanderwaal's radii}}$

(4) $O^{-2} > F^- > Na^+$ (IES)

46. Ans. (3)

S^{-2} Ar K^{+1} $Ca^{+2} \rightarrow$ All are $(18e^-)$ isoelectronic species and for isoelectronic species

$$\text{Atomic radii} \propto \frac{1}{Z_{eff}} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

$$\left[\text{volume} = \frac{4}{3}\pi r^3\right]$$

Radii $\uparrow \Rightarrow$ volume$\uparrow$

So IES have same electronic configuration but different atomic volume.

47. Ans. (2)

An atom & its corresponding ion have the same number of protons (Nuclear charge)

48. Ans. (4)

(1) Cl^-, P^{-3}, Ar $\quad (18e^-)$ IES

(2) N^{-3}, Ne Mg^{+2} $\quad (10e^-)$ IES)

(3) B^{+3}, He, Li^+ $\quad (2e^-)$ IES

(4) N^{-3}, $\quad S^{-2}$, $\quad Cl^- \rightarrow$ not IES

$(10e^-)\ \ (18e^-)\ \ (18e^-)$

49. Ans. (3)

In 3–d series elements

$\underbrace{Fe \approx CO \approx Ni}_{Z_{eff} \approx \text{Screening effect}}$

50. Ans. (1)

$1s^2, 2s^2\ 2p^6, 3s^2 \rightarrow {}_{12}Mg$

$1s^2, 2s^2\ 2p^6, 3s^2\ 3p^1 \rightarrow {}_{13}Al$

$1s^2\ 2s^2\ 2p^6, 3s^2\ 3p^3 \rightarrow {}_{15}P$

$1s^2, 2s^2\ 2p^6, 3s^2\ 3p^5 \rightarrow {}_{17}Cl$

3rd period $\rightarrow$ Mg Al P Cl $\rightarrow$

$Z_{eff}\uparrow$ Size$\downarrow$

51. Ans. (2)

(1) $I^- > I > I^+$

for different ions of an element

$$\text{radii} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

(2) $Mg^{+2} < Na^+ < F^-$ (Isoelectronic species)

$$\text{radii} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

(3) $P^{+5} < P^{+3}$

$$\text{radii} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

(4) Li > Be > B

Moving from left of right in a period Z_{eff} increases so size decreases.

52. Ans. (2)

$_3Li = 1s^2\ 2s^{\overset{\text{Test electron}}{\textcircled{1}}}$

Screening effect is due to inner shell electrons i.e K-shell.

53. Ans. (4)

Ionic radii

(1) $Ti^{+4} > Mn^{+7}$ ($18e^-$) Isoelectronic species

$$\text{radii} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

(2) $^{37}Cl^- = {}^{35}Cl^-$ Isotopes having some radius due to some number of electrons & protons.

(3) $Cl^- > Ar^\circ > K^+$ ($18e^-$)

Isoelectronic

$$\text{radii} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

(4) $P^{+3} > P^{+5}$

For different ions of an element

$$\text{radii} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

54. Ans. (4)

S^{-2} ($18e^-$)

Ar = $18e^-$

Cl^- = $18e^-$

NS^- = $18e^-$

Ti^{+3} = $19e^- \rightarrow$ not isoelectronic

55. Ans. (4)

On moving from left to right is a period effective nuclear charge increases due to increases in number of protons. So attraction on outer most shell electrons increases.

56. Ans. (4)

$r_{anion} > r_{neutral} > r_{cation}$

57. Ans. (3)

Maximum size of first member of a period is due to minimum Z_{eff}.

On moving from L $\rightarrow$ R in a period $Z_{eff} \uparrow$ so size$\downarrow$

58. Ans. (3)

O^{-2} F^- Al^{+3}

$10e^-$ isoelectronic species

Cs^+ (maximum size) due to maximum number of shells.

59. Ans. (1)

(a) $\underbrace{Sc > Ti > V > Cr > Mn}_{Z_{eff} > \text{screenig effect}} > \underbrace{Fe \approx Co \approx Ni}_{Z_{eff} \approx \text{screenig effect}} < \underbrace{Cr < Zn}_{Z_{eff} > \text{screenig effect}}$

(b) F < Cl < Br < I

(c) Al > Si > P > S > Cl

60. Ans. (1)

$$\text{Atomic size} \propto \frac{1}{Z_{eff}} \propto \frac{\text{negative charge}}{\text{positive charge}}$$

61. Ans. (3)

(1) Li > Be $\rightarrow 2^{nd}$ period

$\wedge$

Na $\rightarrow 3^{nd}$ period

(2) Ni < Cu < Zn

Z_{eff} < screening effect

(3) Ti > V > Cr

Z_{eff} > screening effect

(4) Ti 3d

$\wedge$ $\wedge$

Zr 4d

〃 〃 $\rightarrow$ due to lanthanoid contraction

Hf 5d

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62. Ans. (1)

$\underbrace{Br^-}_{(36^-)} > \underbrace{S^{-2} > Cl^-}_{18e^-} > \underbrace{N^{-3}}_{10e^-}$

IES

Size $\propto \frac{-\text{ve charge}}{+\text{ve charge}}$

4th period (4 shell) | 3th period (3 shell) | 2th period (2 shell)

Radius $\propto$ number of shells

63. Ans. (3)

(1) Top to bottom in a group Z_{eff} remains constant

(2) On moving from left to right in a period Zeff increases so size decreases.

(3) Screening effect in a period increases.

(4) In a period atomic radii of inert gas element is maximum as for inert gas we consider Vander Waal's radius.

64. Ans. (4)

$I^- > Se^{-2} > Br^- > O^{-2} > F^{-1}$: Correct order

L → R Size ↓

O^{-2}, S^{-2}, Se^{-2} | F^-, Cl^-, Br^-, I^- | Size ↑

65. Ans. (4)

According to slater's rule

Values of Z_{eff}.

Li = 1.30 — Li

Na = 2.20 — Na

K = 2.20 — K

66. Ans. (4)

IP order for 2nd period

Li	Be	B	C	N	O	F	Ne
$2s^1$	$2s^2$	$2p^1$	$2p^2$	$2p^3$	$2p^4$	$2p^5$	$2p^6$

Due to penetration effect; Half filled stable configuration

Correct order

Li < B < Be < C < O < N < F < Ne

67. Ans. (1)

Ionisation potential depends on nuclear charge i.e. (number of protons)

But in isotopes number of protons is some hence IP of isotopes will be same.

68. Ans. (3)

Sucessive IE is always greater

Ex. $M_{(g)} \xrightarrow{IE_1} M^+_{(g)} + e^-$ order of IE

$M^+_{(g)} \xrightarrow{IE_2} M^{+3}_{(g)} + e^-$ $\boxed{IE_1 < IE_2 < IE_3}$

$M^{+2}_{(g)} \xrightarrow{IE_3} M^{+3}_{(g)} + e)$

69. Ans. (2)

(2) IP : S ($3p^4$) < P ($3p^3$)

Stable half filled configuration

(So removal of electron is difficult)

(1) IP $\propto Z_{eff} \propto \frac{+\text{ve charge}}{-\text{ve charge}}$

of O^- > IP of O^{-2}

(3) IP of N > IP of P

(4) IP $\propto Z_{eff} \propto \frac{+\text{ve charge}}{-\text{ve charge}}$

IP : $Be^+ > Be$

70. Ans. (4)

Sequence of stability

	Al^{+3}	>	Al^+	>	Al^{+2}
EC:	$2s^2\ 2p^6$		$3s^2$		$3s^1$
	(fully filled Complete octet)		(fully filled)		(half filled)

71. Ans. (2)

$\boxed{H} < He^+ < Li^{+2} < Be^{+3}$ (all are isoelectronic)

↓Minimum IP ($1e^-$)

For isoelectronic species

IP $\propto Z_{eff} \propto \frac{+\text{ve charge}}{-\text{ve charge}}$

72. Ans. (2)

Be	B	C	N
$2s^2$	$2p^1$	$2p^2$	$2p^3$

Correct order : B < Be < C < N

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73. Ans. (4)

(1) Size of atom $\propto \frac{1}{IP}$

(2) Effective nuclear charge $\propto$ IP

(3) Electrons present in inner shell $\propto \frac{1}{IP}$

74. Ans. (2)

Highest IP in a period is shown by noble gases because an electron is to be removed from completely filled orbital. This breaks the stable electronic configuration $ns^2\ np^6$ and required a large amount of energy.

75. Ans. (3)

F — Ionic radii ↑
Cl — Atomic radii ↑
Br — Ionisation potential ↓
I

76. Ans. (3)

(1) $1s^2\ 2s^2\ 2p^5 \rightarrow$ F
(2) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^2 \rightarrow$ Si
(3) $1s^2\ 2s^2\ 2p^6\ 3s^1 \rightarrow$ Na
(4) $1s^2\ 2s^2\ 2p^6 \rightarrow$ Ne

Li C F Ne
Na Si
↓
Minimum IP

77. Ans. (2)

IP↑

IP↓	Li	Be	B
	Na	Mg	Al
	K		

Correct order : B > Li > K

78. Ans. (4)

IP_1	IP_2	IP_3	IP_4
165	195	556	595

Maximum difference

In IP_2 & IP_3, Hence e^- in outermost shell = 2

(1) e^- in outermost shell = 4
(2) e^- in outermost shell = 1
(3) e^- in outermost shell = 4
(4) e^- in outermost shell = 2

79. Ans. (1)

	Li	N	O	F
IP_1	$2s^1$	$2p^3$	$2p^4$	$2p^5$
IP_2	$1s^2$ ↓	$2p^2$	$2p^3$	$2p^4$
	max IP			

e^- in 1 shell

80. Ans. (2)

Down the group IE decreases & moving from left to right in a period IE increases.

So K has least IE

81. Ans. (3)

$M_{(g)} \xrightarrow{IP_1} M^+_{(g)} \xrightarrow{IP_2} M^{+2}_{(g)}$

Unipositive ion

82. Ans. (3)

(1) [Ne] $3s^2\ 3p^3$ = P
(2) [Ne] $3s^2\ 3p^4$ = S
(3) [Ne] $3s^2\ 3p^5$ = Cl
(4) [Ar] $3d^{10}\ 4s^2\ 4p^2$ = Ge

IP↑
Si P S Cl
IP↓ Ge
Highest IP

83. Ans. (2)

IP↑
Na Mg Al Si
$3s^1$ $3s^2$ $3p^1$ $3p^2$
Due to penetration effect

Correct order : Si > Mg > Al > Na

84. Ans. (3)

N^{-3} O^{-2} Na^+ Mg^{+2} Isoelectronic species

For isoelectronic species

$IP \propto Z_{eff} \propto \frac{+ve\ charge}{-ve\ charge}$

Minimum IP $\rightarrow N^{-3}$

Maximum IP = Mg^{+2}

85. Ans. (2)

IP↑

IP↓	C	N	O
	Si	P	S

Due to stable half filled EC

So Highest IP among them is Nitrogen

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86. Ans. (2)

(1) Atomic Size $\propto \frac{1}{IP}$

(2) Atomic mass $\rightarrow$ does not depend

(3) Effective nuclear charge $\propto$ IP

(4) shielding effect $\propto \frac{1}{IP}$

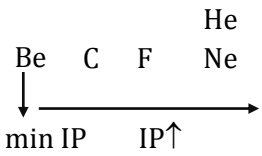
87. Ans. (1)

(1) Be

(2) He

(3) C

(4) F



88. Ans. (2)

(1) $K^+ \rightarrow K^{+2} + e^-$
(like Ar)

(2) $Li^+ \rightarrow Li^{+2} + e^-$
(like He)

(3) $Fe^\circ \rightarrow Fe^+ + e^-$

(4) $Ca^+ \rightarrow Ca^{+2} + e^-$
(like K)

89. Ans. (3)

IP_1	IP_2	IP_3	IP_4
68	370	400	485

Max difference in IP_1 & IP_2

Hence e^- in outermost shell = 1

$1s^2\ 2s^2\ 2s^6\ \underline{3s^1}$

90. Ans. (1)

$$\left[IP \propto Z_{eff} \propto \frac{+ve\ charge}{-ve\ charge}\right]$$

Hence sequence of IP

$M^{+2} > M^{+1} > M > M^-$

$\downarrow$ $\downarrow$

maximum IP minimum IP

91. Ans. (1)

In the plot of the first ionisation energy against atomic number the peaks are obtain by inert gas.

92. Ans. (1)

Li^+ has the highest IP (even though it has same configuration as he since Li^+ is a cation, e^- removal will be more difficult due to increased Z_{eff}.)

93. Ans. (4)

(1) N $>$ P (down the group IP decreases)
$2p^3$ $3p^3$

(2) $Be^+ >$ Be (different ion of an element)

$$\left[IP \propto Z_{eff} \propto \frac{+ve\ charge}{-ve\ charge}\right]$$

(3) N $>$ N^- (different ion of an element)

$$\left[IP \propto Z_{eff} \propto \frac{+ve\ charge}{-ve\ charge}\right]$$

(4) Ne $<$ Ne^+ (different ion of an element)

$$\left[IP \propto Z_{eff} \propto \frac{+ve\ charge}{-ve\ charge}\right]$$

94. Ans. (3)

Successive Ionisation energy is always greater.

$A_1 < A_2 < A_3$

95. Ans. (2)

Different ion of an element

$$\left[IE \propto Z_{eff} \propto \frac{+ve\ charge}{-ve\ charge}\right]$$

$Cl^+ > Cl > Cl^-$

96. Ans. (3)

(1) $Cl \longrightarrow Cl^-$
$3p^5$ $3p^6$

(2) $B \longrightarrow B^-$
$2p^1$ $2p^2$

(3) $N \longrightarrow N^-$
$\boxed{2p^3}$ $2p^4$
$\downarrow$
Half filled extra stable $\Rightarrow$ Minimum energy release

(4) $C \longrightarrow C^-$
$2p^2$ $2p^3$

97. Ans. (2)

$Cl(g) + e^- \xrightarrow{\Delta H} Cl^-(g)$

ΔH = negative $\Rightarrow$ process exothermic.

In general (first electron gain enthalpy) = negative

98. Ans. (3)

(1) $O \rightarrow O^{-2}$

$O(g) + e^- \rightarrow O^-(g) = -141$ KJ/mole

$O^-(g) + e^- \rightarrow O^{-2}(g) = +744$ KJ/mole

$O(g) + 2e^- \rightarrow O^{-2} = +603$ KJ/mole

↓endothermic process

(2) $Mg^+ \xrightarrow{IP} Mg^{+2}$; IP is always an endothermic process

(3) $Cl \xrightarrow{EA} Cl^-$ (max energy release)
$3p^5 \quad 3p^6$

(4) $F \xrightarrow{EA} F^-$ (energy release)

] EA of 3rd period is greater than 2nd period

99. Ans. (4)

(1) $x^- \xrightarrow[endo]{IE} x^\circ + e^-$

(2) $O^-_{(g)} + e^- \xrightarrow[endo]{} O^{-2}$ (addition of 2nd e^- is endothermic)

(3) $O(g) \xrightarrow[endo]{IE} O^+_{(g)} + e^-$

(4) $O + e^- \xrightarrow[exothermic]{} O^-(g)$

100. Ans. (3)

(1) $Cl \xrightarrow[endo]{IE} Cl^+ + e$

(2) $HCl \xrightarrow[endo]{BDE} H^+ + Cl^-$

(3) $Cl + e^- \xrightarrow[exothermic]{EA} Cl^-$

(4) $O^- + e^- \xrightarrow[endo]{EA} O^{-2}$ (addition of 2nd e^- is endothermic)

101. Ans. (2)

(a) Atomic number (35) → Br

(b) Atomic number (17) → Cl

→ maximum electron affinity in periodic table

(c) Atomic number (9) → F

(a) Atomic number (53) → I

102. Ans. (4)

Second electron addition in an isolated gaseous atom is always an endothermic process due to inter electronic repulsion.

103. Ans. (3)

EA order

(1) C < O $\qquad$ $EA \propto Z_{eff}$

(2) F > I

(3) F < Cl

(4) S > O

] EA of 3rd period is greater than 2nd period element.

104. Ans. (3)

Least Electron affinity

$ns^2\ np^3$ → due to half filled stable configuration.

105. Ans. (2)

(1) Fluorine with EC $1s^2\ 2s^2\ 2p^5$ will form a most stable monovalent anion.

(2) Oxygen $1s^2\ 2s^2\ 2p^4$ bivalent anion

(3) Chlorine $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ monovalent anion

(4) Nitrogen $1s^2\ 2s^2\ 2p^3$ trivalent anion

106. Ans. (2)

$M_{(g)} + e^- \xrightarrow{EA_1} M_g^-$

EA_1 = +ve (energy released)

$M^-_{(g)} + e^- \xrightarrow{EA_2} M_g^{-2}$

EA_2 = –ve (energy absorbed)

107. Ans. (4)

$X_{(g)} + e^- \longrightarrow X^-_{(g)}$

Minimum energy released

N → due to half filled ($2s^2\ 2p^3$) stable EC

Maximum energy released

Cl → maximum electron affinity in periodic table.

108. Ans. (2)

$2s^2\ 2p^2$ → after gaming $1e^-$ it attain half filled stable electronic configuration ($2s^2\ 2p^3$)

109. Ans. (2)

Electronegativity scale of pauling is based upon bond energy.

110. Ans. (3)

Polarity of bond ∝ Δ Electronegativity

111. Ans. (4)

Fluorine is the most electronegative element

${}_9F = 1s^2\ 2s^2\ 2p^5$

112. Ans. (4)

(1) Bond energy ∝ Electronegativity difference

(2) Polarity of bond ∝ Electronegativity difference

(3) Nature of oxide : EN ∝ Acidic nature of oxide

113. Ans. (3)

Mulliken scale

$$x_m \text{ or } EN = \frac{\text{Ionisation potential + electron affinity}}{2}$$

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114. Ans. (3)

EN value of P & H is approximately some

i. e. = 2.1

115. Ans. (4)

C, S, I = 2.5

Br = 2.8

Cs = 0.7

116. Ans. (1)

Zeff of Hf is greater than Zr & [EN $\propto$ Zeff] so EN of Hf is greater.

117. Ans. (3)

ICl

I $\rightarrow$ Prefix $\rightarrow$ less EN element

Cl $\rightarrow$ Suffix $\rightarrow$ more EN element

Iodine chloride

118. Ans. (2)

Bond polarity $\propto \Delta EN$

(a) C — I $\Delta EN = 2.5 - 2.5 = 0$ (Least polar)

(b) N—O $\Delta EN = 3.5 - 3 = .05$

(c) C—F $\Delta EN = 4 - 2.5 = 1.5$

(d) P—F $\Delta EN = 4 - 2.1 = 1.9$ (most polar)

119. Ans. (1)

According to mulliken scale

$$EN = \frac{IP + EA}{2}$$

$$X = \frac{IP + EA}{2}$$

$2x - EA - IP = 0$

120. Ans. (2)

Group – 1

Reactivity of metals $\propto \frac{1}{IP}$

$\therefore$ Li < Na < K < Pb < Cs

Group – 17

Reactivity of non-metals $\propto$ EN

$\therefore$ F > Cl > Br > I

121. Ans. (3)

EN can be measured using Pauling's scale and Mulliken's scale.

122. Ans. (4)

When we move from left to right in a period

(1) Ionisation energy $\rightarrow$ increases

(2) Electron affinity $\rightarrow$ increases

(1) Electronegativity $\rightarrow$ increases

(1) Atomic radii $\rightarrow$ decreases

123. Ans. (2)

CO_2 is acidic & K_2O is strongest base among all options so it strongly absorb CO_2.

124. Ans. (4)

H

2.1

O	F
3.5	4
S	Cl
2.5	3

H < S < Cl < O < F

125. Ans. (1)

MgO $\rightarrow$ Basic oxide

SnO, PbO, ZnO $\rightarrow$ Amphoteric oxide

Exercise-II (Previous Year Questions)

1. Ans. (2)

In isoelectronic series

$$\text{size} \propto \frac{1}{Z_{eff}} \propto \frac{-\text{ve charge}}{+\text{ve charge}}$$

Correct order : $Na^+ < F^- < O^{-2}$

2. Ans. (2)

In Be^{+2} $\rightarrow$ Total electrons = 2

In Li^+ $\rightarrow$ Total electrons = 2

3. Ans. (1)

H_2S, H_2Se, H_2Te ↓

Size ↑

Bond length ↑

Bond strength ↓

Ionisation of H^+ ↑

Acidic nature ↑

4. Ans. (1)

Due to poor shielding of f-orbital or negligible screening effect of f-orbital.

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5. Ans. (2)

In isoelectronic series

$$\text{radius} \propto \frac{1}{Z_{eff}} \propto \frac{-\text{ve charge}}{+\text{ve charge}}$$

$Ca^{+2} < K^{+} < Ar$

6. Ans. (1)

No. of d-electrons in Fe^{+2}

$Fe^{+2} = 3d^6$ (d-electrons = 6)

(1) p-electrons in Cl $\rightarrow$ $1s^2 2s^2 \underline{2p^6} 3s^2 \underline{3p^5}$

(p-electrons=11)

(2) d-electrons in Fe$\rightarrow$ $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 \underline{3d^6}$

d-electrons=6)

(3) p-electrons in Ne $\rightarrow$ $1s^2 2s^2 \underline{2p^6}$

(p-electrons=6)

(4) s-electrons in Mg $\rightarrow$ $\underline{1s^2}\ \underline{2s^2}\ 2p^6\ \underline{3s^2}$

(s-electrons=6)

7. Ans. (2)

Due to lanthanoid contraction

Radius of Zr $\simeq$ Radius of H_f

(4d series) $\simeq$ (5d series)

8. Ans. (3)

O$\rightarrow$2nd period (small size) unstable due to anion-electron repulsion.

9. Ans. (2)

Increasing order of energy can be explained by $(n+\ell)$ rule as value of $(n+\ell)$ increases energy increases.

Orbital	n	ℓ	$(n+\ell)$	
3s	3	0	3	
3p	3	1	4	Of value of $(n+\ell)$ is same one having lower value of n will be filled first.
4s	4	0	4	
3d	3	2	5	

Correct order : 3s < 3p < 4s < 3d

10. Ans. (2,3)

(1) For isoelectronic series

$$\text{ionic size} \propto \frac{1}{Z_{eff}} \propto \frac{-\text{ve charge}}{+\text{ve charge}}$$

$Al^{+3} < Mg^{+2} < Na^{+} < F^{-}$

(2) I^{st} IP order

B C N O

$2p^1$ $2p^2$ $\boxed{2p^3}$ $2p^4$

Half filled stable configuration

Correct order

B < C < O < N

(3) Electron gain enthalpy :

EA of 3rd period > EA of 2nd period

Correct order = I < Br < F < Cl

(4) Down the group metallic radius increases.

Li < Na < K < Rb

11. Ans. (1)

Z = 114

From atomic no 104 to 118

Group no. $\rightarrow$ last two digit in atomic no. of element

Z = 114 $\Rightarrow$ 14th group (carbon family)

EC = $[Rn]5f^{14}6d^{10}7s^27p^2$

12. Ans. (2)

Non-metal oxide are acidic in nature BeO is amphoteric in nature and rest are basic in nature.

13. Ans. (4)

Correct order of atomic radii

$B < \underbrace{Ga < Al}_{\text{transition contraction}} < In < Tl$

14. Ans. (4)

$Be(OH)_2$ is an amphoteric hydroxide which react with both acidic & basic oxide/ hydroxide.

15. Ans. (2)

Li Be B C N O F Ne

$2s^1$ $2s^2$ $2p^1$ $2p^2$ $2p^3$ $2p^4$ $2p^5$ $2p^6$ [$IP \propto Z_{eff}$]

Penetration effect

Half-filled stable configuration

Correct order

$\Rightarrow$ Li < B < Be < C < O < N < F < Ne

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16. Ans. (1)

Use $(n+\ell)$ rule $(n+\ell)\uparrow \Rightarrow$ (Energy)$\uparrow$

Orbital	n	ℓ	$(n+\ell)$	
4d	4	2	6	If value of $(n+\ell)$ is same one having lower value of n will be filled first.
5p	5	1	6	
5f	5	3	8	
6p	6	1	7	

Correct decreasing order : 5f > 6p > 5p > 4d

17. Ans. (4)

$Na_2O \rightarrow$ Basic (metal oxide)

$Al_2O_3 \rightarrow$ Amphoteric

$N_2O \rightarrow$ Neutral

$Cl_2O_7 \rightarrow$ Acidic (non-metal)

18. Ans. (1)

Unununium(III) $\rightarrow$ Roentgenium (Uuu)

19. Ans. (3)

$CO \rightarrow$ Neutral

$BaO \rightarrow$ Basic (metal oxide)

$Al_2O_3 \rightarrow$ Amphoteric

$Cl_2O_7 \rightarrow$ Acidic (non-metal oxide)

20. Ans. (2)

$Cu \rightarrow$ Transition metal

$F \rightarrow$ Non-metal

$Si \rightarrow$ Metalloid

$_{58}Ce \rightarrow$ Lanthanoid (58-71)

21. Ans. (1)

$SnO_2 \rightarrow$ Amphoteric

SiO_2, GeO_2, CO_2 $\rightarrow$ Acidic (non – metal oxide)

22. Ans. (1)

HF, HCl, HBr, HI

Size ↑

Bond length ↑

Bond strength ↓

H^+ donating tendency ↑

Acidic strength ↑

23. Ans. (3)

Zr ≈ Hf

(4d) ≈ (5d) due to lanthanoid contraction

Series Series

24. Ans. (2)

HF < HCl < HBr < HI

Top to bottom bond strength ↓ ses so, Acidic ↑ses.

25. Ans. (4)

1 1 9

↓ ↓ ↓

Un un ennium

26. Ans. (1)

X Y

$IP_1 \rightarrow$ 419 590 $\Rightarrow$ removal of e^- from y required more energy

$IP_2 \rightarrow$ 3069 1145 $\Rightarrow$ After removal of $1e^-$ now, removal of e^- is difficult from x means

X is alkali metal y is alkaline earth metal.

27. Ans. (3)

C N O F [IP ∝ Z_{eff}]

$2p^2$ $2p^3$ $2p^4$ $2p^5$

Due to stable half-filled electronic configuration

Correct order :- C < O < N < F

28. Ans. (2)

F^{-1}, N^{-3}, Na^+ & O^{-2}

all ions are isoelectronic containing 10 e^-

$Z_{eff} \rightarrow Na^+ > F^- > O^{-2} > N^{-3}$

order of radius $\rightarrow N^{-3} > O^{-2} > F^- > Na^+$

→ Nitrogen to achieve Noble gas configuration it gain 3 e^-, & form N^{-3}

29. Ans. (3)

Mn_2O_7, SO_2, TeO_3 are acidic oxides.

30. Ans. (4)

NO → neutral CaO → Basic

CO_2 → Acidic ZnO → Amphoteric

31. Ans. (1)

EN order (Pauling scale value)

Si < C < N < O < F

1.8 < 2.5 < 3.0 < 3.5 < 4

32. Ans. (2)

Li < Be > B < C < N

$2s^1$ $2s^2$ $2s^22p^1$ $2s^22p^2$ $2p^22p^3$

Li < B < Be < C < N

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33. Ans. (2)

Atomic radius decreases on moving left to right.

34. Ans. (3)

A–I, B-IV, C-II, D-III

Exercise-III (Analytical Questions)

1. Ans. (1)

S-1:- Electron gain enthalpy of 3rd period element is greater than 2nd period element.

S-2:- Due to small size of F atom electron-electron repulsion occurs so addition of electron is difficult.

2. Ans. (2)

S-1 : Lanthanoid series → Atomic no.-58 to 71 consists of 14 elements.

S-2: f orbital can accommodate maximum 2 electrons

f subshell can accommodate maximum 14 electrons (7-orbitals)

3. Ans. (1)

S : 1 →

G-13

B
∧
Al
∨ → Due to transition contraction
Ga
∧
In
∧
Tl

S : 2 → As Zeff of Ga is grater than Al so atomic radii decreases.

4. Ans. (1)

S-1 : Radii order

3d
∧
4d
≀
5d

S-1 : Due to post shielding of f electrons Zeff increases due to which the outermost shell electrons get highly attracted towards Nucleus.

5. Ans. (1)

S - 1 : Formation of polynegative anion like O^{-2}, N^{-3}, C^{-4} is always an endothermic process.

S - 2 : As we know that

$(EA)_S > (EA)_O$

$O + e^- \longrightarrow O^-$ EA
$O^- \longrightarrow O + e^-$ IE

$(IE)_{S^-} > (IE)_{O^-}$

$EA_{(O)} = |IE|_{O^-}$

$EA_{(S)} = |IE|_{S^-}$

6. Ans. (3)

Electron affinity order of G-17

Cl > F > Br > I

7. Ans. (4)

(A):- On moving from left to right in period Z_{eff} increases

(R):- As [Ionisation energy $\propto Z_{eff}$]

8. Ans. (3)

(A) As know that on moving down in the group electron affinity decreases.

$EA = -\Delta egH$

So on moving down in a group electron gain enthalpy become less negative.

(R) Chlorine has maximum electron gain enthalpy in periodic table.

9. Ans. (2)

Mg	Al	Si
$3s^2$	$3p^1$	$3p^2$

Due to penetration effect

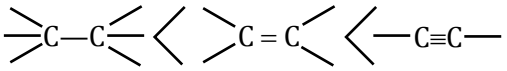
Correct order : Si > Mg > Al

10. Ans. (2)

(A) IE, EA → Both are depend on electronic configuration so regular increasing order was not shown. While EN $\propto Z_{eff}$. So we see a regular increasing order in a period.

(R) Compounds of carbon.

EN order :



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11. Ans. (3)

$$\underset{Z_{eff}\uparrow \text{ Atomic radii}\downarrow}{\overset{Be > C > N > O}{\longrightarrow}}$$

12. Ans. (2)

(1) $1s^2 2s^2 2p^6 \rightarrow Ne$

(2) $1s^2 2s^2 2p^6 3s^1 \rightarrow Na$

(3) $1s^2 2s^2 2p^5 \rightarrow F$

(4) $1s^2 2s^2 2p^4 \rightarrow O$

Correct order

Ne < Na < O < F

+116 −53 −141 −333

↓

Fully filled stable Noble gas EC so least ΔH_{eg}

13. Ans. (3)

(a) B > Tl > Ga > Al > In

→ Ionisation energy of (13th group)

(b) Cl > F > Br > I

→ Electron affinity (17th group)

(c) F > Cl > Br > I

→ Electronegativity (17th group)

(d) B < Ga < Al < In < Tl

→ Atomic radius

14. Ans. (2)

(1) $(n-1)d^5ns^1 \rightarrow$ Block

subshell in which last e^- is received → D block

For D-block

Group → $(n-1)\,de^- + nse^-$

$= 5 + 1$

$= 6^{th}$

(2) $(n-2)f^{14}(n-1)d^0ns^2$ → Block-f

→ For f block elements

Group no. → 3rd

(3) $(n-1)d^{10}ns^2np^3 \rightarrow$ p-block

Group no. → 12 + no of outermost p-elements

$= 12 + 3$

$= 15^{th}$

(4) $(n-1)d^{10}ns^1 \rightarrow$ Block D-block

Group → $(n-1)de^- + nse^-$

$= 10 + 1$

$= 11^{th}$

EXERCISE - I (Conceptual Questions) Build Up Your Understanding

OCTET RULE/VALENCY

1. Which condition favours the bond formation:
(1) Maximum attraction and maximum potential energy
(2) Minimum attraction and minimum potential energy
(3) Minimum potential energy and maximum attraction
(4) None of the above

CCB001

2. Among the following element group number is not equals to valence electrons -
(1) Na (2) N (3) Ne (4) None

CCB002

3. Which one of the following element will never obey octet rule:
(1) Na (2) F (3) S (4) H

CCB003

4. Valency expresses generally :
(1) Total e^- in an atom
(2) Atomicity of an element
(3) Oxidation number of an element
(4) Combining capacity of an element

CCB004

5. Which element do not have valency equals to its group no :
(1) Sodium (2) Aluminium
(3) Oxygen (4) Carbon

CCB005

COVALENT BOND

6. In a triple bond there is sharing of :–
(1) 3–electrons (2) 4–electrons
(3) Several electrons (4) 6–electrons

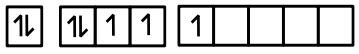
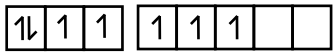
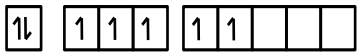
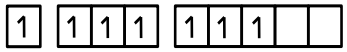
CCB006

7. The strength of bonds by 2s - 2s, 2p - 2p and 2p -2s overlapping has the order :-
(1) s – s > p – p > s – p
(2) s – s > p – s > p – p
(3) p – p > s – p > s – s
(4) p – p > s – s > p – s

CCB007

8. In which excitation state of chlorine ClF_3 is formed :-
(1) In ground state
(2) In third excitation state
(3) In first excitation state
(4) In second excitation state

CCB008

9. Which of the following configuration shows second excitation state of Iodine:-
(1) 
(2) 
(3) 
(4) 

CCB009

10. Variable covalency is exhibited by:-
(1) P and S (2) N and O
(3) N and P (4) F and Cl

CCB010

11. A sigma bond is formed by the overlapping of:–
(1) s-s orbital alone
(2) s and p orbitals alone
(3) s–s, s–p or p–p orbitals along internuclear axis
(4) p–p orbital along the sides

CCB011

12. Overlapping of 2 hybrid orbitals can lead to the formation of :
(1) Ionic bond (2) π –bond
(3) σ –bond (4) both (2) and (3)

CCB012

13. Which of the following statements is not correct ?
(1) Double bonds is shorter than a single bond
(2) σ – bond is weaker than a π bond
(3) Double bond is stronger than a single bond
(4) Covalent bond is stronger than a hydrogen bond

CCB013

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14. Covalent bond is formed by
(1) Combination of ions
(2) Combination of half filled orbitals
(3) Exchange of electrons by atom
(4) None of above

CCB014

15. The type of bond formed between two electronegative atoms would be
(1) Covalent (2) Ionic
(3) Coordinate (4) All the above

CCB015

16. p–p overlapping will be observed in the molecules of :
(1) Hydrogen
(2) Hydrogen bromide
(3) Hydrogen chloride
(4) Chlorine

CCB016

17. Nitrogen does not form NF_5 because
(1) Nitrogen is member of V group
(2) It contains no empty d-orbital
(3) The bond energy of N≡N is very high
(4) E.N. of N is more

CCB017

18. Which compound of xenon is not possible
(1) XeF_2 (2) XeF_4 (3) XeF_5 (4) XeF_6

CCB018

19. The correct order of bond length is
(1) C — C < C = C < C ≡ C
(2) C ≡ C < C = C < C — C
(3) C = C < C ≡ C < C — C
(4) C = C < C — C < C ≡ C

CCB019

20. Which of the following statements regarding covalent bond is not true ?
(1) The electrons are shared between atoms
(2) The bond is non-directional
(3) The strength of the bond depends upon the extent of overlapping
(4) The bond formed may or may not be polar

CCB020

21. Which of the following element does not form diatomic molecule.
(1) Iodine (2) Oxygen
(3) Phosphorus (4) Nitrogen

CCB021

22. Nitrogen forms N_2 but phosphorus do not forms P_2, but it exists as P_4 the reason for this is :
(1) Triple bond is present between phosphorus atoms
(2) pπ - pπ bonding is weak
(3) pπ - pπ bonding is strong
(4) Multiple bond is formed easily

CCB022

23. Which of the following compound does not exist ?
(1) PF_5 (2) PCl_5 (3) PBr_5 (4) PI_5

CCB023

24. The correct order of decreasing bond energy is:-
(1) O–O > S – S > Se – Se
(2) C – C > Si – Si > Ge – Ge
(3) F – F > O – O > N – N
(4) F – F > Cl – Cl > Br – Br

CCB024

HYBRIDISATION

25. The d- orbitals involved in sp^3d hybridisation is:-
(1) $d_{x^2-y^2}$ (2) d_{z^2} (3) d_{xy} (4) d_{xz}

CCB025

26. A sp^3 hybrid orbital contains:-
(1) $\frac{3}{4}$s- character (2) $\frac{1}{4}$p – character
(3) $\frac{3}{4}$p - character (4) $\frac{1}{2}$s - character

CCB026

27. Which set of species have same hybridisation state but different shapes:-
(1) NO_2^+, NO_2, NO_2^- (2) ClO_4^-, SF_4, XeF_4
(3) NH_4^+, H_3O^+, OF_2 (4) $SO_4^{-2}, PO_4^{-3}, ClO_4^-$

CCB027

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28. Which of the following species are expected to be planar:-

(a) NH_3 (b) NH_3^{2+} (c) CH_3^+ (d) PCl_3

The correct answer is:-

(1) b and c (2) c and d

(3) b and d (4) a and d

CCB028

29. The shape of sulphate ion is :-

(1) Hexagonal

(2) Square planar

(3) Trigonal bipyramidal

(4) Tetrahedral

CCB029

30. In which following set of compound/ion has linear shape

(1) CH_4, NH_4^+ , BH_4^- (2) CO_3^{-2} , NO_3^- , BF_3

(3) NO_2^+ , CO_2, XeF_2 (4) $BeCl_2$, BCl_3, CH_4

CCB030

31. Which order of decreasing bond angle is correct:-

(1) $CCl_4 > BF_3 > NO_2^+$ (2) $NH_3 > NCl_3 > NBr_3$

(3) $Br_2O > Cl_2O > OF_2$ (4) $PCl_3 > PBr_3 > PI_3$

CCB031

32. Shape of a molecule having 4 bond pairs and two lone pairs of electrons, will be :-

(1) Square planar (2) Tetrahedral

(3) Linear (4) Octahedral

CCB032

33. Hybridisation in XeF_2, XeO_2F_2 is sp^3d. But shape will be respectively:-

(1) T, 'V' shape

(2) Linear, See-Saw

(3) Both have T shape

(4) T shape, irregular octahedral

CCB033

34. The shape of IF_4^+ will be :-

(1) Square planar

(2) Tetrahedral

(3) Pentagonal bipyramidal

(4) Distorted tetrahedral

CCB034

35. Which of the following compound, has four bond pair and one lone pair :-

(1) NH_4^+ (2) ICl_4^- (3) SF_4 (4) XeF_4

CCB035

36. Which of the following has pyramidal shape ?

(1) BF_3 (2) H_3O^+ (3) NO_3^- (4) CO_3^{2-}

CCB036

37. The geometry of XeO_3F_2 is :–

(1) Triangular planar

(2) Trigonal bipyramidal

(3) Square planar

(4) Tetrahedral

CCB037

38. The type of hybrid orbitals used by chlorine atom in ClO_2^-, ClO_3^- and ClO_4^- is/are :-

(1) sp^2, sp^3 and sp^3d (2) sp and sp^3

(3) Only sp^3 (4) only sp

CCB038

39. Which of the following is having a square planar structure :-

(1) NH_4^+ (2) BF_4^- (3) XeF_4 (4) CCl_4

CCB039

40. Hybridization involves

(1) Addition of an electron pair

(2) Mixing up of atomic orbitals

(3) Removal of an electron pair

(4) Separation of orbitals

CCB040

41. The shape of I_3^- is

(1) Tetrahedral (2) Linear

(3) T-shape (4) Trigonal

CCB041

42. The shape of ClO_3^- ion according to Valence Shell Electron Pair Repulsion (VSEPR) theory will be :

(1) Triangular planar (2) Pyramidal

(3) Tetrahedral (4) Square planar

CCB042

43. The AsF_5 molecule is trigonal bipyramidal. The hybrid orbitals used by the As atoms for bonding are:

(1) $d_{x^2-y^2}, d_{z^2}, s, p_x, p_y$ (2) d_{xy}, s, p_x, p_y, p_z

(3) $s, p_x, p_y, p_z, d_{z^2}$ (4) $d_{x^2-y^2}, s, p_x, p_y$

CCB043

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44. sp^3 hybridisation is found in :

(A) CH_3^+ (B) CH_3^- (C) ClO_3^- (D) SO_3

(1) A & C (2) B & C (3) B & D (4) C & D

CCB044

45. Which of the following molecule has regular geometry –

(1) H_2O (2) PF_3 (3) SF_6 (4) XeF_6

CCB045

46. T-shape molecules are

(1) $C\ell F_3$ (2) $IC\ell_3$

(3) BrF_3 (4) All of these

CCB046

47. Which of the following two are isostructural:-

(1) XeF_2, IF_2^- (2) NH_3, BF_3

(3) CO_3^{2-}, SO_3^{2-} (4) PCl_5, ICl_5

CCB047

48. Select the correct matching :

List I	**List II**
A : XeF_4	1. Pyramidal
B : XeF_6	2. T-shape
C : XeO_3	3. Distorted octahedral
D : $XeOF_2$	4. Square planar

	A	**B**	**C**	**D**
(1)	4	3	1	2
(2)	1	2	3	4
(3)	2	1	3	4
(4)	4	1	3	2

CCB048

49. Which one of the following is a correct pair with respect to molecular formula of xenon compound and hybridization state of xenon in it :

(1) XeF_4, sp^3 (2) XeF_2, sp

(3) XeF_2, sp^3d (4) XeF_4, sp^2

CCB049

COORDINATE BOND

50. Which specie does do not exists:-

(1) $A\ell F_6^{-3}$ (2) BF_4^-

(3) BeF_4^{-2} (4) $CC\ell_6^{-2}$

CCB050

51. Which of the following contains Co-ordinate and covalent bonds:-

(a) $N_2H_5^+$ (b) H_3O^+ (c) HCl (d) H_2O

Correct answer is :-

(1) a & d (2) a & b

(3) c & d (4) Only a

CCB051

52. In Co-ordinate bond, the acceptor atoms must essentially contain in its valency shell an orbital:-

(1) With paired electron

(2) With single electron

(3) With no electron

(4) With three electron

CCB052

53. The bonds present in N_2O_5 are :–

(1) Only ionic

(2) Covalent & coordinate

(3) Only covalent

(4) Covalent & ionic

CCB053

54. Dative bond is present in

(1) SO_3 (2) NH_3 (3) K_2CO_3 (4) BF_3

CCB054

55. Number of coordinate bond present in sulphuric acid are

(1) 1 (2) 2 (3) 3 (4) 4

CCB055

56. The number of coordinate bonds presents in SO_3 molecule are

(1) 1 (2) 2 (3) 3 (4) 4

CCB056

57. In the co-ordinate bond valence :

(1) electrons are equally shared by the atoms

(2) electrons of one atom are shared by two atoms

(3) hydrogen bond is formed

(4) none of the above

CCB057

58. The compound containing co-ordinate bond is :

(1) H_2SO_4 (2) O_3

(3) SO_3 (4) All of these

CCB058

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59. In the protonation of H_2O, change occurs in
(1) Hybridisation state of oxygen
(2) Shape of molecule
(3) Hybridisation and shape both
(4) None

CCB059

60. In the protonation of NH_3 molecule, following statement is true:-
(1) A covalent bond is formed
(2) Hydrogen bond is formed
(3) Hybridisation state of N is changed
(4) Shape of NH_3 molecule is changed

CCB060

DIPOLE MOMENT

61. Which statement is correct:-
(1) All the compounds having polar bonds, have dipole moment
(2) SO_2 is non-polar
(3) H_2O molecule is non-polar, having polar bonds
(4) PH_3 is polar molecule having non-polar bonds

CCB061

62. Which of the following species are symmetrical :-
(a) XeF_4 (b) XeF_6 (c) SO_2 (d) NH_3^{+2}
Correct answer is:-
(1) a and b (2) b and c
(3) c and d (4) a and d

CCB062

63. Which of the following molecule have zero dipole moment:-
(1) BF_3 (2) CH_2Cl_2 (3) NF_3 (4) SO_2

CCB063

64. The dipole moment of NH_3 is :-
(1) Less than dipole moment of NCl_3
(2) Higher than dipole moment of NCl_3
(3) Equal to the dipole moment of NCl_3
(4) None of these

CCB064

65. Which of the following species are polar :-
(a) C_6H_6 (b) XeF_2 (c) SO_2 (d) SF_4
(e) SF_6
Correct answer is :-
(1) (b) and (d) (2) (a), (b) and (e)
(3) (a) and (e) (4) (c) and (d)

CCB065

66. Which set of molecules is polar :-
(1) XeF_4, IF_7, SO_3 (2) PCl_5, C_6H_6, SF_6
(3) $SnCl_2$ SO_2 NO_2 (4) CO_2, CS_2, C_2H_6

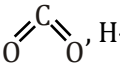
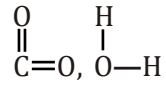
CCB066

67. An example of a polar covalent molecule is
(1) SO_3 (2) H— O—H
(3) Na^+Cl^- (4) F — F

CCB067

68. Which of the following molecules has polar character
(1) CO_2 (2) CH_4 (3) PF_5 (4) NH_3

CCB068

69. Experiment shows that H_2O has a dipole moment where as CO_2 has not. Point out the structures which illustrate these facts –
(1) O=C=O, H—O—H (2)  , H—O—H
(3) O=C=O,  (4) 

CCB069

70. Which of the following has symmetrical structure :
(1) PCl_3 (2) CH_2Cl_2 (3) $CHCl_3$ (4) CCl_4

CCB070

71. Species having zero dipole moment :-
(1) XeF_4
(2) 1,2,4 trichloro benzene
(3) SF_4
(4) CH_2Cl_2

CCB071

72. PCl_5 is non polar because :-
(1) P – Cl bond is non-polar
(2) Its dipole moment is zero
(3) P – Cl bond is polar
(4) P & Cl have equal electronegativity

CCB072

73. Which contains both polar and non-polar bonds ?
(1) NH_4Cl (2) HCN
(3) H_2O_2 (4) CH_4

CCB073

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74. The correct order of dipole moment is :
(1) $CH_4 < NF_3 < NH_3 < H_2O$
(2) $NF_3 < CH_4 < NH_3 < H_2O$
(3) $NH_3 < NF_3 < CH_4 < H_2O$
(4) $H_2O < NH_3 < NF_3 < CH_4$
CCB074

75. Which of the following molecule does not possess a permanent dipole moment :
(1) H_2S (2) SO_2 (3) SO_3^{2-} (4) CS_2
CCB075

76. Which of the following has the highest value of dipole moment :
(1) HCl (2) HF (3) HI (4) HBr
CCB076

MOLECULAR ORBITAL THEORY

77. The ion that is isoelectronic with CO and having same bond order is :-
(1) CN^- (2) O_2^+ (3) O_2^- (4) N_2^+
CCB077

78. Which of the following is paramagnetic:-
(1) O_2^- (2) CN^- (3) CO (4) NO^+
CCB078

79. Which has the bond order in fraction:-
(1) O_2 (2) HeH^+ (3) CO (4) CN
CCB079

80. Bond order in C_2^+ is:-
(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{3}{2}$ (4) 1
CCB080

81. In the following which of the two are paramagnetic
(a) N_2 (b) CO (c) B_2 (d) NO_2
Correct answer is :-
(1) a and c (2) b and c
(3) c and d (4) b and d
CCB081

82. Increasing order of bond length in NO, NO^+ and NO^- is :-
(1) $NO > NO^- > NO^+$ (2) $NO^+ < NO < NO^-$
(3) $NO < NO^+ < NO^-$ (4) $NO < NO^+ = NO^-$
CCB082

83. Bond order of Li_2 is:-
(1) 0 (2) 1 (3) 2 (4) 3
CCB083

84. In which of the following set, the value of bond order will be 2.5 :-
(1) O_2^+, NO, NO^{+2}, CN
(2) CN, NO^{+2}, CN^-, F_2
(3) O_2^+, NO^{+2}, O_2^{+2}, CN^-
(4) O_2^{-2}, O_2^-, O_2^+, O_2
CCB084

85. Which of the following ion is diamagnetic:-
(1) O_2^{-1} (2) O_2^{-2} (3) O_2 (4) O_2^{+1}
CCB085

86. When two atomic orbitals combines, it forms:-
(1) One molecular orbital
(2) Two molecular orbital
(3) Two bonding molecular orbitals
(4) Two anti bonding molecular orbitals
CCB086

87. Of the following species which has the highest bond order and shortest bond length :
NO, NO^+, NO^{2+}, NO^-
(1) NO only
(2) Bond order of NO is highest and bond length of NO^{2+} is shortest
(3) NO^+ only
(4) NO^{2+} only
CCB087

88. Which of the following has fractional bond order ?
(1) O_2^{2+} (2) O_2^{2-} (3) F_2^{2-} (4) H_2^-
CCB088

89. The bond order of CO molecule on the basis of molecular orbital theory is
(1) Zero (2) 2 (3) 3 (4) 1
CCB089

90. Which of the following group of molecules have $2\frac{1}{2}$ bond order:-
(1) N_2^{-2}, O_2^{-2}, CO (2) N_2^+, O_2^+, NO
(3) C_2^{-2}, BN, O_2 (4) CN^-, NO^+, O_2^{+2}
CCB090

91. The molecule having one unpaired electron is
(1) NO (2) CO (3) CN^- (4) O_2
CCB091

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92. On the basis of molecular orbital theory which molecules does not exist
(1) H_2 (2) He_2 (3) HeH (4) Li_2
CCB092

93. Maximum bond energy will be shown by the species
(1) O_2^+ (2) O_2 (3) O_2^- (4) O_2^{-2}
CCB093

94. Which of the following species will have the minimum bond energy
(1) N_2 (2) N_2^- (3) N_2^+ (4) N_2^{-2}
CCB094

95. Which of the following ion has not bond order of 2.5 ?
(1) O_2^- (2) O_2^+ (3) N_2^+ (4) N_2^-
CCB095

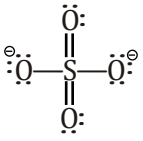
96. In a homonuclear molecule which of the following set of orbitals are degenerate ?
(1) σ_{2s} and σ_{1s} (2) π_{2px} and π_{2py}
(3) π_{2px} and σ_{2pz} (4) σ_{2pz} and π^*_{2px}
CCB096

RESONANCE AND FORMAL CHARGE

97. The weakest Cl–O bond found in.
(1) ClO_4^- (2) ClO_2^- (3) ClO^- (4) ClO_3^-
CCB097

98. In which of the following compounds resonance does not occurs:-
(a) H_2O (b) SiO_2 (c) SO_3 (d) CO_2
Correct answer is-
(1) a and d (2) a and b
(3) c and d (4) b, c and d
CCB098

99. One of the resonating structure of SO_4^{-2} is



Which set of average formal charge on oxygen and bond order is correct
(1) –0.5 and 1.5 (2) –1.5 and 3
(3) –2 and 3 (4) –1.5 and 1.5
CCB099

100. The correct order of the O–O bond length in O_2, H_2O_2 and O_3 is :–
(1) $O_3 > H_2O_2 > O_2$ (2) $O_2 > H_2O_2 > O_3$
(3) $O_2 > O_3 > H_2O_2$ (4) $H_2O_2 > O_3 > O_2$
CCB100

101. The bond length is not affected by:-
(1) Electron affinity (2) Bond order
(3) Hybridisation (4) Resonance
CCB101

102. Resonance is not shown by-
(1) C_6H_6 (2) CO_2 (3) CO_3^{2-} (4) SiO_2
CCB102

103. Higher is the bond order, greater is -
(1) Bond dissociation energy
(2) Covalent character
(3) Bond length
(4) Paramagnetism
CCB103

104. The resonance is shown by
(1) CO_3^{2-} (2) NO_3^-
(3) O_2 (4) Both (1) and (2)
CCB104

105. In PO_4^{3-}, the formal charge on each oxygen atom and the P–O bond order respectively are:
(1) –0.75, 0.6 (2) –0.75, 1.0
(3) –0.75, 1.25 (4) –3, 1.25
CCB105

106. The correct order of increasing C—O bond length of CO, CO_3^{2-}, CO_2 is :
(1) $CO_3^{2-} < CO_2 < CO$ (2) $CO_2 < CO_3^{2-} < CO$
(3) $CO < CO_3^{2-} < CO_2$ (4) $CO < CO_2 < CO_3^{2-}$
CCB106

WEAK FORCES

107. In a molecule of water following bonds are present-
(1) Two hydrogen bond
(2) Two ionic bond
(3) One covalent and one ionic bond
(4) Two covalent bond
CCB107

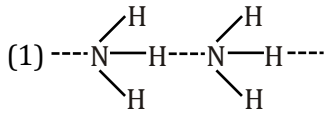
108. In dry ice the bond present between two molecules is
(1) Ionic bond (2) Covalent bond
(3) Hydrogen bond (4) Vander Waal
CCB108

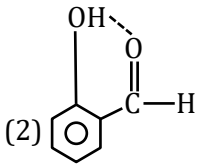
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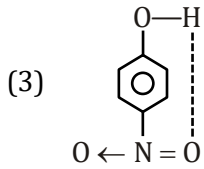
109. The hydrogen bond is strongest in:-
(1) O—H-----S (2) S—H-----O
(3) F—H-----F (4) O—H-----O

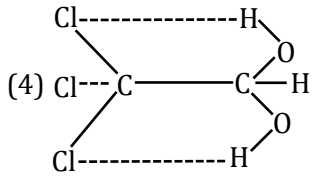
CCB109

110. In which of the following molecule, the shown hydrogen bond is not possible:-

(1) 

(2) 

(3) 

(4) 

CCB110

111. Glycerol is more viscous than glycol the reason is :-
(1) Higher molecular wt.
(2) More covalent
(3) More extent of hydrogen bonding
(4) Complex structure

CCB111

112. The correct order of volatility is:-
(1) $NH_3 < H_2O$
(2) p– nitro phenol < o– nitro phenol
(3) $CH_3OH > CH_3 – O – CH_3$
(4) HF > HCl

CCB112

113. Maximum number of H–bonding is shown by
(1) H_2O (2) H_2Se (3) H_2S (4) HF

CCB113

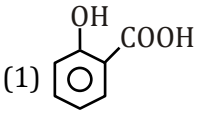
114. Which is the weakest among the following types of bonds ?
(1) Debye force (2) Metallic bond
(3) Dipole-dipole bond (4) Hydrogen bond

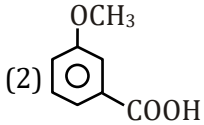
CCB114

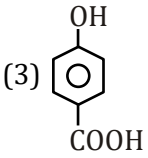
115. The crystal lattice of ice is mostly formed by:-
(1) Ionic forces
(2) Covalent bonds
(3) Inter molecular H–bonds
(4) Covalent as well as H–bonds

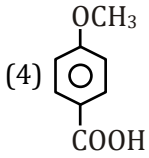
CCB115

116. Which of the following has strongest intra molecular hydrogen bonding :-

(1) 

(2) 

(3) 

(4) 

CCB116

117. Acetic acid exists as dimer in benzene due to:-
(1) Condensation reaction
(2) Hydrogen bonding
(3) Presence of carboxyl group
(4) None of the above

CCB117

118. The high boiling point of water is due to the presence of
(1) Dative bond
(2) Covalant bond
(3) Hydrogen bond
(4) Vander waals bond

CCB118

119. Maximum no. of hydrogen bonds formed by a water molecule in ice is
(1) 4 (2) 3 (3) 2 (4) 1

CCB119

120. Hydrogen bonding is formed in compounds containing hydrogen and :
(1) Highly electro-negative atoms
(2) Highly electro-positive atoms
(3) Metal atoms with d-orbitals occupied
(4) Metalloids

CCB120

121. Which of the following hydrides has the lowest b.p.
(1) H_2O (2) H_2S (3) H_2Se (4) H_2Te

CCB121

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122. The intermolecular force in hydrogen fluoride is due to
(1) Dipole-induced dipole interactions
(2) Dipole-dipole interactions
(3) Hydrogen bond
(4) Disperson interaction

CCB122

123. Strongest hydrogen bond is shown by :
(1) Water
(2) Ammonia
(3) Hydrogen fluoride
(4) Hydrogen sulphide

CCB123

124. NH_3 has abnormally high boiling point because it has :
(1) Alkaline nature
(2) Distorted shape
(3) sp^3 – Hybridisation
(4) Hydrogen bonding

CCB124

125. KF combines with HF to form KHF_2. The compound contains the species :
(1) K^+, F^- and H^+ (2) K^+, F^- and HF
(3) K^+ and $[HF_2]^-$ (4) $[KHF]^+$ and F_2

CCB125

126. In solid argon the atoms are held together
(1) By ionic bonds
(2) By hydrogen bonds
(3) By vander-waals forces
(4) None of these

CCB126

IONIC BOND

127. An atom with atomic number 20 is most likely to combine chemically with the atom whose atomic number is :
(1) 11 (2) 16 (3) 18 (4) 10

CCB127

128. Capacity of solvent to neutralise charge on ionic compound is called:-
(1) Solvation energy
(2) Dielectric constant
(3) Dipole moment
(4) Solubility

CCB128

129. The hydration of ionic compounds involves-
(1) Evolution of heat
(2) Weakening of attractive forces
(3) Dissociation into ions
(4) All

CCB129

130. Born Haber cycle is mainly used to determine
(1) Lattice energy (2) Electron affinity
(3) Ionisation energy (4) Electronegativity

CCB130

131. An ionic compound A^+ B^- is most likely to be formed when –
(1) Ionization energy of A is low
(2) Electron affinity of B is high
(3) Electron affinity of B is low
(4) Both (1) and (2)

CCB131

132. Choose the compound from each of the following pairs that has the more solubility in water :-
(a) $BeSO_4$, $BaSO_4$
(b) LiOH, CsOH
(c) AgCl, AgI
(1) $BeSO_4$, CsOH, AgI (2) $BeSO_4$, CsOH, AgCl
(3) $BaSO_4$, LiOH, AgCl (4) $BaSO_4$, LiOH, AgI

CCB132

133. The electronic configuration of metal 'M' is $1s^2\ 2s^1$ and of the non metal 'X' is $1s^2\ 2s^2\ 2p^3$. When these two elements combines, the formula of the compound will be-
(1) MX_3 (2) M_3X (3) M_2X_3 (4) M_3X_2

CCB133

134. Highest melting point would be of
(1) $AlCl_3$ (2) LiCl (3) NaCl (4) $BeCl_2$

CCB134

135. The bond in which two atoms are bonded by coulombic attraction force is called
(1) Hydrogen bond (2) Covalent bond
(3) Ionic bond (4) Coordinate bond

CCB135

136. The electronic configuration of metal M is $1s^2\ 2s^2\ 2p^6\ 3s^1$. The formula of its oxide will be :
(1) MO (2) M_2O (3) M_2O_3 (4) MO_2

CCB136

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137. Among the following which compound will show the highest lattice energy ?
(1) KF (2) NaF (3) CsF (4) RbF
CCB137

138. Consider two elements with atomic no. 37 and 53, the bond between their atoms would be :
(1) Covalent (2) Ionic
(3) Co-ordinate (4) Metallic
CCB138

139. Which of the following pairs will form the most stable ionic bond ?
(1) Na and Cl (2) Mg and F
(3) Li and F (4) Na and F
CCB139

140. For two ionic solids, CaO and KI. Identify the wrong statement among the following :
(1) Lattice energy of CaO is much larger than that of KI
(2) KI is soluble in benzene
(3) CaO has higher melting point
(4) KI has lower melting point
CCB140

141. The lattice energy of the lithium compounds is in the following order :
(1) LiF > LiCl > LiBr > LiI
(2) LiCl > LiF > LiBr > LiI
(3) LiBr > LiCl > LiF > LiI
(4) LiI > LiBr > LiCl > LiF
CCB141

142. The most covalent halide is:-
(1) AlF_3 (2) $AlCl_3$ (3) $AlBr_3$ (4) AlI_3
CCB142

143. Which bond is least covalent
(1) S – F in SF_2 (2) S – F in SF_4
(3) S – F in SF_6 (4) S – F in SF_8
CCB143

144. Correct order of covalent character of alkaline earth metal chloride in
(1) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2$
(2) $BeCl_2 < CaCl_2 < SrCl_2 < MgCl_2$
(3) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$
(4) $SrCl_2 > BeCl_2 > CaCl_2 > MgCl_2$
CCB144

145. CCl_4 is more covalent than LiCl because :
(1) There is more polarization of Cl in CCl_4
(2) There is more polarization of Cl in LiCl
(3) CCl_4 has more weight
(4) None of above
CCB145

146. Among LiCl, $BeCl_2$, BCl_3 and CCl_4, the covalent bond character follows the order :
(1) $LiCl < BeCl_2 > BCl_3 > CCl_4$
(2) $LiCl > BeCl_2 < BCl_3 < CCl_4$
(3) $LiCl < BeCl_2 < BCl_3 < CCl_4$
(4) $LiCl > BeCl_2 > BCl_3 > CCl_4$
CCB146

147. The correct order of decreasing polarisable ions is:
(1) Cl^-, Br^-, I^-, F^- (2) F^-, I^-, Br^-, Cl^-
(3) F^-, Cl^-, Br^-, I^- (4) I^-, Br^-, Cl^-, F^-
CCB147

148. On the basis of concept of ionic potential (ϕ), the tendency to form covalent bond in a group:
(1) increases
(2) decreases
(3) remains unchanged
(4) shows erratic change
CCB148

149. Ionic conductances of hydrated M^+ ions are in the order –
(1) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) > Cs^+(aq)$
(2) $Li^+(aq) > Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$
(3) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) < Cs^+(aq)$
(4) $Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$
CCB149

150. Which of the following halides has the highest melting point –
(1) NaCl (2) NaI
(3) NaBr (4) NaF
CCB150

151. Which of the following does not give an oxide on heating –
(1) $MgCO_3$ (2) Li_2CO_3
(3) $ZnCO_3$ (4) K_2CO_3
CCB151

152. $MgBr_2$ and MgI_2 are soluble in acetone because of
(1) Their ionic nature
(2) Their covalent nature
(3) Their co-ordinate nature
(4) None is correct

CCB152

153. Which of the following forms metal oxide on heating
(1) Na_2CO_3 (2) Li_2CO_3
(3) K_2SO_4 (4) $NaHCO_3$

CCB153

154. Which of the carbonate will not give oxide on heating.
(1) Na_2CO_3 (2) $CaCO_3$
(3) Li_2CO_3 (4) $BaCO_3$

CCB154

155. Which of the following compound is ionic -
(1) $AlCl_3$ (2) AlF_3
(3) $AlBr_3$ (4) $SnCl_4$

CCB155

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	4	4	3	4	3	3	3	1	3	3	2	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	3	2	2	3	2	4	2	2	3	3	1	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	2	4	3	2	2	3	3	2	2	2	3	2	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	1	1	3	4	2	3	2	1	2	2	2	4	2	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	4	1	2	4	3	2	4	3	4	1	2	3	1	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	1	1	4	3	3	2	2	1	2	2	3	4	3	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	1	2	1	4	1	2	3	2	1	4	1	4	1	4	3
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	4	4	4	3	3	3	2	1	1	4	1	2	3	1	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	2	3	3	4	3	3	2	2	4	1	4	2	2	3	3
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	2	2	2	2	2	1	4	1	3	1	3	4	2	4	4
Question	151	152	153	154	155										
Answer	4	2	2	1	2										

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

AIPMT-2014

1. Which of the following molecules has the maximum dipole moment ?

(1) CO_2 (2) CH_4 (3) NH_3 (4) NF_3

CCB195

2. Which one of the following species has plane triangular shape ?

(1) N_3^- (2) NO_3^- (3) NO_2^- (4) CO_2

CCB196

AIPMT-2015

3. The correct bond order in the following species is:-

(1) $O_2^{2+} < O_2^- < O_2^+$ (2) $O_2^+ < O_2^- < O_2^{2+}$

(3) $O_2^- < O_2^+ < O_2^{2+}$ (4) $O_2^{2+} < O_2^+ < O_2^-$

CCB197

4. Which of the following pairs of ions are isoelectronic and isostructural ?

(1) ClO_3^-, CO_3^{2-} (2) SO_3^{2-}, NO_3^-

(3) ClO_3^-, SO_3^{2-} (4) CO_3^{2-}, SO_3^{2-}

CCB198

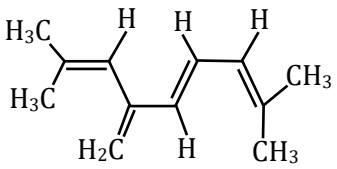
5. Which of the following options represents the correct bond order ?

(1) $O_2^- < O_2 < O_2^+$ (2) $O_2^- > O_2 < O_2^+$

(3) $O_2^- < O_2 > O_2^+$ (4) $O_2^- > O_2 > O_2^+$

CCB199

6. The total number of π-bond electrons in the following structure is :-



(1) 8 (2) 12 (3) 16 (4) 4

CCB200

7. Solubility of the alkaline earth's metal sulphates in water decreases in the sequence:-

(1) Ca > Sr > Ba > Mg (2) Sr > Ca > Mg > Ba

(3) Ba > Mg > Sr > Ca (4) Mg > Ca > Sr > Ba

CCB201

8. Maximum bond angle at nitrogen is present in which of the following ?

(1) NO_2^- (2) NO_2^+ (3) NO_3^- (4) NO_2

CCB202

9. Which of the following species contains equal number of σ- and π- bonds :-

(1) XeO_4 (2) $(CN)_2$

(3) $CH_2(CN)_2$ (4) HCO_3^-

CCB203

Re-AIPMT-2015

10. On heating which of the following releases CO_2 most easily ?

(1) $MgCO_3$ (2) $CaCO_3$

(3) K_2CO_3 (4) Na_2CO_3

CCB204

11. Decreasing order of stability of O_2, O_2^-, O_2^+ and O_2^{2-} is :-

(1) $O_2 > O_2^+ > O_2^{2-} > O_2^-$

(2) $O_2^- > O_2^{2-} > O_2^+ > O_2$

(3) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(4) $O_2^{2-} > O_2^- > O_2 > O_2^+$

CCB205

12. In which of the following pairs, both the species are not isostructural ?

(1) NH_3, PH_3

(2) XeF_4, XeO_4

(3) $SiCl_4$, PCl_4^+

(4) Diamond, silicon carbide

CCB206

13. The variation of the boiling points of the hydrogen halides is in the order

HF > HI > HBr > HCl.

What explains the higher boiling point of hydrogen fluoride ?

(1) The bond energy of HF molecules is greater than in other hydrogen halides

(2) The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule

(3) The electronegativity of fluorine is much higher than for other elements in the group.

(4) There is strong hydrogen bonding between HF molecules

CCB207

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NEET-I 2016

14. Consider the molecules CH_4, NH_3 and H_2O. Which of the given statements is false ?
(1) The H –C–H bond angle in CH_4, the H–N–H bond angle in NH_3, and the H–O–H bond angle in H_2O are all greater than 90°
(2) The H–O–H bond angle in H_2O is larger than the H–C–H bond angle in CH_4.
(3) The H–O–H bond angle in H_2O is smaller than the H–N–H bond angle in NH_3.
(4) The H–C–H bond angle in CH_4 is larger than the H–N–H bond angle in NH_3.

CCB208

15. Which one of the following orders is correct for the bond dissociation enthalpy of halogen molecules?
(1) $I_2 > Br_2 > Cl_2 > F_2$ (2) $Cl_2 > Br_2 > F_2 > I_2$
(3) $Br_2 > I_2 > F_2 > Cl_2$ (4) $F_2 > Cl_2 > Br_2 > I_2$

CCB209

16. Predict the correct order of repulsion among the following :-
(1) lone pair- lone pair > lone pair - bond pair > bond pair - bond pair
(2) lone pair - lone pair > bond pair - bond pair > lone pair - bond pair
(3) bond pair - bond pair > lone pair - bond pair > lone pair - lone pair
(4) lone pair - bond pair > bond pair - bond pair > lone pair - lone pair

CCB210

17. Match the compounds given in **column-I** with the hybridisation and shape given in **column- II** and mark the **correct** option.

Column-I		Column-II	
(a)	XeF_6	(i)	Distorted octahedral
(b)	XeO_3	(ii)	Square planer
(c)	$XeOF_4$	(iii)	Pyramidal
(d)	XeF_4	(iv)	Square pyramidal

Code :-

	(a)	(b)	(c)	(d)
(1)	(i)	(iii)	(iv)	(ii)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(iv)	(i)	(ii)	(iii)

CCB211

NEET-II 2016

18. The correct geometry and hybridization for XeF_4 are:
(1) Planar triangle, sp^3d^3
(2) Square planar, sp^3d^2
(3) Octahedral, sp^3d^2
(4) Trigonal bipyramidal, sp^3d

CCB212

19. Among the following which one is a wrong statement?
(1) SeF_4 and CH_4 have same shape
(2) I_3^+ has bent geometry
(3) PH_5 and $BiCl_5$ do not exist
(4) $p\pi$-$d\pi$ bonds are present in SO_2

CCB213

20. The hybridizations of atomic orbitals of nitrogen in NO_2^+, NO_3^- and NH_4^+ respectively are
(1) sp, sp^2 and sp^3 (2) sp^2, sp and sp^3
(3) sp, sp^3 and sp^2 (4) sp^2, sp^3 and sp

CCB214

21. Which of the following fluoro-compounds is most likely to behave as a Lewis base ?
(1) CF_4 (2) SiF_4 (3) BF_3 (4) PF_3

CCB215

22. Which of the following pairs of ions is isoelectronic and isostructural ?
(1) SO_3^{2-}, NO_3^- (2) ClO_3^-, SO_3^{2-}
(3) CO_3^{2-}, NO_3^- (4) ClO_3^-, CO_3^{2-}

CCB216

NEET(UG) 2017

23. Ionic mobility of which of the following alkali metal ions is lowest when aqueous solution of their salts are put under an electric field ?
(1) K (2) Rb (3) Li (4) Na

CCB217

24. Match the interhalogen compounds of **column-I** with the geometry in **column-II** and assign the correct code.

Column-I		Column-II	
(a)	XX'	(i)	T-shape
(b)	XX'_3	(ii)	Pentagonal bipyramidal
(c)	XX'_5	(iii)	Linear
(d)	XX'_7	(iv)	Square-pyramidal
		(v)	Tetrahedral

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Code :

	(a)	(b)	(c)	(d)
(1)	(iii)	(i)	(iv)	(ii)
(2)	(v)	(iv)	(iii)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(iii)	(iv)	(i)	(ii)

CCB218

25. Which of the following pairs of compounds is isoelectronic and isostructural ?

(1) TeI_2, XeF_2 (2) IBr_2^-, XeF_2

(3) IF_3, XeF_2 (4) $BeCl_2$, XeF_2

CCB219

26. The species, having bond angles of 120° is :-

(1) ClF_3 (2) NCl_3 (3) BCl_3 (4) PH_3

CCB220

27. Which of the following pairs of species have the same bond order ?

(1) O_2, NO^+ (2) CN^-, CO

(3) N_2, O_2^- (4) CO, NO

CCB221

NEET(UG) 2018

28. Among CaH_2, BeH_2, BaH_2, the order of ionic character is

(1) $BeH_2 < CaH_2 < BaH_2$

(2) $CaH_2 < BeH_2 < BaH_2$

(3) $BeH_2 < BaH_2 < CaH_2$

(4) $BaH_2 < BeH_2 < CaH_2$

CCB222

29. Magnesium reacts with an element (X) to form an ionic compound. If the ground state electronic configuration of (X) is $1s^2\ 2s^2\ 2p^3$, the simplest formula for this compound is

(1) Mg_2X_3 (2) MgX_2

(3) Mg_2X (4) Mg_3X_2

CCB223

30. Consider the following species :

CN^+, CN^-, NO and CN

Which one of these will have the highest bond order?

(1) NO (2) CN^- (3) CN^+ (4) CN

CCB224

31. Which one of the following elements is unable to form MF_6^{3-} ion ?

(1) Ga (2) Al (3) B (4) In

CCB225

32. In the structure of ClF_3, the number of lone pairs of electrons on central atom 'Cl' is

(1) one (2) two (3) four (4) three

CCB226

NEET(UG) 2019

33. Which of the following diatomic molecular species has only π bonds according to Molecular Orbital Theory ?

(1) O_2 (2) N_2 (3) C_2 (4) Be_2

CCB227

34. Which of the following species is **not** stable ?

(1) $[SiF_6]^{2-}$ (2) $[GeCl_6]^{2-}$

(3) $[Sn(OH)_6]^{2-}$ (4) $[SiCl_6]^{2-}$

CCB228

35. Identify the **incorrect** statement related to PCl_5 from the following :-

(1) Three equatorial P–Cl bonds make an angle of 120° with each other

(2) Two axial P–Cl bonds make an angle of 180° with each other

(3) Axial P–Cl bonds are longer than equatorial P–Cl bonds

(4) PCl_5 molecule is non–reactive

CCB229

36. Match the Xenon compounds in **Column–I** with its structure in **Column–II** and assign the **correct** code

Column–I		Column–II	
(a)	XeF_4	(i)	pyramidal
(b)	XeF_6	(ii)	square planar
(c)	$XeOF_4$	(iii)	distorted octahedral
(d)	XeO_3	(iv)	square pyramidal

Code :

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(ii)	(iii)	(i)	(iv)
(4)	(iii)	(iv)	(i)	(ii)

CCB230

37. Which is the **correct** thermal stability order for H_2E (E=O, S, Se, Te and Po) ?

(1) $H_2S < H_2O < H_2Se < H_2Te < H_2Po$

(2) $H_2O < H_2S < H_2Se < H_2Te < H_2Po$

(3) $H_2Po < H_2Te < H_2Se < H_2S < H_2O$

(4) $H_2Se < H_2Te < H_2Po < H_2O < H_2S$

CCB231

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NEET(UG) 2019 (ODISHA)

38. Which of the following is paramagnetic ?
(1) N_2 (2) H_2 (3) Li_2 (4) O_2
CCB232

39. Which of the following is the correct order of dipole moment ?
(1) $NH_3 < BF_3 < NF_3 < H_2O$
(2) $BF_3 < NF_3 < NH_3 < H_2O$
(3) $BF_3 < NH_3 < NF_3 < H_2O$
(4) $H_2O < NF_3 < NH_3 < BF_3$
CCB233

40. The number of hydrogen bonded water molecule(s) associated with $CuSO_4.5H_2O$ is :-
(1) 3 (2) 1 (3) 2 (4) 5
CCB234

NEET(UG) 2020

41. Identify a molecule which does not exist.
(1) O_2 (2) He_2 (3) Li_2 (4) C_2
CCB235

42. Which of the following set of molecules will have zero dipole moment ?
(1) Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene
(2) Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene
(3) Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene
(4) Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene
CCB236

NEET(UG) 2020(COVID-19)

43. Among the compounds shown below which one revealed a linear structure ?
(1) NO_2 (2) HOCl
(3) O_3 (4) N_2O
CCB237

44. Match the compounds of Xe in **column-I** with the molecular structure in **column-II**.

Column–I		Column–II	
(a)	XeF_2	(i)	Square planar
(b)	XeF_4	(ii)	Linear
(c)	XeO_3	(iii)	Square pyramidal
(d)	$XeOF_4$	(iv)	Pyramidal

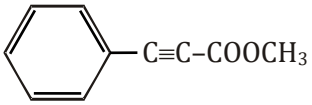
(1) (a)-(ii) (b)-(i) (c)-(iii) (d)-(iv)
(2) (a)-(ii) (b)-(iv) (c)-(iii) (d)-(i)
(3) (a)-(ii) (b)-(iii) (c)-(i) (d)-(iv)
(4) (a)-(ii) (b)-(i) (c)-(iv) (d)-(iii)
CCB238

45. Identify the wrongly matched pair.

Molecule	**Shape or geometry of molecule**
(1) PCl_5	Trigonal planar
(2) SF_6	Octahedral
(3) $BeCl_2$	Linear
(4) NH_3	Trigonal pyramidal

CCB239

46. How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound ?



(1) 7, 5 (2) 8, 6 (3) 7, 6 (4) 8, 5
CCB240

NEET(UG) 2021

47. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are:
(1) sp^3 and 4 (2) sp^3 and 6
(3) sp^2 and 6 (4) sp^2 and 8
CCB241

48. Match **List - I** with **List - II**.

List-I	List-II
(a) PCl_5	(i) Square pyramidal
(b) SF_6	(ii) Trigonal planar
(c) BrF_5	(iii) Octahedral
(d) BF_3	(iv) Trigonal bipyramidal

Choose the **correct** answer from the options given below.
(1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
CCB242

49. Which of the following molecules is non-polar in nature ?
(1) $POCl_3$ (2) CH_2O (3) $SbCl_5$ (4) NO_2
CCB243

50. From the following pairs of ions which one is not an iso-electronic pair ?
(1) O^{2-}, F^- (2) Na^+, Mg^{2+}
(3) Mn^{2+}, Fe^{3+} (4) Fe^{2+}, Mn^{2+}
CCB244

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51. The Cl–C–Cl bond angle in 1, 1, 2, 2-tetrachloroethene and tetrachloromethane respectively are

(1) 120° and 109.5°

(2) 90° and 109.5°

(3) 109.5° and 90°

(4) 109.5° and 120°

CCB245

52. Match the columns.

List I		List II	
a.	$IF_2^{\ominus}$	i.	sp
b.	HCN	ii.	sp^3d
c.	PCl_4^+	iii.	sp^3d^2
d.	XeF_4	iv.	sp^3

(1) a – i, b – iv, c – ii, d – iii

(2) a – ii, b – i, c – iv, d – iii

(3) a – iii, b – ii, c – i, d – iv

(4) a – iv, b – iii, c – ii, d – i

CCB246

53. Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them.

(1) Noble gases are sparingly soluble in water.

(2) Noble gases have very high melting and boiling points.

(3) Noble gases have weak dispersion forces.

(4) Noble gases have large positive values of electron gain enthalpy.

CCB247

54. The correct sequence of bond enthalpy of 'C-X' bond is

(1) $CH_3–F < CH_3–Cl < CH_3–Br < CH_3–I$

(2) $CH_3–F > CH_3–Cl > CH_3–Br > CH_3–I$

(3) $CH_3–F < CH_3–Cl > CH_3–Br > CH_3–I$

(4) $CH_3–Cl > CH_3–F > CH_3–Br > CH_3–I$

CCB248

55. Which of the following has greater bond length?

(1) P – O (2) S – O

(3) Cl – O (4) O = O

CCB370

NEET(UG) 2022

56. Match List-I with List-II

List- I (Hydrides)	List-II (Nature)
(a) MgH_2	(i) Electron precise
(b) GeH_4	(ii) Electron deficient
(c) B_2H_6	(iii) Electron rich
(d) HF	(iv) Ionic

Choose the correct answer from the options given below :

(1) (a)-(iii), (b) - (i), (c) - (ii), (d)- (iv)

(2) (a)-(i), (b) - (ii), (c) - (iv), (d)- (iii)

(3) (a)-(ii), (b) - (iii), (c) - (iv), (d)- (i)

(4) (a) -(iv), (b) - (i), (c) - (ii), (d)- (iii)

CCB249

57. Given below are two statements:

Statement-I : The boiling points of the following hydrides of group 16 elements increases in the order -

$H_2O < H_2S < H_2Se < H_2Te$.

Statement-II : The boiling points of these hydrides increase with increase in molar mass.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and Statement-II are correct

CCB250

58. Which amongst following is incorrect statement ?

(1) C_2 molecule has four electrons in its two degenerate π molecular orbitals.

(2) H_2^+ ion has one electron

(3) O_2^+ ion has diamagnetic.

(4) The bond orders of O_2^+, O_2, O_2^- and O_2^{2-} are 2.5, 2, 1.5 and 1, respectively.

CCB251

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59. Amongst the following which one will have maximum 'lone pair-lone pair' electron repulsions ?

(1) IF_5 (2) SF_4

(3) XeF_2 (4) ClF_3

CCB252

60. The correct order of bond angles in the following compounds/species is:

(1) $H_2O < NH_3 < NH_4^+ < CO_2$

(2) $H_2O < NH_4^+ < NH_3 < CO_2$

(3) $H_2O < NH_4^+ = NH_3 < CO_2$

(4) $CO_2 < NH_3 < H_2O < NH_4^+$

CCB253

61. Match List-I with List-II :

	List-I (Molecules)		List-II (Shape)
(a)	NH_3	(i)	Square pyramidal
(b)	ClF_3	(ii)	Trigonal bipyramidal
(c)	PCl_5	(iii)	Trigonal pyramidal
(d)	BrF_5	(iv)	T-shape

Choose the correct answer from the options given below :

(1) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)

(2) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(3) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

(4) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

CCB254

62. Flourine is a stronger oxidising agent than chlorine because :

(a) F–F bond has a low enthalpy of dissociation.

(b) Flouride ion (F^-) has high hydration enthalpy.

(c) Electron gain enthalpy of flourine is less negative than chlorine.

(d) Flourine has a very small size.

Choose the most appropriate answer from the options given :

(1) (a) and (b) only

(2) (a) and (c) only

(3) (a) and (d) only

(4) (b) and (c) only

CCB255

NEET(UG) 2023

63. Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is

NH_3, $AlCl_3$, $BeCl_2$, CCl_4, PCl_5 :

(1) 2 (2) 4 (3) 1 (4) 3

CCB302

64. The correct order of energies of molecular orbitals of N_2 molecule, is

(1) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

(2) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$

(3) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$

(4) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

CCB295

65. Intermolecular forces are forces of attraction and repulsion between interacting particles that will include:

A. dipole - dipole forces.

B. dipole - induced dipole forces

C. hydrogen bonding

D. covalent bonding

E. dispersion forces

Choose the most appropriate answer from the options given below :

(1) A, B, C, D are correct

(2) A, B, C, E are correct

(3) A, C, D, E are correct

(4) B, C, D, E are correct

CCB296

66. Amongst the given options which of the following molecules/ion acts as a Lewis acid?

(1) H_2O (2) BF_3 (3) OH^- (4) NH_3

CCB297

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67. The stability of Cu^{2+} is more than Cu^{+} salts in aqueous solution due to -
(1) enthalpy of atomization.
(2) hydration energy.
(3) second ionisation enthalpy.
(4) first ionisation enthalpy.

CCB298

68. The correct order of dipole moments for molecules NH_3, H_2S, CH_4 and HF, is:
(1) $CH_4 > H_2S > NH_3 > HF$
(2) $H_2S > NH_3 > HF > CH_4$
(3) $NH_3 > HF > CH_4 > H_2S$
(4) $HF > NH_3 > H_2S > CH_4$

CCB299

69. Which one of the following represents all isoelectronic species ?
(1) Na^+, Cl^-, O^-, NO^+
(2) N_2O, N_2O_4, NO^+, NO
(3) Na^+, Mg^{2+}, O^-, F^-
(4) Ca^{2+}, Ar, K^+, Cl^-

CCB300

70. Which one of the following statements is **incorrect** related to Molecular Orbital Theory?
(1) The π^* antibonding molecular orbital has a node between the nuclei.
(2) In the formation of bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other.
(3) Molecular orbitals obtained from $2P_x$ and $2P_y$ orbitals are symmetrical around the bond axis.
(4) A π-bonding molecular orbital has larger electron density above and below the internuclear axis.

CCB301

NEET(UG) 2024

71. Given below are two statements :
Statement-I : The boiling point of hydrides of Group-16 elements follow the order
$H_2O > H_2Te > H_2Se > H_2S$.
Statement-II : On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O, it has higher boiling point.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both statement-I and Statement-II are true.
(2) Both statement-I and Statement-II are false.
(3) Statement-I is the true but Statement-II is false.
(4) Statement-I is false but Statement-II is true.

CCB303

72. Match List I with List II.

List-I (Compound)		List-II (Shape/geometry)	
(A)	NH_3	I.	Trigonal Pyramidal
(B)	BrF_5	II.	Square Planar
(C)	XeF_4	III.	Octahedral
(D)	SF_6	IV.	Square Pyramidal

Choose the correct answer from the options given below :
(1) A-I, B-IV, C-II, D-III
(2) A-II, B-IV, C-III, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-III, C-IV, D-I

CCB304

73. Identify the **correct** answer.
(1) Three resonance structures can be drawn for ozone
(2) BF_3 has non-zero dipole moment
(3) Dipole moment of NF_3 is greater than that of NH_3
(4) Three canonical forms can be drawn for CO_3^{2-} ion.

CCB305

NEET(UG) 2024 (Re-Examination)

74. Which of the following molecules has "NON ZERO" dipole moment values ?
(1) CCl_4
(2) HI
(3) CO_2
(4) BF_3

CCB306

75. Match List-I with List-II

List-I (Atom/Molecules)		List-II (Property)	
(A)	Nitrogen atom	I.	Paramagnetic
(B)	Fluorine molecule	II.	Most reactive element in group 18
(C)	Oxygen molecule	III.	Element with highest enthalpy in group 15
(D)	Xenon atom	IV.	Strongest oxidising agent

Identify the **correct** answer from the options given below :

(1) A-III, B-I, C-IV, D-II
(2) A-I, B-IV, C-III, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-IV, C-I, D-II

CCB307

76. Identify the **incorrect** statement about PCl_5.

(1) PCl_5 possesses two different Cl–P–Cl bond angles.
(2) All five P–Cl bonds are identical in length.
(3) PCl_5 exhibits sp^3d hybridisation.
(4) PCl_5 consists of five P–Cl (sigma) bonds.

CCB308

77. Match **List-I** with **List-II**

List-I Molecule		List-II Bond enthalpy (kJ mol^{-1})	
(A)	HCl	I.	435.8
(B)	N_2	II.	498
(C)	H_2	III.	946.0
(D)	O_2	IV.	431.0

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-IV, B-III, C-II, D-I
(4) A-IV, B-III, C-I, D-II

CCB309

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**3**	**3**	**1**	**1**	**4**	**2**	**1**	**1**	**3**	**2**	**4**	**2**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**1**	**2**	**1**	**1**	**4**	**2,3**	**3**	**1**	**2**	**3**	**2**	**1**	**4**	**2**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**3**	**2**	**3**	**4**	**4**	**2**	**3**	**4**	**2**	**2**	**2**	**1**	**4**	**4**	**1**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**3**	**3**	**1**	**3**	**4**	**1**	**2**	**2**	**2**	**1**	**4**	**1**	**3**	**3**	**1**
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	**2**	**1**	**4**	**4**	**2**	**2**	**2**	**4**	**4**	**3**	**1**	**1**	**4**	**2**	**4**
Question	76	77													
Answer	**2**	**4**													

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EXERCISE – III (Analytical Questions) Master Your Understanding

1. Given below are two statements:
Statement-I : In the periodic table, the highly electronegative Halogens and the highly electropositive alkali metals are separated by the noble gases.
Statement-II : Formation of an anion from a halogen atom and formation of cation from an alkali metal are stabilized by electro static attraction.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I is incorrect and II is correct.
(2) Statement-I and II both are correct.
(3) Statement-I is correct while II is incorrect.
(4) Both Statement-I and II are incorrect.
CCB256

2. Given below are two statements:
Statement-I : Each covalent bond is formed as a result of unequal sharing of an electron pair between the atoms.
Statement-II : Two or three electron pairs of a multiple bond are treated as single super pair.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and II both are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct while II is incorrect.
(4) Statement-I is incorrect and II is correct.
CCB257

3. Given below are two statements:
Statement-I : Lewis theory explain the shape of a molecules.
Statement-II : It explain the relative stability and energy of the molecules.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and II both are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is incorrect and II is correct.
(4) Statement-I is correct while II is incorrect.
CCB258

4. Given below are two statements:
Statement-I : Most ionic compounds have cations from metallic elements and anion from non-metallic elements.
Statement-II : The ammonium ion NH_4^+ (made up off two non-metallic elements) is not an exception.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and II both are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct while II is incorrect.
(4) Statement-I is incorrect and II is correct.
CCB259

5. Given below are two statements:
Statement-I : Lattice energy of an ionic solid is defined as the energy required to completes separate one mole of ionic solid in gaseous constituent ions.
Statement-II : Stability of an ionic compound depend on its lattice energy.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I is incorrect and II is correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct while II is incorrect.
(4) Statement-I and II both are correct.
CCB260

6. Given below are two statements:
Statement-I : Among the two O–H bonds in H_2O molecule, the energy required to break the first O–H bond and other O–H bond is the same.
Statement-II : In H_2O molecule, electronic environment around oxygen is the same even after breakage of one O–H bond.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I is correct while II is incorrect.
(2) Both Statement-I and II are incorrect.
(3) Statement-I and II both are correct.
(4) Statement-I is incorrect and II is correct.
CCB261

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7. Given below are two statements:

Statement-I : sp Hybridisation is also known as diagonal hybridisation.

Statement-II : Two sp hybrid orbital are oriented in opposite direction forming an angle of 180°.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and II both are correct.

(2) Statement-I is correct while II is incorrect.

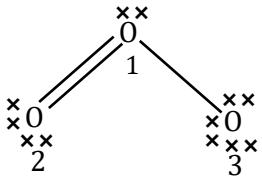
(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect and II is correct.

CCB262

8. Given below are two statements:

Statement-I : The correct Lewis structure of O_3 may be drawn as :



Statement-II : In O_3,the formal charge on atom 1, 2 and 3 are +1, 0 and –1 respectively.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I is incorrect and II is correct.

(2) Statement-I is correct while II is incorrect.

(3) Statement-I and II both are correct.

(4) Both Statement-I and II are incorrect.

CCB263

9. Given below are two statements:

Statement-I : NH_3 molecule is less polar then NF_3 molecule.

Statement-II : 'F' is more electronegative then H element.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I and II both are correct.

(3) Statement-I is correct while II is incorrect.

(4) Statement-I is incorrect and II is correct.

CCB264

10. Given below are two statements:

Statement-I : Promotion of electron is an essential condition prior to Hybridisation.

Statement-II : Half filled, full filled and vacant orbital can participate in the process of Hybridisation.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I is incorrect and II is correct.

(2) Statement-I and II both are correct.

(3) Statement-I is correct while II is incorrect.

(4) Both Statement-I and II are incorrect.

CCB265

11. Given below are two statements:

Statement-I : C_2 molecule has four electrons in it's two degenerate 'π' molecular orbitals

Statement-II : O^{+}_{2} ion is diamagnetic nature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and II both are correct.

(2) Statement-I is incorrect and II is correct.

(3) Statement-I is correct while II is incorrect.

(4) Both Statement-I and II are incorrect.

CCB266

12. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : LiF is sparingly soluble in water.

Reason (R) : LiF has a very high lattice energy.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).

(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) & (R) both are incorrect.

CCB267

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13. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Ionic compounds tend to be no-volatile.
Reason (R) : Intermolecular forces in these compounds are weak.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) (A) is correct but (R) is incorrect.
(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(3) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(4) (A) & (R) both are incorrect.
CCB268

14. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Bond length of (P–Cl) bonds in gaseous PCl_5 and solid PCl_5 are not equal.
Reason (R) : In solid state two PCl_5 molecules are associated.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) (A) is correct but (R) is incorrect.
(2) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(3) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(4) (A) & (R) both are incorrect.
CCB269

15. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : B_2 molecule is diamagnetic.
Reason (R) : The highest occupied molecular orbital in B_2 molecule is of σ-type.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is incorrect.
(3) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(4) (A) & (R) both are incorrect.
CCB270

16. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : SiF_6^{2-} is known but $SiCl_6^{2-}$ is not known.
Reason (R) : Size of F is small and its lone pair of electron is tightly bonded with orbitals.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) (A) is correct but (R) is incorrect.
(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(3) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(4) (A) & (R) both are incorrect.
CCB271

17. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : ClF_3 has bent T-shape structure.
Reason (R) : It has two lone pair arrange at 180° angle.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) (A) & (R) both are incorrect.
(2) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(3) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(4) (A) is correct but (R) is incorrect.
CCB272

18. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Boiling point of o-nitro phenol is greater than point of p-nitro phenol.
Reason (R) : In o-nitro phenol has only inter molecular H-bonding.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) (A) & (R) both are incorrect.
CCB273

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19. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : AgCl is more covalent than NaCl.
Reason (R) : Polarising Power of Ag^+ is greater than Na^+ ion.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) (A) & (R) both are incorrect.
(2) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(3) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(4) (A) is correct but (R) is incorrect.

CCB274

20. Match the column

Column-I (Molecule/Ion)		Column-II (Hybridisation)	
(i)	SF_4	(a)	sp^3d^2
(ii)	IF_5	(b)	d^2sp^3
(iii)	NO_2^+	(c)	sp^3d
(iv)	NH_4^+	(d)	sp^3
		(e)	sp

(1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-a, iii-e, iv-d
(3) i-c, ii-d, iii-e, iv-b (4) i-b, ii-c, iii-d, iv-e

CCB275

21. Match the column

Column-I (Molecule)		Column-II (Bond order)	
(i)	NO	(a)	1.5
(ii)	CO	(b)	2.0
(iii)	O_2^-	(c)	2.5
(iv)	O_2	(d)	3.0

(1) i-a, ii-b, iii-c, iv-d (2) i-b, ii-c, iii-d, iv-a
(3) i-c, ii-d, iii-a, iv-b (4) i-d, ii-a, iii-b, iv-c

CCB276

22. Match the column

Column-I (Ion)		Column-II (Shape)	
(i)	$ICl_2^\ominus$	(a)	Linear
(ii)	$BrF_2^\oplus$	(b)	Pyramidal
(iii)	$ClF_4^\ominus$	(c)	Tetrahedral
(iv)	$AlH_4^\ominus$	(d)	Square planer
		(e)	V shape

(1) i-a, ii-b, iii-c, iv-d (2) i-a, ii-e, iii-d, iv-c
(3) i-c, ii-d, iii-e, iv-b (4) i-b, ii-c, iii-d, iv-e

CCB277

23. Match the column

Column-I (Order)		Column-II (Properties)	
(i)	$H_2O<H_2S<H_2Se<H_2Te$	(a)	Thermal stability
(ii)	HCl<HBr<HI<Hf	(b)	Acidic strength
(iii)	$NH_3>PH_3>AsH_3>SbH_3$	(c)	Boiling point
(iv)	$CH_4>SiH_4>GeH_4>SnH_4>PbH_4$	(d)	Lewis basic strength

(1) i-b, ii-c, iii-d, iv-a (2) i-a, ii-b, iii-d, iv-c
(3) i-c, ii-d, iii-a, iv-b (4) i-d, ii-a, iii-b, iv-c

CCB278

24. Match the column

Column-I (Type's of solid)		Column-II (Example)	
(i)	Covalent solid	(a)	Fe(s)
(ii)	Ionic solid	(b)	Diamond
(iii)	Metallic solid	(c)	LiF
(iv)	Molecular solid	(d)	Dry ice

(1) i-a, ii-b, iii-c, iv-d (2) i-a, ii-b, iii-d, iv-c
(3) i-b, ii-c, iii-a, iv-d (4) i-b, ii-c, iii-d, iv-a

CCB279

25. Match the column

Column-I (Molecule/Ion)		Column-II (Bond angle)	
(i)	NO_2^+	(a)	120°
(ii)	NO_2^-	(b)	118°
(iii)	NO_2	(c)	180°
(iv)	NO_3^-	(d)	134°

(1) i-c, ii-b, iii-d, iv-a (2) i-a, ii-b, iii-c, iv-d
(3) i-c, ii-d, iii-a, iv-b (4) i-d, ii-a, iii-b, iv-c

CCB280

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26. Match the column

Column-I		Column-II	
(i)	ClF_3	(a)	Non-planar and non-polar
(ii)	XeF_4	(b)	Non-planar & polar
(iii)	NF_3	(c)	Planar and polar
(iv)	PCl_5	(d)	Planar and non-polar

(1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-d, iii-b, iv-a
(3) i-c, ii-d, iii-a, iv-b (4) i-d, ii-a, iii-b, iv-c

CCB281

27. Match the column

Column-I [Molecule]		Column-II [Type's of 'π' bond]	
(i)	SO_3	(a)	no π bond
(ii)	XeO_4	(b)	only pπ -pπ
(iii)	C_3O_2	(c)	only pπ -dπ
(iv)	SiO_2	(d)	both pπ -pπ and pπ -dπ

(1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-d, iii-b, iv-a
(3) i-d, ii-c, iii-b, iv-a (4) i-d, ii-a, iii-b, iv-c

CCB282

28. Match the column

Column-I		Column-II	
(i)	HF(s)	(a)	Linear polymeric structure
(ii)	H_3BO_3(s)	(b)	Zig-Zag structure
(iii)	H_2O(s)	(c)	2-D sheet structure
(iv)	NH_3(s)	(d)	Open cage like structure

(1) i-a, ii-b, iii-c, iv-d (2) i-b, ii-c, iii-d, iv-a
(3) i-d, ii-c, iii-a, iv-b (4) i-d, ii-a, iii-b, iv-c

CCB283

29. Match the column

Column-I		Column-II	
(i)	$Li^+(aq) > Na^+(aq) > K^+(aq)$	(a)	ionic mobility in aqueous solution
(ii)	$F^- < Cl^- < Br^- < H^-$	(b)	Hydrated radius of ions
(iii)	$Na^+ < Mg^{+2} < Al^{+3}$	(c)	Polarisability
(iv)	$Ca^{+2}(aq) < Sr^{+2}(aq) < Ba^{+2}(aq)$	(d)	Polarising power

(1) i-b, ii-a, iii-d, iv-c (2) i-b, ii-c, iii-d, iv-a
(3) i-c, ii-d, iii-a, iv-b (4) i-d, ii-a, iii-b, iv-c

CCB284

30. Which of the following molecule(s) is polar ?
(A) $CHCl_3$ (B) CF_4 (C) O_2F_2 (D) C_2F_2
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB285

31. Species having same bond order are –
(A) N_2 (B) N_2^+ (C) N_2^{-2} (D) N_2^-
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB286

32. In which of the following molecule/ion hybridisation of central atom is sp^3d.
(A) $I_3^{\ominus}$ (B) $I_3^{\oplus}$ (C) ClF_3 (D) IF_5
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB287

33. Which of the following species do not exist ?
(A) PH_5 (B) KHF_2
(C) BF_5^{-2} (D) BeF_4^{-2}
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB288

34. Correct order of melting point is/are
(A) LiCl > NaCl
(B) $SnCl_2 > SnCl_4$
(C) $MgF_2 > AlF_3$
(D) Diamond > Al_2O_3
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB289

35. Which of the following combination leads to formation of π bond (Molecular axis is z axis)
(A) $p_y + p_y$ (B) $d_{yz} + d_{yz}$
(C) $p_x + d_{zx}$ (D) $d_{xy} + d_{yz}$
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB290

36. Which of the following molecule exhibit intramolecular hydrogen bonding?
(A) chloral hydrate (B) p-nitrophenol
(C) salicydaldihyde (D) acetic acid
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB291

37. Which of the following gives only SO_2 gas on thermal decomposition.
(A) $ZnSO_4$ (B) $CaSO_3$
(C) $Fe_2(SO_4)_3$ (D) $BaSO_3$
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB292

38. Which of the following order of solubility in water is correct?
(A) AgF > AgCl > AgBr > AgI
(B) $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
(C) LiOH > NaOH > KOH > RbOH > CsOH
(D) $SnCl_4 > SnCl_2$
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB293

39. Which of the following gives paramagnetic gas on heating?
(A) $LiNO_3$ (B) $CaCO_3$
(C) $NaNO_3$ (D) NH_4NO_2
(1) If A, B and C options are correct.
(2) If A & B both are correct.
(3) If B & D both are correct.
(4) If A & C both are correct.

CCB294

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**4**	**2**	**3**	**4**	**2**	**1**	**3**	**4**	**1**	**3**	**1**	**1**	**3**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**2**	**4**	**4**	**2**	**2**	**3**	**2**	**1**	**3**	**1**	**2**	**3**	**2**	**2**	**4**
Question	31	32	33	34	35	36	37	38	39						
Answer	**3**	**4**	**4**	**3**	**1**	**4**	**3**	**2**	**4**						

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Chemical Bonding and Molecular Structure - Chemical Bonding

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (3)

Minimum potential energy and maximum attraction favours bond formation and stability of molecule.

2. Ans. (3)

According to old concept of group number.

Ne has 8 valence electrons but group number is Zero.

Na → Valence electron = 1 Group no. ⇒ I–A

N → Valence electron = 5 Group no. ⇒ V–A

3. Ans. (4)

H can only complete its duplet not octet. While Na, F and S can complete their octet by loosing or sharing of electrons.

4. Ans. (4)

Valency is defined as combining capacity of an element either in ground state or excited state.

5. Ans. (3)

Oxygen → Valency = 2 Group number ⇒ VI–A

While in other element valency is equal to group number.

Na → Valency = 1 Group number ⇒ I–A

Al → Valency = 3 Group number ⇒ III–A

C → Valency = 4 Group number ⇒ IV–A

6. Ans. (4)

6–electrons

For example :- 

In triple bond, there is sharing of 6 electrons.

7. Ans. (3)

p – p > s – p > s – s

2p – 2p > 2p – 2s > 2s – 2s

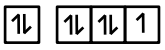
Directional orbitals (2p) overlap more than non-directional (2s) orbitals.

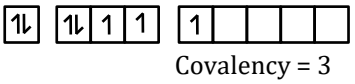
Bond strength ∝ Extent of overlapping

8. Ans. (3)

In first excitation state

Cl — $3s^2$ $3p^5$

Ground state  Covalency = 1 example → ClF

Excited state–I 

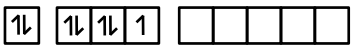
3s 3p 3d

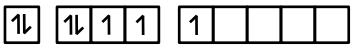
Covalency = 3

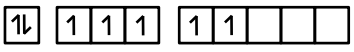
example → ClF_3

9. Ans. (3)

Iodine ⇒ $5s^2$ $5p^5$ $5d^0$

Ground state  Covalency = 1

Excited state - I  Covalency = 3

Excited state - II  Covalency = 5

10. Ans. (1)

P and S exhibits variable covalency.

N, O, F cannot show variable covalency due to unavailability of vacant orbital.

P, S and Cl can show variable covalency due to presence of vacant d-orbital.

11. Ans. (3)

A sigma bond is formed by the overlapping of s – s, p – p and s – p orbitals along internuclear axis by Co–axial overlapping.

12. Ans. (3)

Hybrid orbitals always form sigma bond by overlapping because they are more directional than unhybrid orbitals or atomic orbitals.

13. Ans. (2)

σ–bond is formed by co-axial overlapping so stronger than π–bond which is formed by side way overlapping.

14. Ans. (2)

Covalent bond is formed by overlapping of half filled orbitals.

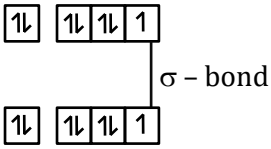
15. Ans. (1)

Covalent bond formed between two electronegative atoms that is non-metals.

16. Ans. (4)

Chlorine molecule

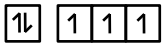
$Cl \Rightarrow 3s^2\ 3p^5$



17. Ans. (2)

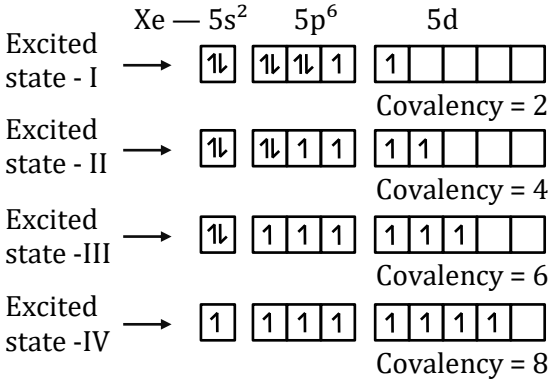
N contain no vacant d orbital.

$N \rightarrow 2s^2\ \ 2p^3$



18. Ans. (3)

XeF_5 is not possible because Xe can show 2, 4, 6, 8 covalencies.



Xe can show covalency – 2, 4, 6 & 8.

19. Ans. (2)

$C \equiv C$	<	$C = C$	<	$C - C$
Triple Bond		Double Bond		Single Bond

20. Ans. (2)

Covalent bond is a directional bond because it forms in a particular direction.

i.e. x–axis/y–axis/z–axis.

21. Ans. (3)

Phosphorus does not form diatomic molecules because 3pπ–3pπ bond is not stable.

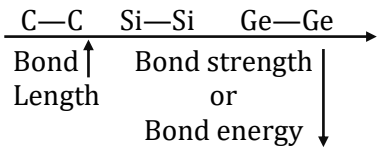
22. Ans. (2)

Because 3pπ–3pπ bond is a weak bond. So phosphorus exists as P_4 molecule and satisfies its covalency by forming 3 single covalent bonds.

23. Ans. (4)

PI_5 does not exist due to steric hindrance as side atom Iodine has a large size.

24. Ans. (2)



25. Ans. (2)

d – orbitals involved in sp^3d hybridisation is dz^2.

26. Ans. (3)

$\frac{3}{4}$ p – character

In sp^3 hybridisation $\Rightarrow \frac{1}{4}$ s – character

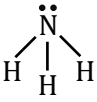
and $\frac{3}{4}$ p – character

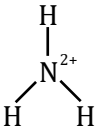
27. Ans. (3)

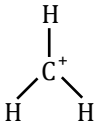
NH_4^+, H_3O^+, OF_2

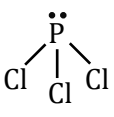
NH_4^+	H_3O^+	OF_2
(4σ)	$(3\sigma + 1L.P.)$	$(2\sigma + 2L.P.)$
sp^3	sp^3	sp^3
Tetrahedral Shape	Pyramidal Shape	Bent shape

28. Ans. (1)

(a) $NH_3 \rightarrow$  $sp^3 \rightarrow$ Pyramidal (non-planar)

(b) $NH_3^{2+} \rightarrow$  $\rightarrow sp^2$ Trigonal planar

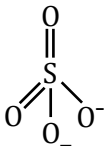
(c) CH_3^+  $\rightarrow sp^2$ Trigonal planar

(d) PCl_3  $\rightarrow sp^3$ Pyramidal

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29. Ans. (4)

$SO_4^{2-} \rightarrow sp^3$–Hybridisation $\rightarrow$ 

$\rightarrow$ Tetrahedral

30. Ans. (3)

$\overset{+}{O}=N=O$ (sp) $\quad O=C=O$ (sp) $\quad F-\underset{sp^3d}{Xe}-F \Rightarrow$ All are Linear

31. Ans. (3)

$Br_2O > Cl_2O > OF_2$

When Hybridisation, central atom and Lone pair on central atom is same then

bond angle $\propto$ size of side atom

32. Ans. (1)

4 bond pairs and 2 Lone pairs

Hybrid Orbital = 4 B.P. + 2L.P. = 6

Hybridisation $\Rightarrow sp^3d^2$

Electronic Geometry $\Rightarrow$ Octahedral

Shape $\Rightarrow$ Square planar

33. Ans. (2)

$XeF_2 \Rightarrow$ 3 Lone pair + 2 sigma bond = 5 sp^3d

shape $\Rightarrow$ linear

$XeO_2F_2 \Rightarrow$ 1 Lone pair + 4 sigma bond = 5 sp^3d

shape $\Rightarrow$ see–saw

34. Ans. (4)

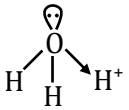
$IF_4^+ \Rightarrow$ 4 sigma-bond + 1 Lone pair = 5 sp^3d

shape $\Rightarrow$ see–saw or Distorted Tetrahedral

35. Ans. (3)

$\left.\begin{matrix}4\text{ Bond Pair}\\1\text{ Lone Pair}\end{matrix}\right] 5 \rightarrow sp^3d \qquad SF_4$

36. Ans. (2)

$H_3O^+ \Rightarrow$ 

$\left.\begin{matrix}3\,\sigma-\text{bond}\\1\text{ Lone Pair}\end{matrix}\right] sp^3 \rightarrow$ Shape – Pyramidal

37. Ans. (2)

$XeO_3F_2 \Rightarrow \left.\begin{matrix}5\,\sigma-\text{bond}\\0\text{ Lone Pair}\end{matrix}\right] 5 \Rightarrow sp^3d$ Trigonal Bipyramidal.

38. Ans. (3)

$\left.\begin{matrix}ClO_2^- \Rightarrow 2\sigma+2\text{ Lone Pair}=4\\ClO_3^- \Rightarrow 3\sigma+1\text{ Lone Pair}=4\\ClO_4^- \Rightarrow 4\sigma+0\text{ Lone pair}=4\end{matrix}\right]$ All are sp^3

39. Ans. (3)

$XeF_4 \Rightarrow 4\sigma$ + 2 Lone Pair $\Rightarrow 6\ sp^3d^2$

Electronic Geometry $\rightarrow$ Octahedral

Shape $\rightarrow$ square planar

40. Ans. (2)

Hybridization involves mixing up of atomic orbitals.

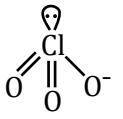
41. Ans. (2)

$I_3^- \Rightarrow$ 2 sigma bond + 3 lone pair = 5 sp^3d

Electronic geometry $\Rightarrow$ Trigonal pyramidal

Shape $\rightarrow$ Linear

42. Ans. (2)

$ClO_3^- \Rightarrow$  $\left.\begin{matrix}3\,\sigma-\text{bond}\\1\text{ Lone Pair}\end{matrix}\right] \rightarrow 4\ sp^3$

Shape – Pyramidal

43. Ans. (3)

$AsF_5 \Rightarrow$ 5 sigma bond + zero Lone pair = 5 Hybrid orbitals

sp^3d Hybridisation $\Rightarrow s + p_x + p_y + p_z + d_{z^2}$

44. Ans. (2)

(a) $CH_3^+ \Rightarrow sp^2$ (3 sigma bond + Zero Lone pair)

(b) $CH_3^- \Rightarrow sp^3$ (3 sigma bond + 1 Lone pair)

(c) $ClO_3^- \Rightarrow sp^3$ (3 sigma bond + 1 Lone pair)

(d) $SO_3 \Rightarrow sp^2$ (3 sigma bond + Zero Lone pair)

45. Ans. (3)

SF_6

$SF_6 \Rightarrow 6\sigma$ + Zero Lone pair $\rightarrow 6\ sp^3d^2$

When central atom has zero Lone pair and all side atoms are same then dipole moment will be zero.

46. Ans. (4)

$\left.\begin{matrix}ClF_3\\ICl_3\\BrF_3\end{matrix}\right]$ All three have 3σ and 2 Lone pair

$3\sigma+2\text{ L.P.}=5\ sp^3d$

So all 3 have T-shape

47. Ans. (1)

XeF_2 and IF_2^- Both have 2 sigma bond and 3 Lone pair. So both have linear shape (sp^3d)

48. Ans. (1)

(a) $XeF_4 \Rightarrow 4\sigma + 2$ L.P. $\Rightarrow sp^3d^2 \rightarrow$ square planar shape.

(b) $XeF_6 \Rightarrow 6\sigma + 1$ L.P. $\Rightarrow sp^3d^3 \rightarrow$ Distorted Octahedral shape.

(c) $XeO_3 \Rightarrow 3\sigma + 1$ L.P. $\Rightarrow sp^3 \rightarrow$ Pyramidal shape.

(d) $XeOF_2 \Rightarrow 3\sigma + 2$ L.P. $\Rightarrow sp^3d \rightarrow$ 'T'-shape.

49. Ans. (3)

$XeF_2 \Rightarrow 2\sigma + 3$ L.P. $\Rightarrow 5\ sp^3d$

$XeF_4 \Rightarrow 4\sigma + 2$ L.P. $\Rightarrow 6\ sp^3d^2$

50. Ans. (4)

CCl_6^{2-} does not exist.

C does not have vacant orbital.

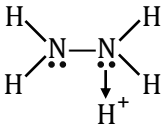
$AlF_6^{3-} \Rightarrow AlF_3 + 3F^-$

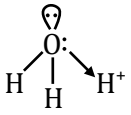
$BF_4^- \Rightarrow BF_3 + F^-$

$BeF_4^{2-} \Rightarrow BeF_2 + 2F^-$

51. Ans. (2)

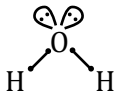
(a) & (b) contains Co-ordinate & covalent bonds.

(a) 

(b) H_3O^+ 

(c) & (d) does not contains co-ordinate bond

(c) H—Cl

(d) 

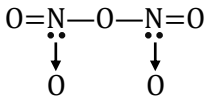
52. Ans. (3)

With no electron

Conditions of co-ordinate bond :- Donor must have lone pair Acceptor must have vacant orbital.

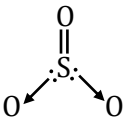
53. Ans. (2)

Covalent & Co-ordinate

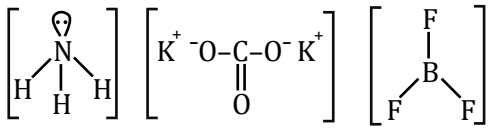
N_2O_5 

54. Ans. (1)

SO_3

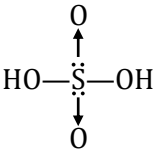


(2), (3) & (4) does not contain dative bond.



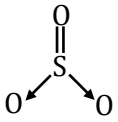
55. Ans. (2)

2 coordinate bonds present in H_2SO_4

H_2SO_4 

56. Ans. (2)

2 coordinate bons present in SO_3.

SO_3 

57. Ans. (2)

Electrons of one atom are shared by 2 atoms.

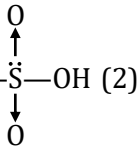
A : ⟶ B

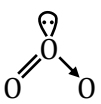
Donor Acceptor

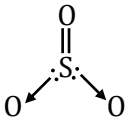
Lone Pair ($2e^-$) of donor atom (A) is shared with acceptor atom (B).

58. Ans. (4)

All of these

(1) H_2SO_4 

(2) O_3 

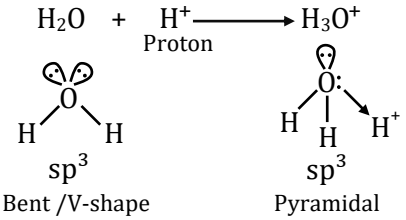
(3) SO_3 

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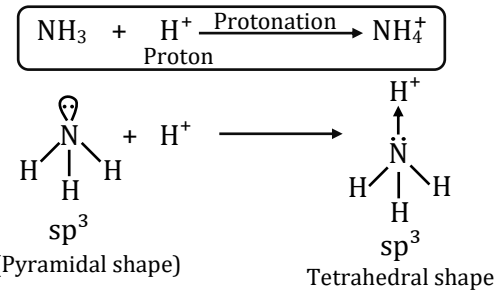
59. Ans. (2)

Shape of molecule.



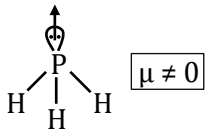
60. Ans. (4)

Shape of NH_3 molecule is changed.

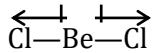


61. Ans. (4)

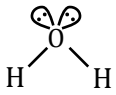
PH_3 is polar molecule having non-polar bonds. In PH_3, P–H bond is non-polar because ΔEN = Zero. But PH_3 molecule is polar due to one lone pair present on P atom.



Option– 1, 2 & 3 are incorrect. explanations are following.

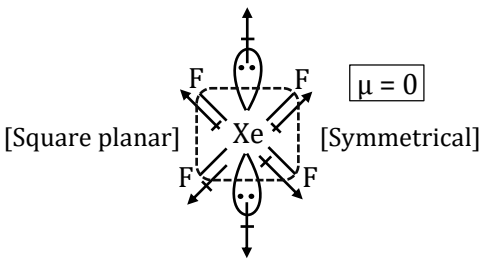
(1)  Bond is polar but molecule is non-polar.

(2)  is a polar molecule due to 1 Lone pair present on S atom.

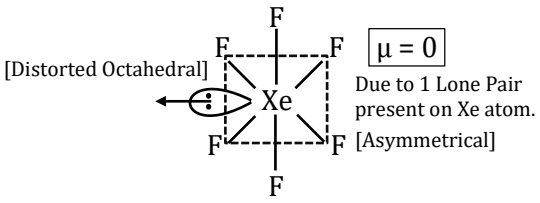
(3)  is a polar molecule because no vectors are cancelled.

62. Ans. (4)

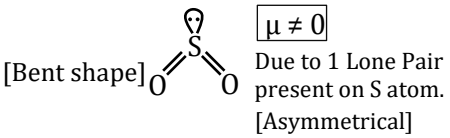
(a) XeF_4



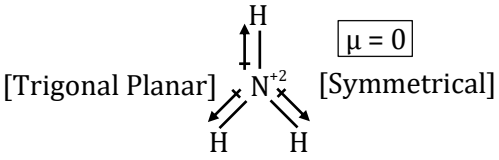
(b) XeF_6



(c) SO_2



(d) NH_3^{+2}

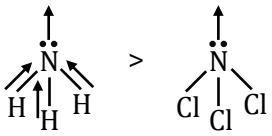


63. Ans. (1)

BF_3

If central atom (B) has no lone pair and all side atoms (F) are same then dipole moment of a molecule will be zero.

64. Ans. (2)



Atomic dipole and bond dipole are in same direction so addition of dipole moment in NH_3.

65. Ans. (4)

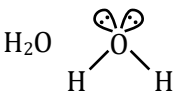
If Central atom has only 1 Lone pair then molecule will be polar always like SO_2 and SF_4.

66. Ans. (3)

$SnCl_2$, SO_2, NO_2

If Central atom has 1 Lone pair or molecule is bent shaped then molecule will be polar.

67. Ans. (2)



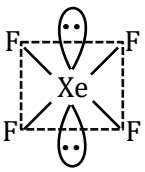
No vectors of dipole moment are cancelled in H_2O. molecule having bent shaped structure are polar.

68. Ans. (4)
If a molecule has only 1 Lone pair on central atom then it will be polar.

69. Ans. (3)
Because CO_2 is linear and Bond Dipole are cancelled in CO_2 so Non-polar. $\overleftarrow{O=C=O}$
and in H_2O bond dipole are not cancelled due to its bent shape so polar.

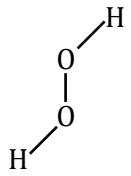
70. Ans. (4)
If a central atom has zero lone pair and all side atoms are same in a molecule then it will be non-polar and symmetrical.

71. Ans. (1)
$XeF_4 \Rightarrow sp^3d^2$ [2 Lone Pair at axis at 180°]
[Square Planar Shape]
[2 F atom at equatorial at 180°]
[2 F atom at equatorial at 180°]



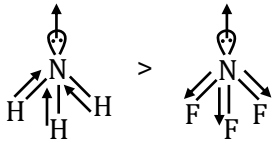
72. Ans. (2)
If a central atom has zero lone pair and all side atoms are same, then molecule will have zero dipole moment and non-polar.

73. Ans. (3)
H_2O_2



O—O bond have ΔEN = 0 so non-polar bond.
O—H bond have ΔEN ≠ 0 so polar bond.

74. Ans. (1)
$CH_4 < NF_3 < NH_3 < H_2O$
⇒ CH_4 have μ = 0 as C has no lone pair and all side atom are same.
⇒ NH_3 have more dipole than NF_3 because in NH_3 all atomic dipole and bond dipole are in same direction but in NF_3 directions of bond dipole and atom dipole opposite.



⇒ In H_2O atomic dipole and bond dipole are in same direction and magnitude on bond dipole in H_2O for O—H bond is more than NH_3 for N—H bond.

75. Ans. (4)
In CS_2, central atom (C) has zero lone pair and all side atoms (S) are same so does not possess a permanent dipole (μ = 0).

76. Ans. (2)
In diatomic molecule
Dipole moment (μ) ∝ ΔEN
HF has maximum ΔEN so maximum dipole moment.

77. Ans. (1)
CN^- and CO are isoelectronic ($14e^-$) so having same bond order (3).

78. Ans. (1)
$O_2^- \Rightarrow 17e^-$ Bond order = 1.5 Paramagnetic
Species having fractional and order are paramagnetic.

79. Ans. (4)
$CN \Rightarrow 13e^-$
$\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2 < \pi2p_x^2 = \pi2p_y^2 < \sigma2p_z^1$

$$\text{Bond order} = \frac{\text{B.M.O } e^-s - \text{A.B.M.O. } e^-s}{2}$$

$$= \frac{9-4}{2} = \frac{5}{2} = 2.5$$

80. Ans. (3)
C_2^+ = 11 electrons
$\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2 < \pi2p_x^2 = \pi2p_y^1$

$$\text{Bond order} = \frac{\text{B.M.O. } e^- - \text{A.B.M.O } e^-}{2}$$

$$= \frac{7-4}{2} = \frac{3}{2}$$

81. Ans. (3)
N_2, CO = $14e^-$ Diamagnetic (All paired electrons)
$B_2 = 10e^- \Rightarrow$ Paramagnetic
$\left[\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2 < \pi2p_x^1 = \pi2p_y^1\right]$
(2 Unpaired electron)
$NO_2 \Rightarrow$ All polyatomic odd electron species are paramagnetic.

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82. Ans. (2)

$$\text{Bond Length} \propto \frac{1}{\text{bond order}}$$

Bond order → NO^+ > NO > NO^-
(3) (2.5) (2)

Bond length → NO^+ < NO < NO^-

83. Ans. (2)

$Li_2 \Rightarrow \quad 6e^- \ \sigma1s^2 < \sigma^*1s^2 < \sigma2s^2$

$$\text{Bond order} = \frac{\text{B.M.O. } e^- - \text{A.B.M.O. } e^-}{2}$$

$$= \frac{4-2}{2} = 1$$

84. Ans. (1)

$O_2^+ \Rightarrow 15e^-$

$NO \Rightarrow 15e^-$

$NO^{+2} \Rightarrow 13e^-$

$CN \Rightarrow 13e^-$

Species having total e^- 13 and 15 have bond order = 2.5 and paramagnetic.

85. Ans. (2)

$O_2^{-2} \Rightarrow 18e^-$

O_2^{2-} have no unpaired e^- so diamagnetic.

86. Ans. (2)

According to molecular orbital theory when 2 atomic orbitals combines, it forms 2 molecular orbitals.

87. Ans. (3)

$NO \rightarrow 15e^- \Rightarrow$ Bond order (B.O.) = 2.5

$NO^+ \rightarrow 14e^- \Rightarrow$ bond order = 3.0

$NO^{+2} \rightarrow 13e^- \Rightarrow$ bond order = 2.5

$NO^- \rightarrow 16e \Rightarrow$ bond order = 2.0

$$\boxed{\text{Bond Length} \propto \frac{1}{\text{Bond order}}}$$

So NO^+ has maximum bond order i.e. 3.0 so minimum bond length.

88. Ans. (4)

$O_2^{2+} \Rightarrow 14e^- \Rightarrow$ bond order $\Rightarrow$ 3.0

$O_2^{2-} \Rightarrow 18e^- \Rightarrow$ bond order $\Rightarrow$ 1.0

$F_2^{-2} \Rightarrow 20e^- \Rightarrow$ bond order $\Rightarrow$ Zero

$H_2^- \Rightarrow 3e^- \Rightarrow$ bond order $\Rightarrow$ 0.5

89. Ans. (3)

$CO \Rightarrow 14e^- \Rightarrow$ bond order = 3

90. Ans. (2)

$N_2^+ \Rightarrow 13e^-$

$O_2^+ \Rightarrow 15e^-$

$NO \Rightarrow 15e^-$

All species have bond order = 2.5

91. Ans. (1)

$NO \Rightarrow 15e^-$ (Bond Order = 2.5)

1 unpaired e^-

$\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2 < \sigma2p_z^2 < \pi2p_x^2$

$= \pi2p_y^2 < \pi2p_x^1$

92. Ans. (2)

$He_2 \Rightarrow 4e^-$ Bond order = Zero

If in a species total electrons are 4, 8 and 20 then bond order will zero and species will not exist.

93. Ans. (1)

$O_2^+ \Rightarrow 15e^-$ Bond Order = 2.5

$O_2 \Rightarrow 16e^-$ Bond Order = 2.0

$O_2^- \Rightarrow 17e^-$ Bond Order = 1.5

$O_2^{-2} \Rightarrow 18e^-$ Bond Order = 1.0

Bond energy $\propto$ Bond order

So O_2^+ has maximum bond energy.

94. Ans. (4)

	Bond order
$N_2 \rightarrow 14e^- \Rightarrow$	3.0
$N_2^- \rightarrow 15e^- \Rightarrow$	2.5
$N_2^+ \rightarrow 13e^- \Rightarrow$	2.5
$N_2^{-2} \rightarrow 16e^- \Rightarrow$	2.0

Bond energy $\propto$ Bond order

So N_2^- has minimum bond order so minimum bond energy.

95. Ans. (1)

	Bond order
$O_2^- \rightarrow 17e^-$	1.5
$O_2^+ \rightarrow 15e^-$	2.5
$N_2^+ \rightarrow 13e^-$	2.5
$N_2^- \rightarrow 15e^-$	2.5

96. Ans. (2)

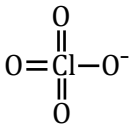
$\pi 2p_x = \pi 2p_y$

Both have same energy so called as degenerate orbitals.

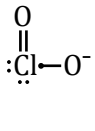
97. Ans. (3)

Bond Order (B.O.)

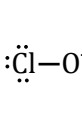
$= \frac{\text{Total number of covalent bonds}}{\text{Number of resonating structure}}$



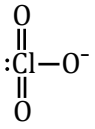
B.O. $= \frac{7}{4} = 1.75$



B.O. $= \frac{3}{2} = 1.5$



B.O. = 1



B.O. $= \frac{5}{3} = 1.66$

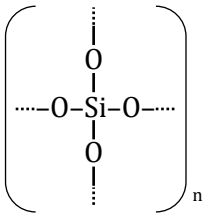
Bond strength $\propto$ Bond Order

So Cl–O$^-$ has weakest bond strength.

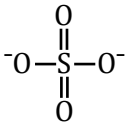
98. Ans. (2)

There is no π-bond present in H_2O and SiO_2 so no resonance will occur.





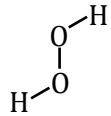
99. Ans. (1)



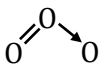
Average formal charge $= \frac{-2}{4} = -0.5$

Average bond order $= \frac{6}{4} = 1.5$

100. Ans. (4)



B.O. = 1



B.O. = 1.5



B.O. = 2.0

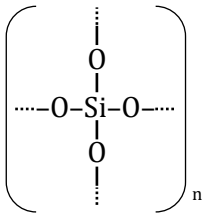
Bond length $\propto \frac{1}{\text{Bond order}}$

Bond length : $O_2 < O_3 < H_2O_2$

101. Ans. (1)

Bond length is affected by bond order, hybridisation and resonance but not affected by electron affinity as it is a property of isolated atom.

102. Ans. (4)

$(SiO_2)_n$ 

$(SiO_2)_n$ has no π-bond

So do not show resonance.

103. Ans. (1)

Bond dissociation energy $\propto$ Bond order

104. Ans. (4)

CO_3^{2-} and NO_3^- show resonance by shifting of π-bond but in O_2 shifting of π-bond is not possible.

105. Ans. (3)

 Average formal charge

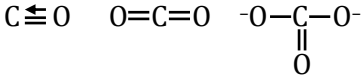
$= \frac{\text{Total charge on side atoms}}{\text{Total side atoms participating resonance}}$

$= \frac{-3}{4} = -0.75$

Average bond order

$= \frac{\text{Total covalent bonds}}{\text{Total resonating structures}} = \frac{5}{4} = 1.25$

106. Ans. (4)



Bond order : 3.0 2.0 1.33

Bond length : CO < CO_2 < CO_3^{2-}

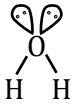
Bond length $\propto \frac{1}{\text{Bond order}}$

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107. Ans. (4)

In a molecular of water, 2 covalent bonds are present.



108. Ans. (4)

Dry Ice is CO_2 (Solid)

O=C=O is a non-polar molecule.

So force between CO_2 molecules is Vander wall force (London Force)

109. Ans. (3)

F—H·····F

F has maximum EN and minimum size so hydrogen bond strength is maximum among all.

110. Ans. (3)

Hydrogen bond is not possible at para position in para nitro phenol as distance is more and hydrogen bond is a weak bond.

111. Ans. (3)



Glycerol **Glycol**

More OH group

More Hydrogen bonding

More viscosity

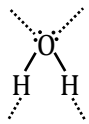
112. Ans. (2)

Inter molecular hydrogen bonding More → Boiling Point More → Volatility Less

p-nitro phenol shows intermolecular H-bonding while o-nitro phenol shows intramolecular H-bonding

113. Ans. (1)

H_2O can form 4 H-bonds



114. Ans. (1)

Debye force is weakest bond among all given bonds.

Metallic force > Hydrogen bond

> Keeson Force (Dipole-Dipole)

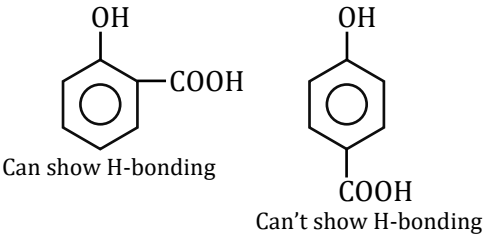
> Debye Force (Dipole-Induced Dipole)

115. Ans. (4)

Crystal lattice of ice is formed by covalent bonds of water molecule and H-bonds formed by water molecule with other water molecules.

116. Ans. (1)

Strongest Intra molecular Hydrogen bonding is formed by molecule having groups at ortho position not by molecule having at para position.

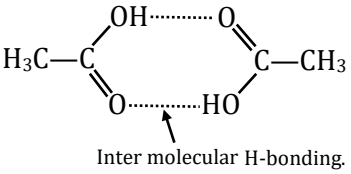


Can show H-bonding

Can't show H-bonding

117. Ans. (2)

Acetic acid exists as dimer in benzene due to H-bonding.

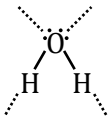


Inter molecular H-bonding.

118. Ans. (3)

The high boiling point of water is due to the presence of hydrogen bonds among water molecules.

119. Ans. (1)



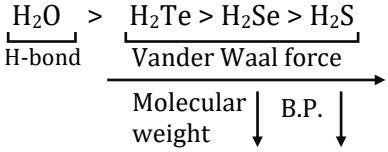
Maximum number of hydrogen bonds formed by water molecule in ice is four.

120. Ans. (1)

Hydrogen bonding is formed in compounds containing hydrogen and highly electronegative atom like F, O and N.

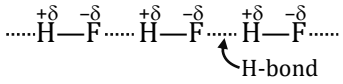
121. Ans. (2)

Order of B.P. ⇒



122. Ans. (3)

Hydrogen bond



123. Ans. (3)

Strength of H–bond $\Rightarrow NH_3 < H_2O < HF$

EN $\Rightarrow N < O < F$

Size $\Rightarrow N > O > F$

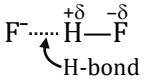
H_2S do not form H–bond because S has low EN and large size.

124. Ans. (4)

NH_3 forms H–bond so more association resulting increase in boiling point.

125. Ans. (3)

$K^+HF_2^- \Rightarrow K^+ + HF_2^-$



126. Ans. (3)

Argon in solid state are held together by Vander Waals force.



127. Ans. (2)

Atomic number 20 is a metal i.e. calcium so it will most likely to combine with a non-metal i.e. atomic number 16 which is a non-metal named sulphur.

Z = 11 $\Rightarrow$ Metal

Z = 18 $\Rightarrow$ Noble Gas

Z = 10 $\Rightarrow$ Noble Gas

128. Ans. (2)

Capacity of solvent to neutralise charge on ionic compound is called Dielectric constant.

129. Ans. (4)

The hydration of ionic compounds involves:-

Evolution of Heat

Weakening of attractive forces

Dissociation into ions

130. Ans. (1)

Born Haber cycle is used to determine lattice energy.

131. Ans. (4)

Ionization energy of metal (A) should be low and electron affinity of non-metal (B) should be high.

132. Ans. (2)

(I) Solubility for groups

If common ion is large like SO_4^{2-} then solubility decreases down the group.

If common ion is small like OH^- then solubility increases down the group.

(II) Solubility of d & p-block compounds

$$\text{Solubility} \propto \frac{1}{\text{Polarisation (Covalent character)}}$$

133. Ans. (2)

Metal (M) $1s^2\ 2s^1 \Rightarrow M^{+1}$

Non-metal (X) $1s^2\ 2s^2\ 2p^3 \Rightarrow X^{-3}$

$M^{+1} \times X^{-3} \Rightarrow M_3X$

134. Ans. (3)

$$\text{M.P.} \propto \frac{1}{\text{Polarisation}}$$

135. Ans. (3)

Ionic Bond is also called as Coulombic attraction force or electrostatic force.

136. Ans. (2)

Metal (M) $1s^2\ 2s^2\ 2p^6\ 3s^1 \Rightarrow M^{+1}$

Oxide (Non-metal) $M^{+1} \times O^{2-} \Rightarrow M_2O$

137. Ans. (2)

$$\text{Lattice energy} \propto \frac{\text{charge on ions}}{\text{size of ions}}$$

NaF KF RbF CsF

Size of cation↑ L.E.↓

138. Ans. (2)

Atomic number (37) is a metal rubidium and atomic number (53) is a non-metal Iodine so the bond between them is ionic bond.

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139. Ans. (2)

Stability of Ionic bond $\propto$ Lattice Energy

$\propto \frac{\text{charge of ions}}{\text{size of ions}}$

$Na^+ Cl^-$ $Mg^{2+} F^{-1}$

$Li^+ F^-$ $Na^+ F^{-1}$

140. Ans. (2)

→ CaO and KI both are ionic solids

→ Lattice energy of $\overset{+2}{Ca}\overset{-2}{O}$ is more than $\overset{+1}{K}\overset{-1}{I}$ so CaO has higher melting point.

→ Both are ionic compounds so soluble in water but not soluble in benzene.

141. Ans. (1)

LiF LiCl LiBr LiI

Size of anion ↑ Lattice Energy ↓

Lattice energy $\propto \frac{\text{charge on ions}}{\text{size of ions}}$

142. Ans. (4)

Covalent character $\propto$ Polarisation

AlF_3 $AlCl_3$ $AlBr_3$ AlI_3

Size of anion ↑ Polarisation ↑ Covalent character ↑

143. Ans. (1)

$\overset{+2}{SF_2}$ $\overset{+4}{SF_4}$ $\overset{+6}{SF_6}$

+ve charge on cation (s) ↑ Polarisation ↑ Covalent character ↑

So SF_2 is least covalent.

144. Ans. (3)

$BeCl_2$ $MgCl_2$ $CaCl_2$ $SrCl_2$

Size of cation ↑ Polarisation ↓ Covalent character ↓

145. Ans. (1)

$\overset{+4}{C}Cl_4 > \overset{+1}{Li}Cl$ (Covalent character)

146. Ans. (3)

$\overset{+1}{LiCl}$ $\overset{+2}{BeCl_2}$ $\overset{+3}{BCl_3}$ $\overset{+4}{CCl_4}$

+ve charge on cation ↑ Polarisation ↑ Covalent character ↑

147. Ans. (4)

I^- Br^- Cl^- F^-

Size of anion ↓ Polarisability ↓

148. Ans. (2)

Ionic potential (ϕ) $\propto \frac{\text{charge of cation}}{\text{size of cation}}$

In a group

Size of cation ↑ ϕ ↓ Covalent nature ↓

149. Ans. (4)

Hydration tendency $\propto$ (ϕ) $\propto \frac{\text{charge of cation}}{\text{size of cation}}$

Hydration ↑ Ionic conductance ↓

Li^+_{aq} Na^+_{aq} K^+_{aq} Rb^+_{aq} Cs^+_{aq}

Hydration ↓ Ionic conductance ↓

150. Ans. (4)

Melting point $\propto$ Lattice energy

NaF NaCl NaBr NaI

Size of anion ↑ Lattice energy ↓ Melting point ↓

151. Ans. (4)

Carbonates of group I–A do not decompose except Li_2CO_3. All I–A (Na, K, Rb, Cs) have low ϕ values.

152. Ans. (2)

Covalent compounds are more soluble in non–polar solvents like acetone so $MgBr_2$ and MgI_2 are soluble in acetone as they are covalent in nature due to polarisation.

153. Ans. (2)

Li_2CO_3 decompose on heating due to high ϕ of Li.

154. Ans. (1)

Na_2CO_3 will not decompose on heating due to low ϕ value of Na.

155. Ans. (2)

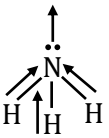
AlF_3 is ionic due to small size of anion (F^-).

Exercise-II (Previous Year Questions)

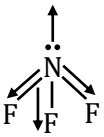
1. Ans. (3)

CO_2, CH_4 Dipole moment = zero

If Central atom has no Lone pair and all side atoms are same.



Addition of Vectors



Substraction of Vectors

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2. Ans. (2)

N_3^- ⟶ Linear (2σ)

NO_3^- ⟶ Trigonal planar (3σ)

NO_2^- ⟶ Bent (2σ + 1 Lone pair)

CO_2 ⟶ Linear (2σ)

3. Ans. (3)

	Bond order
O_2^- ⇒ 17e⁻	⟶ 1.5
O_2^+ ⇒ 15e⁻	⟶ 2.5
O_2^{+2} ⇒ 14e⁻	⟶ 3.0

4. Ans. (3)

ClO_3^-, SO_3^{2-}] 42 electrons, 3σ + 1 Lone pair ⇒ sp^3

Electronic Geometry → Tetrahedral

Shape-Pyramidal

5. Ans. (1)

	Bond order
O_2^- ⇒ 17e⁻	⟶ 1.5
O_2 ⇒ 16e⁻	⟶ 2.0
O_2^+ ⇒ 15e⁻	⟶ 2.5

6. Ans. (1)

4π-bonds in structure ⇒ 8π-electrons in structure

7. Ans. (4)

Mg Ca Sr Ba(sulphates) →

Large common ion (SO_4^{2-})

Down the group solubility ↓

8. Ans. (2)

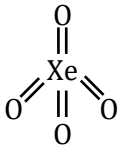
NO_2^- ⟶ sp^2

NO_3^- ⟶ sp^2

NO_2 ⟶ sp^2

NO_2^+ ⟶ sp (Maximum Bond Angle)

9. Ans. (1)



4σ bonds

4π bonds

10. Ans. (1)

$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

Mg has high ionic potential (ϕ) so less thermal stable.

11. Ans. (3)

	Electrons	Bond Order
O_2	16	2.0
O_2^-	17	1.5
O_2^+	15	2.5
O_2^{-2}	18	1.0

Stability ∝ Bond order

12. Ans. (2)

XeF_4	XeO_4
4σ + 2 Lone pair	4σ
sp^3d^2	sp^3
square planar	Tetrahedral

13. Ans. (4)

HF forms strong Hydrogen bonding with other HF molecules but HCl, HBr, HI do not form H-bonding.

14. Ans. (2)

CH_4	NH_3	H_2O	(sp^3)
4σ	3σ	2σ	
	1 L. P.	2 L. P.	

If Hybridisation is same, then

$$\text{Bond Angle} \propto \frac{1}{\text{Number of Lone Pair}}$$

15. Ans. (2)

Order of Bond energy :- $Cl_2 > Br_2 > F_2 > I_2$

Bond energy of $:\ddot{F}-\ddot{F}:$ is less due to lone pair –lone pair repulsion.

16. Ans. (1)

L.P.–L.P. repulsion > L.P.–B.P. repulsion > B.P.–B.P. repulsion

17. Ans. (1)

XeF_6 ⇒ 6σ + 1 L.P. ⇒ sp^3d^3 ⇒ Distorted octahedral

XeO_3 ⇒ 3σ + 1 L.P. ⇒ sp^3 ⇒ Pyramidal

$XeOF_4$ ⇒ 5σ + 1 L.P. ⇒ sp^3d^2 ⇒ Square pyramidal

XeF_4 ⇒ 4σ + 2 L.P. ⇒ sp^3d^2 ⇒ Square planar

18. Ans. (2)

XeF_4 ⇒ 4σ + 2L.P. ⇒ sp^3d^2

Molecular Geometry/shape ⇒ square planar

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19. Ans. (1)

SeF_4	CH_4
$4\sigma + 1$ L.P.	4σ
sp^3d	sp^3
See-saw	Tetrahedral

20. Ans. (1)

NO_2^+	NO_3^-	NH_4^+
2σ	3σ	4σ
sp	sp^2	sp^3

21. Ans. (4)

$\ddot{P}F_3 \Rightarrow PF_3$ has 1 Lone pair so it can donate L.P.

22. Ans. (2, 3)

$ClO_3^- \Rightarrow 42e^-$
$SO_3^{2-} \Rightarrow 42e^-$
$\Big]\ sp^3(3\sigma + 1\text{L.P.})$ Pyramidal

$CO_3^{2-} \Rightarrow 32e^-$
$NO_3^- \Rightarrow 32e^-$
$\Big]\ sp^2(3\sigma)$ Trigonal planar

23. Ans. (3)

Li Na K Rb →

ϕ ↓ Hydration ↓ Hydrated size ↓ Ionic mobility ↑

Hydration $\propto \phi$ (Ionic Potential)

$\propto \dfrac{\text{charge of cation}}{\text{size of cation}}$

24. Ans. (1)

XX^1 ex. → ICl, ClF etc Linear

XX_3^1 ex. ICl_3, ClF_3 etc T-shape

(3σ + 2L.P. $\Rightarrow sp^3d$)

XX_5^1 ex. IF_5, etc square pyramidal

(5σ + 1L.P. $\Rightarrow sp^3d^2$)

XX_7^1 ex. IF_7, etc Pentagonal Bipyramidal

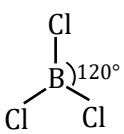
($7\sigma \Rightarrow sp^3d^3$)

25. Ans. (2)

Isoelectronic $\Big[\begin{matrix} IBr_2^- \Rightarrow 2\sigma + 3\text{L.P.} \\ XeF_2 \Rightarrow 2\sigma + 3\text{L.P.} \end{matrix}\Big]\ sp^3d \rightarrow$ Linear shape

26. Ans. (3)

$BCl_3 \Rightarrow 3\sigma + 0$ L.P. $\Rightarrow 3\ sp^2$



27. Ans. (2)

CN^-, $CO \Rightarrow 14e^-$s Bond order = 3.0

28. Ans. (1)

BeH_2 CaH_2 BaH_2 →

size of cation ↑ Polarisation ↓ Ionic character ↑

29. Ans. (4)

Mg^{2+} $X = 1s^22s^22p^3$

Mg^{2+} X^{-3}

Mg_3X_2

30. Ans. (2)

Bond Order

$CN^+ \Rightarrow 12e^- \longrightarrow 2.0$

$CN^- \Rightarrow 14e^- \longrightarrow 3.0$

$NO \Rightarrow 15e^- \longrightarrow 2.5$

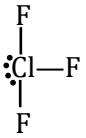
$CN \Rightarrow 13e^- \longrightarrow 2.5$

31. Ans. (3)

B is unable to form MF_6^{-3} because B has no vacant orbital.

32. Ans. (2)

$ClF_3 \Rightarrow \underbrace{3\sigma + 2\text{L.P}}_{sp^3d}$



33. Ans. (3)

$C_2 \Rightarrow 12e^-$

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2$

Net 2 π-bonds

Bond Order = 2.0

34. Ans. (4)

$\left[SiCl_6^{-2}\right]$ Not stable due large size side atoms (Cl) [Steric Hindrance]

35. Ans. (4)

PCl_5 molecule is a reactive molecule.

36. Ans. (2)

$XeF_4 \Rightarrow 4\sigma + 2$ L.P. $\Rightarrow sp^3d^2 \Rightarrow$ Square planar

$XeF_6 \Rightarrow 6\sigma + 1$ L.P. $\Rightarrow sp^3d^3 \Rightarrow$ Distorted octahedral

$XeOF_4 \Rightarrow 5\sigma + 1$ L.P. $\Rightarrow sp^3d^2 \Rightarrow$ Square pyramidal

$XeO_3 \Rightarrow 3\sigma + 1$ L.P. $\Rightarrow sp^3 \Rightarrow$ Pyramidal

37. Ans. (3)

H_2O | Bond length ↑
H_2S | Bond strength ↓
H_2Se | Thermal stability ↓
H_2Te
H_2Po

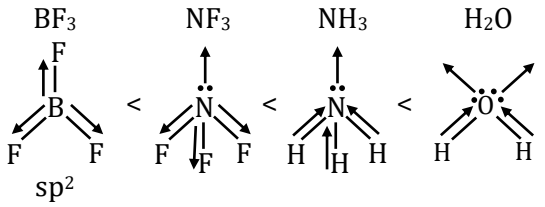
38. Ans. (4)

$\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2 < \sigma2p_z^2$

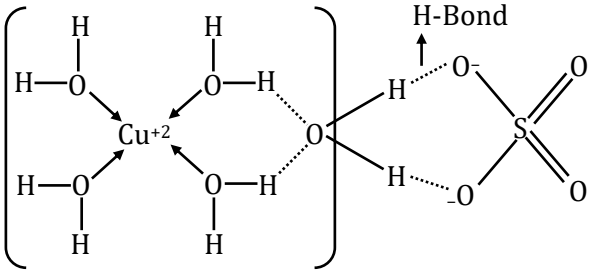
$< \pi2p_x^2 = \pi2p_y^2 < \pi^*2p_x^1 = \pi^*2p_y^1$

2 unpaired e^- (Paramagnetic)

39. Ans. (2)



40. Ans. (2)



$[Cu(H_2O)_4]SO_4.H_2O$

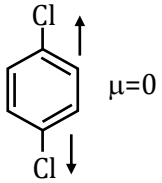
41. Ans. (2)

$He_2 \Rightarrow 4e^-$ $\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2$

Bond order = zero

42. Ans. (1)

BF_3, BeF_2, $CO_2 \Rightarrow$ If central atom has no lone pair and all side atoms are same them dipole moment will be zero.



μ=0

43. Ans. (4)

N_2O $N \equiv N \rightarrow O$ (Linear)

sp

44. Ans. (4)

$XeF_2 \rightarrow 2\sigma + 3L.P. \Rightarrow sp^3d$ Shape → Linear

$XeF_4 \rightarrow 4\sigma + 2L.P. \Rightarrow sp^3d^2$ Shape → square planar

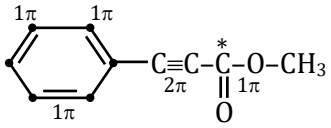
$XeO_3 \rightarrow 3\sigma + 1L.P. \Rightarrow sp^3$ Shape → Pyramidal

$XeOF_4 \rightarrow 5\sigma + 1L.P. \Rightarrow sp^3d^2$ Shape → Square pyramidal

45. Ans. (1)

$PCl_5 \rightarrow 5\sigma \rightarrow sp^3d \rightarrow$ Trigonal Bipyramidal

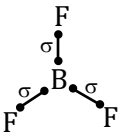
46. Ans. (3)



7 sp^2–C (Dotted & Star marked)

6π-bonds

47. Ans. (3)



$3 + 3 = 6e^-$

$3\sigma \Rightarrow sp^2$ Hybridisation

48. Ans. (1)

$PCl_5 \Rightarrow 5\sigma \Rightarrow sp^3d \Rightarrow$ Shape Trigonal Bipyramidal

$SF_6 \Rightarrow 6\sigma \Rightarrow sp^3d^2 \Rightarrow$ Octahedral

$BrF_5 \Rightarrow 5\sigma + 1L.P. \Rightarrow sp^3d^2 \Rightarrow$ square pyramidal

$BF_3 \Rightarrow 3\sigma \Rightarrow sp^2 \Rightarrow$ Trigonal planar

49. Ans. (3)

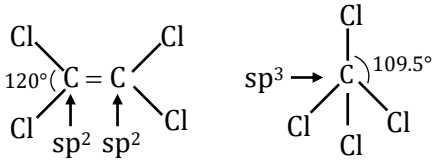
$SbCl_5 \rightarrow$ If central atom has no lone pair and all side atoms are same then molecule will be non-polar.

50. Ans. (4)

$Fe^{+2} \Rightarrow 24$ electrons

$Mn^{+2} \Rightarrow 23$ electrons

51. Ans. (1)



Bond angle $\Rightarrow 120°$ Bond angle $\Rightarrow 109.5°$

52. Ans. (2)

$IF_2^- \Rightarrow 2\sigma + 3$ L.P. $\Rightarrow sp^3d$

$H-C \equiv N \Rightarrow 2\sigma \Rightarrow sp$

$PCl_4^+ \Rightarrow 4\sigma + 0$ L.P. $\Rightarrow sp^3$

$XeF_4 \Rightarrow 4\sigma + 2$ L.P. $\Rightarrow sp^3d^2$

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53. Ans. (2)

Noble gas have low melting and boiling point due weak Vander Waal's force of attraction between noble gas atoms.

54. Ans. (2)

$CH_3–F$ $CH_3–Cl$ $CH_3–Br$ $CH_3–I$

Size of Halogen atom ↑ Bond Length ↑ Bond Enthalpy ↓

55. Ans. (1)

P–O > S–O > Cl–O > O=O

BO=1 BO=1 BO=1 BO=2

On moving from left to right in a period size decreases.

56. Ans. (4)

$GeH_4 \Rightarrow 4 + 4$ electrons $\Rightarrow 8$ electrons

$\Rightarrow$ electron precise hydride in Ge

$B_2H_6 \Rightarrow 3 + 3$ electrons $\Rightarrow 6$ electrons

$\Rightarrow$ Electron deficient hydride

$HF \Rightarrow 7 + 1$ electrons $\Rightarrow 8$ electrons

$\Rightarrow$ Electron rich hydride (F has 3 L.P.)

$MgH_2 \Rightarrow Mg^{2+} + 2H^- \Rightarrow$ Ionic Hydride

57. Ans. (1)

Order of boiling point $\Rightarrow$

$H_2S < H_2Se < H_2Te < H_2O$

Molecular Mass ↑ V.W.F ↑ B.P ↑ H-bonding (maximum B.P)

58. Ans. (3)

$O_2^+ \Rightarrow 15e^-$

$\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2 < \sigma2p_z^2$

$< \pi2p_x^2 = \pi2p_y^2 < \pi^*2p_x^1 = \pi^*2p_y^1 < \sigma^*2p_z$

1 unpaired electron is present in π^*(ABMO) so paramagnetic.

59. Ans. (3)

$IF_5 \longrightarrow$ 1 L.P.

$SF_4 \longrightarrow$ 1 L.P.

$XeF_2 \longrightarrow$ 3 L.P. (Maximum L.P.)

$ClF_3 \longrightarrow$ 2 L.P.

60. Ans. (1)

$CO_2 \Rightarrow 2\sigma \Rightarrow sp$ Bond Angle = 180°

NH_4^+	NH_3	H_2O	
4σ	3σ	2σ	sp^3←same hyb.
0 L.P.	1 L.P.	2 L.P	

L.P. ↑ Bond Angle ↓

61. Ans. (2)

$NH_3 \Rightarrow 3\sigma + 1$ L.P. $\Rightarrow sp^3$ Pyramidal

$ClF_3 \Rightarrow 3\sigma + 2$ L.P. $\Rightarrow sp^3d$ T-shape

$PCl_5 \Rightarrow 5\sigma + 0$ L.P. $\Rightarrow sp^3d$ Trigonal Bipyramidal

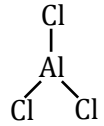
$BrF_5 \Rightarrow 5\sigma + 1$ L.P. $\Rightarrow sp^3d^2$ Square Pyramidal

62. Ans. (1)

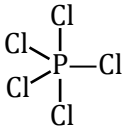
Fluorine is stronger oxidising agent than chlorine because F-F bond has a low enthalpy of dissociation so it breaks easily and accept electron readily due to high hydration enthalpy of fluoride ion.

63. Ans. (4)

Total number of species = 3

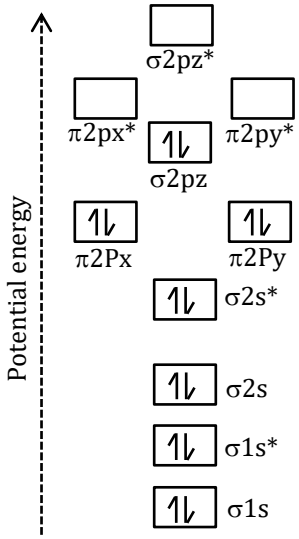
$AlCl_3 \longrightarrow$  Incomplete octet $6e^-$ around Al

$BeCl_2 \longrightarrow Cl—Be—Cl$ Incomplete octet $4e^-$ around Be

$PCl_5 \longrightarrow$  Super/hyper valent $10e^-$ around P.

64. Ans. (4)

Molecular orbital (energy) diagram / sequence of N_2



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65. Ans. (2)

Intermolecular forces means force of attraction between two or more molecules

dipole-dipole (attraction between two or more polar molecules).

Dipole induced dipole (attraction between polar and non polar molecules)

Hydrogen bonding (it is a special type of dipole-dipole and ion-dipole attraction)

Dispersion forces (mainly acts between non polar molecules).

Covalent bonding (acts between atom not between molecules)

66. Ans. (2)

$\left.\begin{matrix} H_2O \\ OH^- \\ NH_3 \end{matrix}\right\}$ can not act as lewis acid because they does not contain vacant orbital

$BF_3 \rightarrow$ Contains vacant orbital on central atom (Boron).

67. Ans. (2)

$Cu(s) \rightarrow Cu(g) \rightarrow Cu^{+}_{(g)} \rightarrow Cu^{+2}_{(g)} \rightarrow Cu^{+2}_{(aq)}$

$\Delta H_{atomisation}$ IE_1 IE_2 Hydration energy

Cu^{+2} is more stable than Cu^{+1} because released hydration energy is more in case of Cu^{+2} than Cu^{+1}.

68. Ans. (4)

$HF > NH_3 > H_2S > CH_4$

(Non-polar)

69. Ans. (4)

Total numbers electrons are same

Ca^{+2}, Ar, K^{+}, $Cl^{-} \rightarrow$ 20 electrons

70. Ans. (3)

In the formation of BMO, the two electron waves of the bonding atoms reinforce each other due to constructive interference.

Molecular orbitals obtained from $2P_x$ and $2P_y$ orbitals are 'unsymmetrical' around bond axis.

71. Ans. (1)

Boiling point order

$H_2O > H_2Te > H_2Se > H_2S$

$\Rightarrow$ Water has max. bpt due to presence of high extent of H-Bonding even though its molecular mass is minimum among these hydrides.

72. Ans. (1)

A-I, B-IV, C-II, D-III

(A) NH_3 – sp^3 + 1LP $\rightarrow$ Trigonal Pyramidal–(I)

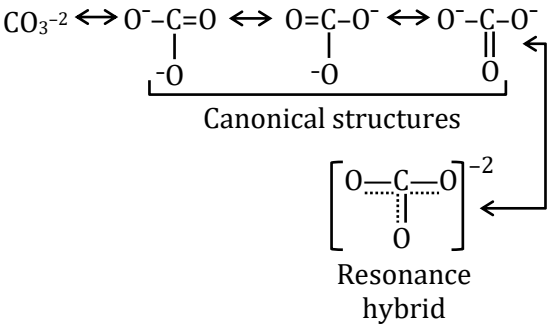
(B) BrF_5–sp^3d^2 + 1LP$\rightarrow$ Square Pyramidal–(IV)

(C) XeF_4 - sp^3d^2 + 2LP $\rightarrow$ Square Planar - (II)

(D) SF_6 - $sp^3d^2 \rightarrow$ Octahedral - (III)

73. Ans. (4)

Sol.



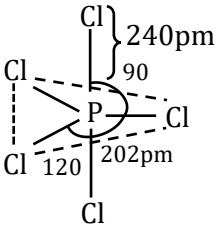
74. Ans. (2)

HI is a polar molecule. Rest other are non-polar.

75. Ans. (4)

(A)	Nitrogen atom	Element with highest IP in group 15
(B)	Fluorine molecule	Strongest oxidising agent
(C)	Oxygen molecule	Paramagnetic (MOT)
(D)	Xenon atom	Most reactive element in group 18

76. Ans. (2)



All P–Cl bond lengths are not equal.

77. Ans. (4)

(A) HCl $\rightarrow$ 431 kJ/mole

(B) $N_2 \rightarrow$ 946 kJ/mole

(C) $H_2 \rightarrow$ 435.8 kJ/mole.

(D) $O_2 \rightarrow$ 498 kJ/mole.

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Exercise-III (Analytical Questions)

1. Ans. (2)

S–I After Halogens noble gases are present in periodic table then alkali metals are present in periodic table.

S–II Halogens form X^- (Halide ion) and alkali metal form M^+ and they are attracted by electrostatic force or ionic bond.

2. Ans. (4)

S–I Covalent bond is formed by equal sharing of electron.

S–II Two or three electron pairs of a multiple bond are treated or known as single super pair.

3. Ans. (2)

S–I Lewis theory explain the complete or incomplete octet of the molecules.

S–II Lewis theory does not explain the relative stability and energy of the molecules.

4. Ans. (3)

S–I Metals form cation and non-metals form anion and the force between them is electrostatic force of attraction or Ionic bond.

S–II Cation is generally formed by metal atom so NH_4^+ (ammonium ion) is an exception because it is made up of 2 non-metal atoms.

5. Ans. (4)

S–I Lattice energy is required to break one mole ionic crystal lattice into its constituent gaseous ions.

S–II $\text{Stability of an Ionic compound} \propto \text{L.E.} \propto \frac{\text{charge of ions}}{\text{size of ions}}$

6. Ans. (2)

S–I Among the 2 O–H bonds in H_2O molecule, the energy required to break the first O–H bond and other O–H bond is different.

S–II Before breaking After breaking



7. Ans. (1)

S–I sp hybridisation is known as diagonal hybridisation.

S–II In sp hybridisation, 2 sp-hybridised orient themselves at 180°.

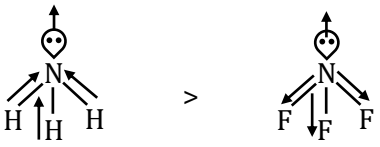
8. Ans. (3)

S–I Yes the given Lewis structure of O_3 is correct.

S–II Formal charge of atom 1, 2, 3 are also correct.

9. Ans. (4)

S–I



More μ Less μ

S–II EN → F > H

4.0 2.1

10. Ans. (1)

S–I Promotion of e^- is not required condition prior to hybridisation.

S–II Yes, its true.

11. Ans. (3)

S–I $C_2 \rightarrow 12$ electrons

$\pi 2p_x^2 = \pi 2p_y^2$

S–II $O_2^+ \rightarrow 15e^-$

Its paramagnetic $\pi^* 2p_x^1 = \pi^* 2p_y$

1 unpaired electron.

12. Ans. (1)

LiF is sparingly soluble in water due to very high lattice energy.

$Li^+ F^- \rightarrow$ Both cation and anion have small size

$\text{L. E.} \propto \frac{1}{\text{size of ions}}$

13. Ans. (1)

Ionic compounds tend to be non-volatile due to strong electrostatic force between them.

14. Ans. (3)

A → Yes, in solid and gaseous PCl_5, P-Cl bond length is different.

R → In solid state $2PCl_5 \rightarrow PCl_4^+ + PCl_6^-$

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15. Ans. (4)

A → B_2 is paramagnetic due to 2 unpaired electrons in $\pi 2p_x^1 = \pi 2p_y^1$

R → The highest occupied molecular orbital in B_2 molecule is of π-type.

16. Ans. (3)

A → SiF_6^{2-} exist $SiCl_6^{2-}$ does not exist

R → Yes, size of side atom F is small so SiF_6^{2-} exist.

17. Ans. (4)

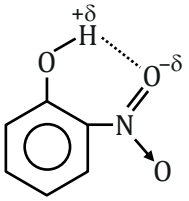
A → ClF_3 has bent T-shape structure.

R → It has 2 L.P. at approximate 120° (equatorial position)

18. Ans. (4)

A → B.P. of O–nitrophenol is less than B.P. of P-nitrophenol because P-nitrophenol has intermolecular H-bonding.

R → Intra molecular H-bonding is present in O-nitrophenol.



19. Ans. (2)

AgCl is more covalent than NaCl because polarising power of Ag^+ is greater than Na^+ ion.

Covalent character ∝ Polarising power of cation

20. Ans. (2)

$SF_4 \rightarrow 4\sigma + 1$ L.P $\Rightarrow 5(sp^3d)$

$IF_5 \rightarrow 5\sigma + 1$ L.P $\Rightarrow 6(sp^3d^2)$

$NO_2^+ \rightarrow 2\sigma + 0$ L.P $\Rightarrow 2(sp)$

$NH_4^+ \rightarrow 4\sigma + 0$ L.P $\Rightarrow 4(sp^3)$

21. Ans. (3)

		Bond Order
NO	$\rightarrow 15e^- \rightarrow$	2.5
CO	$\rightarrow 14e^- \rightarrow$	3.0
O_2^-	$\rightarrow 17e^- \rightarrow$	1.5
O_2	$\rightarrow 16e^- \rightarrow$	2.0

22. Ans. (2)

Shape

$ICl_2^- \rightarrow 2\sigma + 3$L.P. $\rightarrow 5\ sp^3d \rightarrow$ Linear

$BrF_2^+ \rightarrow 2\sigma + 2$L.P. $\rightarrow 4\ sp^3 \rightarrow$ V. Shape

$ClF_4^- \rightarrow 4\sigma + 2$L.P. $\rightarrow 6\ sp^3d^2 \rightarrow$ Square Planar

$AlH_4^- \rightarrow 4\sigma + 0$L.P. $\rightarrow 4\ sp^3 \rightarrow$ Tetrahedral

23. Ans. (1)

H_2O H_2S H_2Se H_2Te

Bond length ↑ Bond strength ↓ Acidic nature or strength ↑

$HCl < HBr < HI$ < HF (H–bond)

Mol. mass ↑ V.W.F. ↑ B.P. ↑

$NH_3 > PH_3 > AsH_3 > SbH_3$

Size of central atom ↑ L.P. density ↓ Lewis basic strength ↓

$\Rightarrow$

$CH_4 > SiH_4 > GeH_4 > SnH_4 > PbH_4$

Atomic Size ↑ Bond length ↑ Bond strength ↓

24. Ans. (3)

Diamond → Covalent bond between carbon atoms so covalent solid.

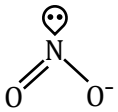
Fe(s) → Metallic bond between Fe atoms so metallic solid.

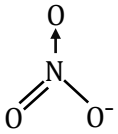
LiF → Ionic bond is present so ionic solid.

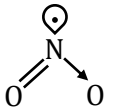
Dry Ice → V.W.F. between molecules of CO_2 so molecular solid.

25. Ans. (1)

$NO_2^+ \Rightarrow O=\overset{+}{N}=O$ $2\sigma \Rightarrow sp \rightarrow 180°$

NO_2^-  $2\sigma + 1$L.P $\Rightarrow sp^2 \rightarrow 118°$

NO_3^-  $3\sigma + 0$ L.P $\Rightarrow sp^2 \rightarrow 120°$

NO_2  Bond angle → 134°

(Value based)

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26. Ans. (2)

$ClF_3 \Rightarrow 3\sigma + 2L.P. \rightarrow sp^3d \rightarrow$ T-shape (Planar)

2L.P. at equatorial at approx. 120° so no cancellation of vectors so polar molecule.

$XeF_4 \rightarrow 4\sigma + 2L.P. \rightarrow sp^3d^2 \rightarrow$ Square Planar

$$\left.\begin{array}{l}\text{2L.P. at } 180°\text{(axis)} \rightarrow \text{vector cancel}\\ \text{4 bond pair at equitorial} \rightarrow \text{vector cancel}\end{array}\right] \text{So non} - \text{polar}$$

$NF_3 \rightarrow 3\sigma + 1L.P. \rightarrow sp^3 \rightarrow$ Pyramidal (Non-Planar)

1L.P. at Central atom $\rightarrow$ So always polar molecule.

$PCl_5 \rightarrow 5\sigma + 0L.P. \rightarrow sp^3d \rightarrow$ Trigonal Bipyramidal (Non-Planar)

No L.P. at Central atom and all side atoms are same so non-polar molecule.

27. Ans. (3)

$SO_3 \rightarrow sp^2 \rightarrow$ 1 is $p\pi - p\pi$ bond

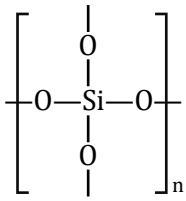
2 are $p\pi - d\pi$ bond

$XeO_4 \rightarrow sp^3 \rightarrow$ All 4 π – bonds are $p\pi - d\pi$

$C_3O_2 \rightarrow$ All C are sp $\rightarrow$ Only $p\pi - p\pi$ (All)

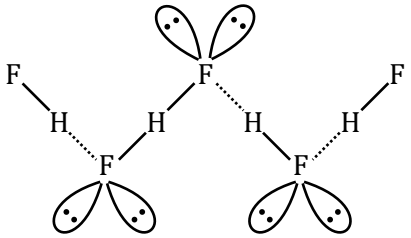
O = C = C = C = O

$(SiO_2)_n \rightarrow$ Polymer $\rightarrow$ No π – bond

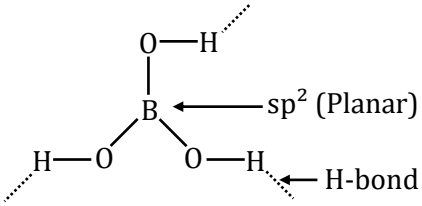


28. Ans. (2)

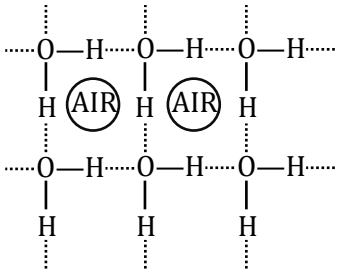
(i) HF (s) $\rightarrow$

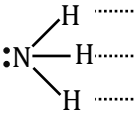


(ii) H_3BO_3 (s)



(iii) $H_2O(s) \rightarrow$ Ice



(iv) $NH_{3(s)}$  (Linear)

29. Ans. (2)

(i) $Li^+_{(aq)} > Na^+_{(aq)} > K^+_{(aq)} \leftarrow$ Hydrated radius of ions

$$\boxed{\phi \propto \frac{\text{charge of cation}}{\text{Size of cation}}} \quad \boxed{\text{Hydration} \propto \phi}$$

(ii) $F^- < Cl^- < Br^- < H^- \leftarrow$ Polarizability

$$\boxed{\text{Polarisability} \propto \text{Size of Anion}}$$

(iii) $Na^+ < Mg^{2+} < Al^{+3} \leftarrow$ Polarising Power (P.P.)

$$\text{P.P.} \propto \frac{\text{Charge of cation}}{\text{Size of cation}}$$

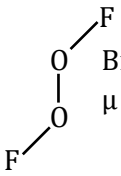
(iv) $Ca^{2+}_{aq} < Sr^{+}_{aq} < Ba^{+2}_{aq} \leftarrow$ Ionic mobility in aq. solution

$$\text{Hydration} \propto \frac{\text{Charg of cation}}{\text{Size of cation}}$$

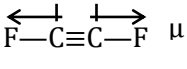
Hydration $\uparrow$ Ionic mobility $\downarrow$

30. Ans. (4)

(A) $CHCl_3 \rightarrow$ If all side atoms are different then molecule will be polar.

(C) $O_2F_2 \rightarrow$  Bihedral $\mu \neq 0$

(B) $CF_4 \rightarrow$ Central atom has no L.P. and all side atoms – same then $\rightarrow$ Non-polar

(D) $C_2F_2 \rightarrow$  $\mu = 0$

Non-polar molecule

31. Ans. (3)

Bond Order

(a) $N_2 \rightarrow 14e^- \rightarrow 3.0$

(b) $N_2^+ \rightarrow 13e^- \rightarrow 2.5$

(c) $N_2^{-2} \rightarrow 16e^- \rightarrow 2.0$

(d) $N_2^- \rightarrow 15e^- \rightarrow 2.5$

32. Ans. (4)

(a) $I_3^- \rightarrow 2\sigma + 3L.P. \rightarrow sp^3d$

(b) $I_3^+ \rightarrow 2\sigma + 2L.P. \rightarrow sp^3$

(c) $ClF_3 \rightarrow 3\sigma + 2L.P. \rightarrow sp^3d$

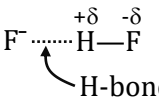
(d) $IF_5 \rightarrow 5\sigma + 1L.P. \rightarrow sp^3d^2$

33. Ans. (4)

(a) $PH_5 \rightarrow$ Not exist $\rightarrow$ d-orbital contraction

H cannot expand the covalency of P due to less EN of H.

(b) $KHF_2 \rightarrow$ Exist $\rightarrow K^+/HF_2^-$



(c) $BF_5^{-2} \rightarrow$ Not exist $\rightarrow$ Due to absence of vacant orbital in B.

(d) $BeF_4^{2-} \rightarrow$ Exist $\rightarrow BeF_2 + 2F^-$

34. Ans. (3)

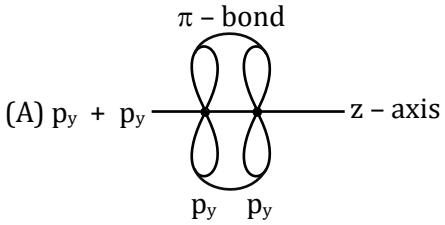
LiCl more covalent less M.P. than NaCl (Polarisation)

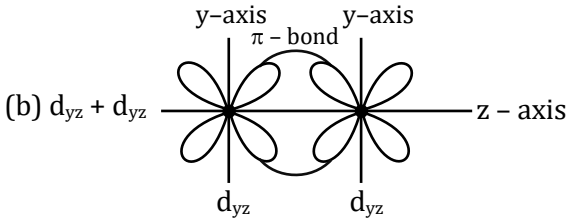
$SnCl_2$ less covalent more M.P. than $SnCl_2$ (Polarisation)

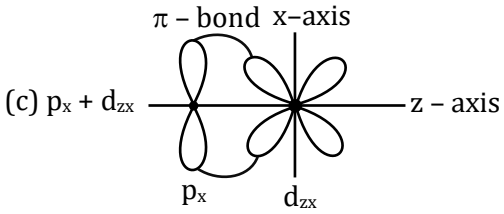
AlF_3 has more lattice energy so more M.P. than MgF_2 (Lattice energy)

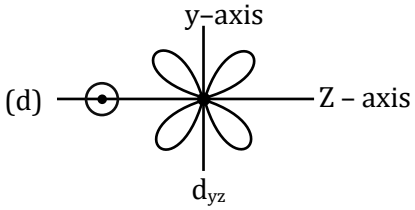
Diamond has more M.P. due to covalent solid. (large number of σ–bonds)

35. Ans. (1)

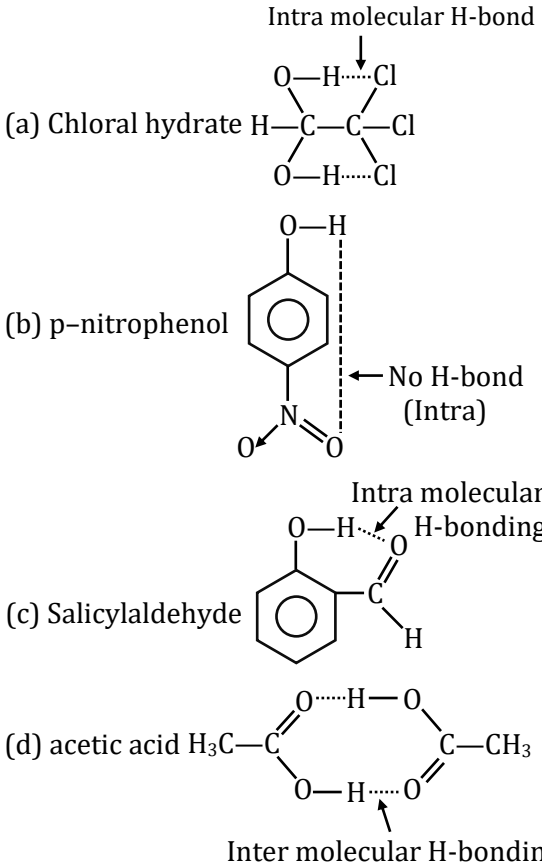








36. Ans. (4)



37. Ans. (3)

(a) $ZnSO_4 \rightarrow ZnO + SO_3$

(b) $CaSO_3 \rightarrow CaO + SO_2$

(c) $Fe_2(SO_4)_3 \rightarrow Fe_2O_3 + SO_3$

(d) $BaSO_3 \rightarrow BaO + SO_2$

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38. Ans. (2)

(a) $AgF > AgCl > AgBr > AgI$

Polarisation ↑ Solubility ↓

(b) $BeSO_4 > - - - - - - - - - - > BaSO_4$

Large Common ion (SO_4^{2-}) Solubility ↓ Down the group

(c) $LiOH < NaOH < KOH < RbOH < CsOH$

Small common Ion (OH^-) Solubility ↑ down the group

(d) $SnCl_4 < SnCl_2$

Polarisation ↓ Solubility ↑

39. Ans. (4)

(a) $LiNO_3 \xrightarrow{\Delta} Li_2O + \underbrace{NO_2 + O_2}_{\text{Paramagnetic}}$

(b) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

(c) $NaNO_3 \longrightarrow NaNO_2 + O_2$

(Paramagnetic gases)

(d) $NH_4NO_2 \longrightarrow N_2 + H_2O$

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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

IMPORTANT CONCEPTS

1. The geometry of the following molecules with respect to central atom is respectively
$N(SiH_3)_3$, Me_3N, $(SiH_3)_3P$
(1) planar, pyramidal, planar
(2) planar, pyramidal, pyramidal
(3) pyramidal, pyramidal, pyramidal
(4) pyramidal, planar, pyramidal
CPB001

2. In which of the following compounds, observed bond angle is found to be greater than expected, but not due to back bonding
(1) $N(SiH_3)_3$ (2) $O(CH_3)_2$
(3) $O(SiH_3)_2$ (4) All of these
CPB002

3. The product formed in the reaction,
$BCl_3 + H_2O \longrightarrow$ Product is -
(1) $H_3BO_3 + HCl$ (2) $B_2O_3 + HOCl$
(3) $B_2H_6 + HCl$ (4) No reaction
CPB003

4. The hybridization of boron in diborane is -
(1) sp (2) sp^2 (3) sp^3 (4) sp^3d^2
CPB004

5. Which is partially hydrolysied :
(1) PCl_3 (2) NCl_3 (3) $AsCl_3$ (4) $SbCl_3$
CPB005

6. Which of the following halides cannot be hydrolysed?
(i) TeF_6 (ii) SF_6 (iii) NCl_3 (iv) NF_3
Choose the correct code
(1) iii and iv (2) i, ii and iii
(3) i, ii and iv (4) ii and iv
CPB006

7. In which of the following dimer empty atomic orbital of central atom of monomer does not involve in hybridisation?
(1) Ga_2H_6 (2) Al_2Br_6 (3) Be_2H_4 (4) Cl_2O_6
CPB007

8. Which of the following molecule is having complete octet
(1) $BeCl_2$(dimer) (2) BeH_2(dimer
(3) BeH_2(s) (4) $BeCl_2$(s)
CPB008

9. Which one of the following oxy acid of fluorine exists?
(1) HOF (2) HFO_3
(3) HFO_4 (4) HFO_2
CPB009

10. Which of the following statements is correct
(1) All form $HOXO_3$ type oxy acid
(2) Only chlorine and bromine form oxyacids
(3) All halogens form oxyacids
(4) Only iodine forms oxyacid
CPB010

11. $2P \xrightarrow{-H_2O} Q \xrightarrow{-[O]} R$
If P is phosphoric acid then according to given information the incorrect statement is
(1) Q is pyro form and R is hypo form of given present oxy acid P
(2) Number of H-atoms present in each given oxy acid is equal to its basicity
(3) In P, Q, R oxy acids, oxidation state of central atom remains same
(4) All given oxy acids have pπ-dπ bond(s) in their structure
CPB011

12. Silicate having one monovalent corner oxygen atom in each tetrahedron unit is
(1) Sheet silicate
(2) Cyclic silicate
(3) Single chain silicate
(4) Double chain silicate
CPB012

13. The silicate anion in the mineral kinoite is a chain of three SiO_4^{-4} tetrahedra, that share corners with adjacent tetrahedra. The charge of the silicate anion is
(1) –4 (2) –8 (3) –6 (4) –2
CPB013

14. Which of the following is an organo silicon polymer?
(1) Silica (2) Silicones
(3) Silicon carbide (4) Silicic acid
CPB014

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15. Which reacts rapidly with oxygen in the air at ordinary temperature :
(1) White P (2) Red P
(3) N_2 (4) C
CPB015

16. Red and yellow phosphorus are :
(1) Allotropes (2) Isobars
(3) Isomers (4) Isotopes
CPB016

17. Which compound does not exist ?
(1) $TlCl_3$ (2) TlI_3
(3) BiF_5 (4) PbI_4
CPB017

18. Graphite conducts electricity because of the -
(1) Highly polarized nature of π-electrons.
(2) Highly delocalized nature of π-electrons
(3) Highly localized nature of π-electrons
(4) None of these
CPB018

BORON AND CARBON FAMILY

19. Alane is chemically –
(1) AlH_3 (2) $(AlH_3)_n$ (3) $LiAlH_4$ (4) None
CPB019

20. Aluminium is not acted upon by pure water as-
(1) Impurities in water are essential for the reaction to occur
(2) It is light metal
(3) It is protected by a film of aluminium oxide
(4) It is not a reactive metal
CPB020

21. The borax bead test is based upon the formation of
(1) Boron oxide (2) Boric acid
(3) Meta borates (4) Elemental boron
CPB021

22. Boric acid polymerizes due to -
(1) The presence of hydrogen bonds
(2) Its acidic nature
(3) Its geometry
(4) Its monobasic nature
CPB022

23. Alum is found to contain hydrated monovalent cation $[M(H_2O)_6]^+$, trivalent cation $[M'(H_2O)_6]^{+3}$ and SO_4^{2-} in the ratio of :
(1) 1 : 1 : 1 (2) 1 : 1 : 2
(3) 1 : 2 : 2 (4) 1 : 2 : 3
CPB023

24. Borax $Na_2B_4O_7.10H_2O$ is actually :-
(1) $Na_2[B_4O_5(OH)_4].8H_2O$
(2) $Na_2[B_4O_4(OH)_6].7H_2O$
(3) $Na_2[B_4O_3(OH)_8].6H_2O$
(4) $Na_2[B_4O_2(OH)_{10}].5H_2O$
CPB024

25. Potash alum is a double salt made up of two salts:
(1) Salt of a (SA + WB) + Salt of a (WA + WB)
(2) Salt of a (SA + SB) + Salt of a (SA + WB)
(3) Salt of a (SA + SB) + Salt of a (WA + WB)
(4) Salt of a (SA + WB) + Salt of a (WA + WB)
CPB025

26. From B_2H_6 all the following can be prepared except :
(1) H_3BO_3 (2) $B_2(CH_3)_4H_2$
(3) $B_2(CH_3)_6$ (4) $NaBH_4$
CPB026

27. The hydrides of group 14 elements are :
(1) Ionic (2) Oxidising
(3) Covalent (4) None of these
CPB027

28. Which gas is responsible for green house effect :
(1) CO_2 (2) SO_2 (3) CO (4) SO_3
CPB028

29. Artificial gem used for cutting glass is :
(1) Graphite (2) Diamond
(3) SiC (4) $CaCN_2$
CPB029

30. Borax and Boric acid are used in manufacturing of-
(1) Pyrex (2) Glass-wool
(3) Fibre glass (4) All of these
CPB030

31. Orthoboric acid acts as -
(1) Antiseptic (2) Antipyretic
(3) Antibiotic (4) None
CPB031

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32. Crucibles made by which material is inert to dilute acids and alkalies.
(1) Graphite (2) Aluminum
(3) Boron (4) Iron
CPB032

33. Alum is not used as -
(1) As a mordant in dyeing
(2) As an insecticide
(3) In the purification of water
(4) In the tanning of leather
CPB033

34. Correct order of IP of Gr 13
(1) B > Al > Ga > In > Tl
(2) B > Al < Ga > In > Tl
(3) B > Al < Ga > In < Tl
(4) B > Al < Ga < In > Tl
CPB035

35. $Al + NaOH + H_2O \rightarrow$ product in aqueous state is
(1) $[Al(H_2O)_6]^{+3}$ (2) $[Al(H_2O)_4]^{3+}$
(3) $[Al(OH)_4]^{-1}$ (4) $Al(OH)_3$
CPB036

36. In Borax bead test, bead of cobalt is-
(1) $Co(BO_2)_2$, Pink (2) $Co(BO_2)_2$, Blue
(3) $Co(BO_3)_2$, Pink (4) $Co(BO_3)_2$, Blue
CPB037

37. Wrong statement is
(1) Boric acid is weak monobasic acid
(2) Boric acid is a protic acid
(3) Boric acid is a lewis acid
(4) Boric acid on heating finally form B_2O_3
CPB038

38. Correct IE order is -
(1) C > Si > Ge > Sn > Pb
(2) C < Si < Ge < Sn > Pb
(3) C > Si > Ge > Sn < Pb
(4) C < Si < Ge < Sn < Pb
CPB039

39. Which stability order is incorrect
(1) $Pb^{+2} > Pb^{+4}$ (2) $Sn^{+2} > Sn^{+4}$
(3) $Sn^{+2} < Pb^{+2}$ (4) $Sn^{+4} > Pb^{+4}$
CPB040

40. Nature of GeO_2 is
(1) Acidic (2) Basic
(3) Amphoteric (4) Neutral
CPB041

41. Which does not have dangling bond
(1) Diamond (2) Graphite
(3) Fullerene (4) All of these
CPB042

42. Match the column

A. Dry ice	i. Soft drinks
B. CO_2 gas	ii. Used as refrigerant for icecream
C. Silicones	iii. Used in softning of water
D. Zeolite	iv. Used for making water proof jackets

	A	B	C	D
(1)	ii	i	iv	iii
(2)	ii	iii	iv	i
(3)	iii	i	ii	iv
(4)	iv	i	ii	iii

CPB043

43. Silicones are-
(1) RSiO- repeating unit polymer
(2) R_2SiO_2- repeating unit polymer
(3) R_2SiO- repeating unit polymer
(4) $RSiO_2$- repeating unit polymer
CPB044

44. In which silicate two oxygen atoms are shared (per unit)
(1) Pyrosilicate (2) Orthosilicate
(3) Chain silicate (4) None of these
CPB045

45. Catenation property is mainly exhibited by carbon in its family but it is not shown by its member
(1) Si (2) Ge (3) Sn (4) Pb
CPB046

46. Which among the following statements is incorrect regarding graphite ?
(1) Each carbon atom is sp^2 hybridised
(2) Fourth electron on each carbon atom forms a π bond.
(3) Layers are held by weak van der Waals forces
(4) Thermodynamically less stable than diamond
CPB047

47. Synthesis gas is -
(1) $CO + H_2$ (2) $CO_2 + H_2O$
(3) $CO + N_2$ (4) None
CPB048

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NITROGEN FAMILY

48. P_4O_{10} is used extensively as a :
(1) Dehydrating agent
(2) Catalytic agent
(3) Reducing agent
(4) Preservative

CPB049

49. PH_3 produces smoky rings when it comes in contact with air. This is because :
(1) It is inflamable
(2) It combines with water vapours
(3) It combines with nitrogen
(4) It contains impurity of P_2H_4

CPB050

50. Which of the following is the correct statement for PH_3
(1) It is less basic than NH_3
(2) It is less poisonous than NH_3
(3) Bond angle of PH_3 > NH_3
(4) It does not show reducing properties

CPB051

51. Concentrated nitric acid reacts with iodine to give:-
(1) HI (2) HOI
(3) $HOIO_2$ (4) $HOIO_3$

CPB052

52. Which of the following statement is true about white phosphorous & red phosphorous?
(1) Both are soluble in CS_2
(2) Both are poisonous
(3) Both can be oxidised by air
(4) Both glow in dark

CPB053

53. A gas which is used as anaesthetic in dental surgery is :
(1) N_2 (2) CO
(3) N_2O (4) NH_3

CPB054

54. On heating, ammonium dichromate and barium azide separately, we get :-
(1) N_2 from ammonium dichromate and NO from barium azide
(2) N_2O from ammonium dichromate and NO_2 from barium azide
(3) N_2O from ammonium dichromate and NO from barium azide
(4) N_2 in both cases

CPB055

55. NO_2 is formed when
(1) Cu reacts with conc. HNO_3
(2) Zn reacts with conc. HNO_3
(3) $Pb(NO_3)_2$ is heated
(4) All

CPB056

56. Which of the following does not produce NO_2 gas with conc. HNO_3 ?
(1) Cu (2) I_2 (3) Ag (4) Au

CPB057

57. Which gas is used to create inert atmosphere in industrial process.
(1) NH_3 (2) N_2
(3) O_2 (4) CO

CPB058

58. Liquid nitrogen is used as a refrigerant to preserve-
(1) Biological material (2) Food items
(3) In cryosurgery (4) All of the above

CPB059

59. Nitrogenous fertilisers are
(1) Ammonium nitrate
(2) Urea
(3) Ammonium phosphate
(4) All of the above

CPB060

60. Which compound is used in the pickling of stainless steel -
(1) Sulphuric acid (2) Nitric acid
(3) Phosphine gas (4) Phosphorous

CPB061

61. The gas which is used in Holme's signals-
(1) SO_2 (2) PH_3 (3) SO_3 (4) NH_3

CPB062

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62. Phosphine gas is used in -
(1) Smoke screen (2) Home screen
(3) Inert atmosphere (4) As a fuel
CPB063

63. Quartz is a crystalline variety of
(1) Silicon (2) SiC
(3) SiO_2 (4) Na_2SiO_3
CPB064

64. Aluminium vessel should not be washed with material containing washing soda because-
(1) Washing soda is easily decomposed
(2) Washing soda is expensive
(3) Washing soda reacts with Aluminium to forms soluble aluminate
(4) Washing soda reacts with Aluminium to forms insoluble aluminium oxide
CPB065

65. Haber's process is used in manufacturing of -
(1) Ammonia (2) Sadium azide
(3) Phosphine (4) None
CPB066

66. NH_3 act as a -
(1) Lewis acid (2) Lewis base
(3) Amphoteric (4) None
CPB067

67. Which of the following group of p-block elements used to form fertilizers.
(1) Carbon family (2) Halogen family
(3) Nitrogen family (4) Oxygen family
CPB068

68. Which does not show Allotropy
(1) N (2) P (3) Sb (4) Bi
CPB069

69. In which compound covalency of N is 4.
(1) N_2O_5 (2) N_2O_4
(3) N_2O_3 (4) All of these
CPB070

70. Impurities responsible for inflammable nature of PH_3 is/are -
(1) P_2H_4 (2) P_4 vapours
(3) Both (4) None
CPB071

71. PCl_3 can be obtained by reaction of
(1) $P_4 + SOCl_2 \rightarrow$ (2) $P_4 + SO_2Cl_2 \rightarrow$
(3) Both (4) None
CPB072

72. Which oxy acid of phosphorous has two P-H bond-
A. Phosphonic acid
B. Phosphinic acid
C. Pyrophosphorous acid
D. Polymetaphosphoric acid
(1) A,D (2) B,C (3) A,C,D (4) A,C
CPB073

73. Which acid can reduce $AgNO_3$ into metallic silver
(1) Hypophosphorous acid
(2) Orthophosphoric acid
(3) Pyrophosphoric acid
(4) Hypophosphoric acid
CPB074

74. What is the Ratio of number of P-P bond in hypophosphoric acid, number of P-H bond in Pyrophosphorous acid and number of P-OH bond in cyclotrimetaphosphoric acid respectively?
(1) 2 : 1 : 3 (2) 1 : 2 : 3
(3) 2 : 2 : 1 (4) 1 : 2 : 2
CPB075

OXYGEN FAMILY

75. SO_2 can acts as -
(1) Reducing agent (2) Oxidising agent
(3) Bleaching agent (4) All
CPB076

76. Ozone acts as
(1) Oxidising agent (2) Disinfectant
(3) Bleaching agent (4) All
CPB077

77. A black sulphide when treated with ozone becomes white. The white compound is :
(1) $ZnSO_4$ (2) $CaSO_4$
(3) $BaSO_4$ (4) $PbSO_4$
CPB078

78. H_2S gas changes a filter paper dipped in lead acetate solution into :
(1) Black (2) Red
(3) Green (4) Yellow
CPB079

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79. The number of S–S bonds in sulphur trioxide trimer (S_3O_9) is :
(1) Three (2) Two
(3) One (4) Zero
CPB080

80. Dry bleaching is done by :
(1) Cl_2 (2) SO_2
(3) O_3 (4) None
CPB081

81. When KBr is treated with conc. H_2SO_4 reddish brown gas is evolved. The gas is :
(1) Br_2 (2) HOBr
(3) NO_2 (4) H_2O_2
CPB082

82. On addition of conc. H_2SO_4 to a chloride salt, colourless fumes are evolved but in case of iodide salt, violet fumes come out. This is because:-
(1) H_2SO_4 reduces HI to I_2
(2) HI is of violet colour
(3) HI gets oxidised to I_2
(4) HI changes to HIO_3
CPB083

83. Which of the following is responsible for turning starch-iodide paper blue when it is brought in contact with O_3?
(1) Liberation of iodine
(2) Liberation of oxygen
(3) Formation of alkali
(4) Reaction of ozone with litmus paper.
CPB084

84. Which one of the following property is not correct for ozone ?
(1) It oxidises lead sulphide
(2) It oxidises potassium iodide
(3) It oxidises mercury
(4) It cannot act as bleaching agent in dry state.
CPB085

85. By which of the following SO_2 is formed ?
(1) Reaction of dilute H_2SO_4 with Zn.
(2) Heating $Fe_2(SO_4)_3$.
(3) Reaction of concentrated H_2SO_4 with Cu.
(4) Heating of Cs_2SO_4.
CPB086

86. Pick out the statement that is wrong :-
(1) Oxygen is paramagnetic in all the three states of matter
(2) Ozone is diamagnetic
(3) Ozone is a linear molecule
(4) The O–O bonds in ozone have considerable double bond character
CPB087

87. SO_2 is used as :
(1) Bleaching agent
(2) Preservative
(3) Refining petroleum and Sugar
(4) All of the above
CPB088

88. Sulphuric acid is used in :
(1) Detergent industry
(2) Storage battery
(3) Manufacture of nitrocellulose product
(4) All the above
CPB089

89. Which 16th group element have only positive oxidation state only
(1) S (2) Se (3) Te (4) Po
CPB090

90. The oxidation state of central atom in the anion of compound NaH_2PO_2 will be :
(1) +3 (2) +5 (3) +1 (4) –3
CPB091

91. Pure ozone is :
(1) Pale blue gas (2) Dark-blue liquid
(3) Violet-black solid (4) All
CPB092

92. Identify the incorrect statement
(1) Ozone absorb U.V. rays from sun.
(2) Depletion of ozone layer is due to C.F.C
(3) Ozone absorb infra-red radiations
(4) Oxides of nitrogen in the atmosphere can cause depletion of ozone layer.
CPB093

93. Arrange the following in decreasing order of their boiling Point ?
I. H_2O II. H_2S
III. H_2Se IV. H_2Te
(1) IV > III > II > I (2) II > III > IV > I
(3) I > IV > III > II (4) I > II >III > IV
CPB094

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94. Large difference in melting and boiling point between O and S is due to -
(1) Non-metallic nature
(2) Difference in electronegativity is high
(3) Atomicity
(4) Very small size
CPB095

95. Which hydride does not shows reducing property
(1) H_2Te (2) H_2Se (3) H_2S (4) H_2O
CPB096

96. True statement regarding halides of group 16 is-
(1) Hexafluorides are the only stable halides
(2) All hexafluorides are gaseous in nature.
(3) All hexafluorides shows octachedral structure
(4) All of these
CPB097

97. Industrially O_2 can be obtained by-
(1) Thermal decomposition of Ag_2O.
(2) Heating $KClO_3$.
(3) From H_2O_2
(4) From liquification and fractional distillation of air.
CPB098

98. Dioxygen does not reacts with
(1) Pt (2) Au
(3) He (4) All of these
CPB099

99. Incorrect about S- allotrope-
(1) At room temperature S-monoclinic is more stable
(2) Both rhombic and monoclinic have S_8 molecules
(3) S_8 ring of both is crown shape
(4) S_2 is paramagnetic like O_2
CPB100

100. Oleum is know as -
(1) $H_2S_2O_7$ (2) $SO_3 + H_2SO_4$
(3) Both (1) and (2) (4) $H_2S_2O_8$
CPB101

101. Which gas is used in oxyacetylene welding ?
(1) Oxygen (2) Phosphine
(3) Carbon dioxide (4) None
CPB102

102. Sulphur show paramagnetic behavior in -
(1) S_8 (2) S_2 (vapour)
(3) S_6 (4) S_2 (Solid)
CPB103

103. Contact process is used for manufacturing of-
(1) HNO_3 (2) H_2SO_4
(3) $H_2S_2O_2$ (4) None
CPB104

HALOGEN FAMILY AND INERT GASES

104. Select the correct statement (s) from the following-
(1) Fluorine displaces other halogens from the corresponding halides
(2) Fluorine reacts slowly with halogens
(3) Fluorine does not decompose water
(4) Except fluorine, other halogens directly combine with carbon
CPB105

105. The halide which does not give a precipitate with $AgNO_3$ is-
(1) F^- (2) Cl^- (3) Br^- (4) I^-
CPB106

106. Volatile nature of halogen is because :
(1) Halogen molecules are bonded by strong forces
(2) Halogen molecules are bonded by electrostatic forces
(3) The forces existing between the discrete molecule are only weak vander Waals force.
(4) Halogen molecules are more reactive
CPB107

107. Iodine gas turns starch iodide paper :
(1) Blue (2) Red
(3) Colourless (4) Yellow
CPB108

108. BrF_5 is a :
(1) Interhalogen compound
(2) Pseudohalogen compound
(3) Both the above
(4) None of the above
CPB109

109. Which of the following does not decolourise iodine?
(1) Na_2SO_3 (2) $Na_2S_2O_3$
(3) NaCl (4) NaOH
CPB110

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110. Which one of the following halogen liberates oxygen on reacting with H_2O

(1) I_2 (2) Cl_2 (3) Br_2 (4) F_2

CPB111

111. Helium is added to oxygen used by deep sea divers because :

(1) It is less soluble in blood than nitrogen under high pressure

(2) It is lighter than nitrogen

(3) It is readily miscible with oxygen

(4) It is less poisonous than nitrogen

CPB112

112. Which of the following is not correct :

(1) XeO_3 has four σ and four π bonds

(2) The hybridization of Xe in XeF_4 is sp^3d^2

(3) Among noble gases, the occurrence of argon is highest in air

(4) Liquid helium is used as cryogenic liquid

CPB113

113. XeF_2 reacts with SbF_5 to form :

(1) $[XeF]^+ [SbF_6]^-$

(2) $[XeF_3]^- [SbF_4]^-$

(3) $Xe^- [PtF_6]^+$

(4) XeF_4

CPB114

114. The compound that cannot be formed by xenon is

(1) XeO_3 (2) XeF_4

(3) $XeCl_4$ (4) XeO_2F_2

CPB115

115. SbF_5 reacts with XeF_4 to form an adduct. The shapes of cation and anion in the adduct are respectively

(1) square planar, trigonal bipyramidal

(2) T-shaped, octahedral

(3) square pyramidal, octahedral

(4) square planar, octahedral

CPB116

116. Which of the following noble gas does not form clathrate compound?

(1) Kr (2) Ne (3) Xe (4) Ar

CPB117

117. An inorganic salt when heated with concentrated H_2SO_4 evolves a colourless pungent smelling gas but with concentrated H_2SO_4 and MnO_2 evolves a coloured pungent smelling gas which bleaches moist litmus paper. The coloured gas is :-

(1) NO_2 (2) Cl_2 (3) Br_2 (4) I_2

CPB118

118. Chlorine gas is not used in

(1) Bleaching agent

(2) Extraction of gold

(3) Refining of sugar

(4) Sterilising drinking water

CPB119

119. Which is used for enrichment of ^{235}U

(1) ClF (2) ICl_3 (3) ClF_3 (4) None

CPB120

120. Which statement is true about Helium

(1) Non-inflammable and heavy gas

(2) Inflammable and light gas

(3) Non-inflamable and light gas

(4) All the above

CPB121

121. Which noble gas is used in gas cooled nuclear reactors :

(1) He (2) Ne (3) Ar (4) Xe

CPB122

122. Which gas is used in discharge tube:

(1) He (2) Ne (3) Ar (4) Xe

CPB123

123. Which type of bulb are used in botanical garden and green houses :

(1) Argon Bulb (2) Neon bulb

(3) Both (4) None

CPB124

124. Which gas is used to provide an inert atmosphere in high temperature metallurgical process :

(1) Ne (2) N_2 (3) Ar (4) He

CPB125

125. Argon is used in arc-welding because of its

(1) Low reactivity with metal

(2) Ability to lower the melting point of metal

(3) High calorific value

(4) Inflammability

CPB126

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126. Salt producing family is -
(1) Nitrogen family (2) Carbon family
(3) Halogen family (4) Inert gas
CPB127

127. Deacon's process is used for manufacturing of
(1) Cl_2 (2) F_2 (3) HCl (4) NaCl
CPB128

128. Correct order of bond dissociation enthalpy is-
(1) HF > HCl > HBr > HI
(2) HF > HBr > HCl > HI
(3) HI > HBr > HCl > HF
(4) None
CPB129

129. Which is used in estimation of CO-
(1) ClO_2 (2) Br_2O
(3) I_2O_5 (4) all of these
CPB130

130. Catalyst used in Deacon's process is -
(1) $CuCl_2$ (2) Cu_2Cl_2
(3) Pt wire (4) Ni powder
CPB131

131. Number of O-O bond in perchloric acid will be-
(1) 4 (2) 3 (3) 2 (4) Zero
CPB132

132. Which one of the following is not formed during hydrolysis of XeF_6?
(1) $XeOF_4$ (2) $XeOF_3$
(3) XeO_3 (4) XeO_2F_2
CPB133

133. Compound of radon have not been isolated but only identified by radiotracer technique, compound is-
(1) RnF_6 (2) RnF_4 (3) RnF_2 (4) RnF
CPB134

134. First inert gas compound was -
(1) $Xe[PtF_6]$ (2) $KrPtF_6$
(3) $XePtCl_6$ (4) $ArPtF_6$
CPB135

135. Which noble gas is not present in atmosphere.
(1) He (2) Ne
(3) Xe (4) Rn
CPB136

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	1	3	4	4	4	4	1	3	3	1	2	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	2	2	3	3	1	2	1	2	3	3	1	3	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	1	2	3	3	2	2	3	2	1	3	1	3	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	1	1	4	1	3	3	3	4	4	4	2	4	4	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	1	3	3	1	2	3	1	4	3	1	2	1	2	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	4	1	4	3	1	3	1	4	3	3	4	4	4	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	4	3	3	3	4	4	4	4	1	3	1	2	2	1	1
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	3	1	1	3	4	1	1	1	3	2	2	2	3	3	3
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	1	2	2	3	1	3	1	1	3	1	4	2	3	1	4

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

AIPMT 2015

1. Nitrogen dioxide and sulphur dioxide have some properties in common. Which property is shown by one of these compounds, but not by the other?
(1) Is a reducing agent
(2) Is soluble in water
(3) Is used as a food-preservative
(4) Forms 'acid-rain'

CPB152

Re-AIPMT 2015

2. Strong reducing behaviour of H_3PO_2 is due to :
(1) High oxidation state of phosphorus
(2) Presence of two –OH groups and one P–H bond
(3) Presence of one –OH group and two P–H bonds
(4) High electron gain enthalpy of phosphorus

CPB153

3. Which of the statements given below is incorrect?
(1) ONF is isoelectronic with O_2N^-
(2) OF_2 is an oxide of fluorine
(3) Cl_2O_7 is an anhydride of perchloric acid
(4) O_3 molecule is bent

CPB154

4. The stability of +1 oxidation state among Al, Ga, In and TI increases in the sequence :
(1) TI < In < Ga < Al (2) In < TI < Ga < Al
(3) Ga < In < Al < TI (4) Al < Ga < In < TI

CPB155

NEET-I 2016

5. Among the following, the correct order of acidity is
(1) $HClO_3 < HClO_4 < HClO_2 < HClO$
(2) $HClO < HClO_2 < HClO_3 < HClO_4$
(3) $HClO_2 < HClO < HClO_3 < HClO_4$
(4) $HClO_4 < HClO_2 < HClO < HClO_3$

CPB156

6. When copper is heated with conc. HNO_3 it produces
(1) $Cu(NO_3)_2$ and NO_2
(2) Cu $(NO_3)_2$ and NO
(3) $Cu(NO_3)_2$, NO and NO_2
(4) $Cu(NO_3)_2$ and N_2O

CPB157

7. Which is the **correct** statement for the given acids?
(1) Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid
(2) Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid
(3) Both are triprotic acids
(4) Both are diprotic acids

CPB158

NEET-II 2016

8. Boric acid is an acid because its molecule
(1) Accepts OH^- from water releasing proton
(2) Combines with proton from water molecule
(3) Contains replaceable H^+ ion
(4) Gives up a proton

CPB159

9. AlF_3 is soluble in HF only in presence of KF. It is due to the formation of
(1) AlH_3 (2) $K[AlF_3H]$
(3) $K_3[AlF_3H_3]$ (4) $K_3[AlF_6]$

CPB160

NEET(UG) 2017

10. In which pair of ions both the species contain S–S bond?
(1) $S_4O_6^{2-}$, $S_2O_3^{2-}$ (2) $S_2O_7^{2-}$, $S_2O_8^{2-}$
(3) $S_4O_6^{2-}$, $S_2O_7^{2-}$ (4) $S_2O_7^{2-}$, $S_2O_3^{2-}$

CPB161

11. It is because of inability of ns^2 electrons of the valence shell to participate in bonding that:-
(1) Sn^{2+} is oxidising while Pb^{4+} is reducing
(2) Sn^{2+} and Pb^{2+} are both oxidising and reducing
(3) Sn^{4+} is reducing while Pb^{4+} is oxidising
(4) Sn^{2+} is reducing while Pb^{4+} is oxidising

CPB162

NEET(UG) 2018

12. Which of the following statements is **not** true for halogens ?
(1) All form monobasic oxyacids.
(2) All are oxidizing agents.
(3) All except fluorine show positive oxidation states.
(4) Chlorine has the highest electron - gain enthalpy.

CPB163

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13. The correct order of N-compounds in its decreasing order of oxidation states is -
(1) HNO_3, NO, N_2, NH_4Cl
(2) HNO_3, NO, NH_4Cl, N_2
(3) HNO_3, NH_4Cl, NO, N_2
(4) NH_4Cl, N_2, NO, HNO_3

CPB164

NEET(UG) 2019

14. Match the following :

(a) Pure nitrogen	(i) Chlorine
(b) Haber process	(ii) Sulphuric acid
(c) Contact process	(iii) Ammonia
(d) Deacon's process	(iv) Sodium azide or Barium azide

Which of the following is the **correct** option ?

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iv)	(i)	(iii)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(iv)	(iii)	(ii)	(i)

CPB165

15. Which of the following is **incorrect** statement?
(1) PbF_4 is covalent in nature
(2) $SiCl_4$ is easily hydrolysed
(3) GeX_4 (X = F, Cl, Br, I) is more stable than GeX_2
(4) SnF_4 is ionic in nature

CPB166

NEET(UG) 2019 (ODISHA)

16. The liquified gas that is used in dry cleaning along with a suitable detergent is :-
(1) Water gas (2) Petroleum gas
(3) NO_2 (4) CO_2

CPB167

17. Which of the following compounds is used in cosmetic surgery?
(1) Silica (2) Silicates
(3) Silicones (4) Zeolites

CPB168

18. A compound 'X' upon reaction with H_2O produces a colorless gas 'Y' with rotton fish smell. Gas 'Y' is absorbed in a solution of $CuSO_4$ to give Cu_3P_2 as one of the products. Predict the compound 'X'
(1) Ca_3P_2 (2) NH_4Cl
(3) As_2O_3 (4) $Ca_3(PO_4)_2$

CPB169

19. Which of the following oxoacids of phosphorus has strongest reducing property?
(1) $H_4P_2O_7$ (2) H_3PO_3 (3) H_3PO_2 (4) H_3PO_4

CPB170

20. Identify the correct formula of oleum from the following
(1) $H_2S_2O_7$ (2) H_2SO_3 (3) H_2SO_4 (4) $H_2S_2O_8$

CPB171

NEET(UG) 2020

21. Which of the following oxoacid of sulphur has –O–O– linkage ?
(1) $H_2S_2O_7$, pyrosulphuric acid
(2) H_2SO_3, sulphurous acid
(3) H_2SO_4, sulphuric acid
(4) $H_2S_2O_8$, peroxodisulphuric acid

CPB172

22. Identify the **correct** statements from the following:
(a) $CO_2(g)$ is used as refrigerant for ice-cream and frozen food.
(b) The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
(c) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
(d) CO is colorless and odourless gas.
(1) (c) and (d) only
(2) (a) and (b) and (c) only
(3) (a) and (c) only
(4) (b) and (c) only

CPB173

23. Which of the following is **not** correct about carbon monoxide?
(1) It is produced due to incomplete combustion
(2) It forms carboxyhaemoglobin
(3) It reduce oxygen carrying ability of blood
(4) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.

CPB174

NEET(UG) 2020(COVID-19)

24. Which one of the following reactions does not come under hydrolysis type reaction ?
(1) $SiCl_{4(l)} + 2H_2O_{(l)} \rightarrow SiO_{2(s)} + 4HCl_{(aq)}$
(2) $Li_3N_{(s)} + 3H_2O_{(l)} \rightarrow NH_{3(g)} + 3LiOH_{(aq)}$
(3) $2F_{2(g)} + 2H_2O_{(l)} \rightarrow 4HF_{(aq)} + O_{2(g)}$
(4) $P_4O_{10(s)} + 6H_2O_{(l)} \rightarrow 4H_3PO_{4(aq)}$

CPB175

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NEET(UG) 2021

25. Noble gases are named because of their inertness towards reactivity. Identify an **incorrect** statement about them.
(1) Noble gases are sparingly soluble in water.
(2) Noble gases have very high melting and boiling points.
(3) Noble gases have weak dispersion forces.
(4) Noble gases have large positive values of electron gain enthalpy.

CPB176

26. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?
(1) $HF < HCl < HBr < HI$: Increasing acidic strength
(2) $H_2O < H_2S < H_2Se < H_2Te$: Increasing pK_a values
(3) $NH_3 < PH_3 < AsH_3 < SbH_3$: Increasing acidic character
(4) $CO_2 < SiO_2 < SnO_2 < PbO_2$: Increasing oxidizing power

CPB177

NEET(UG) 2022

27. Match List-I with List-II

List- I (Hydrides)	**List-II (Nature)**
(a) MgH_2	(i) Electron precise
(b) GeH_4	(ii) Electron deficient
(c) B_2H_6	(iii) Electron rich
(d) HF	(iv) Ionic

Choose the correct answer from the options given below :
(1) (a)-(iii), (b) - (i), (c) - (ii), (d)- (iv)
(2) (a)-(i), (b) - (ii), (c) - (iv), (d)- (iii)
(3) (a)-(ii), (b) - (iii), (c) - (iv), (d)- (i)
(4) (a) -(iv), (b) - (i), (c) - (ii), (d)- (iii)

CPB218

28. Choose the correct statement :
(1) Diamond is covalent and graphite is ionic.
(2) Diamond is sp^3 hybridised and graphite is sp^2 hybridized.
(3) Both diamond and graphite are used as dry lubricants.
(4) Diamond and graphite have two dimensional network.

CPB219

29. Which of the following statement is not correct about diborane ?
(1) The four terminal B-H bonds are two centre two electron bonds.
(2) The four terminal Hydrogen atoms and the two Boron atoms lie in one plane.
(3) Both the Boron atoms are sp^2 hybridised
(4) There are two 3-centre-2-electron bonds.

CPB220

NEET(UG) 2023

30. Match **List-I** with **List-II** :

List-I (Oxoacids of Sulphur)	**List-II (Bonds)**
A. Peroxodisulphuric acid	I. Two S-OH, Four S=O, One S-O-S
B. Sulphuric acid	II. Two S-OH, One S=O
C. Pyrosulphuric acid	III. Two S-OH, Four S=O, One S-O-O-S
D. Sulphurous acid	IV. Two S-OH, Two S=O

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-III, C-II, D-IV

CPB221

NEET(UG) 2024

31. Among Group 16 elements, which one does **NOT** show –2 oxidation state ?
(1) O (2) Se
(3) Te (4) Po

CPB222

32. Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.
A. Al^{3+} B. Cu^{2+}
C. Ba^{2+} D. Co^{2+}
E. Mg^{2+}
Choose the correct answer from the options given below:
(1) B, A, D, C, E (2) B, C, A, D, E
(3) E, C, D, B, A (4) E, A, B, C, D

CPB223

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33. During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), Which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

(1) dilute hydrochloric acid

(2) concentrated sulphuric acid

(3) dilute nitric acid

(4) dilute sulphuric acid

CPB224

NEET(UG) 2024 (Re-examination)

34. Identify the **incorrect** statement from the following:

(1) The acidic strength of HX(X = F, Cl, Br and I) follows the order : HF > HCl > HBr > HI

(2) Fluorine exhibits –1 oxidation state whereas other halogens exhibit +1, +3, +5 and +7 oxidation states also.

(3) The enthalpy of dissociation of F_2 is smaller than that of Cl_2

(4) Fluorine is stronger oxidising agent than chlorine.

CPB225

35. Identify the **incorrect** statement

(1) PEt_3 and $AsPh_3$ as ligands can form $d\pi$-$d\pi$ bond with transition metals

(2) The N-N single bond is as strong as the P-P single bond.

(3) Nitrogen has unique ability to form $p\pi$-$p\pi$ multiple bonds with nitrogen, carbon and oxygen

(4) Nitrogen cannot form $d\pi$-$p\pi$ bond as other heavier elements of its group.

CPB226

36. Match **List-I** with **List-II** :

List-I (Test/reagent)	**List-II (Radical identified)**
A. Lake Test	I. NO_3^-
B. Nessler's Reagent	II. Fe^{3+}
C. Potassium sulpho cyanide	III. Al^{3+}
D. Brown Ring Test	IV. NH_4^+

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-III, D-I

(2) A-II, B-IV, C-III, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-III, B-IV, C-II, D-I

CPB227

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**3**	**2**	**4**	**2**	**1**	**2**	**1**	**4**	**1**	**4**	**1**	**1**	**4**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**4**	**3**	**1**	**3**	**1**	**4**	**1**	**4**	**3**	**2**	**2**	**4**	**2**	**3**	**1**
Question	31	32	33	34	35	36									
Answer	**4**	**1**	**4**	**1**	**2**	**4**									

EXERCISE – III (Analytical Questions) Master Your Understanding

1. Given below are two statements :
Statement-I : BCl_3 is more stable than $TlCl_3$.
Statement-II : Boric acid is a protic acid.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CPB178

2. Given below are two statements :
Statement-I : Lead is known but not to form an iodide PbI_4.
Statement-II : Conc. HNO_3 can be transported in Al Container.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CPB179

3. Given below are two statements :
Statement-I : Aqueous solution of Borax is alkaline to litmus paper.
Statement-II : Diamond is thermo-dynamically more stable than graphite.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CPB180

4. Given below are two statements :
Statement-I : Chain length of Silicon polymer can be controlled by adding $(CH_3)_2SiCl_2$.
Statement-II : Silica gel is used as a drying agent.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CPB181

5. Given below are two statements :
Statement-I : Graphite has layered structure in which layers are held by Vander wall's force.
Statement-II : In fullerene there are six and five member ring.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CPB182

6. Given below are two statements :
Statement-I : In B_2H_6 bridge bonds are stronger than terminal bonds.
Statement-II : In Diborane all atoms can lie in the same plane.

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In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

CPB183

7. Given below are two statements :

Statement-I : First inert gas compound was $Xe[PtF_6]$ formed by Neil Bartlett.

Statement-II : Xenon fluorides are powerful fluorinating agent.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

CPB184

8. Given below are two statements :

Statement-I : ClF_3 and BrF_3 are used for the production of UF_6 in the enrichment of U_{235}.

Statement-II : Inter halogen compound's are covalent molecules and are paramagnetic in nature.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

CPB185

9. Given below are two statements :

Statement-I : With cold & dilute alkalies Cl_2 produces a mixture of chloride and chlorite.

Statement-II : Chlorine is a powerful bleaching agent and bleaching action is due to reduction.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

CPB186

10. Given below are two statements :

Statement-I : NH_3 can be dried by concentrated. H_2SO_4.

Statement-II : Decomposition of O_3 into O_2 result in the liberation of heat.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

CPB187

11. Given below are two statements :

Statement-I : Oxy acids of phosphorous contain (P–H) bond act as a reducing agent.

Statement-II : H_3PO_4 gives disproportionation reaction on heating.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

CPB188

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12. Given below are two statements :

Statement-I : PCl_3 hydrolyses in the presence of moisture gives fumes of HCl.

Statement-II : White phosphorous has discrete units of P_4 molecules.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

CPB189

13. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : SF_6 cannot be hydrolysed but SF_4 can be.

Reason : Six F atoms in SF_6 prevent the attack of H_2O on S-atom of SF_6.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB190

14. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : HI cannot be prepared by the reaction of KI with conc. H_2SO_4.

Reason : HI has lowest H–X bond strength among halogen acids.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB191

15. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : HNO_3 makes iron passive.

Reason : HNO_3 forms a protective layer of ferric nitrate on the surface of iron.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB192

16. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : H_3PO_2 is a dibasic acid.

Reason : Two H-atoms are directly attached to phosphorous.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB193

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17. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : Boric acid is considered as a weak acid.

Reason : It is not able to release $H^{\oplus}$ ions on its own.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB194

18. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : Aluminium wire is used to make transmission cables.

Reason : Aluminium has a high electrical and thermal conductivity and due to its low cost & light weight.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB195

19. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : (B–F) Bond lengths in BF_3 and BF_4^- is different.

Reason : In both BF_3 and BF_4^- $P_\pi - P_\pi$ back bonding is present.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB196

20. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion : Stability of +5 O.S. state decreases and that of +3 oxidation state increases down the group in group 15.

Reason : Inert pair effect is more prone in case of heavier members of p-block.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

CPB197

21. Match the column :

Column-I		Column-II	
(A)	Borax $\xrightarrow{\Delta}$	(i)	BN
(B)	$B_2H_6 + H_2O \rightarrow$	(ii)	B_2H_6
(C)	$B_2H_6 + NH_3 \rightarrow$ (excess)	(iii)	H_3BO_3
(D)	$BCl_3 + LiAlH_4 \rightarrow$	(iv)	$NaBO_2 + B_2O_3$

(1) A - iii, B - iv, C - ii, D - i

(2) A - ii, B - i, C - iv, D - iii

(3) A - i, B - ii, C - iii, D - iv

(4) A - iv, B - iii, C - i, D - ii

CPB198

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22. Match the column :

Column-I		Column-II	
(A)	He	(i)	It's partial hydrolysis does not change oxidation state of central atom.
(B)	XeF_6	(ii)	It is used in modern diving apparatus.
(C)	XeF_4	(iii)	It is used to provide inert atmosphere for filling electrical bulbs.
(D)	Ar	(iv)	Its central atom has sp^3d^2 hybridisation

(1) A - iv, B - iii, C - ii, D - i
(2) A - ii, B - i, C - iv, D - iii
(3) A - iii, B - iv, C - i, D - ii
(4) A - i, B - ii, C - iii, D - iv

CPB199

23. Match the column :

Column-I		Column-II	
(A)	$H_4P_2O_7$	(i)	Peroxy disulphuric acid
(B)	$H_2S_4O_6$	(ii)	Pyro sulphurous acid
(C)	$H_2S_2O_5$	(iii)	Pyro phosphoric acid
(D)	$H_2S_2O_8$	(iv)	Tetra thionic acid

(1) A - ii, B - iii, C - iv, D - i
(2) A - iii, B - iv, C - ii, D - i
(3) A - iii, B - iv, C - i, D - ii
(4) A - i, B - ii, C - iii, D – iv

CPB200

24. Match the column :

Column-I		Column-II	
(A)	Ostwald process	(i)	in the manufacturing of NH_3
(B)	Deacon's process	(ii)	in the manufacturing of H_2SO_4
(C)	Contact process	(iii)	in the manufacturing of Cl_2
(D)	Haber process	(iv)	in the manufacturing of HNO_3

(1) A - i, B - iii, C - ii, D - iv
(2) A - iv, B - iii, C - ii, D - i
(3) A - iii, B - ii, C - iv, D - i
(4) A - iv, B - i, C - iii, D - ii

CPB201

25. Match the column :

Column-I		Column-II (Basic unit)	
(A)	Ortho Silicate	(i)	SiO_2
(B)	Pyro Silicate	(ii)	$(Si_2O_5^{-2})_n$
(C)	2-D Silicate	(iii)	$Si_2O_7^{-6}$
(D)	3-D Silicate	(iv)	(SiO_4^{-4})

(1) A - iv, B - iii, C - ii, D - i
(2) A - ii, B - iii, C - i, D - iv
(3) A - iii, B - ii, C - iv, D - i
(4) A - iv, B - i, C - iii, D - ii

CPB202

26. Match the column :

Column-I		Column-II	
(A)	Br_2	(i)	used in estimation of CO
(B)	HF	(ii)	powerful bleaching agent
(C)	O_3	(iii)	liquid at room temperature
(D)	I_2O_5	(iv)	used in permanent marking on glass surface

(1) A - i, B - iii, C - ii, D - iv
(2) A - ii, B - i, C - iii, D - i
(3) A - iii, - B - iv, C - ii, D- i
(4) A - iv, B - ii, C - i, D - iii

CPB203

27. Match the column :

Column-I		Column-II	
(A)	Anhydride of HNO_3	(i)	P_4O_{10}
(B)	Anhydride of HNO_2	(ii)	P_4O_6
(C)	Anhydride of H_3PO_4	(iii)	N_2O_3
(D)	Anhydride of H_3PO_3	(iv)	N_2O_5

(1) A - i, B - iv, C - iii, D - ii
(2) A - iv, B - iii, C - i, D - ii
(3) A - ii, B - i, C - iv, D - iii
(4) A - iii, B - ii, C - iv, D – i

CPB204

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28. Match the column :

Column-I		Column-II	
(A)	Red Lead	(i)	$CoCl_2$
(B)	Carborandum	(ii)	$Cl_3C.NO_2$
(C)	Tear gas	(iii)	SiC
(D)	Phosgene	(iv)	Pb_3O_4

(1) A - iii, B - iv, C - i, D - ii
(2) A - i, B - ii, C - iv, D - iii
(3) A - iv, B - iii, C - ii, D - i
(4) A - ii, B - i, C - iii, D - iv

CPB205

29. Match the column :

Column-I		Column-II	
(A)	Bleaching powder	(i)	$Na_2S_2O_3.5H_2O$
(B)	Pseudo halogen	(ii)	ICl_3
(C)	Inter halogen compound	(iii)	$(CN)_2$
(D)	Hypo solution	(iv)	$CaOCl_2$

(1) A - i, B - ii, C - iii, D - iv
(2) A - ii, B - iv, C - i, D - iii
(3) A - iii, B - i, C - iv, D - ii
(4) A - iv, B - iii, C - ii, D - i

CPB206

30. Match the column :

Column-I		Column-II	
(A)	Borax	(i)	Form complex with haemoglobin
(B)	Diborane	(ii)	used as a fire extinguisher
(C)	Carbon mono oxide	(iii)	Swell's up on heating
(D)	Carbon dioxide	(iv)	Catches fire spontaneously

(1) A - i, B - iii, C - ii, D - iv
(2) A - ii, B - i, C - iv, D - iii
(3) A - iii, B - iv, C - i, D - ii
(4) A - iv, B - ii, C - iii, D – i

CPB207

31. Which of the following statements are correct?
(A) P_4 molecule in white phosphorous have angular strain therefore white phosphorous is very reactive.
(B) All the three (N–O) bond lengths are equal in HNO_3.
(C) PCl_5 is ionic in solid state in which cation is tetrahedral and anion is octahedral.
(D) All (P–Cl) B.L. in PCl_5 mol. in gaseous state are equal.
(1) If (A), (B) and (C) options are correct.
(2) If (A) and (B) both options are correct.
(3) If (B) and (D) both options are correct.
(4) If (A) and (C) both options are correct.

CPB208

32. In which of the following reactions conc. H_2SO_4 is not used as an oxidising agent :-
(1) $Zn + 2H_2SO_4 \rightarrow ZnSO_4 + SO_2 + 2H_2O$
(2) $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$
(3) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$
(4) $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

CPB209

33. In which of the following terminal ($2C - 2e^-$) bond and bridge bond are not lying in the same plane ?
(1) B_2H_6 (2) Solid $BeCl_2$
(3) Al_2Cl_6 (4) I_2Cl_6

CPB210

34. PH_3 can not be obtained by :-
(1) Heating of H_3PO_3
(2) Reacting white P_4 with hot & conc. NaOH
(3) Hydrolysis of Ca_3P_2
(4) Heating of H_3PO_4

CPB211

35. Acids of sulphur which contain peroxy linkage :-
(1) $H_2S_2O_8$ (2) $H_2S_2O_7$
(3) $H_2S_2O_6$ (4) $H_2S_2O_3$

CPB212

36. Which of the following will not give N_2 when heated :

(1) NaN_3 (2) NH_4NO_2

(3) $(NH_4)_2Cr_2O_7$ (4) NH_4NO_3

CPB213

37. Heating of oxalic acid with conc. H_2SO_4 not evolves:-

(1) CO (2) CO_2

(3) Both (1) and (2) (4) SO_2

CPB214

38. SiO_2 reacts with :-

(1) HF (2) NaOH

(3) Both (1) and (2) (4) CO_2

CPB215

39. Which of the following compound produce oxy acid of central atom on hydrolysis ?

(1) NF_3 (2) SF_6 (3) PCl_5 (4) NCl_3

CPB216

40. Which of the following cation can not give borax bead test ?

(1) Cu^{2+} (2) Co^{+2} (3) Zn^{+2} (4) Mn^{+2}

CPB217

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**1**	**2**	**4**	**1**	**2**	**1**	**2**	**3**	**4**	**2**	**1**	**1**	**2**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**4**	**1**	**1**	**3**	**1**	**4**	**2**	**2**	**2**	**1**	**3**	**2**	**3**	**4**	**3**
Question	31	32	33	34	35	36	37	38	39	40					
Answer	**4**	**3**	**1**	**4**	**1**	**4**	**4**	**3**	**3**	**3**					

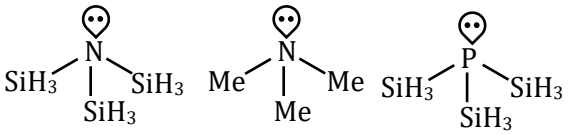
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p-Block Elements - Group 13 to 18

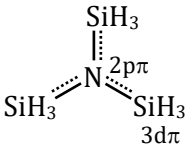
SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (2)

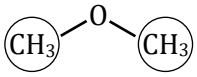


In $N(SiH_3)_3$ Back bonding is possible whereas in $N(Me)_3$ & $P(SiH_3)_3$ No back bonding is there because $N(SiH_3)_3$ $2p\pi$-$3d\pi$ back bonding occurs so shape because trigonal planar.



2. Ans. (2)

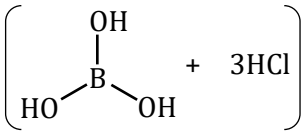
In $N(SiH_3)_3$ back bonding occurs same in $O(SiH_3)_2$ but in $O(CH_3)_2$ No vacant orbital is there so No back bonding and bond angle increases due to large size of side molecule.



Inter Molecular Repulsion

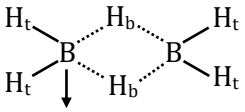
3. Ans. (1)

$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$



4. Ans. (3)

Hybridisation of diborane

$B_2H_6 \rightarrow$ 

$[sp^3]$

5. Ans. (4)

Partial hydrolysis occurs in

$SbCl_3 + H_2O \longrightarrow [SbOCl]$

↓

Ionic character increases as we move down the group so reactivity decreases hence partial hydrolysis occurs.

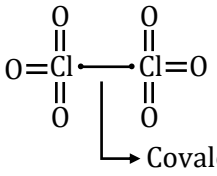
6. Ans. (4)

SF_6 cannot be hydrolysed because of steric repulsion is much more in SF_6.

Whereas in NF_3 No vacant orbital is there.

7. Ans. (4)

In the formation of Cl_2O_6 from ClO_3



Covalent Bond is formed

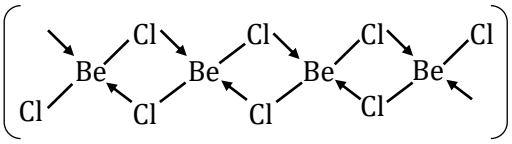
So covalent bond is formed by half-filled orbital no empty atomic orbital in volved.

8. Ans. (4)

$[BeCl_2]_n \rightarrow$ (Solid)

↓

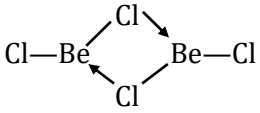
(Polymer)



$\rightarrow 4\sigma + 0$ LP

$\rightarrow sp^3$

$\rightarrow [8e^-]$



$[6e^-]$ $[6e^-]$

$(BeH_2) \rightarrow [3C\text{–}2e^-]$

↓

Electron deficient

9. Ans. (1)

$\rightarrow$ HOF $\rightarrow$ As a (Intermidiate)

$\rightarrow$ EN of F $\rightarrow$ Highest $\rightarrow$ (+ve)x

[HOF] [(–1) Oxidising state]

↓

(Highest EN)

$\begin{bmatrix} HFO_3 \\ HFO_2 \end{bmatrix} \rightarrow +ve \rightarrow [x]$

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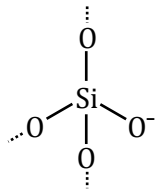
10. Ans. (3)
All Halogens F, Cl, Br, I form oxyacids but not of all type as F only form HOF.

11. Ans. (3)
It P is H_3PO_4

$$\underset{ie}{H_3PO_4} \xrightarrow{-1H_2O} \underset{Ous}{HPO_3} \xrightarrow{-O} \underset{Hypo}{HPO_2}$$

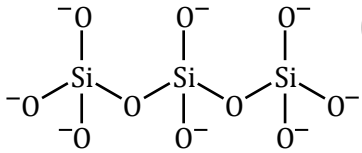
In All these oxyacids we have different oxidation state.

12. Ans. (1)
One monovalent corner oxygen atom in each tetrahedron i.e. 3 oxygen atoms are shared.



→ At forms sheet silicate $[Si_2O_5^{-2}]$

13. Ans. (2)
In Kinonite $3SiO_4^{-4}$ is attached
$Ca_2Cu_2Si_3O_{10}$



(–8)

14. Ans. (2)
Organo silicon polymer are silicones

15. Ans. (1)
White phosphorous reacts with oxygen at ordinary room temperature.

16. Ans. (1)
Red & Yellow phosphorous are allotropes of phosphorous.

17. Ans. (4)
PbI_4 do not exist due to inert pair effect Pb^{+2} is much more stable than Pb^{+4} and EN of I is very less. So Higher Oxidising state of Pb cannot be achieved.

18. Ans. (2)
Graphite conduct electricity, due to the presence of Delocalisation of π electrons.

19. Ans. (2)
Alane is polymer of $(AlH_3)_n$ chemical name of $(AlH_3)_n$

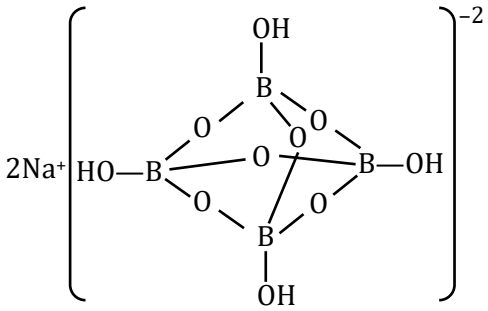
20. Ans. (3)
Aluminium is protected by a film of Aluminium oxide (Al_2O_3) this layer protect Aluminium from acid attack & water attack.

21. Ans. (3)
In borax based test different coloured meta borates has been formed from which we can identify metal ions.
Ex. → $Cu(BO_2)_2$ → Blue Bead
↓
Copper Meta Borate

22. Ans. (1)
Boric acid $H_3BO_3[B(OH)_3]$ Polymerises due to hydrogen bonding.

23. Ans. (2)
$M_2SO_4.M_2'(SO_4)_3.24H_2O$
$2M:2M':4(SO_4)$
[1 : 1 : 2] Ratio for M, M', SO_4.

24. Ans. (1)
$[Na_2B_4O_7.10H_2O]$
$Na_2[B_4O_5.(OH)_4].8H_2O$
↓ ↓
Cation Anionic part



25. Ans. (2)
Double salt is made up of 2 salts
(SA + SB) + (SA + WB)

26. Ans. (3)
B_2H_6 all terminal H can be displaced but in $[B_2(CH_3)_6]$ all hydrogen is removed so $B_2(CH_3)_6$ cannot be formed.

27. Ans. (3)
Group 14 element
NH_3, PH_3 ------ Hydrides are covalent in nature.

28. Ans. (1)
Gas responsible for green house is CO_2

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29. Ans. (3)

Artificial Gem used for diamond cutting is SiC

30. Ans. (4)

Borax & Boric Acid is used in the manufacture of all of these Pyrex Glass, Fibre, Glass wool.

31. Ans. (1)

Orthoboric acid is used as antiseptic.

32. Ans. (1)

Crucibles is made up of Graphite.

33. Ans. (2)

Alum is used as mordant, purification of water, fanning of leather not as an insecticide.

34. Ans. (3)

B
∨
Al ┐
∧ ├→ Due to contraction Z_{eff} ↑ IP↑
Ga ┘
∨
In ┐
∧ ├→ Lanthanide Contraction
Tl ┘

35. Ans. (3)

$Al + NaOH + H_2O \rightarrow$ In Aq sol

It produces sodium tetra hydroxoaluminate

$Na[Al(OH)_4]$

36. Ans. (2)

With cobalt, Borax Bead test bead of cobalt is

$[CO(BO_2)_2]$

[Blue Bead]

37. Ans. (2)

$B(OH)_3$

$H_3BO_3 \rightarrow$ Lewis acid

Weak mono basic acid.

$2(H_3BO_3) \rightarrow (B_2O_3)$

Protic acid ×

38. Ans. (3)

Correct order of I.E.

C > Si > Ge > Sn < Pb

Sn → Lanthanide contraction → Z_{eff} ↑, IP↑

39. Ans. (2)

$Sn^{+2} < Sn^{+4}$ (Is more stable)

Do not posses inert pair effect so Higher Oxidising state is more stable.

40. Ans. (1)

Some of the p block oxides are amphoteric & neutral but maximum are acidic. GeO_2 is acidic by nature.

41. Ans. (3)

Diamond & Graphite have dangling bond but due to closed soccer Bucky ball like structure Buck Minister fullerene do not have dangling bond.

42. Ans. (1)

Dry ice → Refrigerant in ice-cream

CO_2 gas → Soft Drinks

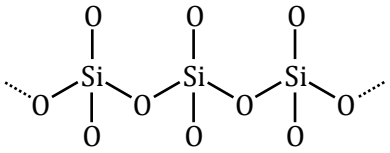
Silicones → Polymer it is so used in water proof Jacket

Zeolite → Used in softening of Hard water with impurity of ($CaCl_2$, $MgCl_2$)

43. Ans. (3)

Basic unit of silicones is $(R_2SiO)_n$

44. Ans. (3)



$SiO_{2+0.5+0.5} \rightarrow \left(SiO_3^{-2}\right)_n$

Linear chain silicate.

45. Ans. (4)

As we move from top to bottom in a group catenation property decreases due to large size self-linking tendency decreases.

46. Ans. (4)

Graphite is thermodynamically more stable than diamond.

47. Ans. (1)

Syn gas & Synthesis gas

$[CO + H_2]$ → formed by $C + H_2O \xrightarrow{\Delta} [CO + H_2]$

48. Ans. (1)

P_4O_{10} is used as Dehydrating agent for many complexes.

49. Ans. (4)

PH_3 Produces smoky ring when it has impurity of P_2H_4.

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50. Ans. (1)
PH_3 is less basic than NH_3 as Lone Pair donation tendency is less due to larger size.

51. Ans. (3)
$HNO_3 + I_2 \rightarrow HIO_3$
$\downarrow$
Which can be written were as $\rightarrow HOIO_2$

52. Ans. (3)
Both red & white phosphorous can be oxidised be air.

53. Ans. (3)
N_2O is used in anaesthetic dental surgery.

54. Ans. (4)
Ammonium dichromate & barium azide on heating both of then we get N_2 gas

55. Ans. (4)
NO_2 is formed
Cu + conc. $HNO_3 \rightarrow$ Metal Nitrate + NO_2
Zn + conc. $HNO_3 \rightarrow$ Metal Nitrate + NO_2
$Pb(NO_3)_2 \xrightarrow{\Delta} PbO + NO_2 + O_2$
By all these 3 reaction NO_2 is evolved.

56. Ans. (4)
Au being noble metal directly do not react with conc. HNO_3.

57. Ans. (2)
N_2 is used to create inert atmospheric in steel industry & many industrial process.

58. Ans. (4)
In the preservation of biological material, food items, in cryogenic surgery.

69. Ans. (4)
Ammonium nitrate, urea, ammonium phosphate and many ammonium complex is used in fertilizers.

60. Ans. (2)
Nitric acid is used in the pickling of steel.

61. Ans. (2)
PH_3 is used in Holme's signal for ships.

62. Ans. (1)
Phosphine gas is used in creation of smoke screen.

63. Ans. (3)
Quartz is a crystalline variety of silicates (SiO_2)

64. Ans. (3)
Aluminium vessel should not be washed with washing soda because it reacts with aluminium to form soluble aluminate.

65. Ans. (1)
Haber's Process
$N_2 + 3H_2 \xrightarrow{\text{condition}} \underset{\text{Ammonia}}{2NH_3}$

66. Ans. (2)
NH_3 can act as a Lewis base Lone Pair donor.

67. Ans. (3)
Nitrogen family is used in the formation of fertilizers.

68. Ans. (1)
Nitrogen do not show allotropy.

69. Ans. (4)
Covalency of N is = 4 in all of present oxides N_2O_5, N_2O_4, N_2O_3.
No of Covalent bond 4 is show by Nitrogen.

70. Ans. (3)
PH_3 can be inflammable in the presence of impurity of P_2H_4, P_4 vapours.

71. Ans. (1)
$P_4 + SOCl_2 \rightarrow [PCl_3] + SO_2$

72. Ans. (2)
Pyrophosphorous acid & Phosphinic acid both have 2 P-H links

73. Ans. (1)
Acid can reduce $AgNO_3$ into Metallic silver
Hypophosphorous acid
H_3PO_2

74. Ans. (2)
Ratio of

P-P Bond	:	P-H Bond	:	P-OH Bond
Hypo-phosphoric		Pyro-phosphorous		Cyclotrimeta-phosphoric acid
1	:	2	:	3

75. Ans. (4)
SO_2 can act as Reducing agent & Oxidising agent and as a Bleaching agent for oxidation.

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76. Ans. (4)

Ozone after decomposition release oxygen due to which it is used as oxidising agent, disinfectant, Bleaching agent.

77. Ans. (4)

Black sulphide (Galena)

$PbS + O_3 \rightarrow [PbSO_4]$

white

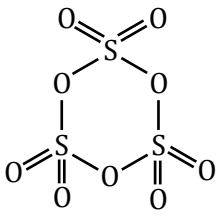
78. Ans. (1)

When H_2S is tested with filter paper moistened with lead acetate paper. It terms silvery black due to the formation of PbS.

$(CH_3COO)_2Pb + H_2S \rightarrow 2CH_3COOH + PbS.$

(Silvery black)

79. Ans. (4)



$S_3O_9 \rightarrow (SO_3)_3 \rightarrow$ S–O–S Links

80. Ans. (3)

Dry bleaching action is done with bleaching powder & O_3 ozone.

81. Ans. (1)

$KBr + \text{conc. } H_2SO_4 \rightarrow KHSO_4 + Br_2 + SO_2$

Bisulphate

Bromine gas is Evolved.

82. Ans. (3)

Conc. H_2SO_4 presence converts or oxidises $HI \longrightarrow I_2$ Due to which its colour comes out that is iodine.

$NaI + H_2SO_4 \longrightarrow Na_2SO_4 + HI$

$HI \xrightarrow[H_2SO_4]{\text{conc.}} H_2O + SO_2 + \underset{\substack{\text{Voilet}\\ \text{Colour}}}{I_2}$

83. Ans. (1)

KI $\longrightarrow$ (I_2)

Starch Iodide Reaction — Liberation of Iodine

84. Ans. (4)

Ozone is a good oxidising agent it can act as oxidising agent and also used as bleaching agent in dry state.

85. Ans. (3)

(1) $Zn + \underset{\text{Dil}}{H_2SO_4} \longrightarrow ZnSO_4 + H_2\uparrow$

(2) $Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + SO_3$

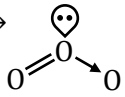
(3) $Cu + H_2SO_4 \longrightarrow CuSO_4 + [SO_2] + H_2O$

$\downarrow$

Sulphur Dioxide

(4) $Cu_5SO_4 \xrightarrow{\Delta}$ Do not Decompose

86. Ans. (3)

$O_3 \rightarrow$ 

(V Bent)

(1) O_2 is paramagnetic proved by MOT & O—O BO in ozone is between 1 & 2

(2) O_3 is Diamagnetic

87. Ans. (4)

(1) SO_2 has bleaching action & temporary bleaching due to its reducing action [S oxidises to higher Oxidising state]

(2) It's used as preservative & Anti oxidation is food industry.

(3) Liquid SO_2 is used to extract aromatics from kerosene pool. So all of above.

88. Ans. (4)

→ Sulphuric acid is used in formation of detergent in Neutralisation process of lauryl alcohol.

→ In storage battery in lead acid batteries.

→ In Nitrocellulose process sulphuric acid act as reaction catalyst. All of above.

89. Ans. (4)

As we move top to bottom in a group only +ve OS as metallic nature increases.

S

Se

Te

Po → + ve → B

T ↓ Metallic nature ↑

90. Ans. (3)

NaH_2PO_2

$\left[Na^{+1}H_2^{+1+1}\overset{+1}{P}/O_2^{-2-2}\right]$

[P = +1]

91. Ans. (4)
Pure ozone colour is Pale Blue in gas as Molecule are less dense & becomes dark blue liq & then become violet black in solid form.

92. Ans. (3)
Ozone absorb UV radiation from solution protect atmosphere by restrain the entry of UV rays but do not stop infra red radiation.
Main reason of ozone depletion is By CFC's & By Nitrogen oxides. Which very fast reacts with ozone.

93. Ans. (3)
$H_2O \rightarrow$ H bond
$\vee$
H_2S
H_2Se } VWFA [VWFA $\propto$ Size $\propto$ SA]
H_2Te

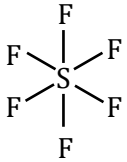
H_2O > $H_2T > H_2Se > H_2S$
↓ ↓
H Bond ↑ VWFA ↑

Bpt ↑

94. Ans. (3)
O_2 Exist as diatomic where as S_8 exist as polyatomic. The Vander Waal of after in S_8 is very much higher than that why difference in m.p & B.pt is much higher.

95. Ans. (4)
H_2O cannot show reducing property due to high bond energy of O-H bond it cannot easily break.
On moving down the group reducing nature increases.

96. Ans. (4)
(1) Hexafluorides are only stable halides due to small size of fluorine less steric repulsion.
(2) Hexafluorides are non toxic, colourless, Odourless gas. (SF_6).
(3) All Hexafluorides (SF_6)



$6\sigma + 0LP \rightarrow$ Octahedral
All of above

97. Ans. (4)
Industrial oxygen (O_2) can be obtained by fractional distillation of air as both N_2 & O_2 have different (B. Pt)
So, from here we get large amount of pure (O_2) at 183°C

98. Ans. (4)
As Pt & Au (Gold) both of them are noble metals where as He is noble gas so all these of them are non-reactive to oxygen.

99. Ans. (1)
There are different allotropes of 's' called monoclinic, plastic and orthorhombic sulphur.

100. Ans. (3)
Oleum → is also known as Pyro sulphuric acid → $[H_2S_2O_7]$
→ $H_2SO_4 + SO_3 \rightarrow [H_2S_2O_7]$
Both 1 & 2

101. Ans. (1)
Oxygen/Acetylene welding or gas welding is a process which relies an combustion of oxygen & acetylene mixed together in a proportion (oxygen).

102. Ans. (2)
In vapour phase sulphurs exist as S_2 molecule & S_2 molecule has 2 unpaired e^- in ABMO 3P orbital & hence exhibits paramagnetism.

103. Ans. (2)
Contacts process is used in the manufacture of $[H_2SO_4]$

104. Ans. (1)
(1) Due to High EN of F it can replace all halogen.
(2) Florine reacts fast with halogens to form interhalogen compounds.
(3) Fluorine reacts readily with H_2O to form Hydrofluoric acid.
(4) Halogens & Oxygen do not combine directly with carbon.

105. Ans. (1)
AgF forms ionic by nature so no ppt, colourless.

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106. Ans. (3)

Volatile nature totally depends upon the force of attraction. Between 2 dif. Molecules. As halogens have weak Vander Waal's force of attraction. it because volatile.

107. Ans. (1)

Iodine gas turns starch iodide paper blue/black colour.

As they oxidises iodine to KI & free chlorine gets reduce to chloride ion.

108. Ans. (1)

BrF_5 is Interhalogen

109. Ans. (3)

As iodine do not react with NaCl directly so no reaction takes place so it cannot decolarise iodine.

110. Ans. (4)

$$F_2 + H_2O \longrightarrow H^+_{(aq)} + F^-_{(aq)} + \left(O_2 \uparrow\right)$$

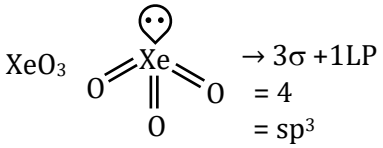
F_2 reacts violently with water to form Hydrogen fluoride $O_2 \uparrow$

111. Ans. (1)

Helium is added to oxygen used be deep sea divers as it is less soluble in blood then nitrogen under high pressure and also maintain the pressure of oxygen in derivers cylinder.

112. Ans. (1)

Which is not correct



113. Ans. (1)

$XeF_2 + SbF_5 \longrightarrow [XeF]^+ + [SbF_6]^-$

Displacement reaction

114. Ans. (3)

$XeCl_4$ can not be formed because Cl has Low EN unable to excite Xe at +4

115. Ans. (2)

$$SbF_5 + XeF_4 \longrightarrow \underset{\text{Anion Octahedral}}{\left[SbF_6^-\right]} + \underset{\text{Cation Tshape}}{\left[XeF_3^+\right]}$$

116. Ans. (2)

Ne due to its small size cannot form clathrate compound.

117. Ans. (2)

Conc. H_2SO_4

118. Ans. (3)

Chlorine gas is not used in refining of sugar

→ Cl is used in Bleaching powder to maintain its quality approx. 33% of dry bleaching powder consist of chlorine.

→ Water chlorination is done with sodium hypochlorite to sterilise water to kill Virus, Bacteria, Microbes.

→ $Au + Cl_2 \xrightarrow{\text{Moisture}} \left[AuCl_3\right]$

Which is soluble in water the pure gold is then precipitated by ferrous sulphate & SO_2

119. Ans. (3)

$CIF_3 + U_3O_8 \longrightarrow [UF_6]$

and further we get pure uranium

U_{238}	U_{235}	
(Major)	0.7%	Before enrichment
	3–5%	After enrichment

120. Ans. (3)

He is non inflammable & light gas it do not support combustion.

121. Ans. (1)

He is used in gas cooled nuclear reactor as a coolant as it condense at very low temperature.

122. Ans. (2)

Ne when used in Discharge tube produces an orange-red glow.

123. Ans. (2)

Ne bulb is used in botanical of green house garden cause it develop orang-red glow helps to reef cool.

124. Ans. (3)

Argon is used to provide inert atmosphere even at high tamp. Metallurgical process to agitate the metal in refining process.

125. Ans. (1)

Ar is used in are welding because it has lowest reactivity with metal.

126. Ans. (3)

Halogen are salt forming or producing family.

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127. Ans. (1)
Deacon's process is used in manufacture of NaCl. By electrolysis

$\left[4HCl + O_2 \xrightarrow[400°C]{CuCl_2} 2Cl_2 + H_2O\right]$

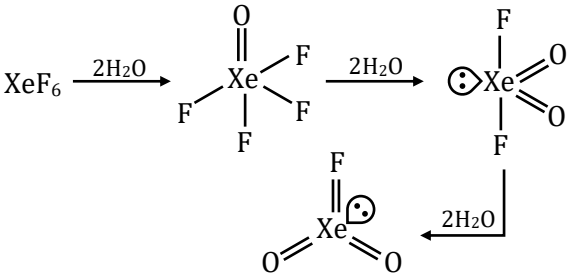
128. Ans. (1)
[HF > HCl > HBr > HI]
[ΔEN↑, BE↑, BDE↑]

129. Ans. (3)
I_2O_5 is used
$I_2O_5 + CO \longrightarrow CO_2 + I_2$
[I_2O_5 reduced to I_2]
[CO is oxidised to CO_2]

130. Ans. (1)
Catalyst used in Deacon's process is $CuCl_2$ it is a type of positive catalyst

131. Ans. (4)
No of O–O bond in perchloric acid is 0

132. Ans. (2)



$XeOF_3$ cannot formed

133. Ans. (3)
RnF_2 it cannot be isolated due its high stability.

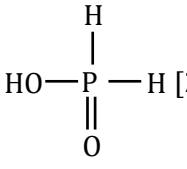
134. Ans. (1)
First inert gas compound was
$Xe[PtF_6]$

135. Ans. (4)
Rn is not found in earth atmosphere.

Exercise-II (Previous Year Questions)

1. Ans. (3)
NO_2 is used for food preservation where as we can't use SO_2 as food preservative.

2. Ans. (3)
Strong reducing behaviour of H_3PO_2 is due to



[2→P-H bond] 1 P–OH bond
High tendency to release proton H^+

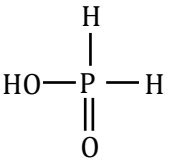
3. Ans. (2)
ONF is iso e^- with O_2N^-
8+7+9 = 8+8+8
= 24 = 24
OF_2 is a fluoride of oxygen not oxide of fluorine.

4. Ans. (4)
+1 is Lower Oxidising state here so Al is more stable in +3 as per Inert pair effect
Al < Ga < In < Tl → Most stable
+1

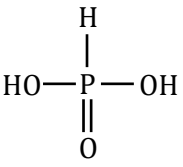
5. Ans. (2)
$HClO < HClO_2 < HClO_3 < HClO_4$
As oxygen atom increases attached to Cl, More e^- pulled away from O–H bonds becomes weak hence acidic strength ↑

6. Ans. (1)
$Cu + Conc.\ HNO_3 \rightarrow Cu(NO_3)_2 + NO_2$
Metal Nitrate + NO_2
every metal

7. Ans. (2)



Phosphinic
Hypo Phosphorous
It is mono protic due to the presence of 1 P–OH Bond



Phosphonic
Diprotic due to the presence of 2 P–OH bond.

8. Ans. (1)
Boric acid is a Lewis acid as it accepts OH^- from water releasing proton.

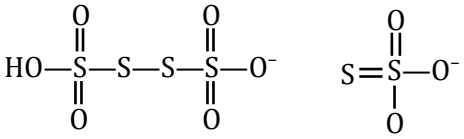
9. Ans. (4)
AlF_3 is soluble in HF only in presence of KF it is due to
$K_3[AlF_6]$
↓
Hexa Dentate

10. Ans. (1)

S–S Bond

$\left[S_4O_6^{-2}, S_2O_3^{-2}\right]$

$[H_2S_4O_6]$ $[H_2S_2O_3]$



Thio Sulphate ion.

11. Ans. (4)

Due to Inert Pair Effect inability of ns^2e^- to participate in valence shell.

$\underline{Sn^{+2}}$ is reducing while $\underline{Pb^{+7}}$ is oxidising

↓ ↓

Gain e^- Loss e^-

12. Ans. (1)

All Halogen oxyacid can be monobasic but HOF is Not as it break giving oxygen.

13. Ans. (1)

HNO_3	NO	N_2	$\overset{+1}{NH_4}/Cl^{-1}$
+5	+2	0	Here nitrogen have -ve OS of –3

14. Ans. (4)

(a) Pure Nitrogen is obtained from heating of barium nitride

$$Ba_3N_2 \xrightarrow{\Delta} 3Ba + \underset{100\%\text{Pure}}{[N_2]}$$

(b) Haber's process

$$N_2 + 3H_2 \xrightarrow{\text{Condition}} 2NH_3$$

(c) Contacts process

For H_2SO_4 $SO_2 + O_2 \rightarrow SO_3$

$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$

$(2H_2SO_4) \leftarrow \overset{+}{H_2}O$

(d) Deacon's process

$$HCl + O_2 \xrightarrow[723K]{CuCl_2} [Cl_2] + H_2O$$

15. Ans. (1)

PbF_4 is ionic by nature due to law of polarisation as size of anion is small.

$SiCl_4$ can be hydrolysed because it have less steric repulsion and also VO. So H_2O can attack easily.

GeX_4 is way more stable than GeX_2 because In GeX_4 it has Higher Oxidising state

$[Ge^{+4} > Ge^{+2}]$

SnF_4 is ionic

16. Ans. (4)

CO_2 is used in Dry cleaning as super efficient natural solvent to clean textiles particular for oil based stains.

17. Ans. (3)

Silicones are used in cosmetic surgery.

Specially	Siloxane
	Silanol
	Decamethylcyclopentane
	Siloxane

All the polymer of silicones are cyclic

18. Ans. (1)

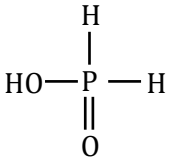
$Ca_3P_2 + H_2O \longrightarrow Ca(OH)_2 + PH_3$

$PH_3 + CuSO_4 \longrightarrow Cu_3P_2 + H_2SO_4$

19. Ans. (3)

The oxyacid which has highest no. of P-H bond

H_3PO_2



(2P-H Bonds)

20. Ans. (1)

Oleum is also known as Pyro-sulphuric acid.

$[H_2S_2O_7]$

21. Ans. (4)

Peroxy-linkage is present is marshals acid $H_2S_2O_8$ Peroxy-disulphuric acid

22. Ans. (1)

(1) CO_2(gas) is not used as refrigerant but CO_2(S) dry ice is used as refrigerant.

(2) C_{60} → Buckminister fullerene

has $\begin{bmatrix} 12 \rightarrow 5\text{Membered ring} \\ 20 \rightarrow 6\text{Membered ring} \end{bmatrix}$

(3) ZSM-5 Zeolite is used as catalyst to convert alcohols to petrol (gasoline)

(4) CO is colorless and odourless gas.

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23. Ans. (4)

(CO) carboxy heamoglobin is way more stable than oxy heamoglobin.

24. Ans. (3)

$2F_2 + H_2O \rightarrow HF + O_2 \uparrow$

25. Ans. (2)

Noble gases bound with weak Vander Waal forces of attraction that's why have low melting point & boiling point.

26. Ans. (2)

$H_2O < H_2S < H_2Se < H_2Te$

Acidic strength depends on bond dissociation enthalpy and acidity of hydride increases down the group.

27. Ans. (4)

Electron deficient hydride → Less than $8e^-$ (B_2H_6)

Electron precise hydride → having $8e^-$ without l.p. (GeH_4)

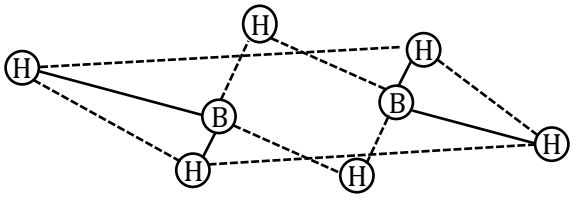
Electron rich hydride → having $8e^-$ with l.p. (HF)

28. Ans. (2)

In diamond each carbon is bonded with four other carbon atoms. So hybridisation of carbon atom is sp^3.

In graphite each carbon is bonded with three other carbon atoms. So hybridisation of carbon atom is sp^2.

29. Ans. (3)

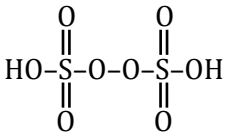


B has sp^3 Hybridisation

Non- planar

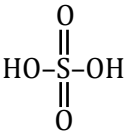
30. Ans. (1)

A → Peroxodisulphuric acid

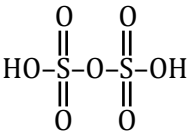


B → Sulphuric acid

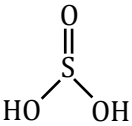
H_2SO_4



C → Pyrosulphuric acid $H_2S_2O_7$



D → Sulphurous acid H_2SO_3



31. Ans. (4)

Po does not show –2 oxidation state due to its metallic nature

32. Ans. (1)

Basic Radicals	Group Number
B. Cu^{+2}	II
A. Al^{+3}	III
D. Co^{+2}	IV
C. Ba^{+2}	V
E. Mg^{+2}	VI

Ans. (1) B, A, D, C, E

33. Ans. (4)

Dil. H_2SO_4 do not acts as oxidising agent which prevent oxidation of Fe^{+2} into Fe^{+3}. While concentrated H_2SO_4 will oxidise Fe^{+2} into Fe^{+3}

34. Ans. (1)

Correct order of acidic strength of HF, HCl, HBr, HI is HF < HCl < HBr < HI

35. Ans. (2)

Due to more l_p– l_p repulsion, Bond energy of N-N single bond is lesser than bond energy P-P single bond.

36. Ans. (4)

(A) Lake Test → Al^{+3}

(B) Nessler's Reagent → NH_4^+

(C) Potassium sulphocyanide → Fe^{+3}

(D) Brown ring Test → NO_3^-

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Exercise-III (Analytical Questions)

1. Ans. (2)

Statement -I :- BCl_3 is more stable than $TlCl_3$ in Higher Oxidising state Tl is more stable in +1

Statement – II :- Boric acid is Not protic acid it cannot donate or release H^+ it accepts (OH^-) in aq. Solution.

Statement-I is correct but statement-II is wrong.

2. Ans. (1)

PbI_4 does not exist as it reduces to PbI_2 so yes PbI_4 is Not stable.

Al protects itself by protective layer of Al_2O_3 hence it renders passive by nitric acid.

3. Ans. (2)

Aq. solutions of borax is alkaline to litmus paper. Because borax is a salt of sodium hydroxide and boric acid which is SB & WA that why solution is alkaline by nature.

Statement-II

No graphite is more thermodynamics stable than diamond.

4. Ans. (4)

Statement-I :- To stop chain reaction R_3SiCl is used.

Statement-II :- Silica gel is used as drying agent as it adsorb moisture on its surface to stope growth of

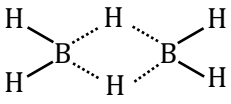
5. Ans. (1)

(1) In Graphite layer there is a Vander wall force of attraction.

(2) In fullerene 6 membering → 5 & 6 both ring.

5 Member ring to fuse with 6 membered ring.

6. Ans. (2)



Only terminal hydrogen lie in same plane

7. Ans. (1)

(1) $Xe[PtF_6]$

(2) XeF_6 are powerful fluorinating agent as they can donate fluorine on large scale.

8. Ans. (2)

2 Type of isotopes are there in which U_{238}, U_{235} we get U_{235} from 0.7% → 3.5% by laser tech.

⇒ Inter halogens are covalent but they are not paramagnetic.

9. Ans. (3)

(1) $Cl_2 + NaOH \xrightarrow{Cold} \underset{\text{chloride}}{NaCl} + \underset{\text{hypochloride}}{NaOCl} + H_2O$

(2) It is a powerful bleaching agent but due to nascent oxygen it can produce in the reaction with H_2O.

10. Ans. (4)

$\ddot{N}H_3$ is basic in nature so it will react with H_2SO_4 to form salt.

$O_2 + O \xrightarrow{Endo} O_3$

$O_3 \xrightarrow{Exo} O_2 + [O]$

11. Ans. (2)

(1) P–H bonds

(2) $H_3PO_4 \xrightarrow{\Delta} P_2O_5 + H_2O$

No disproportionation

12. Ans. (1)

(1) $PCl_3 + H_2O \rightarrow H_3PO_3 + HCl$

(2) White (P_4) has discrete unit of P_4 molecules.

13. Ans. (1)

(A) SF_6 cannot be hydrolysed but SF_4 can be hydrolysed.

(R) 6 F atom in SF_6 are prevent by the attack of H_2O due to steric repulsion.

14. Ans. (2)

(A) HI cannot be prepared by

$KI + H_2SO_4 \rightarrow K_2SO_4 + SO_2 + I_2 + H_2O$

(R) Bond strength $\propto \Delta EN$

15. Ans. (3)

(A) HNO_3 makes iron passive (By making layer of oxides)

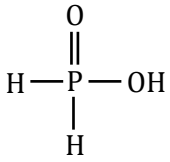
(R) HNO_3 forms protective layer of oxide on the surface of iron.

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16. Ans. (4)

(A) H_3PO_2 is monobasic, not dibasic as only 1 OH present.

(R) There are 2P–H bonds



17. Ans. (1)

(A) Yes boric acid is a weak acid

(B) It cannot release H^+ ion on its own as it accepts OH^- ion in aq. Solution.

18. Ans. (1)

(A) Al cables are used in transmission cables

(B) As Al has high electrical and thermal conductivity and also of light weight.

19. Ans. (3)

(A) B–F bond length in BF_3 & BF_4^- are different

(R) In both BF_3 & BF_4^- $p\pi$-$p\pi$ bond is not present.

20. Ans. (1)

(A) In N family stability of higher oxidising state ie (+5 Oxidising state) decreases as we move down group an +3 Oxidising state increases down the group 15.

(R) All, due to inert pair effect which is more prone in case of heavier member of p- block.

21. Ans. (4)

(a) $\underset{\text{Borax}}{Na_2B_4O_7} \xrightarrow{\Delta} NaBO_2 + B_2O_3$

(b) $B_2H_6 + H_2O \longrightarrow H_3BO_3$

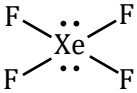
(c) $B_2H_6 + \underset{\text{(excess)}}{NH_3} \xrightarrow{\Delta} (BN)_x$

(d) $BCl_3 + LiAlH_4 \longrightarrow B_2H_6$

22. Ans. (2)

(a) He → It is used in modern diving apparatus.

(b) XeF_6 → Its partial hydrolysis does not change oxidation state of central atom.

(c) XeF_4 →  → 4B.P. + 2LP → sp^3d^2 hybd.

(d) Ar → It is used to provide inert atmosphere for electrical bulbs.

23. Ans. (2)

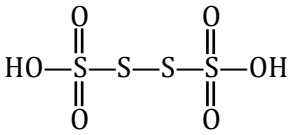
(a) $H_4P_2O_7$ → Pyro Phosphoric acid

(b) $H_2S_4O_6$ → Tetra thionic acid

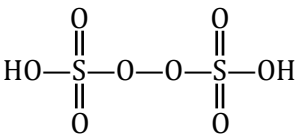
(c) $H_2S_2O_5$ → Pyro sulphurous acid

(d) $H_2S_2O_8$ → Peroxy disulphuric acid

$2(H_3PO_4) = H_6P_2O_8$ & $H_6P_2O_8 - H_2O = H_4P_2O_7$



$2(H_2SO_3) - H_2O \rightarrow H_2S_2O_5$



(Peroxy disulphuric acid)

24. Ans. (2)

(a) Ostwald → HNO_3

$$NH_3 + 5O_2 \xrightarrow[\substack{850°C \\ 5\text{ atom}}]{Pt/Rh} NO + 6H_2O$$

$$NO + \frac{1}{2}O_2 \longrightarrow NO_2$$

$NO_2 + H_2O \rightarrow HNO_3$

(b) Deacon → (Cl_2)

$$4HCl + O_2 \xrightarrow[400°C]{CuCl_2} 2Cl_2 + H_2O$$

(c) Contact → (H_2SO_4)

$$SO_2 + O_2 \xrightarrow[\substack{450°C \\ 1-2\text{atom}}]{V_2O_5} SO_3$$

$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$

(d) Haber → $N_2 + 3H_2 \xrightarrow[\text{Iron catalyst}]{200} (2NH_3)$

25. Ans. (1)

(a) Ortho silicate → SiO_4^{-4} (Basic)

(b) Pyrosilicate → $Si_2O_7^{-6}$

(c) 2D Silicate → $(Si_2O_5^{-2})_n$

(d) 3D Silicate → $(SiO_2)_n$

26. Ans. (3)

(a) Br_2 → Liq. at room temperature

(b) HF → Used in permanent marking on glass surface.

(c) O_3 → Powerful bleaching agents (by giving nascent oxygen)

(d) I_2O_5 → In Estimation of CO

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27. Ans. (2)

(a) Anhydride of $HNO_3 \rightarrow H_2N_2O_6 - H_2O \rightarrow N_2O_5$

(b) Anhydride of $HNO_2 \rightarrow H_2N_2O_4 - H_2O \rightarrow N_2O_3$

(c) Anhydride of $4(H_3PO_4) \rightarrow H_{12}P_4O_{16} - 6H_2O \rightarrow P_4O_{10}$

(d) Anhydride of $4(H_3PO_3) \rightarrow 4H_3PO_3 - 6H_2O \rightarrow P_4O_6$

28. Ans. (3)

(a) Red Lead $\rightarrow Pb_3O_4$

(b) Carborandum $\rightarrow SiC$

(c) Tear gas $\rightarrow Cl_3C.NO_2$

(d) Phosgene $\rightarrow COCl_2$

29. Ans. (4)

(a) Bleaching $\rightarrow CaOCl_2$

(b) Pseudo halogen $\rightarrow (CN)_2$

(c) Inter halogen compound $\rightarrow ICl_3$

(d) Hypo Solution $\rightarrow Na_2S_2O_3.5H_2O$

30. Ans. (3)

(a) Borax $\rightarrow$ Swell's up on heating

(b) Diborane $\rightarrow$ Catches fire spontaneously

(c) CO $\rightarrow$ Form complex with haemoglobin

(d) $CO_2 \rightarrow$ Fire extinguisher

31. Ans. (4)

(a) P_4 in white Phosphorous have angular strain that's why it is more reactive

(c) PCl_5 Solid $\rightarrow$ Ions

$\underset{\text{Tetra}}{\left[PCl_4^+\right]}\ \underset{\text{Octa}}{\left[PCl_6^-\right]}$

32. Ans. (3)

In 3rd option

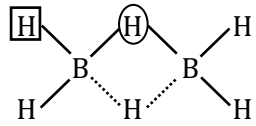
$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

Oxidising state remains same

No Reduction & oxidation

Hence cannot act as oxidising agent.

33. Ans. (1)



$\boxed{H}$ & $\textcircled{H}$ are not in same plane.

34. Ans. (4)

(1) $2(H_3PO_3) \xrightarrow{\Delta} PH_3 + H_3PO_4$

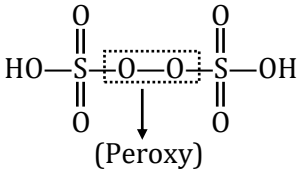
(2) P_4 + con. NaOH $\xrightarrow{\Delta}$ $NaH_2PO_2 + PH_3$

(3) $Ca_3P_2 \rightarrow \xrightarrow{H_2O} Ca(OH)_2 + PH_3$

(4) $H_3PO_4 \xrightarrow{\Delta} P_2O_5 + H_2O\uparrow$

35. Ans. (1)

$H_2S_2O_8 \rightarrow$ Marshal's acid



(Peroxy)

36. Ans. (4)

$NH_4NO_3 \xrightarrow{\Delta} N_2O + H_2O$

No Nitrogen

Azide on heating given purest form of N_2 i.e. 100% Pure.

37. Ans. (4)

$C_2H_2O_4 \xrightarrow[\text{Con. } H_2SO_4]{\Delta} CO + CO_2 + H_2O$

$(COOH)_2$

But Does not evolve SO_2

38. Ans. (3)

SiO_2 is very stable, hence unreactive but still reacts with HF

$\left[\begin{array}{l} SiO_2 + HF \longrightarrow H_2SiF_6 \\ SiO_2 + NaOH \longrightarrow Na_2SiO_3 \end{array}\right]$

39. Ans. (3)

$PCl_5 + H_2O \longrightarrow H_3PO_4 + HCl$

$\downarrow$

Phosphoric acid

40. Ans. (3)

Zn^{+2} cannot give as it is diamagnetic by nature.

EXERCISE - I (Conceptual Questions) Build Up Your Understanding

d-BLOCK ELEMENTS

1. For a catalyst which condition is not essential:
(1) Variable oxidation states
(2) High ionisation energy
(3) Vacant orbitals
(4) Free valency on the surface
CDF001

2. To which of the following series the transition element from Z = 39 to Z = 48 belong :
(1) 3d series (2) 4d series
(3) 5d series (4) 6d series
CDF002

3. $KMnO_4$ on treatment with conc. H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively
(1) Mn_2O_7, MnO_2 (2) Mn_2O, Mn_2O_3
(3) $MnSO_4$, Mn_2O_3 (4) Mn_2O_3, MnO_2
CDF003

4. Which pair of ions is colourless :
(1) Mn^{3+}, Co^{3+} (2) Fe^{3+}, Cr^{3+}
(3) Zn^{2+}, Sc^{3+} (4) Ti^{2+}, Cu^{2+}
CDF004

5. $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is :
(1) Cr_2O_3 (2) CrO_2
(3) Cr_2O_4 (4) CrO_3
CDF005

6. $Cr_2O_7^{2-} \underset{Y}{\overset{X}{\rightleftharpoons}} 2CrO_4^{2-}$, X and Y are respectively
(1) $X = OH^-$, $Y = H^+$ (2) $X = H^+$, $Y = OH^-$
(3) $X = OH^-$, $Y = H_2O_2$ (4) $X = H_2O_2$, $Y = OH^-$
CDF006

7. The higher oxidation states of transition elements are found to be the combination with A and B, which are:
(1) F, O (2) O, N
(3) O, Cl (4) F, Cl
CDF007

8. Magnetic moment of x^{n+} is $\sqrt{24}$ B.M. Hence No. of unpaired electron and value of 'n' respectively. (Atomic number = 26)
(1) 4, 3 (2) 3, 5 (3) 4, 2 (4) 4, 1
CDF008

9. The product of oxidation of I^- with MnO_4^- in weak alkaline medium is :
(1) IO_3^- (2) I_2 (3) IO^- (4) IO_4^-
CDF009

10. Which of the following oxide of chromium is amphoteric in nature
(1) CrO (2) Cr_2O_3
(3) CrO_3 (4) CrO_5
CDF010

11. Compared to Cu^{2+} having $3d^9$ configuration, Cu^+ having $3d^{10}$ configuration (aq. solution) :-
(1) Is more stable
(2) Is equally stable
(3) Is less stable
(4) Stability depends upon nature of copper salt
CDF011

12. CrO_3 is red orange in colour. The nature of oxide is:-
(1) Acidic (2) Basic
(3) Amphoteric (4) Neutral
CDF012

13. Maximum spin magnetic moment is shown by:
(1) d^5 (2) d^6 (3) d^7 (4) d^8
CDF013

14. Disproportion can be shown by
(1) MnO_4^{2-} in acidic medium
(2) Cu^+ in aqueous medium
(3) Cl_2 in alkaline medium
(4) All of these
CDF014

15. Which of the following configurations of 3d series exhibits the largest number of oxidation states –
(1) [Ar] $3d^8\ 4s^2$ (2) [Ar] $3d^{10}\ 4s^1$
(3) [Ar] $3d^5\ 4s^2$ (4) [Ar] $3d^7\ 4s^2$
CDF015

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16. Which of the following pair of ions has same value of "spin-only" magnetic moment
(1) Cu^{+}, Cu^{2+} (2) Co^{3+}, Fe^{2+}
(3) Ti^{2+}, V^{2+} (4) Sc^{2+}, Zn^{+2}
CDF016

17. Which of the following transition metal do not form amalgam with Hg :-
(1) Fe (2) Co
(3) Ni (4) All of the above
CDF017

18. Highest fluoride and oxide of Mn are respectively:-
(1) MnF_7, Mn_2O_7 (2) MnF_5, Mn_2O_7
(3) MnF_4, Mn_2O_7 (4) MnF_6, Mn_2O_7
CDF018

19. Which is not a property of intestitial compound ?
(1) Non stiochiometric
(2) Harder than pure metal
(3) Chemically inert
(4) Having low melting point
CDF019

20. Which halide of copper does not exist ?
(1) CuF_2 (2) $CuCl_2$
(3) Cu_2I_2 (4) CuI_2
CDF020

21. Which order is correct for oxidising power ?
(1) $Cr_2O_7^{-2} < VO_2^{+}$ (2) $VO_2^{+} > MnO_4^{-}$
(3) $VO_2^{+} < Cr_2O_7^{-2}$ (4) $Cr_2O_7^{-2} > MnO_4^{-}$
CDF021

22. Permangnate titrations in presence of Hydrochloric acid are not satisfactory because :-
(1) HCl is a weak acid
(2) HCl is a volatile acid
(3) $KMnO_4$ will oxidise HCl into Cl_2
(4) $KMnO_4$ will show disproportionation in presence of HCl
CDF022

23. Which compound is used in manufacturing of $K_2Cr_2O_7$?
(1) Na_2CrO_4 (2) CrO_5
(3) $FeCr_2O_4$ (4) CrO_3
CDF023

24. Which of the following does not give chromyl chloride test ?
(1) NaCl (2) KCl
(3) NH_4Cl (4) $HgCl_2$
CDF024

25. Which is correct match
(1) Brass ; Cu + Zn
(2) Bronze ; Cu + Sn
(3) Stainless steel ; Cr + Ni
(4) All of the above
CDF025

26. Which of the following 3d series metal has positive value of $E°(M^{+2}/M)$?
(1) Zn (2) Cu
(3) Mn (4) Cr
CDF026

f-BLOCK ELEMENTS

27. Which of the following actinide element does not found in nature ?
(1) $_{90}Th$ (2) $_{91}Pa$
(3) $_{92}U$ (4) $_{103}Lr$
CDF027

28. Which element does not show +4 oxidation state:-
(1) Ce (2) Ti
(3) Pt (4) Eu
CDF028

29. Which of the following ion is diamagnetic and colourless ?
(1) La^{+3}
(2) Lu^{+3}
(3) Zn^{+2}
(4) All of the above
CDF029

30. The general electronic configuration of lanthanoid elements is :
(1) [Xe] $4f^{0-14}$ $5d^{0-1}$ $6s^2$
(2) [Xe] $4f^{0-14}$ $5d^{1-2}$ $6s^1$
(3) [Xe] $4f^{0-14}$ $5d^{0-1}$ $6s^{1-2}$
(4) None of these
CDF030

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31. Cerium can show the oxidation state of +4 because
(1) It resemble alkali metals
(2) It has very low value of I.E.
(3) Of its tendency to attain noble gas configuration of xenon
(4) Of its tendency to attain $4f^7$ configuration
CDF031

32. The highest oxidation state shown by actinides is:
(1) +6 (2) +7
(3) +5 (4) +4
CDF032

33. The electronic configuration of gadolinium (At. No. 64) is :
(1) [Xe] $4f^7 5d^1 6s^2$ (2) [Xe] $4f^8 5d^0 6s^2$
(3) [Xe] $4f^8 5d^1 6s^1$ (4) [Xe] $4f^7 5d^0 6s^2$
CDF033

34. Which oxidation state is not shown by lanthanoids?
(1) +2 (2) +3
(3) +4 (4) +6
CDF034

35. Which of the following species acts as a reducing agent ?
(1) Ce^{+4} (2) Lu^{+3}
(3) Eu^{+2} (4) Ce^{+3}
CDF035

36. Which of the following pair have almost similar size?
(1) $_{22}Ti$ and $_{40}Zr$ (2) $_{41}Nb$ and $_{73}Ta$
(3) $_{39}Y$ and $_{57}La$ (4) $_{20}Ca$ and $_{77}Ir$
CDF036

37. An increase in both atomic and ionic radii with atomic number occurs in any group of the periodic table. In accordance of this the ionic radii of Ti (IV) and Zr (IV) ions are 0.68 A^0 and 0.74 A^0 respectively but for Hf (IV) ion the ionic radius is 0.75 A^0, which is almost the same as that for Zr (IV) ion. This is due to :-
(1) greater degree of covalency in compounds of Hf^{4+}
(2) Lanthanide contraction
(3) Difference in the co-ordination number of Zr^{+4} and Hf^{+4} in their compounds
(4) Actinide contraction
CDF037

38. When lanthanoid element is heated with sulphur then compound obtained is :-
(1) LnS_2 (2) Ln_2S_3
(3) Ln_2S (4) Ln_3S_4
CDF038

39. Formula of halide of lanthanoid may be
(1) LnX_3 (2) LnX_5
(3) LnX_2 (4) LnX_6
CDF039

40. What is the composition of misch metal alloy?
(1) Lanthanoid metal (95%) + Iron (5%)
(2) Cu (69%) + Zn (31%)
(3) Al (20%) + Cu (80%)
(4) Cr (50%) + Zn (30%) + Cu (20%)
CDF040

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	1	3	1	1	1	3	1	2	3	1	1	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	3	4	4	3	3	3	4	4	2	4	4	4	4
Question	31	32	33	34	35	36	37	38	39	40					
Answer	3	2	1	4	3	2	2	2	1	1					

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

AIPMT 2015

1. Magnetic moment 2.84 B.M. is given by :-
(At. no.), Ni = 28, Ti = 22, Cr = 24, Co = 27)
(1) Ti^{3+} (2) Cr^{2+}
(3) Co^{2+} (4) Ni^{2+}

CDF055

2. Which of the following processes does not involve oxidation of iron ?
(1) Decolourization of blue $CuSO_4$ solution by iron
(2) Formation of $Fe(CO)_5$ from Fe
(3) Liberation of H_2 from steam by iron at high temperature
(4) Rusting of iron sheets

CDF056

3. Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii ? (Number in the parenthesis are atomic numbers).
(1) Zr (40) & Nb (41) (2) Zr (40) & Hf (72)
(3) Zr (40) & Ta (73) (4) Ti (22) & Zr (40)

CDF057

NEET-I 2016

4. Which one of the following statements is correct when SO_2 is passed through acidified $K_2Cr_2O_7$ solution ?
(1) The solution turns blue
(2) The solution is decolourized
(3) SO_2 is reduced
(4) Green $Cr_2(SO_4)_3$ is formed

CDF058

5. The electronic configurations of Eu(Atomic No 63), Gd(Atomic No 64) and Tb (Atomic No. 65) are
(1) $[Xe]4f^7 6s^2$, $[Xe]4f^8 6s^2$ and $[Xe]4f^8 5d^1 6s^2$
(2) $[Xe]4f^6 5d^1 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^9 6s^2$
(3) $[Xe]4f^6 5d^1 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^8 5d^1 6s^2$
(4) $[Xe]4f^7 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^9 6s^2$

CDF059

NEET-II 2016

6. Which one of the following statements related to lanthanons is **incorrect** ?
(1) All the lanthanons are much more reactive than aluminium
(2) Ce(+4) solutions are widely used as oxidizing agent in volumetric analysis
(3) Europium shows +2 oxidation state.
(4) The basicity decreases as the ionic radius decreases from Pr to Lu.

CDF060

NEET(UG) 2017

7. Name the gas that can readily decolourise acidified $KMnO_4$ solution :
(1) SO_2 (2) NO_2
(3) P_2O_5 (4) CO_2

CDF061

8. $HgCl_2$ and I_2 both when dissolved in water containing I^- ions the pair of species formed is
(1) HgI_2, I^- (2) HgI_4^{2-}, I_3^-
(3) Hg_2I_2, I^- (4) HgI_2, I_3^-

CDF062

9. The reason for greater range of oxidation states in actinoids is attributed to :-
(1) Actinoid contraction
(2) 5f, 6d and 7s levels having comparable energies
(3) 4f and 5d levels being close in energies
(4) The radioactive nature of actinoids

CDF063

NEET(UG) 2018

10. Which one of the following ions exhibits d–d transition and paramagnetism as well ?
(1) CrO_4^{2-} (2) $Cr_2O_7^{2-}$
(3) MnO_4^- (4) MnO_4^{2-}

CDF064

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11. Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the ***correct*** code :

Column I	**Column II**
a. Co^{3+}	i. $\sqrt{8}$ B.M.
b. Cr^{3+}	ii. $\sqrt{35}$ B.M.
c. Fe^{3+}	iii. $\sqrt{3}$ B.M.
d. Ni^{2+}	iv. $\sqrt{24}$ B.M.
	v. $\sqrt{15}$ B.M.

	a	**b**	**c**	**d**
(1)	iv	v	ii	i
(2)	i	ii	iii	iv
(3)	iv	i	ii	iii
(4)	iii	v	i	ii

CDF065

NEET(UG) 2019

12. Which of the following reactions are disproportionation reaction ?

(a) $2Cu^{+} \rightarrow Cu^{2+} + Cu^{0}$

(b) $3MnO_4^{2-} + 4H^{+} \rightarrow 2MnO_4^{-} + MnO_2 + 2H_2O$

(c) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

(d) $2MnO_4^{-} + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^{\oplus}$

Select the **correct** option from the following :-

(1) (a) and (b) only

(2) (a), (b) and (c)

(3) (a), (c) and (d)

(4) (a) and (d) only

CDF066

13. The manganate and permanganate ions are tetrahedral, due to

(1) The π–bonding involves overlap of p–orbitals of oxygen with d–orbitals of manganese

(2) There is no π–bonding

(3) The π–bonding involves overlap of p–orbitals of oxygen with p–orbitals of managanese

(4) The π–bonding involves overlap of d–orbitals of oxygen with d–orbitals of manganate

CDF067

NEET(UG) 2019 (ODISHA)

14. Match the catalyst with the process :-

Catalyst	**Process**
(i) V_2O_5	(a) The oxidation of ethyne to ethanal
(ii) $TiCl_4+Al(CH_3)_3$	(b) Polymerisation of alkynes
(iii) $PdCl_2$	(c) Oxidation of SO_2 in the manufacture of H_2SO_4
(iv) Nickel complexes	(d) Polymerisation of ethylene

Which of the following is the correct option ?

(1) i-c, ii-d, iii-a, iv-b (2) i-a, ii-b, iii-c, iv-d

(3) i-a, ii-c, iii-b, iv-d (4) i-c, ii-a, iii-d, iv-b

CDF068

15. When neutral or faintly alkaline $KMnO_4$ is treated with potassium iodide, iodide ion is converted into 'X'. 'X' is-

(1) I_2 (2) IO_4^- (3) IO_3^- (4) IO^-

CDF069

16. The oxidation state of Cr in CrO_5 is :

(1) –6 (2) +12 (3) +6 (4) +4

CDF070

NEET(UG) 2020

17. The calculated spin only magnetic moment of Cr^{2+} ion is :

(1) 2.84 BM (2) 3.87 BM

(3) 4.90 BM (4) 5.92 BM

CDF071

18. Identify the incorrect statement.

(1) The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are not the same

(2) Cr^{2+} (d^4) is a stronger reducing agent than Fe^{2+}(d^6) in water.

(3) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.

(4) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.

CDF072

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NEET(UG) 2020(COVID-19)

19. The oxidation number of the underlined atom in the following species

(1) $Cu_2\underline{O}$ is –1 (2) $\underline{Cl}O_3^-$ is +5

(3) $K_2\underline{Cr}_2O_7$ is + 6 (4) $H\underline{Au}Cl_4$ is +3

Identify the incorrect option.

CDF073

20. Match the following aspects with the respective metal.

Aspects	Metal
(a) The metal which reveals a maximum number of oxidation states	(i) Scandium
(b) The metal although placed in 3d-series is considered not as a transition element	(ii) Copper
(c) The metal which does not exhibit variable oxidation states	(iii) Manganese
(d) The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	(iv) Zinc

Select the correct option :

(1) (a)-(i) (b)-(iv) (c)-(ii) (d)-(iii)

(2) (a)-(iii) (b)-(iv) (c)-(i) (d)-(ii)

(3) (a)-(iii) (b)-(i) (c)-(iv) (d)-(ii)

(4) (a)-(ii) (b)-(iv) (c)-(i) (d)-(iii)

CDF074

NEET(UG) 2021

21. The **incorrect** statement among the following is :

(1) Actinoid contraction is greater for element to element than Lanthanoid contraction.

(2) Most of the trivalent Lanthanoid ions are colorless in the solid state.

(3) Lanthanoids are good conductors of heat and electricity.

(4) Actinoids are highly reactive metals, especially when finely divided.

CDF075

22. Zr (Z = 40) and Hf (Z = 72) have similar atomic and ionic radii because of :

(1) Belonging to same group

(2) Diagonal relationship

(3) Lanthanoid contraction

(4) Having similar chemical properties

CDF076

NEET(UG) 2022

23. Given below are two statements :

Statement-I : Cr^{2+} is oxidising and Mn^{3+} is reducing in nature.

Statement-II : Sc^{3+} compounds are repelled by the applied magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are correct

(2) Both Statement-I and Statement-II are incorrect

(3) Statement-I is correct but Statement-II is incorrect

(4) Statement-I is incorrect but Statement-II is correct

CDF097

NEET(UG) 2023

24. Which of the following statements are **incorrect** ?

A. All the transition metals except scandium form MO oxides which are ionic.

B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7.

C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5.

D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.

E. CrO is basic but Cr_2O_3 is amphoteric.

Choose the **correct** answer from the options given

below:

(1) B and D only (2) C and D only

(3) B and C only (4) A and E only

CDF098

25. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Ionisation enthalpy increases along each series of the transition elements from left to right. However, small variations occur.

Reason (R) : There is corresponding increase in nuclear charge which accompanies the filling of electrons in the inner d-orbitals.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **(A)** is correct but **(R)** is not correct.

(2) **(A)** is not correct but **(R)** is correct.

(3) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

(4) Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**.

CDF099

NEET(UG) 2024

26. 'Spin only' magnetic moment is same for which of the following ions ?

A. Ti^{3+} B. Cr^{2+}

C. Mn^{2+} D. Fe^{2+}

E. Sc^{3+}

Choose the most appropriate answer from the options given below :

(1) B and D only (2) A and E only

(3) B and C only (4) A and D only

CDF100

27. The E° value for the Mn^{3+}/Mn^{2+} couple is more positive than that of Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+} due to change of

(1) d^5 to d^4 configuration

(2) d^5 to d^2 configuration

(3) d^4 to d^5 configuration

(4) d^3 to d^5 configuration

CDF101

28. The pair of lanthanoid ions which are diamagnetic is

(1) Ce^{4+} and Yb^{2+} (2) Ce^{3+} and Eu^{2+}

(3) Gd^{3+} and Eu^{3+} (4) Pm^{3+} and Sm^{3+}

CDF102

NEET(UG) 2024 (Re-examination)

29. Which of the following pairs of ions will have same spin only magnetic moment values within the pair ?

(A) Zn^{2+}, Ti^{2+} (B) Cr^{2+}, Fe^{2+}

(C) Ti^{3+}, Cu^{2+} (D) V^{2+}, Cu^{+}

Choose the **correct** answer from the options given below :

(1) C and D only (2) A and D only

(3) A and B only (4) B and C only

CDF103

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**2**	**4**	**4**	**1**	**1**	**2**	**2**	**4**	**1**	**1**	**1**	**1**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	**3**	**3**	**1**	**1**	**2**	**2**	**3**	**4**	**2**	**3**	**1**	**3**	**1**	**4**	

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EXERCISE – III (Analytical Questions) Master Your Understanding

1. Given below are two statements :
Statement-I : Ag is a transition element.
Statement-II : Ag atom has completely filled d-orbital ($4d^{10}$) in it's ground state.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct while II is incorrect.
(3) Statement-I and II both are correct.
(4) Statement-I is incorrect and II is correct.
CDF077

2. Given below are two statements :
Statement-I : All the transition elements show typical metallic properties such as high tensile strength, ductility but low thermal and electrical conductivity.
Statement-II : Zn, Cd & Hg are very hard and have low volatility.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I is correct while II is incorrect.
(2) Both Statement-I and II are incorrect.
(3) Statement-I and II both are correct.
(4) Statement-I is incorrect and II is correct.
CDF078

3. Given below are two statements :
Statement-I : Transition elements exhibit higher enthalpies of atomisation.
Statement-II : Transition elements have stronger inter atomic interaction.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I is correct while II is incorrect.
(2) Statement-I and II both are correct.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect and II is correct.
CDF079

4. Given below are two statements :
Statement-I : Cr^{+2} act as reducing agent and Mn^{+3} act as oxidising agent.
Statement-II : $E^0(M^{+2}/M)$ value for all 3d-series element's are negative.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I is incorrect and II is correct.
(2) Statement-I and II both are correct.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is correct while II is incorrect.
CDF080

5. Given below are two statements :
Statement-I : All Cu(II) halides are known except the iodide.
Statement-II : Increasing order of oxidising power is $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both Statement-I and II are incorrect.
(2) Statement-I is incorrect and II is correct.
(3) Statement-I and II both are correct.
(4) Statement-I is correct while II is incorrect.
CDF081

6. Given below are two statements :
Statement-I : Ce^{+4} is a strong oxidant but Eu^{+2} is a strong reducing agent.
Statement-II : All lanthanoid ion in +3 oxidation state are paramagnetic.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Statement-I and II both are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct while II is incorrect.
(4) Statement-I is incorrect and II is correct.
CDF082

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7. Match the column :

Column-I (Catalysis)		Column-II (Process)	
(i)	Ni in the presence of Hydrogen	(a)	Zieglar-Natta catalyst
(ii)	$CuCl_2$	(b)	Contact process
(iii)	V_2O_5	(c)	Vegetable oil to vegetable ghee
(iv)	$TiCl_4+Al(C_2H_5)_2$	(d)	Decon process

(1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-d, iii-b, iv-a
(3) i-c, ii-d, iii-a, iv-b (4) i-b, ii-c, iii-d, iv-a

CDF083

8. Match the column :

Column-I (Property)		Column-II (Element)	
(i)	Lanthanoid which show +4 oxidation state	(a)	Pm
(ii)	Lanthanoid which can show +2 oxidation state	(b)	Gd
(iii)	Radioactive lanthanoid	(c)	Ce
(iv)	Lanthanoid which has $4f^7$ configuration in +3 oxidation state	(d)	Eu

(1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-d, iii-b, iv-a
(3) i-c, ii-d, iii-a, iv-b (4) i-b, ii-c, iii-d, iv-a

CDF084

9. Match the column :

Column-I		Column-II	
(a)	$FeSO_4.7H_2O$	(p)	Green vitriol
(b)	$MgSO_4.7H_2O$	(q)	Blue vitriol
(c)	$ZnSO_4.7H_2O$	(r)	Epsom salt
(d)	$CuSO_4.5H_2O$	(s)	White vitriol

(1) a-p, b-r, c-s, d-q (2) a-q, b-r, s-s, d-p
(3) a-r, b-s, c-p, d-q (4) a-q, b-p, c-s, d-r

CDF085

10. Match the column :

Column-I (Composition)		Column-II (Alloy)	
(a)	Cu + Ni	(p)	Silver UK coins
(b)	Ni+Cu+ Zn	(q)	German silver
(c)	Cu+ Zn+ Sn	(r)	Gun metal
(d)	Ln(95%)+Fe	(s)	Misch metal

(1) a-p, b-q, c-r, d-s
(2) a-q, b-r, c-s, d-p
(3) a-s, b-q, c-r, d-p
(4) a-q, b-p, c-s, d-r

CDF086

11. Match the column :

Column-I (Compound)		Column-II (Properties)	
(i)	K_2MnO_4	(a)	Amphoteric
(ii)	$KMnO_4$	(b)	Paramagnetic & green coloured
(iii	$K_2Cr_2O_7$	(c)	Diamagnetic & purple coloured
(iv)	V_2O_5	(d)	Diamagnetic & orange coloured

(1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-d, iii-b, iv-a
(3) i-b, ii-c, iii-d, iv-a (4) i-d, ii-a, iii-b, iv-c

CDF087

12. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : $K_2Cr_2O_7$ is used as a primary standard solution in Titration.

Reason (R) : It is not a hygroscopic in nature.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).
(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(3) (A) is not correct but (R) is incorrect.
(4) (A) & (R) both are correct.

CDF088

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13. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The 3rd ionisation energy of Gd is abnormally low.

Reason (R) : Exchange energy of 4f subshell is high.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).

(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(3) (A) is not correct but (R) is incorrect.

(4) (A) & (R) both are correct.

CDF089

14. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Ce^{+4} can oxidise water and it is a good analytical reagent.

Reason (R) : The E^0 value for Ce^{+4}/Ce^{+3} is +1.74V.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).

(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(3) (A) is not correct but (R) is incorrect.

(4) (A) & (R) both are correct.

CDF090

15. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Eu^{+2} & Yb^{+2} are reducing agent.

Reason (R) : Both ions have strong tendency to convert into stable +3 in its aq. solution.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).

(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(3) (A) is not correct but (R) is incorrect.

(4) (A) & (R) both are correct.

CDF091

16. Which of the following oxides are Amphoteric in nature.

(A) V_2O_5 (B) Cr_2O_3

(C) CrO_5 (D) Mn_2O_7

(1) If A, B and C options are correct.

(2) If A & B both options are correct.

(3) If B & D both are options correct.

(4) If A & C both are options correct.

CDF092

17. Which of the following statement is correct.

(A) In Lanthanide series metallic radius regularly decreases from La to Lu.

(B) La is actually an elements of transition series rather than lanthanide.

(C) $La(OH)_3$ is less basic than $Lu(OH)_3$

(D) The atomic radius of Zr & Hf are nearly similar because of lanthanoid contraction.

(1) If A, B and C options are correct.

(2) If A & B both options are correct.

(3) If B & D both are options correct.

(4) If A & C both are options correct.

CDF093

18. Which of the following element does not $[Ar]3d^5$ in its electronic configuration.

(A) Cr (B) Fe (C) Mn (D) Ni

(1) If A, B and C options are correct.

(2) If A & B both options are correct.

(3) If B & D both are options correct.

(4) If A & C both are options correct.

CDF094

19. Which of the following order of given properties are correct.

(A) CrO > Cr_2O_3 > CrO_3 : Melting point

(B) Mn_2O_7 > Mn_2O_3 > MnO : Acidic nature

(C) CrO_3>MoO_3>WO_3: Oxidising nature character

(D) V_2O_5 > V_2O_4 > V_2O_3 : Basic nature

(1) If A, B and C options are correct.

(2) If A & B both options are correct.

(3) If B & D both are options correct.

(4) If A & C both are options correct.

CDF095

20. MnO_4^{-2} can be converted to MnO_4^-

(A) By oxidation of O_3

(B) By electrolytic oxidation

(C) By the addition of dil. H_2SO_4

(D) By the addition of HCl

(1) If A, B and C options are correct.

(2) If A & B both options are correct.

(3) If B & D both are options correct.

(4) If A & C both are options correct.

CDF096

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**2**	**4**	**3**	**3**	**2**	**3**	**1**	**1**	**3**	**1**	**1**	**1**	**1**
Question	16	17	18	19	20										
Answer	**2**	**3**	**3**	**1**	**1**										

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d & f block Elements

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (2)
Higher I.E. is not the essential condition.

2. Ans. (2)
Elements in the 4d series start from Y(39) to Cd(48).

3. Ans. (1)
$KMnO_4$ on treatment with conc. H_2SO_4 forms a compound which is –

$KMnO_4 + H_2SO_4 \rightarrow H_2O + K_2SO_4 + \underset{(x)}{Mn_2O_7}$

x on heating gives following –

$\underset{(x)}{Mn_2O_7} \xrightarrow{heat} \underset{(y)}{MnO_2}$

4. Ans. (3)
Zn^{+2}, Sc^{+3} are diamagnetic and colourless.

5. Ans. (1)
Cr_2O_3 is green in colour.

6. Ans. (1)

$\underset{(orange)}{Cr_2O_7^{-2}} \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} \underset{(yellow)}{2CrO_4^{-2}}$

7. Ans. (1)
Higher oxidation states of transition metals are found with oxygen and fluorine highest fluoride – MnF_4, highest oxide Mn_2O_7

8. Ans. (3)
Atomic number – 26 (Fe)

$Fe \rightarrow [Ar]4s^2\ 3d^6$ ⇅ ⇅ | 1 | 1 | 1 | 1

$Fe^{+2} \rightarrow [Ar]4s^0\ 3d^6$ □ ⇅ | 1 | 1 | 1 | 1

4 unpaired electrons

$X^{n+} \Rightarrow n = 2$

So, magnetic moment $\Rightarrow \sqrt{n(n+2)} \Rightarrow \sqrt{24}$.

9. Ans. (1)
Oxidation of I^- with $KMnO_4$ is given by –

$MnO_4^- + I^- \longrightarrow IO_3^- + Mn^{+2}$

10. Ans. (2)
Chromium (III) oxide is amphoteric in nature.

11. Ans. (3)
Cu^+ is less stable than Cu^{+2} in aqueous solution because Cu^{+2} has greater hydration energy.

12. Ans. (1)
$InCrO_3$ Cr is in (+6) oxidation state which is acidic in nature.

13. Ans. (1)
d^5 configuration will have maximum spin magnetic moment.

14. Ans. (4)
Disproportion is shown by –
MnO_4^{-2} in acidic medium
Cu^+ in aqueous medium
Cl_2 in alkaline medium

15. Ans. (3)
Mn shows maximum oxidation states from +2 to + 7.

16. Ans. (2)
Co^{+3} – $[Ar]\ 4s^0\ 3d^6$
Fe^{+2} – $[Ar]\ 4s^0\ 3d^6$
They will have same value of spin only magnetic moment.

17. Ans. (4)
These metals do not from amalgam with Hg.

18. Ans. (3)
MnF_4 (maximum fluoride)
Mn_2O_7 (maximum oxide)

19. Ans. (4)
Interstitial compounds have high melting point.

20. Ans. (4)
Iodine ions are strong reducing agents therefore Cu (II) iodide reduces to copper (I) iodide.

21. Ans. (3)
Oxidising power refers to how a compound readily accepts electron.

$\underset{+5}{VO_2^+} < \underset{+6}{Cr_2O_7^{-2}}$

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22. Ans. (3)
In permagnate titrations $KMnO_4$ will oxidise HCl into Cl_2.
$2KMnO_4+16\ HCl \rightarrow 2KCl+2MnCl_2+8H_2O+5Cl_2$

23. Ans. (3)
$K_2Cr_2O_7$ is prepared from chromite ore ($FeCr_2O_4$)

24. Ans. (4)
$HgCl_2$ does not give chromyl chloride test due to its covalent character.

25. Ans. (4)
Brass – Cu + Zn
Bronze – Cu + Sn
Stainless steel – Cr + Ni

26. Ans. (2)
Cu has positive of (M^{+2}/M) E° in the 3d series due to high enthalpy of atomization.

27. Ans. (4)
$_{103}Lr$ is a transuranic element.

28. Ans. (4)
Europium does not show +4 oxidation state.

29. Ans. (4)
All the options are diamagnetic and colourless.

30. Ans. (4)
General electronic configuration of Lanthanoids are – $[(n-2)f^{1-14}\ (n-1)d^{0-1}ns^2]$

31. Ans. (3)
Ce shows +4 oxidation state due to its tendency to attain noble gas configuration.

32. Ans. (2)
+7 oxidation state is shown by Neptunium and plutonium.

33. Ans. (1)
$[Xe]\ 4f^7\ 5d^1\ 6s^2$ is the configuration of gadolinium

34. Ans. (4)
+6 Oxidation state is not shown by lanthanoids, common O.S. are – +2, +3, +4.

35. Ans. (3)
Eu^{+2} will loose electron hence it acts as a reducing agent.

36. Ans. (2)
Radius 4d ≈ 5d (V B)

37. Ans. (2)
Due to Lanthanoid contraction.

38. Ans. (2)
Lanthanoid element heated with sulphar then Ln_2S_3 is formed.

39. Ans. (1)
Formula of halide of Lanthanoid will become LnX_3 as Lanthanoids generally have +3 (oxidation sate)

40. Ans. (1)
Misch Metal alloy is a mixture of Lanthanoid metal (95%) + Iron (5%).

Exercise-II (Previous Year Questions)

1. Ans. (4)
$\sqrt{n(n+2)}=2.84$
Value of n = 2
So, Ni^{+2}

2. Ans. (2)
$Fe(CO)_5$ from Fe does not involve oxidation of iron.

3. Ans. (2)
Due to lanthanoid contraction
Radius : $\boxed{Zr \approx Hf}$

4. Ans. (4)
When SO_2 is passed through acidified $K_2Cr_2O_7$ solution turns green from orange.
$K_2Cr_2O_7 + 2H_2SO_4 + 3SO_2 \rightarrow$
$2Cr_2(SO_4)_3 + K_2SO_4 + H_2O$

5. Ans. (4)
$_{63}Eu \rightarrow [Xe]4f^7 6s^2$
$_{64}Gd \rightarrow [Xe]4f^7 5d^1 6s^2$
$_{65}Tb \rightarrow [Xe]4f^9 6s^2$

6. Ans. (1)
Some lanthanoids are much more reactive than aluminium.

7. Ans. (1)
SO_2 decolourise $KMnO_4$ from purple to colourless.
$2KMnO_4+2H_2O+5SO_2 \rightarrow$
$K_2SO_4+2MnSO_4+2H_2SO_4$
(purple) (colourless)

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8. **Ans. (2)**

$HgCl_2$ and I_2 dissolved in water forms HgI_4^{-2}, I_3^-

$\Rightarrow HgCl_2 + 4I^- \longrightarrow HgI_4^{-2} + 2Cl^-$

$\Rightarrow I_2 + I^- \longrightarrow I_3^-$

9. **Ans. (2)**

5f, 6d and 7s levels having comparable energy. So, actinoids show variable O.S.

10. **Ans. (4)**

MnO_4^{-2} has +6 oxidation state. So, 1d electron is present. So, paramagnetic as well as d-d transition exist.

11. **Ans. (1)**

$Co^{+3} = [Ar]4s^0 3d^6$ = 4 unpaired e^-

$Cr^{+3} = [Ar]4s^0 3d^3$ = 3 unpaired e^-

$Fe^{+3} = [Ar]4s^0 3d^5$ = 5 unpaired e^-

$Ni^{+2} = [Ar]4s^0 3d^8$ = 2 unpaired e^-

12. **Ans. (1)**

1. $2\overset{\text{Oxidation}}{\underbrace{Cu^+}} \longrightarrow \overset{\text{Reduction}}{\overbrace{Cu^{+2} + Cu^0}}$

2. $3\overset{\text{Oxidation}}{\underbrace{MnO_4^{-2}}} + 4H^+ \longrightarrow \overset{\text{Reduction}}{\overbrace{2MnO_4^- + MnO_2}} + 2H_2O$

13. **Ans. (1)**

Manganate and permanganate ions are tetrahedral due to formation π bond involving p orbital of oxygen with d orbital of manganese.

14. **Ans. (1)**

V_2O_5 → Used in contact process

$TiCl_4 + Al(CH_3)_3$ → Polymerisation of ethylene.

$PdCl_2$ → Oxidation of ethyne to ethanal

Nickel complex → Polymerisation of alkynes.

15. **Ans. (3)**

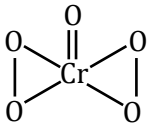
$2KMnO_4 + H_2O + KI \rightarrow 2MnO_2 + 2KOH + KIO_3$

16. **Ans. (3)**

In CrO_5 it has four O atoms as peroxide bearing –1 (charge) and one oxide having –2 (charge) So,

O.S. $\Rightarrow x + 4(-1) + 1 \times (-2) = 0$

$[x = +6]$



17. **Ans. (3)**

$Cr^{+2} = [Ar]4s^0 3d^4 \Rightarrow$ 4 unpaired e^{-1}s

Magnetic moment $= \sqrt{n(n+2)}$

$= \sqrt{4(6)} = \sqrt{24}$ = 4.90 BM

18. **Ans. (1)**

Oxidation state $\Rightarrow$

$CrO_4^{-2} \Rightarrow x + 4(-2) = -2 \Rightarrow [x = +6]$

$Cr_2O_7^{-2} \Rightarrow 2x + 2(-7) = -2 \Rightarrow [x = +6]$

19. **Ans. (1)**

$Cr_2\underline{O} \Rightarrow$ charge $\Rightarrow -2$

Cu atom will have +1

20. **Ans. (2)**

Maximum oxidation states	– Mn
Metal of 3d series not considered transition metal	– Zn
+1 O.S. undergoes disproportionation	– Cu
Does not show variable oxidation state	– Sc

21. **Ans. (2)**

Some trivalent lanthanoids are colourless in the solid state.

Like → La^{+3}, Ce^{+3}, Gd^{+3}, Yb^{+3} & Lu^{+3} are colourless rest all are coloured.

22. **Ans. (3)**

Due to lanthanoid contraction Zr and Hf have similar atomic radius.

23. **Ans. (4)**

$Cr^{+2} \rightarrow Cr^{+3}$

$3d^4$ $3d^3$ (t_{2g} Half filled)

(R.A.)

$Mn^{+3} \rightarrow Mn^{+2}$

$3d^4$ $3d^5$ (Half filled 'd' orbital).

(O.A.)

24. **Ans. (2)**

$C \rightarrow \overset{+3}{V_2}O_3 \rightarrow \overset{+4}{V_2}O_4 \rightarrow \overset{+5}{V_2}O_5$

Acidic Nature ↑

D → V_2O_5 dissolve in acid to give VO_4^{-3} salts

This doesn't shown by V_2O_4

25. **Ans. (3)**

Reason is the correct explanation of Assertion.

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26. Ans. (1)

μ_{spin} only = $\sqrt{n(n+2)}$

where n = no. of unpaired e^-

(A) $Ti^{+3} \rightarrow 3d^1$ n = 1

(B) $Cr^{+2} \rightarrow 3d^4$ n = 4

(C) $Mn^{+2} \rightarrow 3d^5$ n = 5

(D) $Fe^{+2} \rightarrow 3d^6$ n = 4

(E) $Sc^{+3} \rightarrow 3d^0$ n = 0

Since (B) and (D) contain same number of unpaired e^- so they have same spin only magnetic moment.

27. Ans. (3)

Mn^{+3}/Mn^{+2} E° = +ve

$_{25}Mn \rightarrow 3d^5 4s^2$

$Mn^{+2} \rightarrow 3d^5$

$Mn^{+3} \rightarrow 3d^4$

28. Ans. (1)

$Ce^{+4} = {}_{54}[Xe]$

$Yb^{+2} = {}_{54}[Xe]4f^{14}5d^0$

Both species do not contain any unpaired electron. Hence, they are diamagnetic.

29. Ans. (4)

In option (B) and (C), species contain equal number of unpaired electron. Hence will have same value of spin only magnetic moment.

Exercise-III (Analytical Questions)

1. Ans. (3)

Statement-I :- All d block elements except Zn, Cd, Hg & Cn are transition elements.

Statement-II:- Electronic configuration of Ag :– $_{36}[Kr]4d^{10}5s^1$

2. Ans. (2)

Transition metals other than Zn, Cd & Hg are hard and have high m. p. because of unpaired e^- which form metallic bonds.

3. Ans. (2)

Statement-I :- stronger is the metallic bond more will be heat of atomisation.

Statement-II :- More number of unpaired electron will favour the formation of metallic bond.

4. Ans. (4)

Statement-I :- $E^\circ_{Cr^{+2}/Cr^{+3}} \rightarrow +0.41$

$E^\circ_{Mn^{+3}/Mn^{+2}} \rightarrow +1.57$

Statement-II :- Positive only for Cu among 3d elements because hydration energy of Cu^{+2} is not enough to compensate for sublimation energy, IE_1 and IE_2 for Cu.

5. Ans. (3)

Statement-I :- Because Cu^{+2} oxidises I^- to I_2

$2Cu^{+2} + 4I^- \longrightarrow Cu_2I_2(s) + I_2$

Statement-II :- Oxidizing power $\propto$ positive oxidation state of the central metal.

$VO_2^+ \rightarrow +3$

$Cr_2O_7^{-2} \rightarrow +6$

$MnO_4^- \rightarrow +7$

6. Ans. (3)

Statement-I :- Ce^{+4} is a strong oxidant and comes to Ce^{+3}

Eu^{+2} is strong reducing agent changing to common +3 oxidation state.

Statement-II:- La^{+3} & Lu^{+3} ions does not shows any color.

7. Ans. (2)

(i) Ni in the presence of Hydrogen → (c) Vegetable oil to vegetable ghee (Hydrogenation of oil)

(ii) $CuCl_2$ → (d) Decon process (Manufacture of Cl_2)

(iii) V_2O_5 → (b) Contact Process (Manufacture of H_2SO_4)

(iv) $TiCl_4 + Al(CH_3)_3$ → (a) Zieglar-Natta catalyst

8. Ans. (3)

(i) Ce → +4

(ii) Eu → +2

(iii) Promethium is radioactive

(iv) Gd^{+3} :- $_{54}[Xe]4f^7$

9. Ans. (1)

(a) $FeSO_4.7H_2O$ → (p) Green vitriol

(b) $MgSO_4.7H_2O$ → (r) Epsom salt

(c) $ZnSO_4.7H_2O$ → (s) White vitriol

(d) $CuSO_4.5H_2O$ → (q) Blue vitriol

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10. Ans. (1)

(a) Cu + Ni → (p) Silver UK coins

(b) Ni+Cu+ Zn → (q) German silver

(c) Cu+ Zn+ Sn → (r) Gun metal

(d) Ln(95%)+Fe → (s) Misch metal

11. Ans. (3)

(i) $K_2MnO_4 \rightarrow Mn^{+6}$ $(3d^1)$ Paramagnetic and green coloured

(ii) $KMnO_4 \rightarrow Mn^{+7}$ $(3d^0)$ Purple colour due to charge transfer

(iii) $K_2Cr_2O_7 \rightarrow Cr^{+6}$ $(3d^0)$ orange colour

(iv) V_2O_5 → Amphoteric (reacts with both acids & bases)

12. Ans. (1)

$K_2Cr_2O_7$ is preferred over $Na_2Cr_2O_7$ because $Na_2Cr_2O_7$ is Hygroscopic in nature.

13. Ans. (1)

Gd :- $_{54}[Xe]4f^75d^16s^2$

Gd^{+3} :- $_{54}[Xe]4f^7$

14. Ans. (1)

Analytical reagent :- A chemical compound of known high standard purity.

Now $E^0_{Ce^{+4}/Ce^{+3}}$ = 1.74V this shows that it has sufficient kinetic stability due to which the reaction rate is very slow hence it is a good analytical agent.

15. Ans. (1)

Eu^{+2} & Yb^{+2} have strong tendency to convert in +3 oxidation state therefore they act as reducing agent.

16. Ans. (2)

V_2O_5 & Cr_2O_3 (Cr^{+3}) is amphoteric in nature.

17. Ans. (3)

(A) Decrease in ionic radii of M^{+3} comparatively more regular.

(B) La :- $[Xe]5d^16s^2$ → d block element

(C) Covalent character of hydroxides increases from lanthanum to Lutetium so basic character decreases.

(D) $\left.\begin{matrix} Ti \\ \wedge \\ Zr \\ \approx \\ Hf \end{matrix}\right]$ due to lanthanoid contraction

18. Ans. (3)

Fe :- $[Ar]3d^64s^2$

Ni :- $[Ar]3d^84s^2$

19. Ans. (1)

(A) $\overset{+2}{Cr}O$ $\overset{+3}{Cr_2}O_3$ $\overset{+6}{Cr}O_3$

As the oxidation number of transition metal in transition metal oxide increases the ionic character decreases.

$$\text{Oxidation Number} \propto \frac{1}{\text{Ionic character}}$$

$$\text{Ionic Character} \propto \text{Melting Point}$$

(B) The oxide in lower oxidation state of a metal is basic and in higher oxidation state of the metal is acidic

$Mn_2O_7 \rightarrow +7$, $Mn_2O_3 \rightarrow +3$, $MnO \rightarrow +2$

(C) In CrO_3 chromium in its higher oxidation state '+6' so it will tend to reduce itself to oxidation state +3 but as we go down the group stability of heavier transition metals in their higher oxidation state increases. So Molybdenum and Tungsten have little tendency to get reduced to lower oxidation state.

20. Ans. (1)

$$MnO_2 + K_2CO_3/KOH \rightarrow K_2MnO_4 \xrightarrow[\text{Oxidation}]{\text{Electrolytic}} KMnO_4$$

K_2MnO_4 $\xrightarrow{Cl_2}$ $KMnO_4 + KCl$

K_2MnO_4 $\xrightarrow{O_3}$ $KMnO_4 + O_2$

K_2MnO_4 $\xrightarrow{H^+/CO_2}$ $KMnO_4 + MnO_2$

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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

TERMINOLOGY OF COORDINATION COMPOUND

1. In the complex ion $[Fe(EDTA)]^-$ the coordination number and oxidation state of central metal ion is :-

(1) C.N. = 6 O.N. = +3
(2) C.N. =1 O.N. = –1
(3) C.N. = 4 O.N. = +2
(4) C.N. = 3 O.N. = +3

CCO001

2. The coordination number and oxidation number of the central metal ion in the complex $[Pt(en)_2]^{+2}$ is :-

(1) C.N. = 2, O.N. = +2 (2) C.N. = 6, O.N. = +4
(3) C.N. = 4, O.N. = +4 (4) C.N. = 4, O.N. = +2

CCO002

3. Select bidentate or didentate ligand from the following.

(1) CO (2) SCN^-
(3) CH_3COO^- (4) $C_2O_4^{2-}$

CCO003

4. The oxidation and coordination number of Pt in $[Pt(C_2H_4)Cl_3]^-$ is respectively :-

(1) + 1, 3 (2) + 2, 4
(3) + 3, 6 (4) + 2, 5

CCO004

5. The CN and ON of X in the complex compound $[X(SO_4)(NH_3)_5]$ will be respectively :-

(1) 10 and +3 (2) 1 and +6
(3) 6 and +4 (4) 6 and +2

CCO005

6. What is the oxidation number of Fe in $[Fe(H_2O)_5(NO)]^{2+}$ ion?

(1) +2 (2) +3 (3) +1 (4) 0

CCO006

7. Which one is a heteroleptic complex ?

(1) Ferrocene (2) Chromocene
(3) Prussian blue (4) Zeise's salt

CCO007

8. Select the set representing correct statements only.

(I) A coordination entity constitutes a central metal atom or ion bonded to a fixed number of ions or molecules.
(II) Ligand having two different or similar donor atoms in the complex is called bidentate ligand.
(III) All bidentate ligands are chelating ligands.
(IV) Ni^{+2}, Co^{+3} and Fe^{+3} are acting as Lewis acids in $[NiCl_2(H_2O)_4]$, $[CoCl(NH_3)_5]^{+2}$ and $[Fe(CN)_6]^{-3}$.

(1) I, II, III and IV (2) II and III
(3) I, III and IV (4) I, II and IV

CCO008

9. Oxidation number of Cobalt in $K[Co(CO)_4]$ is :-

(1) +1 (2) –1 (3) 0 (4) +2

CCO009

10. Which of the following species is not expected to be ligand :-

(1) NO (2) NH_4^+
(3) $NH_2CH_2CH_2NH_2$ (4) CO

CCO010

11. Identify the statement which is not correct :-

(1) Coordination compounds are mainly known for transition metals.
(2) Coordination number and oxidation state of central metal atom in a complex is same.
(3) Ligand donates at least one electron pair to CMA
(4) $[Co(NH_3)_4Cl_2]^+$ is a heteroleptic complex.

CCO011

12. Which of the following ligands will not show chelation?

(1) EDTA
(2) DMG
(3) Ethane-1, 2 diamine
(4) SCN^-

CCO012

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IUPAC - NOMENCLATURE

13. $K_3[Fe(CN)_6]$ is :-
(a) Potassium hexacyano ferrate (II)
(b) Potassium hexacyano ferrate (III)
(c) Potassium ferri-cyanide
(d) Hexacyano ferrate (III) potassium
Correct answer is :-
(1) Only (a) and (b) (2) Only (b) and (c)
(3) Only (a) and (c) (4) Only (b) and (d)

CCO013

14. Give the IUPAC name of the complex compound $[Co(NH_3)_4(H_2O)Br](NO_3)_2$.
(1) Bromoaquotetraamine Cobalt (III) nitrate
(2) Bromoaquotetraaminocobalt (III) nitrate
(3) Bromoaquatetraammine cobaltate (III) nitrate
(4) Tetraammineaquabromido cobalt (III) nitrate

CCO014

15. Which of the following complex is anion :-
(1) Fluoro pentaammine cobalt(III)
(2) Trioxalato ferrate(III)
(3) Penta Carbonyl iron(0)
(4) Dichloro diammine platinum(II)

CCO015

16. The chloro-bis (ethylenediamine) nitro cobalt (III) ion is :-
(1) $[Co\ (NO_2)_2\ (en)_2\ Cl_2]$
(2) $[CoCl\ (NO_2)_2\ (en)_2]^+$
(3) $[CoCl\ (en)_2\ NO_2]^+$
(4) $[Co\ (en)\ Cl_2\ (NO_2)_2]^-$

CCO016

17. Which of the following complexes is not a chelate
(1) Bis (dimethylglyoximato) nickel(II)
(2) Potassium ethylenediaminetetra thiocyanatochromate(III)
(3) Tetrammine dichlorocobalt(III) nitrate
(4) Trans–diglycinatoplatinum(II)

CCO017

18. The correct IUPAC name of the complex $[Fe(\eta^5\text{–}C_5H_5)_2]$ is
(1) Cyclopentadienyl iron (II)
(2) Bis (cyclopentadienyl) iron (II)
(3) Dicyclopentadienyl ferrate (II)
(4) Ferrocene

CCO018

19. The correct name of $[Pt(NH_3)_4Cl_2]\ [PtCl_4]$ is :-
(1) Tetraammine dichloro platinum (IV) tetrachloro platinate (II)
(2) Dichloro tetra ammine platinum (IV) tetrachloro platinate (II)
(3) Tetrachloro platinum (II) tetraammine platinate(IV)
(4) Tetrachloro platinum (II) dichloro tetraamine platinate (IV)

CCO019

20. The IUPAC name of $K_2[Cr(CN)_2O_2(O)_2(NH_3)]$ is:-
(1) Potassiumamminedicyano dioxoperoxochromate(VI)
(2) Potassium-ammine-cyano-peroxodioxo chromium (VI)
(3) Potassiumamminecyanoperoxodioxo chromium (VI)
(4) Potassiumamminecyanoperoxodioxo chromate (IV)

CCO020

21. The IUPAC name for $[Co(NH_3)_6]\ [Cr(CN)_6]$ is :-
(1) Hexaammine cobalt (III) hexacyanochromate (III)
(2) Hexacyanochromium cobalt hexaammine (VI)
(3) Hexaammine cobalt (III) hexacyanochromium (VI)
(4) Hexacyanochromium (III) hexaammine cobalt (III)

CCO021

22. The IUPAC name for $[Co(NCS)\ (NH_3)_5]Cl_2$ is :-
(1) Pentaammine (thiocyanato-N) cobalt (III) chloride
(2) Pentaammine (thiocyanato-S) cobalt (III) chloride
(3) Pentaammine (isothiocyanato-N,S)cobalt (III) chloride
(4) Pentaammine (mercapto-N) cobalt (III) chloride

CCO022

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23. Which rule is incorrect regarding IUPAC nomenclature of complex ions ?
(1) Cation is named first and than anion is named
(2) In the coordination sphere, the ligands are named alphabetically.
(3) Positively charged ligands have suffix-ate.
(4) More than one same type of ligands are indicated by di, tri, tetra etc.

CCO023

WERNER'S THEORY & EAN

24. Which of the following has least conductivity in aqueous solution.
(1) $CoCl_3 \cdot 4NH_3$ (2) $CoCl_3 \cdot 3NH_3$
(3) $CoCl_3 \cdot 5NH_3$ (4) $CoCl_3 \cdot 6NH_3$

CCO024

25. If EAN of a central metal ion X^{+2} in a complex is 34. and atomic number of X is 28. The number of monodentate ligands present in complex are:-
(1) 3 (2) 4 (3) 6 (4) 2

CCO025

26. The EAN of cobalt in the complex ion $[Co(en)_2Cl_2]^+$ is :-
(1) 27 (2) 36 (3) 33 (4) 35

CCO026

27. The effective atomic number of Cr (atomic no. 24) in $[Cr(NH_3)_6]\ Cl_3$ is
(1) 35 (2) 27
(3) 33 (4) 36

CCO027

28. Which gives only 25% mole of AgCl, when reacts with $AgNO_3$:-
(1) $PtCl_2.4NH_3$ (2) $PtCl_4.5NH_3$
(3) $PtCl_4.4NH_3$ (4) $PtCl_4.3NH_3$

CCO028

29. In the metal carbonyls of general formula $M(CO)_x$ (Which follows EAN rule) if M is Ni, Fe and Cr the value of x will be respectively:-
(1) 6, 5, 6 (2) 4, 5, 6
(3) 4, 4, 5 (4) 4, 6, 6

CCO029

30. A compound has the empirical formula $CoCl_3.5NH_3$. When an aqueous solution of this compound is mixed with excess silver nitrate, 2 mol of AgCl precipitate per mol of compound. On reaction with excess HCl, no NH_4^+ is detected. Hence it is
(1) $Co(NH_3)_5Cl_2]Cl$
(2) $[Co(NH_3)_5Cl]Cl_2$
(3) $[Co(NH_3)_5Cl_3]$
(4) $[Co(NH_3)_4Cl_2]Cl.NH_3$

CCO030

31. Which is incorrect regarding Werner theory :-
(1) Co-ordination compounds metals show two types of valancy.
(2) Primary valency is satisfied by negative ions.
(3) Secondary valency is satisfied by only neutral molecules.
(4) Secondary valency is responsible for geometry of complex.

CCO031

32. Number of electrolytes in the complex $CoCl_3.5NH_3$
(1) 3 (2) 2 (3) 4 (4) 5

CCO032

VALENCE BOND THEORY

33. Which of the following compound is paramagnetic
(1) Tetracyanonickelate (II) ion
(2) Tetraamminezinc (II) ion
(3) Hexaamine chromium (III) ion
(4) Diammine silver (I) ion

CCO033

34. The shape of the complex $[Ag(NH_3)_2]^+$ is :
(1) Octahedral (2) Square planar
(3) Tetrahedral (4) Linear

CCO034

35. Hexafluoroferrate(III) ion is an outer orbital complex. The number of unpaired electrons are
(1) 1 (2) 5 (3) 4 (4) 3

CCO035

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36. The shape of $[Cu(NH_3)_4]SO_4$ is :
(1) Square planar (2) Pyramidal
(3) Octahedral (4) Tetrahedral
CCO0036

37. Among the following ions, which one has the highest paramagnetism ?
(1) $[FeF_6]^{3-}$ (2) $[Fe(H_2O)_6]^{2+}$
(3) $[Cu(H_2O)_6]^{2+}$ (4) $[Zn(H_2O)_6]^{2+}$
CCO0037

38. In the complex ion ML_6^{n+}, M^{n+} has five d-electrons and L is weak ligand then the magnetic properties of the complex ion correspond to how many unpaired electrons
(1) 0 (2) 5
(3) 2 (4) 3
CCO0038

39. A magnetic moment of 1.73 BM will be shown by one among of the following compounds .
(1) $[Cu(NH_3)_4]^{2+}$ (2) $[Ni(CN)_4]^{2-}$
(3) $TiCl_4$ (4) $[CoCl_6]^{3-}$
CCO0039

40. The magnetic property and the shape of $[Cr(NH_3)_6]^{3+}$ complex ions are :
(1) Paramagnetic, Octahedral
(2) Diamagnetic, square planar
(3) Paramagnetic, tetrahedral
(4) None of the above
CCO0040

41. Among the following ions which one is high spin complex.
(1) $[Cr(NH_3)_6]^{2+}$ (2) $[Fe(H_2O)_6]^{3+}$
(3) $[Cu(H_2O)_6]^{2+}$ (4) $[Zn(H_2O)_6]^{2+}$
CCO0041

42. Which of the following complexes is an inner orbital complex ?
(1) $[CoF_6]^{3-}$ (2) $[FeF_6]^{3-}$
(3) $[Cr(NH_3)_6]^{3+}$ (4) $[Fe(H_2O)_6]^{2+}$
CCO0042

43. What is Incorrect for $K_4[Fe(CN)_6]$
(1) O.N of Iron is +2
(2) It exhibit diamagnetic character
(3) It exhibit paramagnetic character
(4) It involves d^2sp^3 hybridisation
CCO0043

44. In which of the following molecules, central atom used orbitals of different principle quantum number in the hybridisation :-
(1) $[Fe(CO)_5]$ (2) $[Zn(NH_3)_6]^{+2}$
(3) $[Ni(CO)_4]$ (4) $[Ni(NH_3)_6]^{+2}$
CCO0044

45. What are the geometric shape and the oxidation number of the copper atom, respectively, for the complex ion, $[Cu(NH_3)_4(OH_2)_2]^{2+}$?
(1) Tetrahedral; + 2 (2) Square planar; –2
(3) Linear; + 3 (4) Octahedral; + 2
CCO0045

46. For $[FeF_6]^{3-}$ and $[Fe(CN)_6]^{3-}$ magnetic moment of the fluoride complex is expected to be :-
(1) The same as the magnetic moment of the cyanide complex
(2) Larger than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
(3) Smaller than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
(4) Larger than the magnetic moment of the cyanide complex because there are fewer unpaired electrons in the fluoride complex
CCO0046

47. Which of the following contains one unpaired electron in the 4p orbitals :-
(1) $[Cu(NH_3)_2]^+$ (2) $[Cu(NH_3)_4]^{2+}$
(3) $[Cu(CN)_4]^{3-}$ (4) $[Ni(CN)_4]^{2-}$
CCO0047

48. How many unpaired electrons are present in the Brown Ring complex $[Fe(H_2O)_5(NO)]SO_4$
(1) 4 (2) 3 (3) 0 (4) 5
CCO0048

49. Which of the following statements about $[Cr(CO)_6]$ is correct ?
(1) It is paramagnetic and inner orbital complex
(2) It is diamagnetic and inner orbital complex
(3) It is diamagnetic and outer orbital complex
(4) It is paramagnetic and outer orbital complex
CCO0049

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50. Which is true for complex $[Ni(en)_2]^{2+}$?
(1) paramagnetism, dsp^2, square planar, C.N. of Ni =2
(2) diamagnetism, dsp^2, square planar, C.N. of Ni = 4
(3) diamagnetism, sp^3, tetrahedral, C.N. of Ni = 4
(4) paramagnetism, sp^3, tetrahedral, C.N. of Ni = 4

CCO050

51. Arrange the following in order of decreasing number of unpaired electrons :

I : $[Fe(H_2O)_6]^{2+}$ II : $[Fe(CN)_6]^{3-}$
III : $[Fe(CN)_6]^{4-}$ IV : $[Fe(H_2O)_6]^{3+}$
(1) IV, I, II, III (2) I, II, III, IV
(3) III, II, I, IV (4) II, III, I, IV

CCO051

52. Spin magnetic moment of $[MnBr_4]^{-2}$ is 5.9 BM its geometry would be :-
(1) Square planar (2) Tetrahedral
(3) Square pyramidal (4) Octahedral

CCO052

53. The hybridisation in $[Co(C_2O_4)_3]^{-3}$
(1) sp^3d^2 (2) sp^3d^3
(3) dsp^3 (4) d^2sp^3

CCO053

54. Which complex is diamagnetic ?
(1) $[MnO_4]^-$ (2) $[Co(NH_3)_6]^{+2}$
(3) $[Fe(CN)_6]^{-3}$ (4) $[Cr(H_2O)_6]^{+3}$

CCO054

CFT, COLOUR OF THE COMPLEX AND STABILITY

55. Which one of these ions absorbs energy from visible spectrum :-
(1) $[Cu(NH_3)_4]^+$
(2) $[Cu(NH_3)_4]^{+2}$
(3) $[Zn(H_2O)_6]^{+2}$
(4) $[CdCl_4]^{-2}$

CCO055

56. In the complex $[Ni(H_2O)_2(NH_3)_4]^{+2}$ the magnetic moment (μ) of Ni is :-
(1) Zero (2) 2.83 BM
(3) 1.73 BM (4) 3.87 BM

CCO056

57. Which of the following system has maximum number of unpaired electrons :-
(1) d^5 (Octahedral, low spin)
(2) d^8 (Tetrahedral)
(3) d^6 (Octahedral, low spin)
(4) d^3 (Octahedral)

CCO057

58. In an octahedral crystal field, the t_{2g} orbitals are
(1) Raised in energy by 0.4 Δ_0
(2) Lowered in energy by 0.4 Δ_0
(3) Raised in energy by 0.6 Δ_0
(4) Lowered in energy by 0.6 Δ_0

CCO058

59. If $\Delta_0 < P$, the correct electronic configuration for d^4 system will be :-
(1) $t_{2g}^4\ e_g^0$ (2) $t_{2g}^3\ e_g^1$
(3) $t_{2g}^0\ e_g^4$ (4) $t_{2g}^2\ e_g^2$

CCO059

60. Match List-I (Complex ions) with List-II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :-

List-I (Complex ions)	**List-II (Number of Unpaired Electrons)**
A. $[CrF_6]^{4-}$	i. One
B. $[MnF_6]^{4-}$	ii. Two
C. $[Cr(CN)_6]^{4-}$	iii. Three
D. $[Mn(CN)_6]^{4-}$	iv. Four
	v. Five

Code :	**A**	**B**	**C**	**D**
(1)	iv	i	ii	v
(2)	ii	v	iii	i
(3)	iv	v	ii	i
(4)	ii	i	iii	v

CCO060

61. Consider the following complex formation reactions and comment on their formation constant value

(i) $Fe^{2+}(aq) + 6NH_3 \overset{K_f}{\rightleftharpoons} \left[Fe(NH_3)_6\right]^{2+}$

(ii) $Fe^{2+}(aq) + 3en \overset{K_f'}{\rightleftharpoons} \left[Fe(en)_3\right]^{2+}$

(1) $K_f > K'_f$
(2) $K_f < K_f'$
(3) $K_f = K_f'$
(4) Can not be compared

CCO061

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62. Select most stable complex :-

(1) $[Co(H_2O)_6]^{3+}$ (2) $[Co(NH_3)_2(en)_2]^{+3}$

(3) $[Co(en)_3]^{+3}$ (4) $[Co(NH_3)_4(en)]^{+3}$

CCO062

63. Red color of gem stone ruby is due to impurities of :-

(1) Al_2O_3 (2) Cr^{+3}

(3) Al^{+3} ions (4) All of these

CCO063

ISOMERISM

64. A square planar complex is cis platin $[Pt(NH_3)_2Cl_2]$ shows :

(1) Geometrical isomerism

(2) Optical isomerism

(3) Linkage isomerism

(4) Ionisation isomerism

CCO064

65. The two compounds pentaamminesulphato cobalt (III) bromide and pentaamminebromo cobalt (III) sulphate represent :

(1) Linkage isomerism

(2) Ionisation isomerism

(3) Co-ordination isomerism

(4) Hydrate isomerism

CCO065

66. Which of the following complex can not exhibit geometrical isomerism :-

(1) $[Pt(NH_3)_2Cl\ NO_2]$

(2) $[Pt(gly)_2]$

(3) $[Cu(en)_2]^{+2}$

(4) $[Pt(H_2O)(NH_3)BrCl]$

CCO066

67. Which one of the following compounds will exhibit linkage isomerism :-

(1) $[Pt\ (NH_3)_2\ Cl_2]$

(2) $[Co\ (NH_3)_3\ NO_2]Cl_2$

(3) $[Co\ (NH_3)_4\ Cl_2]Cl$

(4) $[Co\ (en)_2Cl_2]Cl$

CCO067

68. The compound $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ represent

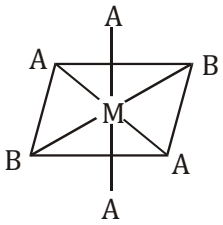
(1) Linkage isomerism

(2) Hydrate isomerism

(3) Ligand isomerism

(4) Ionisation isomerism

CCO068

69. Compound - 

can be marked as -

(1) Cis form (2) Leavo form

(3) Dextro form (4) Trans form

CCO069

70. A Planar Complex [Mabcd] gives :-

(1) Two Optical isomer

(2) Two geometrical isomer

(3) Three optical isomer

(4) Three geometrical isomers

CCO070

71. No. of geometrical isomers expected for octahedral complex [Mabcdef] is :-

(1) Zero (2) 30 (3) 15 (4) 9

CCO071

72. Which of the following complex produce ppt with $AgNO_3$ and exist in two geometrical isomeric form

(1) $PtCl_2.4NH_3$ (2) $PtCl_2.3NH_3$

(3) $PtCl_4.4NH_3$ (4) $PtCl_4.2NH_3$

CCO072

73. Cis$[Pt(en)_2Br_2]Cl_2.2H_2O$ and trans$[Pt(en)_2Br_2]Cl_2.2H_2O$ are :-

(1) Ionisation isomer

(2) Geometrical isomer

(3) Hydrate isomer

(4) Optical isomers

CCO073

74. Which of the following octahedral complex have maximum stereoisomers :-

(1) $[Fe(CN)_6]^{-4}$ (2) $[Co(en)(CN)_4]^{-1}$

(3) $[Co(en)_2(Cl)_2]^{+}$ (4) $[Co(en)_3]^{+3}$

CCO074

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75. Which of the following complex does not show geometrical isomerism ?
(1) $[Co(NH_3)_4Cl_2]^+$
(2) $[Co(NH_3)_3(NO_2)_3]$
(3) $[Cr(en)_3]^{3+}$
(4) $[Pt(gly)_2]$

CCO075

76. Which of the following complex compounds does not exhibits geometrical isomerism ?
(1) $[PtCl_2(NH_3)_2]$
(2) $[PdCl_2BrI]$
(3) $[Pt(NH_3)(py)(Cl)(Br)]$
(4) $[Pt(NH_3)_3(Br)]$

CCO076

77. The pair of molecules that exhibit geometrical isomerism are
(1) $[Cr(NH_3)_4Cl_2]^+$ and $[Co(NH_3)_3Cl_3]$
(2) $[Cr(NH_3)_5Cl]^{2+}$ and $[Co(NH_3)_5Cl]^{2+}$
(3) $[Cr(NH_3)_6]^{3+}$ and $[Co(NH_3)_6]^{3+}$
(4) All of the above show geometric isomerism

CCO077

78. Complex $[Co(NH_3)_3(NO_2)_3]$ show which isomerism:-
(1) Ionisation and linkage
(2) Linkage and geometrical
(3) Geometrical and ligand
(4) Coordination and linkage

CCO078

79. Which of the following represents incorrect match?
(1) $[Co(NH_3)_5(NO_2)](NO_3)_2$: Linkage isomerism
(2) $[Pt(NH_3)(H_2O)Cl_2]$: Geometrical isomerism
(3) $K[Cr(H_2O)_2(C_2O_4)_2]$: Geometrical and optical
(4) $[Cr(H_2O)_5Cl]Cl_2H_2O$: Hydrate and linkage isomerism

CCO079

80. $CrCl_3.6H_2O$ show 3 different coloured compounds due to :-
(1) Ionisation isomerism
(2) Coordination isomerism
(3) Optical isomerism
(4) Hydrate isomerism

CCO080

ORGANOMETALLIC COMPOUNDS

81. Which of the following is π –acid ligand
(1) NH_3 (2) CO
(3) glycinato (4) ethylene diamine

CCO081

82. Which of the following statement is/are wrong:-
(a) Al_4C_3 is an organometallic compound
(b) Metal carbonyls are organometallic compounds
(c) TEL is π bonded organometallic compound
(d) Frankland reagent is σ - bonded organometallic compound
The answer is:-
(1) c & d (2) a & c (3) a & b (4) b & d

CCO082

83. Compounds which contain direct metal carbon bonds are called :
(1) Organic compound
(2) Complex compound
(3) Metal carbides
(4) OMC compounds.

CCO083

84. Which one of the following is a π bonded OMC?
(1) Wilkinson's catalyst
(2) Tetraethyl lead
(3) Zeise's salt
(4) Grignard's reagent

CCO084

85. Which of the following statement is true
(1) $FeCO_3$ and Fe_3C are organometallic compounds.
(2) In Ferrocene ligand is cyclopentadienyl.
(3) $Pb(C_2H_5)_4$ is π-bonded OMC
(4) In Zeise's salt central metal is sp^3 hybridised.

CCO085

86. Which of the following is not an organo metallic compound :-
(1) $(C_2H_5)_2Zn$ (2) $CH_3B(OCH_3)_2$
(3) $B(OCH_3)_3$ (4) $Ni(CO)_4$

CCO086

87. Number of metal -metal bonds in $[Co_2(CO)_8]$ will be :-
(1) 0 (2) 3 (3) 2 (4) 1

CCO087

APPLICATION OF COORDINATION CHEMISTRY

88. In a ferric salt on adding $K_4[Fe(CN)_6]$ a prussian blue is obtained which is :-
(1) $K_3[Fe(CN)_6]$ (2) $Fe_3[Fe(CN)_6]_4$
(3) $FeSO_4[Fe(CN)_6]$ (4) $Fe_4[Fe(CN)_6]_3$

CCO088

89. Hypo is used in photography because it is :-
(1) A reducing agent
(2) An oxidising agent
(3) A Complexing agent
(4) Photo sensitive Compound

CCO089

90. The solubility of AgBr in hypo solution is due to the formation of :-
(1) Ag_2SO_3 (2) $Ag_2S_2O_3$
(3) $[Ag(S_2O_3)]^-$ (4) $[Ag(S_2O_3)_2]^{3-}$

CCO090

91. Which of the following is related to Nessler's reagent?
(1) $PtCl_4 + KCl \rightarrow$ (2) $AgCl + NH_3 \rightarrow$
(3) $AgBr + Na_2S_2O_3 \rightarrow$ (4) $HgI_2 + KI \rightarrow$

CCO091

92. $K_4[Fe(CN)_6]$ reacts with $FeCl_3$ to form :-
(1) $K_3[Fe(CN)_6]$ (2) $K_4[Fe(CN)_3Cl_3]$
(3) $K_3[Fe(CN)_5Cl]$ (4) $KFe[Fe(CN)_6]$

CCO092

93. A reagent used for identifying nickel ion is :-
(1) Potassium ferrocyanide
(2) Phenolphthalein
(3) Dimethyl glyoxime
(4) EDTA

CCO093

94. Which one of the following statement is false for nickel-dimethylglyoximate complex ?
(1) The stability of complex is only due to the presence of intra-molecular hydrogen bonding
(2) The complex is stable, only because dimethyl glyoxime ligand is a stronger ligand
(3) The complex is stable as it has five membered chelate rings as well as intra molecular hydrogen bonding
(4) Both (1) and (2)

CCO094

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**4**	**4**	**2**	**4**	**3**	**4**	**1**	**2**	**2**	**2**	**4**	**2**	**4**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**3**	**2**	**1**	**1**	**1**	**1**	**3**	**2**	**2**	**2**	**3**	**4**	**2**	**2**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**3**	**1**	**3**	**4**	**2**	**1**	**1**	**2**	**1**	**1**	**2**	**3**	**3**	**1**	**4**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**2**	**2**	**2**	**2**	**2**	**1**	**2**	**4**	**1**	**2**	**2**	**4**	**2**	**2**	**3**
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	**2**	**3**	**2**	**1**	**2**	**3**	**2**	**2**	**4**	**4**	**3**	**3**	**2**	**3**	**3**
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	**4**	**1**	**2**	**4**	**4**	**2**	**2**	**4**	**3**	**2**	**3**	**4**	**4**	**3**	**4**
Question	91	92	93	94											
Answer	**4**	**4**	**3**	**4**											

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which of the following complexes is used to be as an anticancer agent?
(1) mer-$[Co(NH_3)_3Cl_3]$ (2) cis-$[PtCl_2(NH_3)_2]$
(3) cis-$K_2[PtCl_2Br_2]$ (4) Na_2CoCl_4
CCO117

AIPMT 2015

2. Cobalt (III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test of chloride ions with silver nitrate at 25°C?
(1) $CoCl_3 \cdot 4NH_3$ (2) $CoCl_3 \cdot 5NH_3$
(3) $CoCl_3 \cdot 6NH_3$ (4) $CoCl_3 \cdot 3NH_3$
CCO118

3. Which of these statements about $[Co(CN)_6]^{3-}$ is true:-
(1) $[Co(CN)_6]^{3-}$ has four unpaired electrons and will be in a low-spin configuration.
(2) $[Co(CN)_6]^{3-}$ has four unpaired electrons and will be in a high spin configuration.
(3) $[Co(CN)_6]^{3-}$ has no unpaired electrons and will be in a high-spin configuration.
(4) $[Co(CN)_6]^{3-}$ has no unpaired electrons and will be in a low-spin configuration.
CCO119

Re-AIPMT 2015

4. The name of complex ion, $[Fe(CN)_6]^{3-}$ is :-
(1) Tricyanoferrate (III) ion
(2) Hexacyanidoferrate (III) ion
(3) Hexacyanoiron (III) ion
(4) Hexacyanitoferrate (III) ion
CCO120

5. The hybridization involved in complex $[Ni(CN)_4]^{2-}$ is (At.No. Ni = 28)
(1) d^2sp^2 (2) d^2sp^3 (3) dsp^2 (4) sp^3
CCO121

6. The sum of coordination number and oxidation number of the metal M in the complex $[M(en)_2(C_2O_4)]Cl$ (where en is ethylenediamine) is:-
(1) 7 (2) 8 (3) 9 (4) 6
CCO122

7. Number of possible isomers for the complex $[Co(en)_2Cl_2]$ Cl will be : (en = ethylenediamine)
(1) 3 (2) 4 (3) 2 (4) 1
CCO123

NEET-I 2016

8. Which of the following has longest C–O bond length? (Free C–O bond length in CO is 1.128Å).
(1) $Ni(CO)_4$ (2) $[Co(CO)_4]^-$
(3) $[Fe(CO)_4]^{2-}$ (4) $[Mn(CO)_6]^+$
CCO124

NEET(UG) 2017

9. An example of a sigma bonded organometallic compound is :
(1) Grignard's reagent (2) Ferrocene
(3) Cobaltocene (4) Ruthenocene
CCO125

10. Pick out the correct statement with respect to $[Mn(CN)_6]^{3-}$:-
(1) It is sp^3d^2 hybridised and tetrahedral
(2) It is d^2sp^3 hybridised and octahedral
(3) It is dsp^2 hybridised and square planar
(4) It is sp^3d^2 hybridised and octahedral
CCO126

11. Correct increasing order for the wavelengths of absorption in the visible region the complexes of Co^{3+} is :-
(1) $[Co(H_2O)_6]^{3+}$, $[Co(en)_3]^{3+}$, $[Co(NH_3)_6]^{3+}$
(2) $[Co(H_2O)_6]^{3+}$, $[Co(NH_3)_6]^{3+}$, $[Co(en)_3]^{3+}$
(3) $[Co(NH_3)_6]^{3+}$, $[Co(en)_3]^{3+}$, $[Co(H_2O)_6]^{3+}$
(4) $[Co(en)_3]^{3+}$, $[Co(NH_3)_6]^{3+}$, $[Co(H_2O)_6]^{3+}$
CCO127

NEET(UG) 2018

12. The type of isomerism shown by the complex $[CoCl_2(en)_2]$ is
(1) Geometrical isomerism
(2) Coordination isomerism
(3) Ionization isomerism
(4) Linkage isomerism
CCO128

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13. The geometry and magnetic behaviour of the complex $[Ni(CO)_4]$ are
(1) Square planar geometry and diamagnetic
(2) Tetrahedral geometry and diamagnetic
(3) Square planar geometry and paramagnetic
(4) Tetrahedral geometry and paramagnetic

CCO129

14. Iron carbonyl, $Fe(CO)_5$ is
(1) Tetranuclear (2) Mononuclear
(3) Trinuclear (4) Dinuclear

CCO130

NEET(UG) 2019

15. What is the **correct** electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory ?
(1) $t_{2g}^4e_g^2$ (2) $t_{2g}^6e_g^0$
(3) $e^3t_2^3$ (4) $e^4t_2^2$

CCO131

NEET(UG) 2019 (ODISHA)

16. The Crystal Field Stabilisation Energy (CFSE) for $[CoCl_6]^{4-}$ is 18000 cm^{-1}. The CFSE for $[CoCl_4]^{2-}$ will be-
(1) 6000 cm^{-1} (2) 16000 cm^{-1}
(3) 18000 cm^{-1} (4) 8000 cm^{-1}

CCO132

NEET(UG) 2020

17. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds ?
(1) $CN^- < C_2O_4^{2-} < SCN^- < F^-$
(2) $SCN^- < F^- < C_2O_4^{2-} < CN^-$
(3) $SCN^- < F^- < CN^- < C_2O_4^{2-}$
(4) $F^- < SCN^- < C_2O_4^{2-} < CN^-$

CCO133

18. Urea reacts with water to form A which will decompose to form B. B when passed through Cu^{2+} (aq), deep blue colour solution C is formed. What is the formula of C from the following ?
(1) $CuCO_3 \cdot Cu(OH)_2$ (2) $CuSO_4$
(3) $[Cu(NH_3)_4]^{2+}$ (4) $Cu(OH)_2$

CCO134

NEET(UG) 2021

19. Ethylene diaminetetraacetate (EDTA) ion is:
(1) Hexadentate ligand with four "O" and two "N" donor atoms
(2) Unidentate ligand
(3) Bidentate ligand with two "N" donor atoms
(4) Tridentate ligand with three "N" donor atoms

CCO135

20. Match List-I with List-II

List-I		List-II	
(a)	$[Fe(CN)_6]^{3-}$	(i)	5.92 BM
(b)	$[Fe(H_2O)_6]^{3+}$	(ii)	0 BM
(c)	$[Fe(CN)_6]^{4-}$	(iii)	4.90 BM
(d)	$[Fe(H_2O)_6]^{2+}$	(iv)	1.73 BM

Choose the **correct** answer from the options given below
(1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
(3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
(4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

CCO136

NEET(UG) 2022

21. The IUPAC name of the complex - $[Ag(H_2O)_2][Ag(CN)_2]$ is:
(1) diaquasilver(II) dicyanidoargentate(II)
(2) dicyanidosilver(I) diaquaargentate(I)
(3) diaquasilver(I) dicyanidoargentate(I)
(4) dicyanidosilver(II) diaquaargentate(II)

CCO167

NEET(UG) 2023

22. Homoleptic complex from the following complexes is :
(1) Diamminechloridonitrito-N-platinum (II)
(2) Pentaamminecarbonatocobalt (III) chloride
(3) Triamminetriaquachromium (III) chloride
(4) Potassium trioxalatoaluminate (III)

CCO168

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23. Which complex compound is most stable?

(1) $\left[Co(NH_3)_3(NO_3)_3\right]$

(2) $[CoCl_2(en)_2]NO_3$

(3) $\left[Co(NH_3)_6\right]_2(SO_4)_3$

(4) $\left[Co(NH_3)_4(H_2O)Br\right](NO_3)_2$

CCO169

24. Which of the following forms a set of complex and a double salt, respectively ?

(1) $CuSO_4.5H_2O$ and $CuCl_2,4NH_3$

(2) $PtCl_2.2NH_3$ and $PtCl_4.2HCl$

(3) $K_2PtCl_2.2NH_3$ and $KAl(SO_4)_2.12H_2O$

(4) $NiCl_2.6H_2O$ and $NiCl_2(H_2O)_4$

CCO170

25. Type of isomerism exhibited by compounds $[Cr(H_2O)_6]Cl_3$, $[Cr(H_2O)_5Cl]Cl_2.H_2O$, $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ and the value of coordination number (CN) of central metal ion in all these compounds, respectively is :

(1) Geometrical isomerism, CN = 2

(2) Optical isomerism, CN = 4

(3) Ionisation isomerism, CN = 4

(4) Solvate isomerism, CN = 6

CCO171

NEET(UG) 2024

26. Given below are two statements :

Statement-I : Both $[Co(NH_3)_6]^{+3}$ and $[CoF_6]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.

Statement-II : $[Co(NH_3)_6]^{3+}$ is diamagnetic whereas $[CoF_6]^{3-}$ is paramagnetic.

In the light of the above statements, choose the *correct* answer from the options given below :

(1) Both statement-I and Statement-II are true

(2) Both Statement-I and Statement-II are false

(3) Statement-I is true but Statement-II is false

(4) Statement-I is false but Statement-II is true

CCO172

27. Match List I with List II.

	List-I (Complex)		**List-II (Type of isomerism)**
A.	$\left[Co(NH_3)_5(NO_2)\right]Cl_2$	I.	Solvate isomerism
B.	$\left[Co(NH_3)_5(SO_4)\right]Br$	II.	Linkage isomerism
C.	$\left[Co(NH_3)_6\right]\left[Cr(CN)_6\right]$	III.	Ionization isomerism
D.	$\left[Co(H_2O)_6\right]Cl_3$	IV.	Coordination isomerism

Choose the correct answer from the options given below :

(1) A-II, B-III, C-IV, D-I

(2) A-I, B-III, C-IV, D-II

(3) A-I, B-IV, C-III, D-II

(4) A-II, B-IV, C-III, D-I

CCO173

28. Given below are two statements :

Statement-I : $\left[Co(NH_3)_6\right]^{3+}$ is a homoleptic complex whereas $\left[Co(NH_3)_4Cl_2\right]^{+}$ is a heteroleptic complex.

Statement-II : Complex $\left[Co(NH_3)_6\right]^{3+}$ has only one kind of ligands but $\left[Co(NH_3)_4Cl_2\right]^{+}$ has more than one kind of ligands.

In the light of the above statements, choose the *correct* answer from the options given below.

(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

(3) Statement-I is true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

CCO174

NEET(UG) 2024 (Re-examination)

29. Ethylene diaminetetraacetate ion is a/an :

(1) hexadentate ligand

(2) ambidentate ligand

(3) monodentate ligand

(4) bidentate ligand

CCO175

30. Which of the following is **not** an ambidentate ligand?

(1) $C_2O_4^{2-}$ (2) SCN^- (3) NO_2^- (4) CN^-

CCO176

31. $[Mn_2(CO)_{10}]$ and $[Co_2(CO)_8]$ structures have :

A. Metal - Metal linkage

B. Terminal CO groups

C. Bridging CO groups

D. Metal in zero oxidation state

Choose the **correct** answer from the options given below :

(1) Only A, B, C (2) Only B, C, D

(3) Only A, C, D (4) Only A, B, D

CCO177

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	4	2	3	3	1	3	1	2	4	1	2	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	3	1	4	3	4	2	3	4	1	1	1	1	1
Question	31														
Answer	4														

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EXERCISE – III (Analytical Questions) Master Your Understanding

1. Given below are two statements :

Statement-I : Complex $[Fe(H_2O)_5(NO)]SO_4$ is paramagnetic.

Statement-II : The Fe in Complex $[Fe(H_2O)_5(NO)]SO_4$ has three unpaired electrons.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO137

2. Given below are two statements :

Statement-I : All the geometrical formers of the complex $[M(NH_3)_4Cl_2]$ are optically inactive.

Statement-II : Both geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ does not posses plane of symmetry.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO138

3. Given below are two statements :

Statement-I : $[Cr(H_2O)_6]Cl_2$ and $[Fe(H_2O)_6]Cl_2$ are oxidising in nature.

Statement-II : $[Fe(CN)_6]^{4-}$ ion is an outer orbital complex.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO139

4. Given below are two statements :

Statement-I : Cis-platin is used in the treatment of Cancer.

Statement-II : EDTA is used for treatment of lead poisoning.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO140

5. Given below are two statements :

Statement-I : The metal to ligand bonding creates a synergic effect which strength the bond between CO and the metal.

Statement-II : The complex with d^0 and d^{10} configuration show d–d transition.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO141

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6. Given below are two statements : (According to Werner)

Statement-I : The primary valency are normally ionisable and are satisfied by negative ions.

Statement-II : The secondary valency are non-ionisable are satisfied by neutral molecules or negative ions.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO142

7. Given below are two statements :

Statement-I : Valence bond theory explain the colour exhibited by co-ordination compounds.

Statement-II : CFT considers the metal-ligand bond to be ionic from electrostatic interactions between the metal ion and the ligand.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO143

8. Given below are two statements :

Statement-I : d^4 to d^7 co-ordination entities are less stable for strong field as compared to weak field cases.

Statement-II : Ruby is aluminium oxide (Al_2O_3) containing about 0.5 – 1% Cr^{+2} ions (d^4).

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO144

9. Given below are two statements :

Statement-I : Decacarbonyl diamangnese (0) is made up of two square pyramidal $[Mn(CO)_5]$ units joined by a Mn–Mn bond.

Statement-II : Zero oxidation state of central metal cannot be stabilised by CO groups.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO145

10. Given below are two statements :

Statement-I : Excess of copper and iron from plant/animal systems are removed by the chelating ligands D-penicillamine and deferrioxime B via the formation of Co-ordination compounds.

Statement-II : Crystal field splitting energy in tetrahedral complex is less as compared to the octahedral complexes.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CCO146

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11. Given below are two statements :
Assertion (A) : EDTA ligand forms complex with metal ion of 3d series in the ratio of 1 : 1
Reason (R) : EDTA ligand has four –COOH group's
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is not correct.
(4) (A) is not correct but (R) is correct.

CCO147

12. **Assertion (A) :** The correct order of the wavelength of absorption in the visible vision is $[Ni(NO_2)_6]^{4-} < [Ni(NH_3)_6]^{+2} < [Ni(H_2O)_6]^{+2}$
Reason (R) : The stability of different complexes depends on the strength of the ligand field of the various ligand
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is not correct.
(4) (A) is not correct but (R) is correct.

CCO148

13. **Assertion (A) :** $KMnO_4$ & $K_2Cr_2O_7$ both are coloured compound.
Reason (R) : The colour of the compound is only due to d–d transition.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is not correct.
(4) (A) is not correct but (R) is correct.

CCO149

14. **Assertion (A) :** Cu^{+2} ion never form complex with CN^- & I^- ions.
Reason (R) : Cu^{+2} is a stronger oxidising while CN^- & I^- both are stronger reducing agent.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is not correct.
(4) (A) is not correct but (R) is correct.

CCO150

15. **Assertion (A) :** The colour of $[Co(NH_3)_5NO_2]^{+2}$ and $[Co(NH_3)_5ONO]^{2+}$ is different.
Reason (R) : Both are linkage isomers.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is not correct.
(4) (A) is not correct but (R) is correct.

CCO151

16. Match the column :

Column-I [Complex Ion]		**Column-II [Hybridisation & no. of unpaired electron]**	
(1)	$[Co(H_2O)_6]^{+3}$	(a)	dsp^2, 0
(2)	$[Fe(CN)_6]^{3-}$	(b)	dsp^2, 1
(3)	$[Ni(CN)_4]^{2-}$	(c)	d^2sp^3, 0
(4)	$[Co(CN)_4]^{2-}$	(d)	d^2sp^3, 1

(1) 1 - c, 2 - a, 3 - d, 4 - b
(2) 1 - c, 2 - d, 3 - b, 4 - a
(3) 1 - c, 2 - d, 3 - a, 4 - b
(4) 1 - a, 2 - c, 3 - d, 4 - b

CCO152

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17. Match the column :

Column-I [Complex Ion]		Column-II [No. of geometrical Isomer]	
(1)	$[Co(NH_3)_6]^{+3}$	(a)	5
(2)	$[Co(NH_3)_2Cl_2Br_2]^-$	(b)	0
(3)	$[Pt(NH_3)_3Cl_3]^+$	(c)	15
(4)	$[Pt(NH_3)(H_2O)(Py)ClBr\ I]^+$	(d)	2

(1) 1 - b, 2 - d, 3 - a, 4 - c
(2) 1 - b, 2 - a, 3 - d, 4 - c
(3) 1 - a, 2 - b, 3 - d, 4 - c
(4) 1 - c, 2 - a, 3 - d, 4 - b

CCO153

18. Match the column :

Column-I (Complexion)		Column-II (Type of complex with magnetic moment)	
(1)	$[Co(ox)_3]^{3-}$	(a)	Inner orbital, complex, $\mu = \sqrt{3}$
(2)	$[Fe(EDTA)]^-$	(b)	Outer orbital, complex, $\mu = \sqrt{24}$
(3)	$[MnF_6]^{3-}$	(c)	Outer orbital, complex, $\mu = \sqrt{8}$
(4)	$[Ni(H_2O)_6]^{+2}$	(d)	Inner orbital, complex, $\mu = 0$

(1) 1 - d, 2 - a, 3 - b, 4 - c
(2) 1 - c, 2 - a, 3 - b, 4 - d
(3) 1 - d, 2 - b, 3 - a, 4 - c
(4) 1 - a, 2 - d, 3 - b, 4 – c

CCO154

19. Match the column :

Column-I [Configuration in octahedral complex]		Column-II [Magnetic moment]	
(1)	d^4 [Low spin]	(a)	$\mu = \sqrt{3}$ BM
(2)	d^4 [High spin]	(b)	$\mu = \sqrt{35}$ BM
(3)	d^5 [Low spin]	(c)	$\mu = \sqrt{8}$ BM
(4)	d^5 [High spin]	(d)	$\mu = \sqrt{24}$ BM

(1) 1 - b, 2 - d, 3 - a, 4 - c
(2) 1 - a, 2 - d, 3 - d, 4 - b
(3) 1 - d, 2 - c, 3 - a, 4 - b
(4) 1 - c, 2 - d, 3 - a, 4 - b

CCO155

20. Match the column :

Column-I [Complex Ion]		Column-II [Total no. of chelate ring]	
(1)	$[Fe(EDTA)]^-$	(a)	2
(2)	$[Ni(dmg)_2]$	(b)	3
(3)	$[Fe(en)_3]^{+3}$	(c)	4
(4)	$[Co(en)_2Cl_2]^+$	(d)	5

(1) 1 - a, 2 - c, 3 - b, 4 - d
(2) 1 - d, 2 - c, 3 - b, 4 - a
(3) 1 - b, 2 - c, 3 - d, 4 - a
(4) 1 - c, 2 - d, 3 - b, 4 - a

CCO156

21. Match the column :

Column-I [Complex Ion]		Column-II [Value of CFSE]	
(1)	$[Cr(H_2O)_6]^{+2}$	(a)	Zero
(2)	$[Cr(NH_3)_6]^{+2}$	(b)	$-0.4\ \Delta_0$
(3)	$[Fe(H_2O)_6]^{+3}$	(c)	$-0.6\ \Delta_0$
(4)	$[CoF_6]^{3-}$	(d)	$-1.6\ \Delta_0$

(1) 1 - a, 2 - c, 3 - b, 4 - d
(2) 1 - b, 2 - d, 3 - c, 4 - a
(3) 1 - c, 2 - d, 3 - a, 4 - b
(4) 1 - d, 2 - a, 3 - c, 4 – b

CCO157

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22. Match the column :

Column-I (Complex compound)		Column-II (Element)	
(1)	Chlorophyll	(a)	Rhodium
(2)	Blood pigment	(b)	Cobalt
(3)	Vitamin B_{12}	(c)	Magnesium
(4)	Wilkinson catalyst	(d)	Iron

(1) 1 - a, 2 - d, 3 - b, 4 - c

(2) 1 - c, 2 - d, 3 - b, 4 - a

(3) 1 - c, 2 - b, 3 - d, 4 - a

(4) 1 - d, 2 - a, 3 - b, 4 – c

CCO158

23. Which of the following is/are inner orbital complex(es) ?

(A) $[Fe(NH_3)_6]^{+2}$ (B) $[Co(H_2O)_6]^{+3}$

(C) $[Mn(NH_3)_6]^{+2}$ (D) $[Co(C_2O_4)_3]^{-3}$

(1) If (A), (B) and (C) options are correct.

(2) If (A) and (B) both options are correct.

(3) If (B) and (D) both options are correct.

(4) If (A) and (C) both options are correct.

CCO159

24. The compound(s) belonging to the alum family is(are)

(A) $K.Al(SO_4)_2.12H_2O$

(B) $K_2SO_4.Cr_2(SO_4)_3.24H_2O$

(C) $FeSO_4.(NH_4)_2SO_4.6H_2O$

(D) $KCl.MgCl_2.6H_2O$

(1) If (A), (B) and (C) options are correct.

(2) If (A) and (B) both options are correct.

(3) If (B) and (D) both options are correct.

(4) If (A) and (C) both options are correct.

CCO160

25. Which of the following complexes is/are have square planar in shape

(A) $[Cu(H_2O)_4]^{+2}$ (B) $[Ni(CN)_4]^{-2}$

(C) $[Pt(Gly)_2]$ (D) $[Zn(Cl)_4]^{-2}$

(1) If (A), (B) and (C) options are correct.

(2) If (A) and (B) both options are correct.

(3) If (B) and (D) both options are correct.

(4) If (A) and (C) both options are correct.

CCO161

26. Which statement is/are correct :-

(A) $[Ni(CO)_4]$ tetrahedral & diamagnetic

(B) $[Ni(CN)_4]^{-2}$ square planar & diamagnetic

(C) $[Cu(NH_3)_4]^{+2}$ tetrahedral and diamagnetic

(D) $[NiCl_4]^{-2}$ tetrahedral and paramagnetic

(1) If (A), (B) and (C) options are correct.

(2) If (A) and (B) and (D) options are correct.

(3) If (B) and (C) both options are correct.

(4) If (A) and (C) both options are correct.

CCO162

27. Which of the following complexes are paramagnetic in nature ?

(A) $[Ni(CO)_4]$ (B) $[Mn(CN)_6]^{-3}$

(C) $[Co(H_2O)_6]^{+3}$ (D) $[Ni(NH_3)_6]^{+2}$

(1) If (A), (B) and (C) options are correct.

(2) If (A) and (B) both options are correct.

(3) If (B) and (D) both options are correct.

(4) If (A) and (C) both options are correct.

CCO163

28. Correct match is/are -

(A) Nessler's reagent – $K_2[HgI_4] + KOH$

(B) Ferrocene – $[Fe\{\eta^5\text{–}C_5H_5\}_2]$

(C) Chromocene – $[Cr(\eta^6 - C_6H_6)_2]$

(D) Zeigler Natta catalyst – $[(Ph_3P)_3RhCl]$

(1) If (A), (B) and (C) options are correct.

(2) If (A) and (B) both options are correct.

(3) If (B) and (D) both options are correct.

(4) If (A) and (C) both options are correct.

CCO164

29. Which is true for $EDTA^{-4}$ ligand :-
(A) Chelating ligand
(B) Flexidentate ligand
(C) Polydentate ligands
(D) Ambidentate ligands
(1) If (A), (B) and (C) options are correct.
(2) If (A) and (B) both options are correct.
(3) If (B) and (D) both options are correct.
(4) If (A) and (C) both options are correct.
CCO165

30. Which of the following are organometallic compounds
(A) Zeise's salt
(B) Ferrocene
(C) Grignard reagent
(D) Cis-platin
(1) If (A), (B) and (C) options are correct.
(2) If (A) and (B) both options are correct.
(3) If (B) and (D) both options are correct.
(4) If (A) and (C) both options are correct.
CCO166

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EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**2**	**3**	**1**	**2**	**1**	**4**	**3**	**2**	**1**	**3**	**2**	**3**	**1**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**2**	**1**	**4**	**2**	**3**	**2**	**3**	**2**	**1**	**2**	**3**	**1**	**1**	**1**

Important Notes

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Coordination Compounds

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (1)

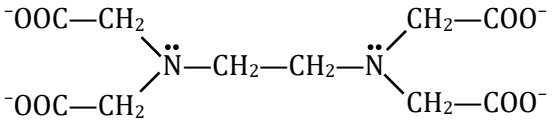
$[Fe(EDTA)]^-$

$x - 4(i) = -$

$x = +4$

Oxidation state of Fe = +3

EDTA is hexadentate ligand hence CN = 6



2. Ans. (4)

$[Pt(en)_2]^{+2}$

$x + 20(0) = +2$

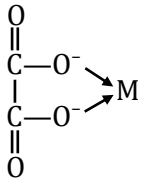
$x = +2$ Oxidation state of Pt = +2

en or Ethylene diamine is a bidentate ligand hence CN = 2 × 2=4

3. Ans. (4)

Bidentate or didentate means, ligands have 2 donor atoms.

$C_2O_4^{-2}$



4. Ans. (2)

$[Pt(C_2H_4)Cl_3]^-$

$x + 0 + 3(-1) = -1$

$x = +2$ Oxidation state of Pt = +2

All ligands are monodentate.

Hence CN = 4

5. Ans. (4)

$[x(SO_4)(NH_3)_5]$

$x + 1(-2) + 5(0) = 0$

$x = +$ Oxidation state of x = +2

SO_4^{-2} is behaving as monodentate ligand

NH_3 is also monodentate.

Hence CN = 6

6. Ans. (3)

Fe shows +1 oxidation state in brown ring complex as NO transfer e^- to convert +1.

$[Fe(H_2O)_5NO]^{+2}$

Fe = +1

7. Ans. (4)

Zeise's salt $K[Pt(C_2H_4)Cl_3]$ has more than one type of ligands, hence it's hetroleptic.

(1) $(C_5H_5)Fe$

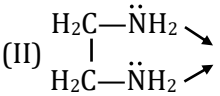
(2) $(C_5H_5)_2Cr$

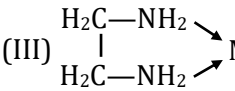
(3) $Fe_4[Fe(CN)_6]_3$

8. Ans. (1)

Statement

(I) $[M(L)_x]^{0/+/-}$ Correct

(II)  two donor atoms Correct

(III)  M Form a ring, Correct

(IV) $[NiCl_2(H_2O)_4]$ $\rightarrow Ni^{+2}$

$[CoCl(NH_3)_5]^{+2}$ $\rightarrow Co^{+3}$

$[Fe(CN)_6]^{-3}$ $\rightarrow Fe^{+3}$

All are e^- deficient and accepting ℓp from ligands. (correct)

9. Ans. (2)

$K[Co(CO_4)]$ CO is a neutral ligand.

$K + [Co(CO)_4]^{-1}$

$x + (4)(0) = -1$

$x = -1$ Oxidation state & Co

10. Ans. (2)

Ligand donate lone pair to CMA in NH_4^+ species does not have lone pair to donate.

11. Ans. (2)

CN is no. of co-ordinate bonds are formed by ligands with CMA. → i.e. secondary valenes.

While oxidation state is the amount of change on CMA:- i.e. primary valency.

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12. Ans. (4)
Chelation means formation of ring, which can be done by bidented or polydented ligands while SCN^- is monodentate.

13. Ans. (2)
$K_3[Fe(CN)_6]$
$K^+[Fe(CN)_6]^{-3}$
$X + 6(-1) = -3$
$x = +3$ Fe(III)
Potassium hexacynate ferrate(III)-IUPAC
Potassium ferri-cyanide → Common name.

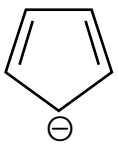
14. Ans. (4)
$[Co(NH_3)_4(H_2O)Br](NO_3)_2$
$[Co(NH_3)_4(H_2O)Br]^{+2}$
$X + 4(0) + 1(0) + 1(-1) = +2$
$x = +3$ Co in +3 oxidation state
NH_3 → ammine
H_2O → aqua
Br^- → Bromido
Hence IUPAC Name –
tetrammineaquabromido cobalt(III) nitrate.

15. Ans. (2)
Trioxalato ferrate (III) – $(C_2O_4^{-2})$
$[Fe(C_2O_4)_3]^{-3}$ Fe = +3

16. Ans. (3)
Chloro-bis(ethylenediamine) nitro cobalt (III) ion
$(Cl^-)(en)_2(NO_2^-)Co^{+3}$
$[CoCl(en)_2NO_2]^+$

17. Ans. (3)
Complex which does not have bi or polydented ligands will not have ring formation or chelate.
Tetraamine dichloro cobalt (III) nitrate (NH_3) Cl both are monodentate ligands.
$[Co(NH_3)_4Cl_2]NO_3$

18. Ans. (2)
$\left[Fe\left(\eta^5 - C_2H_5\right)_2\right]$
$x + 2(-1) = 0$
$x = +2$
$(C_5H_5)^-$
↓
Cyclopentadienyl



hence Bis(Cyclopentadienyl iron (II)

19. Ans. (1)
$[Pt(NH_3)_4Cl_2]$ $[PtCl_4]$
Pt = +4 Pt = +2
CN 6 in +4 oxidation
CN 4 in +2 oxidation
(state)
Tetramminedichloro platinum (IV) tetrachloroplatenate(II)

20. Ans. (1)
$K_2[Cr(CN)_2O_2(NH_3]$
$K^+[Cr(CN)_2O_2(O)_2(NH_3)]^{-2}$
$x + (2)(-1) + (1)(-2) + (2)(-2) + 1(0) = 2$
$x - 2 - 2 - 4 = -2$
$x = +6$
Cr in +6 oxidation state
CN → Cyano
O_2^{-2} → Peroxo
O^{-2} → OXO
NH_3 → Ammine
Hence, Potassium ammine dicyanodioxoperoxochromate(VI)

21. Ans. (1)
$[Co(NH_3)_6]$ $[Cr(CN)_6]$ Co^{+3} CN = 6
Cr^{+3} CN=6
Hexaamminocobalt(III)
hexacyanochromate(III)

22. Ans. (1)
$[Co(NCS)(NH_3)_5]Cl_2 \rightarrow [Co(NCS)(NH_3)_5]^{+2}Cl$
$x + 1(-1) + 5(0) = +2$
$x = +3$ Co in +3 oxidation state
NCS → thicyanato-N
NH_3 → ammine
Pentaammine (thiocyanato-N) cobalt(III) chloride

23. Ans. (3)
Positively changed ligands have suffix as-ium-not-ate.
–ate is used for negatively changed

24. Ans. (2)
$COCl_3.3NH_3$ CO^{+3} CN = 6
Hence $[CO(NH_3)_3Cl_3]$ is a neutral complex hence least ion and least conductivity in aq. Solution.

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25. Ans. (2)

Given EAN = 34 Oxidation state = +2 (X^{+2})

Z = 28

EAN = Z–O.S + 2(CN)

34 = 28–(2) + 2(CN)

CN = 4

26. Ans. (2)

$[Co(en)_2Cl_2]^+$ Z of W = 27

x + (2)(0) + 2(–1) = +1 CN

x – 2 = +1

x = +3

EAN = Z–OS + 2(CN)

= 27–(3) + 2(6)

= 27 – 3 + 12

EAN = 36

27. Ans. (3)

$[Cr(NH_3)_6]Cl_3$ $[Cr(NH_3)_6]^{+3}$ Cl^-

Cr = +3 oxidation state Z of Cr = 24

EAN = Z–O.S. + 2(CN)

EAN = 24–(+3) + 2(6)

EAN = 24–3 + 12

EAN = 33

28. Ans. (4)

25% Note or 1/4th of Cl atoms = 1

$PtCl_4.3NH_3$–O.S. of Pt = +4 CN = 6

Hence $[Pt(NH_3)_3Cl_3]Cl$

$PtCl_4.3NH_3$

29. Ans. (2)

$M(CO)_x$ follows EAN Rule Means it will attain nearest noble gas cont., which is 36e^- CO is a neutral ligand

Z of Ni = 28Fe = 26 Cr = 24

For 36

Ni 36 = 28–0 + 2(CN)

CN = 4

Fe 36=26–0+2 (CN)

CN = 5

Cr 36 = 24–0 + 2(CN)

CN = 6

30. Ans. (2)

$CoCl_3.5NH_3$ Co O.S. = +3 CN = 6

2 mole of AgCl ppt hence 2 Cl atoms needs to be ionised. Hence formula is

$[Co(NH_3)_5Cl]Cl_2$ →

31. Ans. (3)

Secondary valency can be satisfied by neutral cation anion which acts as ligands.

32. Ans. (1)

$CoCl_3.5NH_3$

O.S. of CO = +3 CN = 6

$[Co(NH_3)_5Cl]Cl_2 \equiv [Co(NH_3)_5Cl]^{+2}$ $2Cl^-$

Hence 3 ions.

33. Ans. (3)

In Hexaammine chromium(III) ion $[Cr(NH_3)_6]^{+3}$ Cr^{+3} hence $3d^3$ configuration and will have 3 unpaired e^- so paramagnetic.

34. Ans. (4)

$[Ag(NH_3)_2]^+$

sp hyb. hence linear shape

35. Ans. (2)

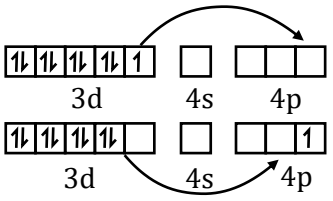
$[FeF_6]^{-3}$ O.S. of Fe = +3 hence $Fe^{+3} \rightarrow [Ar]3d^5$ and F^- is a W.F.L. so CFSE is low, no pairing so

$t_{2g}^3 e_g^2$

hence 5 unpaired e^-

36. Ans. (1)

$[Cu(NH_3)_4]SO_4$ $Cu^{+2} \rightarrow [Ar]3d^9$ & $NH_3 \rightarrow$ SFL



$dsp^2 \rightarrow$ square planar

37. Ans. (1)

Highest paramagnetism = Maximum no. of unpaired e^-

$Fe^{+3} \rightarrow [Ar]3d^5$ (5)

$Fe^{+2} \rightarrow [Ar]3d^6$ (4)

$Cu^{+2} \rightarrow [Ar]3d^9$ (1)

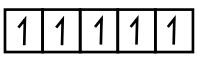
$Zn^{+2} \rightarrow [Ar]3d^{10}$ (0)

Hence $[FeF_6]^{-3}$ has highest paramagnetism

38. Ans. (2)

ML_6^{n+} M^{n+} $d^5 \rightarrow$ WFL meane no. pairing.

Hence 5 unpaired e^-



39. Ans. (1)

$Cu^{+2} \Rightarrow [Ar]3d^9$ 1 unpaired (S.F.L)

$Ni^{+2} \Rightarrow [Ar]3d^8$ 0 unpaired (W.F.L)

$Ti^{+4} \Rightarrow [Ar]$ 0 unpaired

$Co^{+3} \Rightarrow [Ar]3d^6$ 4 unpaired (W.F.L.)

μ = 1.73 BM Means 1 unpaired e^-

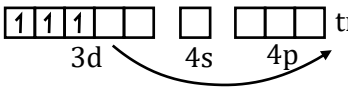
hence $\mu = \sqrt{n(n+2)}$ n=1 μ = 1.73 BM

(1) $[Cu(NH_3)_4]^{+2}$

40. Ans. (1)

$[Cr(NH_3)_6]^{+3} \rightarrow CN = 6$

Cr^{+3} $[Ar]3d^3$



NH_3 d^2sp^3-Octahedral

3 unpaired $e^- \Rightarrow$ Paramagnetic.

41. Ans. (2)

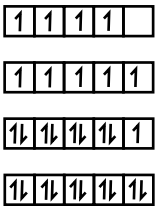
High spin complex = Highest no. of unpaired e^-

$Cr^{+2} = [Ar]3d^4$

$Fe^{+3} = [Ar]3d^5$

$Cu^{+2} = [Ar]3d^9$

$Zn^{+2} = [Ar]3d^{10}$



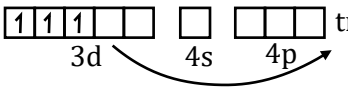
$[Fe(H_2O)_6]^{+3}$ is high spin complex

42. Ans. (3)

Inner orbital complex means inner d orbital is used.

It occurs in presence of SFL or (n-1) as pairing will take place

$[Cr(NH_3)_6]^{+3}$ $Cr^{+3} \rightarrow [Ar]3d^3$



$NH_3 \rightarrow SFL \rightarrow d^2sp^3$

43. Ans. (3)

$K_4[Fe(CN)_6]$

$Fe^{+2} \rightarrow [Ar]3d^6$ $CN^- \rightarrow SFL \rightarrow$ Pairing occurs

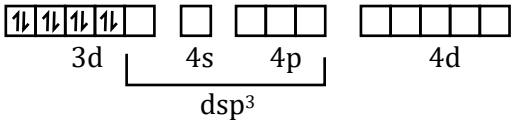
eg 

t_{2g}  -Zero unpaired e^- diamagnetic

3rd option is wrong.

44. Ans. (1)

$[Fe(CO)_5]$ $\underline{Fe}^0$ $[Ar]3d^6$ $CO \rightarrow SFL$



3d 4s 4p are used with different Q.N.

$[Fe(CO)_5]$

45. Ans. (4)

$[Cu(NH_3)_4(OH_2)_2]^{+2}$

$Cu^{+2} \rightarrow [Ar]3d^9$ O.S.= +2

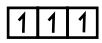
Coordination number = 6

So octahedral.

$x + (4\times0) + (2\times0) = 2$

$x = 2$

46. Ans. (2)

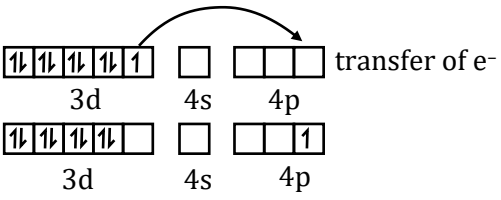
$[FeF_6]^{-3}$	$[Fe(CN)_6]^{-3}$
$Fe^{+3} \rightarrow [Ar]3d^5$	$Fe^{+3} \rightarrow [Ar]3d^5$
W.F.L (F^-)	SFL (CN^-)
No pairing	Pairing
eg 	
t_{2g} 	
5 unpaired	1 unpaired

$[FeF_6]^{-3}$ more magnetic moment than $[Fe(CN)_6]^{-3}$

47. Ans. (2)

$[Cu(NH_3)_4]^{+2}$

$Cu^{+2} \rightarrow [Ar]3d^9$ $NH_3 \rightarrow SFL$



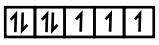
$dsp^2 \rightarrow$ square planar

48. Ans. (2)

$[Fe(H_2O)_5(NO)]SO_4$

$[Fe(H_2O)_5(NO)]^{+2}$ SO_4^{-2}

Fe^+ $[Ar]4s^13d^6 \rightarrow [Ar]4s^03d^7$



3 unpaired e^-

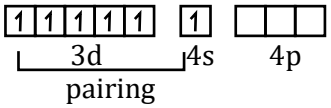
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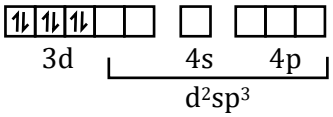
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49. Ans. (2)

$[Cr(CO)_6]$

$Cr \rightarrow [Ar]3d^5 4s^1$ CO → SFL-Pairing occurs





→ Inner orbital complex

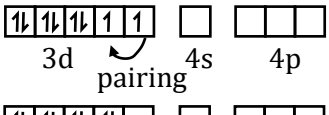
→ Zero unpaired e⁻ SO, diamagnetic.

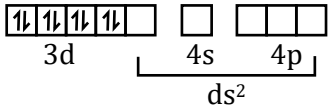
50. Ans. (2)

$[Ni(en)_2]^{+2}$

$Ni^{+2} \rightarrow [Ar]3d^8$ en → SFL → pairing occurs

CN = 4





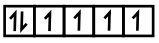
→ zero unpaired → Diamagnetic

→ d^0sp^2 → square planer

→ CN = 4

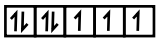
51. Ans. (1)

(I) $[Fe(H_2O)_6]^{+2}$ $Fe^{+2} \rightarrow [Ar]3d^6$



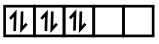
4 unpaired

(II) $[Fe(CN)_6]^{-3}$ $Fe^{+3} \rightarrow [Ar]3d^5$ SFL → Pairing



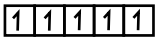
1 unpaired

(III) $[Fe(CN)_6]^{-4}$ $Fe^{+2} \rightarrow [Ar]3d^6$ SFL → Pairing



0 unpaired

(IV) $[Fe(H_2O)_6]^{+3}$ $Fe^{+3} \rightarrow [Ar]3d^5$



5 unpaired

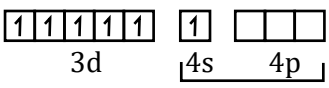
(IV) > (I) > (II) > (III)

52. Ans. (2)

$[MnBr_4]^{-2}$

$Mn^{+2} \rightarrow [Ar]3d^5$ WFL → No pairing

CN = 4



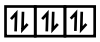
sp^3 → Tetrahedral

53. Ans. (4)

$[Co(C_2O_4)_3]^{-3}$ CN = 6

$Co^{+3} \rightarrow [Ar]3d^6$ Ox-WFL → but for Co^{+3} it act as SFL so pairing will be these

eg 

t_{2g} 

→ Inner d will be used so inner orbital complex

→ d^2sp^3

54. Ans. (1)

$[MnO_4]^-$ ion Mn^{+7}

$Mn = [Ar]3d^5 4s^2$

Mn^{+7} = [Ar] Zero unpaired e^- so diamagnetic

55. Ans. (2)

If unpaired e⁻ are present than they will absorb the energy.

(1) $Cu^+ \rightarrow [Ar]3d^{10}$ Paired

(2) $Cu^{+2} \rightarrow [Ar]3d^9$ Unpaired

(3) $Zn^{+2} \rightarrow [Ar]3d^{10}$ Paired

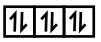
(4) $Cd^{+2} \rightarrow [Ar]4d^{10}$ Paired

(2) $[Cu(NH_3)_4]^{+2}$

56. Ans. (2)

$[Ni(H_2O)_2(NH_3)_4]^{+2}$

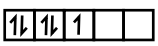
$Ni^{+2} \rightarrow [Ar]3d^8$ eg 

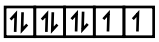
t_{2g} 

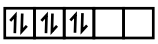
hence 2 unpaired e⁻ n = 2

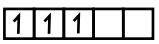
$\mu = \sqrt{n(n+2)}$ = 2.83 BM

57. Ans. (4)

(I) d^5  low spin 1 unpaired

(II) d^8  2 unpaired

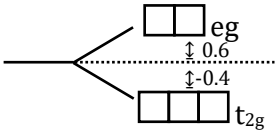
(III) d^6  low spin 0 unpaired

(IV) d^3  3 unpaired

58. Ans. (2)

$$CFSE = -\frac{2}{5}t_{2g} + \frac{3}{5}e_g$$

$$= -0.4t_{2g} + 0.6e_g$$

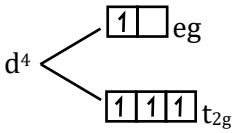


(2) lowered by 0.4 Δ_0

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59. Ans. (2)

Δ_0 < P than pairing will not happen as energy gap is less.



(2) $t_{2g}^3 eg^1$

60. Ans. (3)

(A) $[CrF_6]^{-4} \Rightarrow Cr^{+2}$ = [Ar]$3d^4$ (WFL)
= 4 unpaired e^- (iv)

(B) $[MnF_6]^{-4} \Rightarrow Mn^{2+}$ = [Ar]$3d^5$ (WFL)
= 5 unpaired e^- (v)

(C) $[Cr(CN)_6]^{-4} \Rightarrow Cr^{2+}$ = [Ar]$3d^4$ SFL
= 2 unpaired e^- (ii)

(D) $[Mn(CN)_6]^{-4} \Rightarrow Mn^{+2}$ = [Ar]$3d^5$ (SFL)
= 1unpaired e^- (i)

61. Ans. (2)

(i) $Fe^{+2}_{(aq)} + 6NH_3 \rightleftharpoons[K_f] [Fe(NH_3)_6]^{+2}$

(ii) $Fe^{+2}_{(aq)} + 3en \rightleftharpoons[K_f^1] [Fe(en)_3]^{+2}$

Higher the stability of complex; higher will be the formation constant value.

Due to chelation $[Fe(en)_3]^{+2}$ more stable the $[Fe(NH_3)_6]^{+2}$

(2) $K_f < K_f^1$

62. Ans. (3)

Stability of complex order

Synergic bonding > chelation > SFL/Z_{eff} of CMA $[Co(en)_3]^{+3}$ is the most stable complex as maximum no. of chelate ring (3)

63. Ans. (2)

Al^{+3} ion is replaced by Cr^{+3} ion impurity which causes the red colour of Ruby.

(3) Cr^{+3}

64. Ans. (1)

$[Pt(NH_3)_2Cl_2]$ cis platin square planer complex only shows geometrical isomerism.

65. Ans. (2)

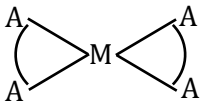
$[Co(NH_3)_5SO_4]Br$

$[Co(NH_3)_5Br]\ SO_4$

→ are ionisation isomers

66. Ans. (3)

$[Cu(en)_2]^{+2}$ can not show geometrical isomers (symmetric bidentate ligand)



67. Ans. (2)

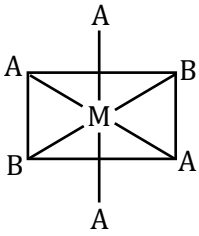
$[Co(NH_3)_3NO_2]Cl_2$ exhibit linkage isomerism.

As NO_2^- can acts as ONO^- also different doner atom.

68. Ans. (2)

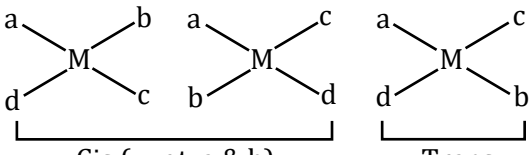
$[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ are hydrate isomers due to water molecules in complex.

69. Ans. (4)



Same functional group are at 180°, have it is trans form.

70. Ans. (4)



[Mabcd] gives 3 geometrical isomerism 2 Cis 1 Trans

71. Ans. (3)

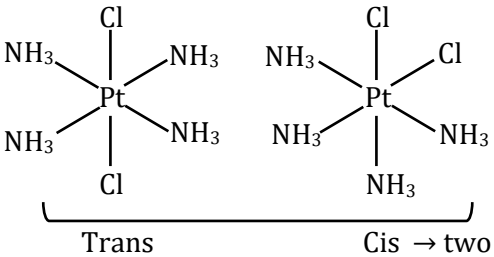
[M abcdef] shows 15 geometrical isomers.

72. Ans. (3)

Two Geometrical isomerism = Cis and trans

$PtCl_4.4NH_3 \Rightarrow [Pt(NH_3)_4Cl_2]Cl_2 \rightarrow$

$2Cl^-$ ions produces 2 moles of AgCl.



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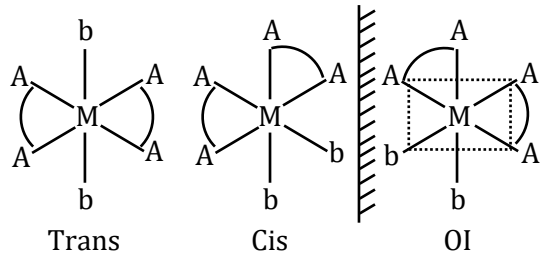
73. Ans. (2)

The given compound is $[M(AA)_2b_2]$ type and cis and trans, geometrical isomers.

74. Ans. (3)

$[Co(en)_2Cl_2]^+$ is $[M(AA)_2b_2]$

Type of complex :



Total stereoisomer = 3

75. Ans. (3)

$[Cr(en)_3]^{+3}$

is $[M(AA)_3]$ type of complex, and have same type of doner atom, so will not show geometrical isomerism.

76. Ans. (4)

$[Pt(NH_3)_3Br]^+$ will have dsp^2 Hybridisation and square planer geometry.

$[Ma_3b]$ will not show geometrical isomerism.

77. Ans. (1)

$[Cr(NH_3)_4Cl_2]^+$ is $[M(A)_4B_2]$ (Cis & trans)

$[Co(NH_3)_3Cl_3]$ is $[M\ A_3B_3]$ (Fec & Mer)

Type of complex both will show geometrical.

78. Ans. (2)

$[Co(NH_3)_3(NO_2)_3]$ show both linkage and geometrical isomer as NO_2 is an ambidente linkage and $[Ma_3b_3]$ type of complex.

79. Ans. (4)

$[Cr(H_2O)_5Cl]Cl_2.H_2O$ can not show linkage isomerism as no ambidente ligand is present. It can only show hydrate isomerism.

80. Ans. (4)

$CrCl_3.6H_2O \Rightarrow$ $[Cr(H_2O)_6]Cl_3$

$[Cr(H_2O)_5Cl]Cl_2.H_2O$

$[Cr(H_2O)_4Cl_2]Cl.2H_2O$

Three hydrate isomers.

81. Ans. (2)

CO is non classical or π-acid ligand as it accepts ℓp from CMA in its ABMO orbital.

82. Ans. (2)

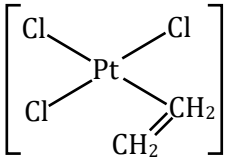
Al_4C_3 is not an OMC it is ionic compound and TEL $Pb(C_2H_5)_4$ is π bonded OMC.

83. Ans. (4)

Organometallic compound contain direct M–C bonds (the carbon must belong to a hydrocarbon group.)

84. Ans. (3)

Zeise's salt is a π-bonded OMC.



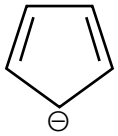
(i) Wilkinson's catalyst $[RhCl(PPH_3)_3]$

(ii) T & L $Pb(C_2H_5)_4$

(iv) R-Mg-X

85. Ans. (2)

$(C_5H_2)_2Fe$ $(C_5H_5)^-$



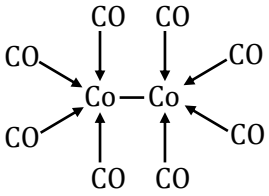
86. Ans. (3)

$B(OCH_3)_3 = CH_3–O–B(–O–CH_3)$ (O–CH_3)

It has no direct M-C bond. So not can OMC

87. Ans. (4)

$[CO_2(CO)_8]$



1-Metal-Metal bond

88. Ans. (4)

$Fe^{+3} + K_4[Fe(CN)_6] \longrightarrow Fe_4[Fe(CN)_6]_3$

salt 1,2-drops Prussian blue

89. Ans. (3)

Undecomposed AgBr (at the time of film development) is removed from photographic film by Hypo solution it forms soluble complex.

$AgBr + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2]^- + NaBr$

Soluble complex

90. Ans. (4)

$AgBr + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2] + NaBr$

$[Ag(S_2O_3)_2]^{-3}$

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91. Ans. (4)

Nessler's Reagent is $K_2[HgI_4]$

$HgI_2 + 2KI \longrightarrow K_2[HgI_4]$

92. Ans. (4)

$FeCl_3 + K_4[Fe(CN)_6] \longrightarrow 4KFe[Fe(CN)_6]$

test of Fe+3 ion Colloidal solution

93. Ans. (3)

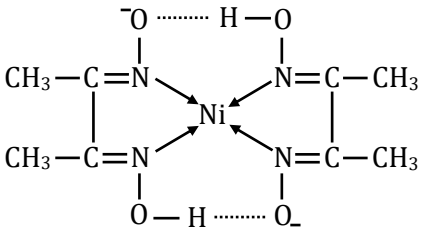
Test of Ni^{+2} ion

$Ni^{+2} + 2dmg \longrightarrow [Ni(dmg)_2]$

Dimethyl glyoxime Rosy Red ppt----

94. Ans. (4)

$[Ni(dmg)_2]$



It is stable due to both H-bonding as well dmg chelation. More number of rings are formed which weakes it is more stable.

Exercise-II (Previous Year Questions)

1. Ans. (2)

Cis-Platin is used as anticancer agent

Cis $[PtCl_2(NH_3)_2]$

2. Ans. (4)

The complex does not have Cl^- as counter ions will not give test of Cl^- ion

(I) $CoCl_3.4NH_3 \rightarrow Co^{+3}$ CN = 6 $[Co(NH_3)_4Cl_2]Cl$

(II) $CoCl_3.5NH_3 \rightarrow [Co(NH_3)_5]Cl_2$

(III) $CoCl_3.6NH_3 \rightarrow [Co(NH_3)_6]Cl_3$

(IV) $CoCl_3.3NH_3 \rightarrow [Co(NH_3)_3Cl_3]$ No Cl^- ion

3. Ans. (4)

$[Co(CN)_6]^{-3} \Rightarrow Co^{+3} \rightarrow [Ar]3d^6$

SFL so pairing occurs

eg ☐☐ ← 0 unpaired electron

t_{2g} ↿⇂ ↿⇂ ↿⇂ ← hence low spin complex

4. Ans. (2)

$[Fe(CN)_6]^{-3}$ Fe^{+3} $CN^- \rightarrow$ cyanido

$x + 6(-1) = -3$ Anionic part hence ferrate

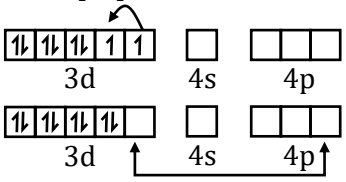
$x = +3$

→ Hexacyanido ferrate (III) ion

5. Ans. (3)

$[Ni(CN)_4]^{-2}$ CN = 4

Ni^{+2} – $[Ar]3d^8$ $CN^- \rightarrow$ SFL pairing occurs



$dsp^2 \rightarrow$ square plane

6. Ans. (3)

$[M(en)_2(C_2O_4)]Cl$

$[M(en)_2(C_2O_4)]^+ Cl^-$

en & C_2O_4 both are bidentate ligand hence

CN = 6

$x + 2(0) + 1(-2) = +1$

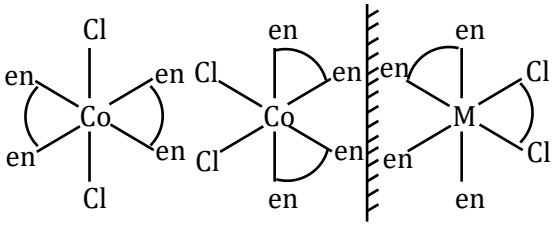
$x = +3$ O.S. of M = +3

total = 6 + 3

= 9

7. Ans. (1)

$[Co(en)_2Cl_2]Cl$ is $[M(AA)_2b_2]$ type complex



Trans

2-GI

2-OI

Total 3 → SI

8. Ans. (3)

For C-O bond

If synergic bond ↑es, Bond order ↓, Bond length ↑es and synergic bonding order is

$M^{-n} > M > M^{+n}$

Hence in

$[Fe(CO)_4]^{-2} \rightarrow Fe^{-2}$ hence will have extra electron to donate CO and synergic bond increases So C-O bond order ↓es and bond length ↑es

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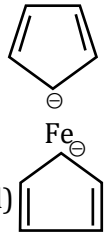
9. Ans. (1)

(I) R–Mg–X — Carbon and Metal are bonded by σ-bond.

(II) $Fe(C_5H_5)_2$

(Iii) $Co(C_5H_5)_2$ — All are π - doner

(IV) $Ru(C_5H_5)_2$

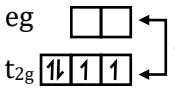


10. Ans. (2)

$[Mn(CN)_6]^{-3}$

$Mn^{+3} \rightarrow [Ar]3d^4$

SFL → Pairing occurs



d^2sp^3-octahedral

11. Ans. (4)

$\Delta = E_{absorb} = \frac{hc}{\lambda_{absorb}}$

For SFL $\Delta\uparrow$ $\lambda\downarrow$es

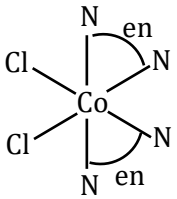
For WFL $\Delta\downarrow$ λ es

So, Order of $\lambda_{absorbed}$

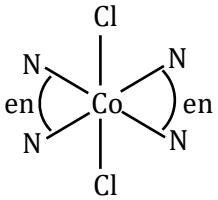
$[Co(en)_3]^{+3} < [Co(NH_3)_6]^{+3} < [Co(H_2O)_6]^{+3}$

12. Ans. (1)

$[CoCl_2(en)_2]$



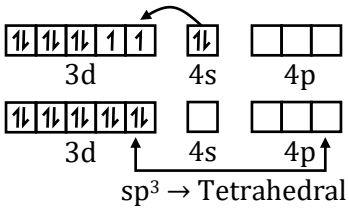
Cis



Trans (GI)

13. Ans. (2)

$[Ni(CO)_4]$ $Ni \rightarrow [Ar]3d^84s^2$ SFL pairing



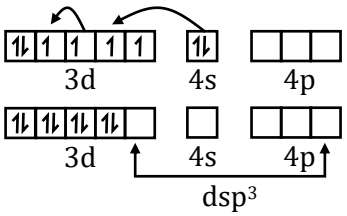
0 → unpaired so diamagnetic

sp^3 → Tetrahedral

14. Ans. (2)

$Fe(CO)_5$ → No. of metal atom present is one is mononuclear compound with dsp^3 hybridisaton.

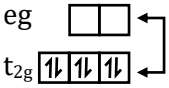
$Fe(CO)_5$ Fe^0 $[Ar]3d^64s^2$



15. Ans. (2)

$K_4[Fe(CN)_6]$

$Fe^{+2} \rightarrow [Ar]3d^6$ CN^- SFL so pairing occurs



t_{2g}^6 eg^0

16. Ans. (4)

Δ_0 of $[CoCl_6]^{-4} = 18000\ cm^{-1}$

$$\Delta_t = \frac{4}{9}\Delta_0$$

$$\Delta_t \text{of } [CoCl_4]^{-2} = \frac{4}{9} \times 18000$$

$= 8000\ cm^{-1}$

17. Ans. (2)

Synergic ligand > Chelating ligand

in SCN^- → S doner atom < F^-

$SCN^- < F^- < C_2O_4^{-2} < CN^-$

18. Ans. (3)

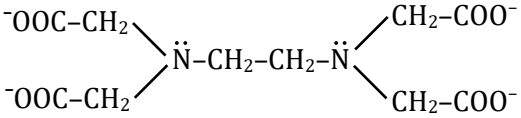
$$\underset{\text{Urea}}{NH_2CONH_2} + H_2O \longrightarrow \underset{(A)}{NH_4OH} + CO_2$$

$$\underset{(A)}{NH_4OH} \longrightarrow \underset{(B)}{NH_3} + H_2O$$

$$Cu^{+2}(aq) + 4NH_3 \longrightarrow \underset{(C)}{[Cu(NH_3)_4]^{+2}} \text{ (deep blue)}$$

19. Ans. (1)

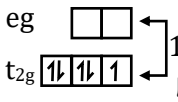
EDTA (ion)



20. Ans. (4)

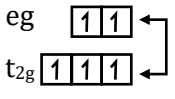
(a) $[Fe(CN)_6]^{-3} \rightarrow Fe^{+3} \rightarrow [Ar]3d^5\ CN^- \rightarrow$ SFL

pairing occurs



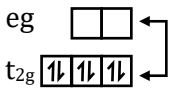
(b) $[Fe(H_2O)_6]^{+3} \rightarrow Fe^{+3} \rightarrow [Ar]3d^5\ H_2O \rightarrow$ WFL

no pairing



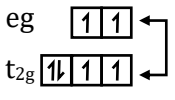
(c) $[Fe(CN)_6]^{-4} \rightarrow Fe^{+2} \rightarrow [Ar]3d^6\ CN^- \rightarrow$ SFL

pairing occurs



(d) $[Fe(H_2O)_6]^{+2} \rightarrow Fe^{+2} \rightarrow [Ar]3d^6 \rightarrow H_2O$ WFL

no pairing



$\boxed{\mu = \sqrt{n(n+2)}}$

21. (3)

IUPAC

$[Ag(H_2O)_2]\ [Ag(CN)_2]$

Coordination number = 2,

Oxidation state = Ag^{+1}

Diaquasilver(I) dicyanidoargentate(I)

22. (4)

$K_3[Al(C_2O_4)_3]$ contain all ligands are of same type i.e. why complex will be homoleptic.

23. (2)

Due to Chelation effect of (en).

24. (3)

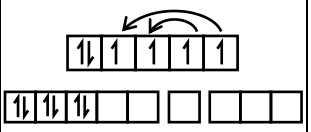
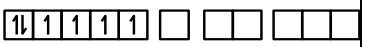
Complex salt is $K_2[Pt(NH_3)_2Cl_2]$

Double salt is $KAl(SO_4)_2.12H_2O$ (potash alum)

25. (4)

Given complex compounds exhibit solvate isomerism having co-ordination number = 6.

26. Ans. (1)

$[Co(NH_3)_6]^{+3}$	$[CoF_6]^{-3}$
$_{27}Co \rightarrow 3d^7\ 4s^2$ $Co^{+3} \rightarrow 3d^6$ $(NH_3 \rightarrow$ SFL) so pairing of electron  d^2sp^3 dia. $\boxed{\mu = 0}$	$_{27}Co \rightarrow 3d^7 4s^2$ $Co^{+3} \rightarrow 3d^6$ $(F^- \rightarrow$ WFL) no pairing of electron  sp^3d^2 Para $\boxed{\mu = 4.9}$

27. Ans. (1)

(A) $[Co(NH_3)_5(NO_2)]Cl_2$ Linkage

NO_2^- can be converted into ONO^-

(B) $[Co(NH_3)_5SO_4]Br$ / $[Co(NH_3)_5Br]\ SO_4$ Ionisation

(C) $[Co(NH_3)_6]\ [Cr(CN)_6]$ Coordination

(D) $[Co(H_2O)_6]Cl_3$ Solvate

28. Ans. (1)

In Homoleptic complex only one kind of ligands are present. While in Hetereoleptic complex more than one kind of ligands are present in coordination sphere.

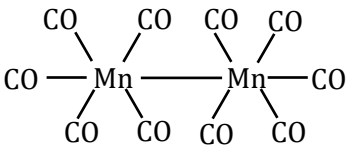
29. Ans. (1)

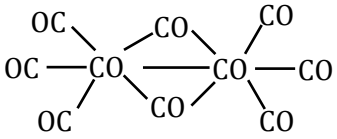
EDTA^{-4} is a hexadentate ligand.

30. Ans. (1)

Except $C_2O_4^{-2}$, all the given, ligands are ambidentate.

31. Ans. (4)





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Exercise-III (Analytical Questions)

1. Ans. (1)

$[Fe(H_2O)_5(NO)]SO_4$

$Fe \rightarrow Fe^{+} \rightarrow [Ar]3d^6 4s^1$

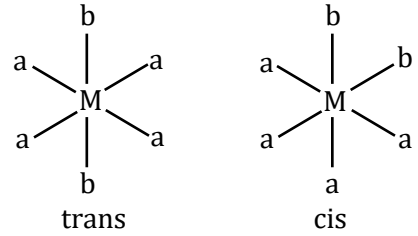
Due to filling of d-orbital, energy gap decreases between 3d & 4s, hence 4s e^- moves to 3d and H_2O & NO both are W.F.L.

d^7 ⇅ ⇅ 1 1 1 3 unpaired e^- and it is paramagnetic.

2. Ans. (2)

$[M(NH_3)_4Cl_2]$ is $[Ma_4b_2]$ type complex

It would have



both will have plane of symmetry, hence only geometrical isomers.

3. Ans. (3)

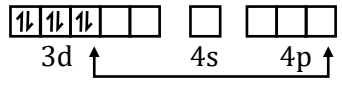
(I) $[Cr(H_2O)_6]Cl_2$ & $[Fe(H_2O)_6]Cl_2$

Cr^{+2} Fe^{+2}

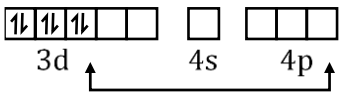
both central metal atom are in their lower oxidation state, so they tend to be good reducing agent, not oxidising agents

(II) $[Fe(CN)_6]^{-4}$

$Fe^{+2} \rightarrow [Ar]3d^6$, $CN^- \rightarrow$ SFL so pairing occurs



d^2sp^3-Inner orbital complex



d^2sp^3-Inner orbital complex

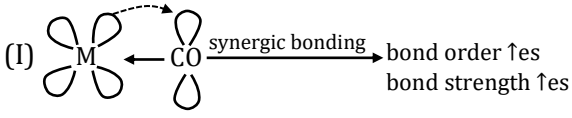
4. Ans. (1)

(I) Cis-Platin is an anticancer drug.

(II) $[Ca(EDTA)]^{+2} + Pb^{+2} \rightarrow [Pb(EDTA)]^{+2} + Ca^{+2}$

Form complex which passes through urine.

5. Ans. (2)

(I) 

(II) d° ☐☐☐☐☐] no. d-d transition possible

d^{10} ⇅ ⇅ ⇅ ⇅ ⇅]

6. Ans. (1)

(I) $CoCl_3.6NH_3 \rightarrow Co^{+3}$ $3Cl^-$

$+3 \rightarrow$ Primary valency

(II) Secondary valency is co-ordination number

For above CN = 6 for Co^{+3}

hence complex $[Co(NH_3)_6]Cl_3$

7. Ans. (4)

(I) VBT does not explain about the colour exhibition it is explained by CFT.

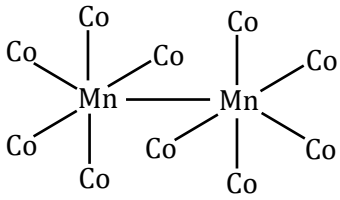
(II) w.r.t. Postulates of CFT, CMA is considerd as point charge and ligand as opposite charge and attraction between them is ionic.

8. Ans. (3)

(I) CFSE explain about the stability of complexes, not depends only for d^4 and d^7 configuration.

(II) Ruby contain Cr^{+3} ion (d^3) not Cr^{+2} ion.

9. Ans. (2)

(I) 

$[Mn_2(CO)_{10}]$

(II) $Fe(CO)_5 \rightarrow Fe^0$

Compound is stable.

10. Ans. (1)

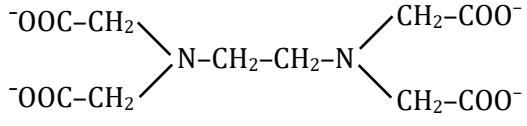
(I) correct statement → biological uses of coordination compound.

(II) $\Delta_t = \frac{4}{9}\Delta_0$

11. Ans. (3)

(A) EDTA is a hexadented ligand hence will form 6 co-ordination bonds with CMA. In general 3d CMA will show 6 CN

So ratio would be 1 : 1

(R) 

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12. Ans. (2)

(A) Strength of the ligand ∝ Energy (CFSE)

$\boxed{E = \frac{hc}{\lambda}}$ strong the ligand; lower the $\lambda_{absorbed}$.

Strength $NO_2^- > NH_3 > H_2O$

Hence λ order would be reverse

$[Ni(NO_2)_6]^{-4} < [Ni(NH_3)_6]^{+2} < [Ni(H_2O)_6]^{+2}$

(R) Strength of ligand ∝ E ∝ Stability.

13. Ans. (3)

(A) $KMnO_4$ $K_2Cr_2O_7$ both are coloured

Purple Orange Red

Mn^{+7} Cr^{+6} both have d^0 configuration so no-d-d transition.

the colour of above compounds is due to change transfer by oxygen to metal.

14. Ans. (1)

Cu^{+2} → It is in it's highest oxidation state, hence it acts as oxidising agent.

CN^- & I^- both are reducing agent. So both will react to redox reaction.

$KCN + CuSO_4 \rightarrow Cu(CN)_2 \rightarrow K_3[Cu(CN)_4]$

unstable Cu in +1 O.S.

$KI + CuSO_4 \rightarrow CuI_2 \rightarrow CuI + I_2 + K_2SO_4$

Unstable

15. Ans. (1)

$[Co(NH_3)_5NO_2]^{+2}$ $[Co(NH_3)_5ONO]^{+2}$

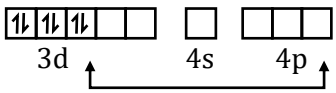
NO_2^- can show ONO^- as ambidentate ligand so above are linkage isomers.

Hence both are different.

16. Ans. (3)

(I) $[Co(H_2O)_6]^{+3}$ Co^{+3} with H_2O acts as SFL hence pairing occurs.

$Co^{+3} \rightarrow [Ar]3d^6$



d^2sp^3– 0 unpaired e^-

(II) $[Fe(CN)_6]^{-3}$

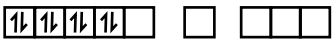
$Fe^{+3} \rightarrow [Ar]3d^5$ $CN^- \rightarrow$ SFL so pairing



d^2sp^3 & 1 unpaired e^-

(III) $[Ni(CN)_4]^{-2}$

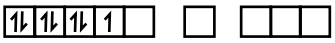
$Ni^{+2} \rightarrow [Ar]3d^8$ $CN^- \rightarrow$ SFL so pairing



dsp^2 & 0 unpaired e^-

(IV) $[Co(CN)_4]^{-2}$

$Ni^{+2} \rightarrow [Ar]3d^7$ $CN^- \rightarrow$ SFL so pairing occurs.



dsp^2 & 1 unpaired e^-



dsp^2 & 1 unpaired e^-

17. Ans. (2)

(i) $[Co(NH_3)_6]^+ \rightarrow [Ma_6] \rightarrow$ No isomers (0)

(ii) $[Co(NH_3)_2Cl_2Br_2]^- \rightarrow [Ma_2b_2c_2]$ type. 4 trans and 1-Cis isomers total-(5)

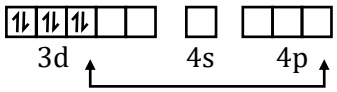
(iii) $[Pt(NH_3)Cl_3]^+ \rightarrow [Ma_3b_3]$ type fac & mer isomers

(iv) $[Pt(NH_3)(H_2O)(Py)ClBrI]^+ \rightarrow [Ma_3b_3]$ type has-(15) GI

18. Ans. (1)

$\mu = \sqrt{n(n+2)}$ n = no. of unpaired e^-

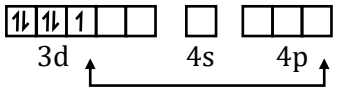
(i) $[Co(O_x)_3]^{-3}$ $Co^{+3} \rightarrow [Ar]3d^6$ O_x is SFL so pairing occurs



d^2sp^3 inner orbital complex

n=0 μ=0

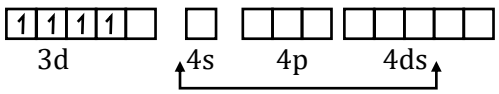
(ii) $[Fe(EDTA)]^-$ $Fe^{+3} \rightarrow [Ar]3d^5$ EDTA SFL so pairing occurs



d^2sp^3-inner orbital complex

n=1, $\mu = \sqrt{3}$

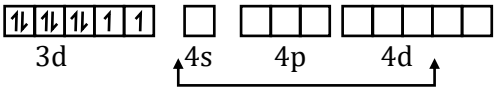
(iii) $[MnF_6]^{-3}$ $Mn^{+3} \rightarrow [Ar]3d^4$ F^- is WFL so no pairing



sp^3d^2-outer orbital

n = 4 $m = \sqrt{24}$

(iv) $[Ni(H_2O)_6]^{+2}$ $Ni^{+2} \rightarrow [Ar]3d^8$ H_2O WFL so no pairing



sp^3d^2 outer orbital

n=2 $m=\sqrt{8}$

19. Ans. (4)

$\mu = \sqrt{n(n+2)}$ BM

(i) d^4 (low spin) n = 2 m = $\sqrt{8}$ BM

(ii) d^4(High spin) n = 4 m = $\sqrt{24}$ BM

(iii) d^5 low spin n = 1 m = $\sqrt{3}$ BM

(iv) d^5 high spin n = 5 m = $\sqrt{35}$ BM

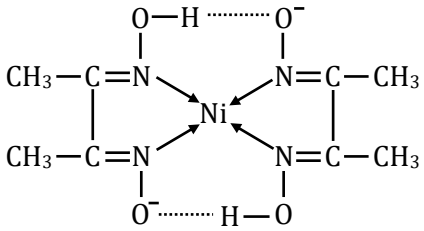
20. Ans. (2)

(i) $[Fe(EDTA)]^-$

no. of chelate ring = denticity–1

EDTA-hexadentate ring = 5

(ii) $[Ni(dmg)_2]$ dmg → bidentate ligand so 2 dmg will give 2 rings but in this compound due to H-bonding 2 more rings are formed so total = 4



(iii) $[Fe(en)_3]^{+3}$ – en → is bidentate ligand

1en = 1 ring

3en = 3 ring

(iv) $[Co(en)_2Cl_2]^+$ –en → is bidentate ligand

1en = 1 ring

2en = 2 rings

21. Ans. (3)

(i) $[Cr(H_2O)_6]^{+2}$ $Cr^{+2} \rightarrow [Ar]3d^4$ WFL so no pairing

$$\Delta = \left(-\frac{2}{5}n_{t_{2g}} + \frac{3}{5}n_{eg}\right)\Delta_0 + x.p$$

$$\Delta_0 = \left(-\frac{6}{5} + \frac{3}{5}\right)\Delta_0 + 0 \quad [x = P.E.]$$

$$\boxed{\Delta_0 = -0.6\Delta_0}$$

(ii) $[Cr(NH_3)_6]^{+2}$ $Cr^{+2} \rightarrow [Ar]3d^4$ SFL so pairing occurs

Conf. → $t_{2g}^4\, eg^0$ $\Delta_0 = -\frac{2}{5}\times 4 + 1P$

$$\boxed{\Delta_0 = -1.6\Delta_0 + P}$$

(iii) $[Fe(H_2O)_6]^{+3}$ $Fe^{+3} \rightarrow [Ar]3d^5$ WFL so no pairing

Conf. → $t_{2g}^3\, eg^2$ $\Delta_0 = -\frac{2}{5}(3) + \frac{3}{5}(2) + 0$

$$\boxed{\Delta_0 = 0}$$

(iv) $[CoF_6]^{-3}$ $Co^{+3} \rightarrow [Ar]3d^6$ WFL so no pairing

Conf. → $t_{2g}^4\, eg^2$ $\Delta_0 = -\frac{2}{5}\times 4 + \frac{3}{5}\times 2 + 0$

$$\boxed{\Delta_0 = -0.4\Delta_0}$$

22. Ans. (2)

(i) Chlorophyll → is compound of Mg

(ii) Blood pigment blood has haemoglobin → which is Fe complex

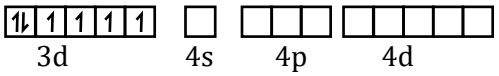
(iii) Vitamin B_{12} → also known as Cobalamin cobalt complex.

(iv) Wilkinson catalyst → $RhCl(PPh_3)_3$

23. Ans. (3)

(A) $[Fe(NH_3)_6]^{+2}$ $Fe^{+2} \rightarrow NH_3$ does not act as SFL so no pairing

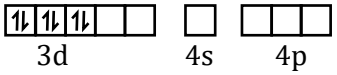
$Fe^{+2} \rightarrow [Ar]3d^6$



sp^3d^2 → outer orbital

(B) $[Co(H_2O)_6]^{+3}$ Co^{+3} H_2O act as SFL so pairing

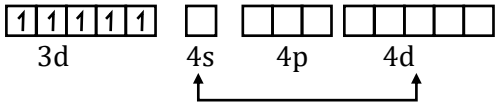
$Co^{+3} \rightarrow [Ar]3d^6$



d^2sp^3 Inner orbital

(C) $[Mn(NH_3)_6]^{+2}$ Mn^{+2} NH_3→WFL-No pairing

$Mn^{+2} \rightarrow [Ar]3d^5$

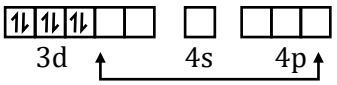


sp^3d^2 outer orbital complex

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(D) $[Co(C_2O_4)_3]^{-3}$ Co^{+3} $C_2O_4^{-2}$→ SFL so pairing occurs

Co^{+3}-$[Ar]3d^6$



d^2sp^3-Inner orbital complex

24. Ans. (2)

Alum are M_2SO_4. $M'_2(SO_4)_3.24H_2O$ or

$MM^1(SO_4)_2.12H_2O$

M is → +1 cation

M' is → +3 cation

(A) $K.Al(SO_4)_2.12H_2O$ $K \rightarrow K^+$, $Al \rightarrow Al^{+3}$

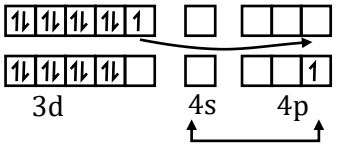
(B) $K_2SO_4.Cr_2(SO_4)_3.24H_2O$ K^+ → cation, Cr^{+3}→ cation

(C) $K_2SO_4.(NH_4)_2SO_4.6H_2O$ Fe^{+2}→cation, NH_4^+→cation

(D) $KCl.MgCl_2.6H_2O$

25. Ans. (1)

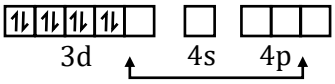
(A) $[Cu(H_2O)_4]^{+2}$ $Cu^{+2} \rightarrow [Ar]3d^9$



dsp^2-square planer

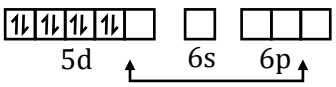
Shifting of e^- does not depend on strength of ligand

(B) $[Ni(CN)_4]^{-2}$ $Ni^{+2} \rightarrow [Ar]3d^8$ CN^- SFL pairing occurs



dsp^2-square planer

(C) $[Pt(Gly)_2] \rightarrow Pt^{+2} \rightarrow [Xe]5d^8$ SFL so pairing Occurs

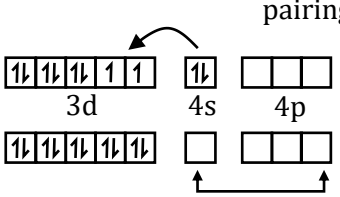


dsp^2-square planer

(D) $[Zn(Cl)_4]^{-2}$ $Zn^{+2} \rightarrow [Ar]3d^{10}$ so sp^3 tetrahedral.

26. Ans. (2)

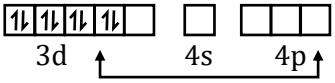
(A) $[Ni(CO)_4]$ $Ni^0 \rightarrow [Ar]3d^84s^2$ CO is SFL so pairing occurs



sp^3-tetrahedral

0-unpaired-Diamagnetic

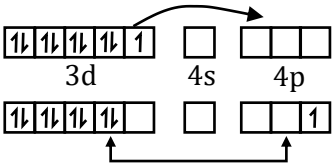
(B) $[Ni(CN)_4]^{-2}$ $Ni^{+2} \rightarrow [Ar]3d^8$ CN^- SFL so pairing occurs



dsp^2→square planer

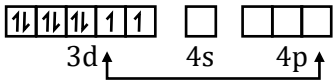
0 unpaired-Diamagnetic

(C) $[Cu(NH_3)_4]^{+2}$ $Cu^{+2} \rightarrow [Ar]3d^9$ NH_3-SFL e^- shifted to 4p



dsp^2-square planer-1unpaired e^- so paramagnetic

(D) $[NiCl_4]^{-2}$ $Ni^{+2} \rightarrow [Ar]3d^8$ Cl^- WFL no pairing



sp^3→tetrahedral & paramagnetic

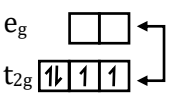
27. Ans. (3)

(A) $[Ni(CO)_4]$ $Ni^0 \rightarrow [Ar]3d^84s^2$ due to SFL CO pairing occurs

Ni → $[Ar]3d^{10}4s^0$

→ 0 unpaired e^- Diamagnetic

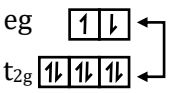
(B) $[Mn(CN)_6]^{-3}$ $Mn^{+3} \rightarrow [Ar]3d^4$ CN^- SFL pairing occurs



2 unpaired so paramagnetic

(C) $[Co(H_2O)_6]^{+3}$ $Co^{+3} \rightarrow [Ar]3d^6$ Co^{+3}, H_2O acts as SFL so pairing occurs

(D) $[Ni(NH_3)_6]^{+2}$ $Ni^{+2} \rightarrow [Ar]3d^8$ NH_3-SFL, pairing occurs



2 unpaired e^- paramagnetic

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28. Ans. (1)

(A) Nesseler's reagent → $K_2[HgI_4]$ + KOH

(B) Ferrocene → $[Fe(\eta^5\text{-}C_5H_5)_2]$

(C) Chromocene → $[Cr(\eta^5\text{-}C_5H_5)_2]$

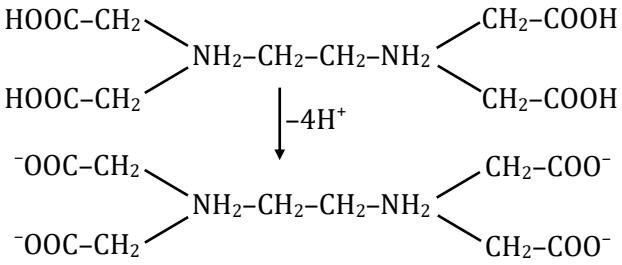
(D) Ziegler Natta catalyst → $(CH_3)_3Al$, $TiCl_4$

$RhCl(PPh_3)_3$ → Wilkinson catalyst.

29. Ans. (1)

$EDTA^{-4}$ is derived from EDTA

EDTA-



30. Ans. (1)

(A) Zeise's salt → $K[PtCl_3(\eta^2\text{–}C_2H_4)]$

π-bonded OMC

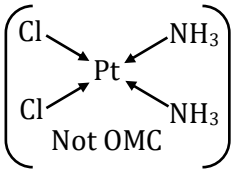
(B) Ferrocence→ $[Fe(\eta^5\text{–}C_5H_5)_2]$

π-bonded OMC

(C) Grignard reagent – R-Mg-X

(σ-bonded OMC)

(D) Cis platin



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